

Initiating Search

October 13, 2025, 7:48 PM

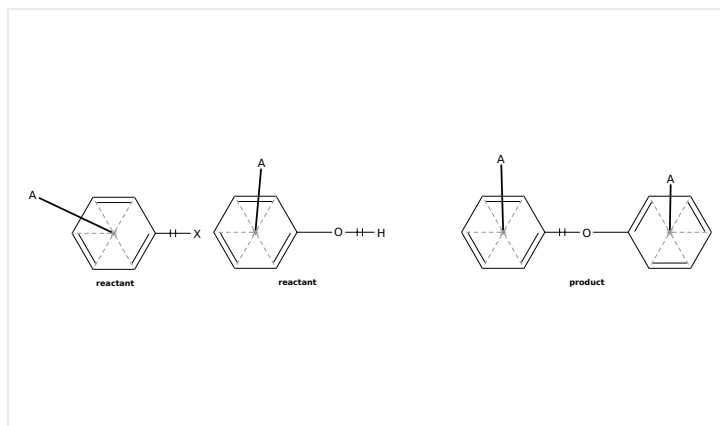
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Yield: 90-100%, 80-89%, 70-79%, <10%

Reaction Mapping: Mapping Data Available

Document Type: Journal



Structure Match: Substructure

Search Tasks

Task	Result Type	View
Exported: Returned Reaction Results + Filters (6,395)	 Reactions	View Results

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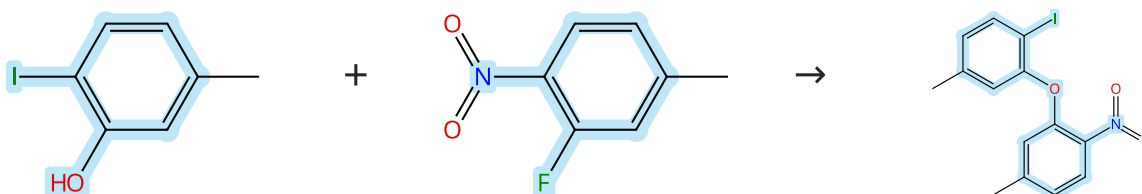
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Reactions (500)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (47)

 Suppliers (81)

31-614-CAS-42623270

Steps: 1 Yield: 100%

Photostability of Phenoxazine Derivatives

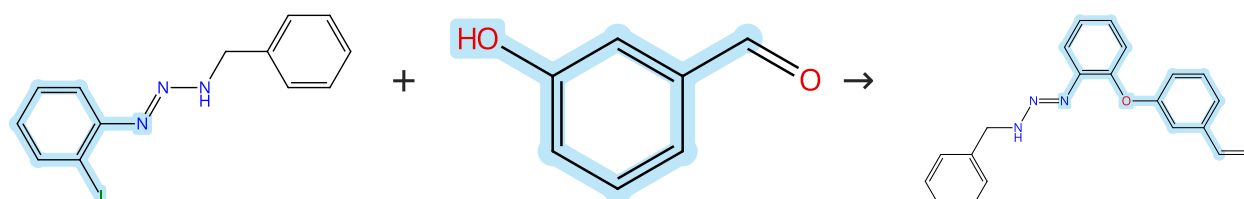
1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; overnight, 100 °C

By: Goll, Felix D.; et al
ChemPhysChem (2024), 25(21), e202400506.

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



resin bound

 Suppliers (97)

resin bound

31-170-CAS-15056603

Steps: 1 Yield: 100%

Solid phase synthesis of selectively deuterated arenes

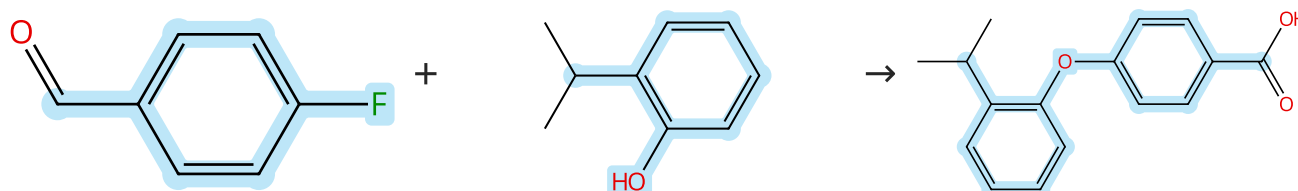
1.1 Reagents: Sodium carbonate, Bromo[1,1'-thiobis[methane]] copper
Solvents: Acetonitrile, Pyridine; 48 h, 75 °C

By: Vanderheiden, Sylvia; et al
Chemical Communications (Cambridge, United Kingdom) (2011), 47(32), 9063-9065.

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (93)

 Suppliers (76)

 Suppliers (21)

31-170-CAS-11010811

Steps: 1 Yield: 100%

Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist

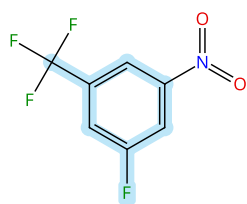
1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 100 °C

1.2 Reagents: 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate
Solvents: *tert*-Butanol, Tetrahydrofuran, Water

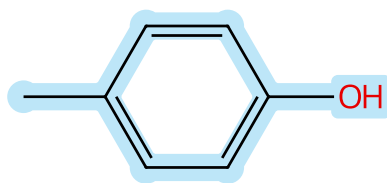
By: Nakamura, Tsuyoshi; et al
Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.

Scheme 4 (1 Reaction)

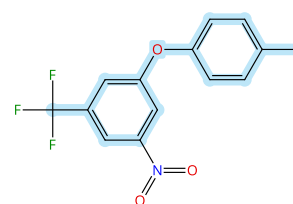
Steps: 1 Yield: 100%



+



→



Suppliers (70)

Suppliers (89)

Supplier (1)

31-170-CAS-14391909

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide

Experimental Protocols

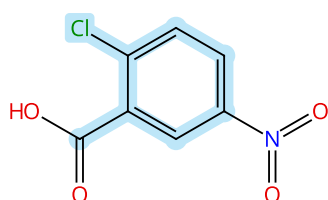
Effect of the nucleophile structure on the relative mobility of the nitro group and fluorine atom in reactions of 3,5-dinitro- and 3-fluoro-5-nitrobenzotrifluorides with phenols in the presence of potassium carbonate

By: Khalfina, I. A.; et al

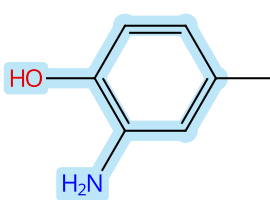
Russian Journal of Organic Chemistry (Translation of Zhurnal Organicheskoi Khimii) (2002), 38(1), 47-58.

Scheme 5 (1 Reaction)

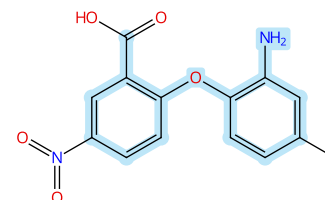
Steps: 1 Yield: 100%



+



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Suppliers (102)

Suppliers (81)

Suppliers (2)

31-170-CAS-6347121

Steps: 1 Yield: 100%

No Data Available

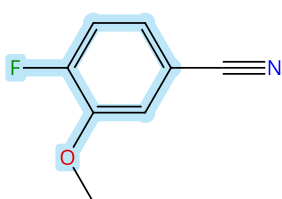
Synthesis of 5H-benzoxazolo[3,2-a]quinazolin-5-ones

By: Kim, Dong Han

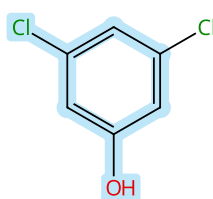
Journal of Heterocyclic Chemistry (1981), 18(2), 287-91.

Scheme 6 (1 Reaction)

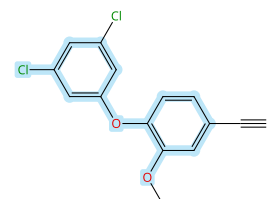
Steps: 1 Yield: 100%



+



→



Suppliers (83)

Suppliers (92)

Suppliers (3)

31-170-CAS-8522631

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; 5 h, 130 °C

Experimental Protocols

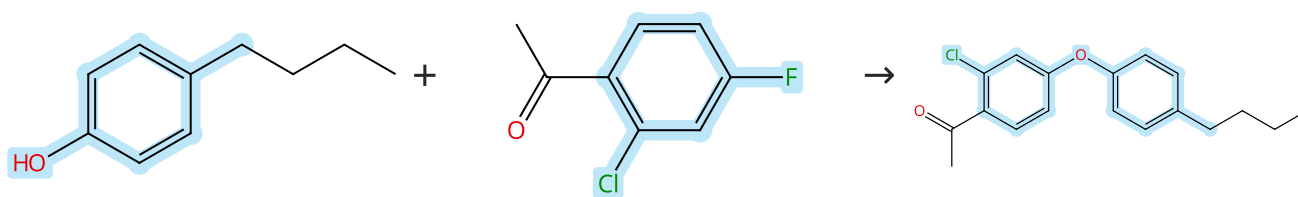
Computationally-Guided Optimization of a Docking Hit to Yield Catechol Diethers as Potent Anti-HIV Agents

By: Bollini, Mariela; et al

Journal of Medicinal Chemistry (2011), 54(24), 8582-8591.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (71)

Suppliers (79)

Supplier (1)

31-170-CAS-19212482

Steps: 1 Yield: 100%

Synthesis of N-Substituted Iminosugar Derivatives and Evaluation of Their Immunosuppressive Activities

By: Lv, Zhuo; et al

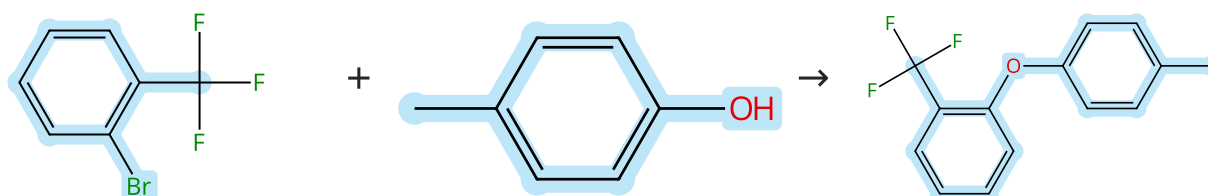
ChemMedChem (2018), 13(4), 338-351.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 30 min, 50 °C
- 1.2 overnight, 115 °C

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (78)

Suppliers (89)

Suppliers (3)

31-614-CAS-36696313

Steps: 1 Yield: 100%

Synthesis of Fluorenones and Xanthenes through Intramolecular C-F Arylation

By: Hamada, Shusuke; et al

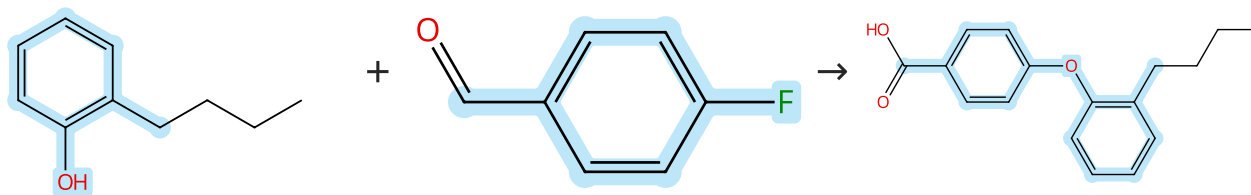
Bulletin of the Chemical Society of Japan (2023), 96(5), 401-405.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Copper, Potassium iodide
Solvents: Pyridine; 24 h, reflux

Experimental Protocols

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (21)

Suppliers (93)

31-170-CAS-8882912

Steps: 1 Yield: 100%

Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist

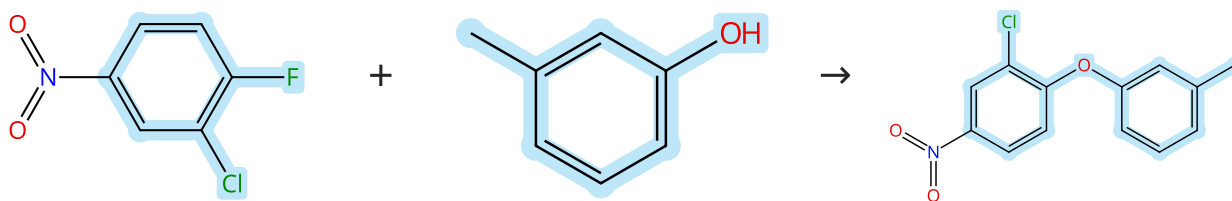
By: Nakamura, Tsuyoshi; et al

Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 100 °C
- 1.2 **Reagents:** 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate
Solvents: *tert*-Butanol, Tetrahydrofuran, Water

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (91)

Suppliers (11)

31-170-CAS-19152390

Steps: 1 Yield: 100%

Surmounting the resistance against EGFR inhibitors through the development of thieno[2,3-d]pyrimidine-based dual EGF R/HER2 inhibitors

By: Milik, Sandra N.; et al

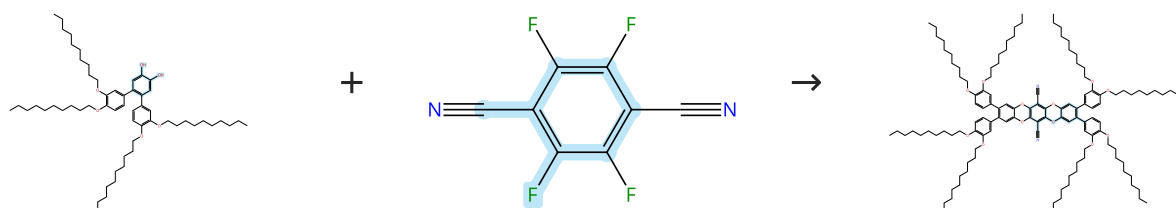
European Journal of Medicinal Chemistry (2018), 155, 316-336.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 50 °C; cooled
- 1.2 **Reagents:** Water; cooled

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

31-170-CAS-19755528

Steps: 1 Yield: 100%

Synthesis and Characterization of Liquid-Crystalline Tetraoxapentacene Derivatives Exhibiting Aggregation-Induced Emission

By: Hiscock, Lana K.; et al

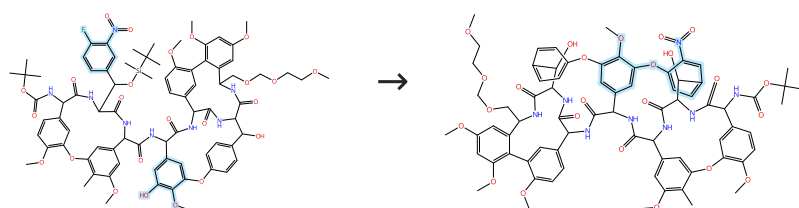
Chemistry - A European Journal (2019), 25(4), 1018-1028.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 3 h, rt → 65 °C

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



31-049-CAS-9562144

Steps: 1 Yield: 100%

Total Synthesis of the Ristocetin Aglycon

By: Crowley, Brendan M.; et al

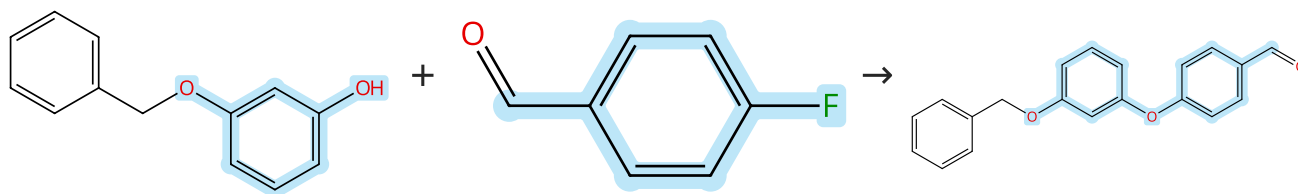
Journal of the American Chemical Society (2004), 126(13), 4310-4317.

- 1.1 **Reagents:** Cesium fluoride
Solvents: Dimethylformamide; 8 h, 25 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water; 25 °C

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

Suppliers (93)

Suppliers (15)

31-614-CAS-34652761

Steps: 1 Yield: 100%

Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

By: Chen, Tao; et al

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

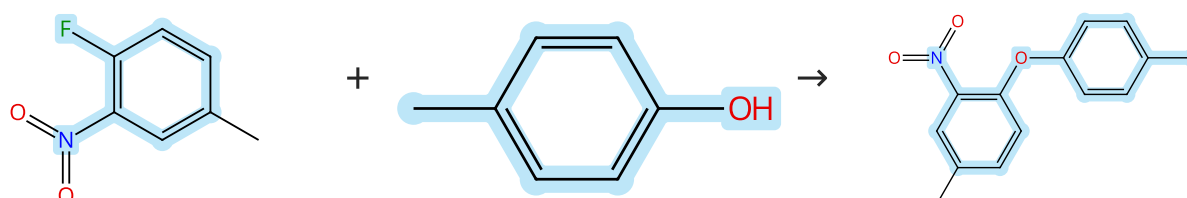
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

Suppliers (89)

Suppliers (19)

31-170-CAS-3426458

Steps: 1 Yield: 100%

Four- and Sixfold Tandem-Domino Reactions Leading to Dimeric Tetrasubstituted Alkenes Suitable as Molecular Switches

By: Tietze, Lutz F.; et al

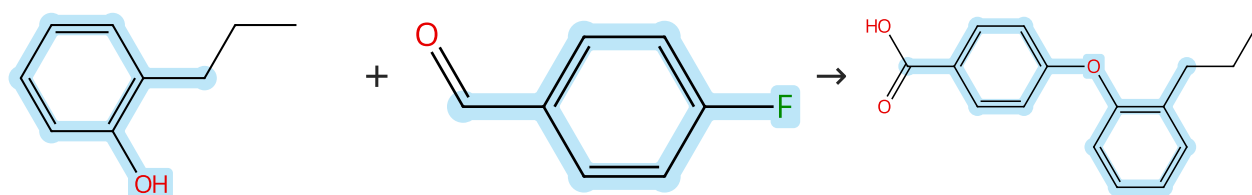
Angewandte Chemie, International Edition (2015), 54(35), 10317-10321.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; 24 h, 95 °C

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (63)

Suppliers (93)

31-170-CAS-14991911

Steps: 1 Yield: 100%

Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist

By: Nakamura, Tsuyoshi; et al

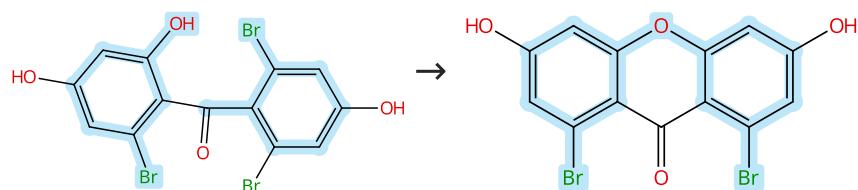
Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 100 °C

1.2 **Reagents:** 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate
Solvents: *tert*-Butanol, Tetrahydrofuran, Water

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-27425881

Steps: 1 Yield: 100%

1.1 Reagents: Sodium bis(trimethylsilyl)amide
Solvents: Dimethylformamide; 1 h, 100 °C

1.2 Reagents: Hydrochloric acid
Solvents: Water; 0 °C

Experimental Protocols

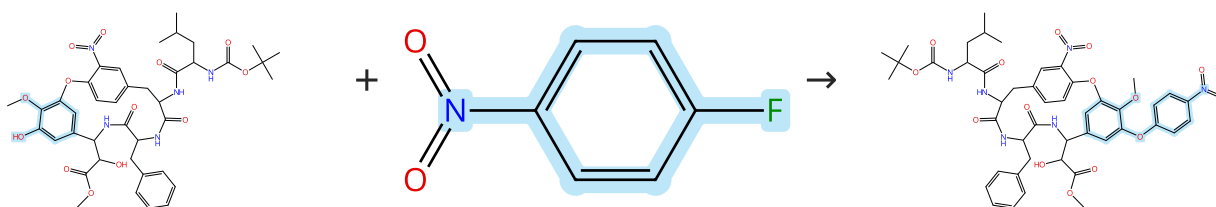
1,8-Disubstituted Xanthyldene-Based Remote Carbenes: Photolytic Generation and Isolation of Low-Coordinate Palladium(II) Complex

By: Sugawara, Shun; et al

European Journal of Inorganic Chemistry (2015), 2015(3), 534-541.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (100)

31-170-CAS-3581617

Steps: 1 Yield: 100%

1.1 Reagents: Cesium fluoride
Solvents: Dimethyl sulfoxide; 2 h, rt

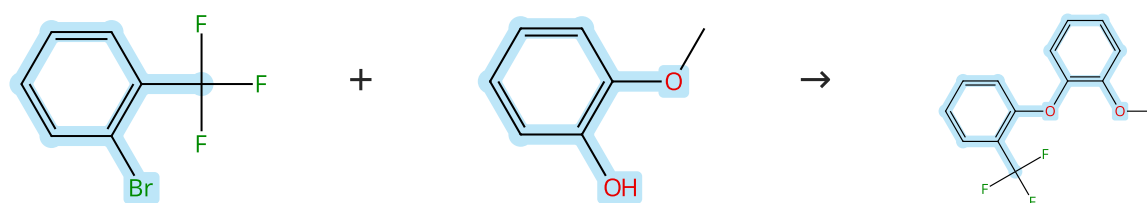
Design and synthesis of simple macrocycles active against vancomycin-resistant Enterococci (VRE)

By: Jia, Yanxing; et al

Chemistry - A European Journal (2006), 12(20), 5334-5351.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (78)

Suppliers (125)

31-614-CAS-36696316

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Catalysts: Copper, Potassium iodide
Solvents: Pyridine; 24 h, reflux

Experimental Protocols

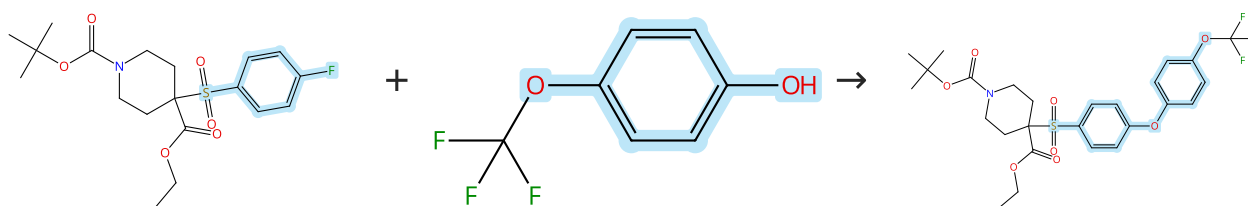
Synthesis of Fluorenones and Xanthenes through Intramolecular C-F Arylation

By: Hamada, Shusuke; et al

Bulletin of the Chemical Society of Japan (2023), 96(5), 401-405.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

Suppliers (93)

Supplier (1)

31-170-CAS-10405304

Steps: 1 Yield: 100%

Orally Active MMP-1 Sparing α -Tetrahydropyranyl and α -Piperidiny Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease

1.1 Reagents: Cesium carbonate

Solvents: Dimethylformamide; 20 h, 90 °C; 90 °C → rt

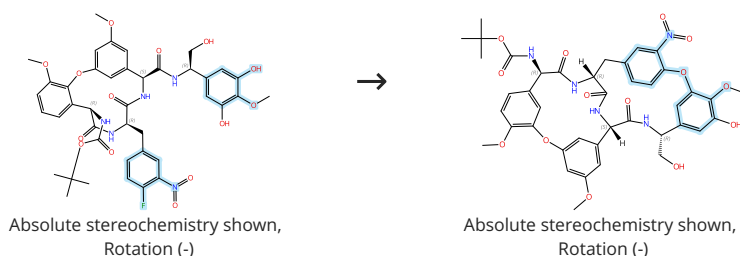
Experimental Protocols

By: Becker, Daniel P.; et al

Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



31-170-CAS-13620913

Steps: 1 Yield: 100%

Alternative synthesis and thermal atropisomerism of a fully functionalized DEFG ring system of teicoplanin

1.1 Reagents: Cesium fluoride

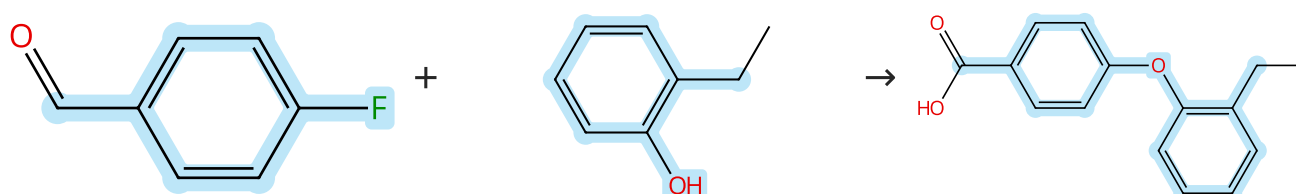
Solvents: Dimethylformamide

By: Mori, Y.; et al

Tetrahedron Letters (2001), 42(35), 6061-6064.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (73)

Suppliers (18)

31-170-CAS-12861407

Steps: 1 Yield: 100%

Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist

1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 100 °C

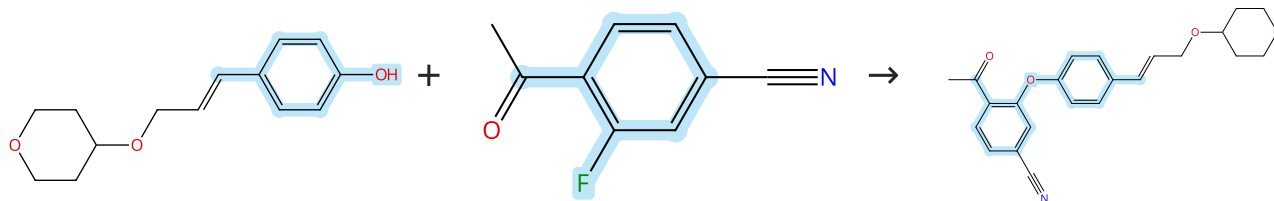
By: Nakamura, Tsuyoshi; et al

1.2 Reagents: 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate

Solvents: *tert*-Butanol, Tetrahydrofuran, Water

Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.

Steps: **1** Yield: **100%**



31-614-CAS-42451362

Steps: **1** Yield: **100%**

Synthesis and evaluation of a novel series of diphenyl amine and diphenylether derivatives with osteoblastogenic and osteogenic effects via CDK8 inhibition

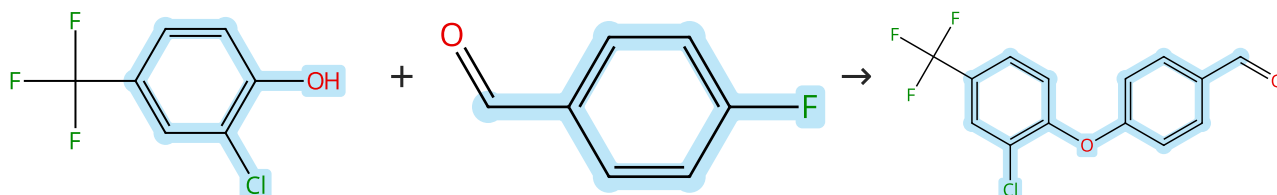
By: Morishita, Ko; et al

Chemical & Pharmaceutical Bulletin (2024), 72(10), 862-883.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 11 h, 100 °C

Experimental Protocols

Steps: 1 Yield: 100%



 Suppliers (93)

 Suppliers (3)

31-614-CAS-34652763

Steps: 1 Yield: 100%

Discovery of novel cytochrome bc₁ complex inhibitor based on natural product neopeltolide

By: Chen, Tao; et al

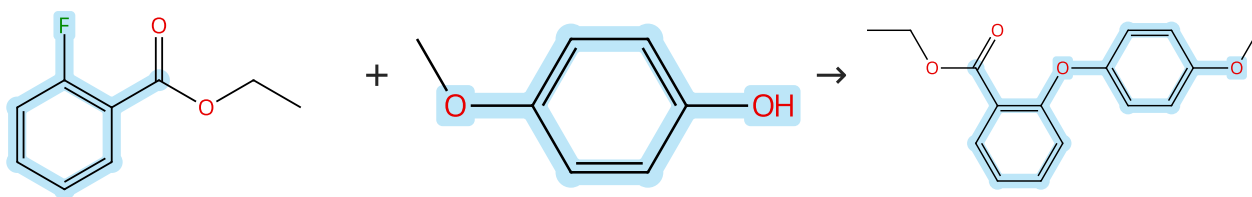
Letters in Drug Design & Discovery (2022), 19(4), 263-268.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt

1.2 Reagents: Water; rt

Experimental Protocols

Steps: 1 Yield: 100%



 Suppliers (121)

 Suppliers (7)

31-170-CAS-6014787

Steps: 1 Yield: 100%

Synthesis of dictyomedins using phosphazene base catalyzed diaryl ether formation

By: Ebisawa, Masaru; et al

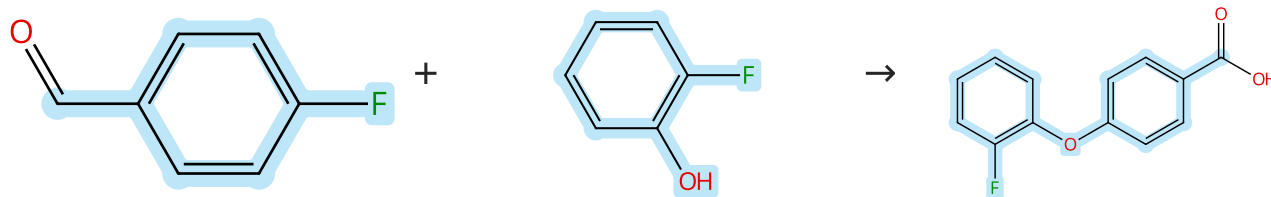
Tetrahedron Letters (2007), 48(50), 8918-8921.

1.1 **Reagents:** (Diethylamino)trimethylsilane
Catalysts: *N*''-(1,1-Dimethylethyl)-*N,N,N*'-tris[tris(dimethylamino)phosphoranylidene]phosphorimidic triamide
Solvents: Dimethylformamide; 2 h, 100 °C

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (101)

Suppliers (48)

31-170-CAS-6471690

Steps: 1 Yield: 100%

Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist

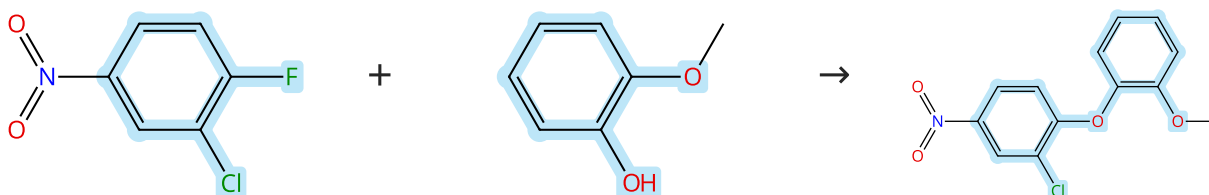
By: Nakamura, Tsuyoshi; et al

Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 100 °C
- 1.2 **Reagents:** 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate
Solvents: *tert*-Butanol, Tetrahydrofuran, Water

Scheme 26 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (125)

Suppliers (11)

31-614-CAS-31980147

Steps: 1 Yield: 100%

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt

31-170-CAS-23043758

Steps: 1 Yield: 100%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

By: Fujii, Shinya; et al

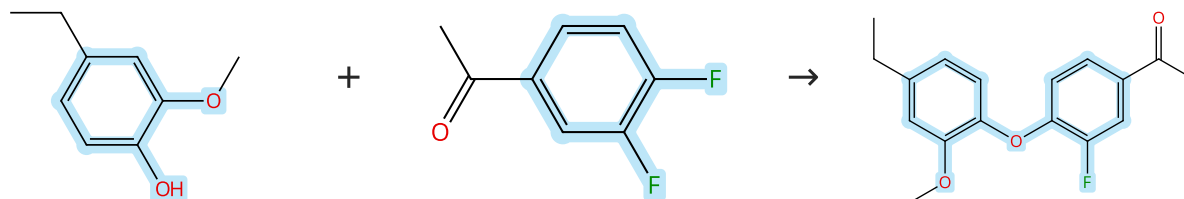
Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 17 h, rt

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (81)

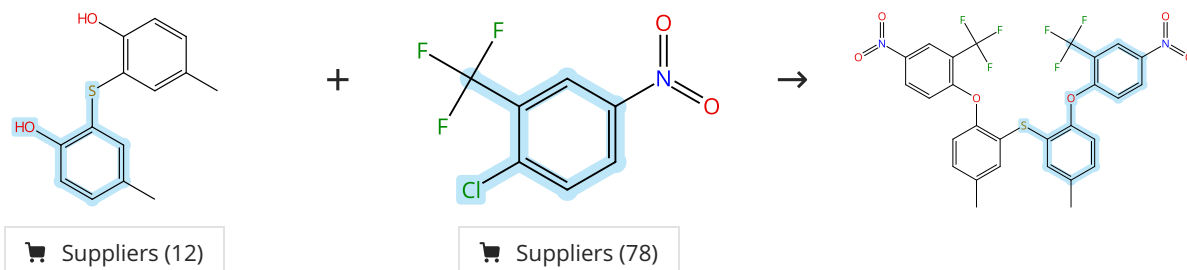
Suppliers (80)

Suppliers (2)

31-170-CAS-7275776 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Solvents: Dimethylformamide; 20 h, 110 °C 1.2 Reagents: Sodium hydroxide Solvents: Water Experimental Protocols	From Triclosan toward the Clinic: Discovery of Nonbiocidal, Potent FabI Inhibitors for the Treatment of Resistant Bacteria By: Gerusz, Vincent; et al Journal of Medicinal Chemistry (2012), 55(22), 9914-9928.
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Scheme 28 (1 Reaction)

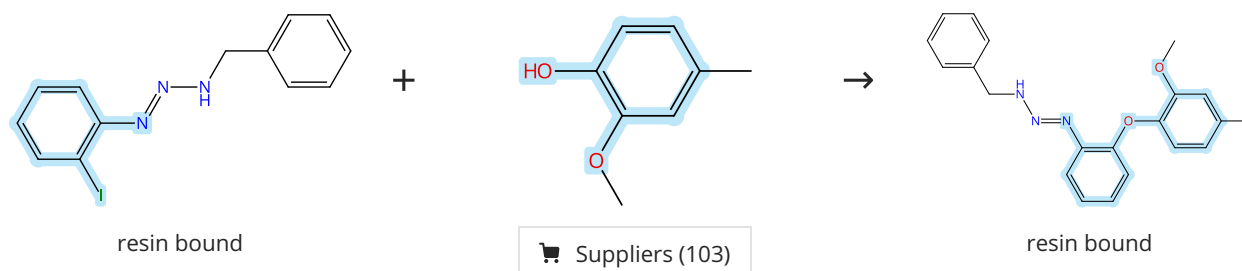
Steps: 1 Yield: 100%



31-170-CAS-8417810 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 12 h, 90 °C; 90 °C → rt	Synthesis and properties of organosoluble fluorinated polyamides bearing 2-2'-thio-bis(4-methyl phenoxy) groups By: Shockravi, Abbas; et al Designed Monomers and Polymers (2009), 12(2), 119-128.
--	--

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



31-170-CAS-10799853 Steps: 1 Yield: 100% 1.1 Reagents: Sodium carbonate, Bromo[1,1'-thiobis[methane]] copper Solvents: Acetonitrile, Pyridine; 48 h, 75 °C Experimental Protocols	Solid phase synthesis of selectively deuterated arenes By: Vanderheiden, Sylvia; et al Chemical Communications (Cambridge, United Kingdom) (2011), 47(32), 9063-9065.
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Scheme 30 (1 Reaction)

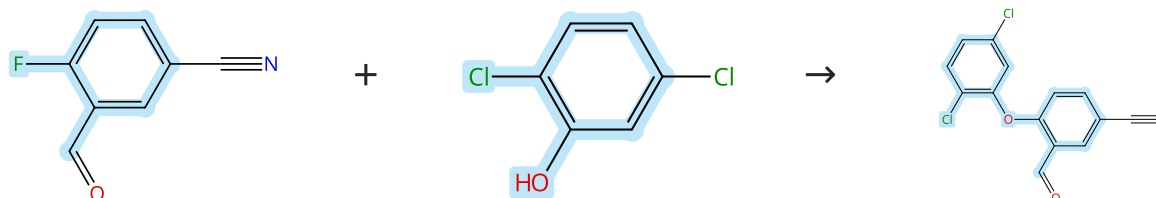
Steps: 1 Yield: 100%



31-170-CAS-23442286	Steps: 1 Yield: 100%	Curved Perylene Diimides Fused with Seven- Membered Rings
1.1 Reagents: Potassium carbonate, 18-Crown-6 Solvents: Dimethylformamide; rt; 3 h, 50 °C		By: Fujimoto, Keisuke; et al
1.2 Reagents: Water		Chemistry - An Asian Journal (2021), 16(6), 690-695.
Experimental Protocols		

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



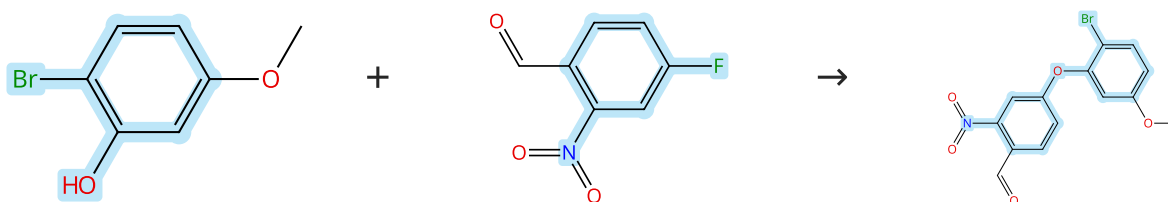
Suppliers (81)

Suppliers (71)

31-170-CAS-6681563	Steps: 1 Yield: 100%	Computationally-Guided Optimization of a Docking Hit to Yield Catechol Diethers as Potent Anti-HIV Agents
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 4 h, 50 °C		By: Bollini, Mariela; et al
Experimental Protocols		Journal of Medicinal Chemistry (2011), 54(24), 8582-8591.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



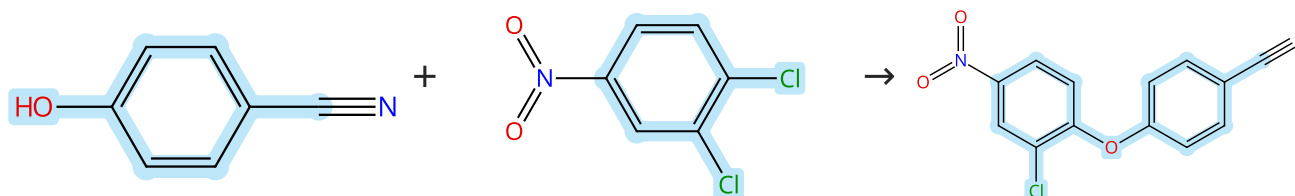
Suppliers (65)

Suppliers (70)

31-170-CAS-21519190	Steps: 1 Yield: 100%	Methoxy-Substituted Nitrodibenzofuran-Based Protecting Group with an Improved Two-Photon Action Cross-Section for Thiol Protection in Solid Phase Peptide Synthesis
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 6 h, rt; 12 h, 50 °C; 50 °C → rt		By: Bader, Taysir K.; et al
1.2 Reagents: Ammonium chloride Solvents: Water; rt		Journal of Organic Chemistry (2020), 85(3), 1614-1625.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

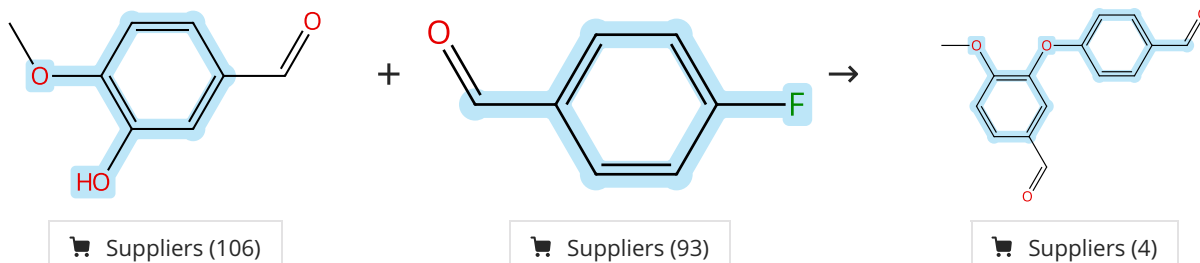
Suppliers (63)

Suppliers (5)

31-170-CAS-15221476 Steps: 1 Yield: 100%	Translation of microwave methodology to continuous flow for the efficient synthesis of diaryl ethers via a base-mediated S_NAr reaction By: Wiles, Charlotte; et al Beilstein Journal of Organic Chemistry (2011), 7, 1360-1371.
1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Solvents: Acetonitrile; 10 min, 25 bar, 195 °C	

Scheme 34 (1 Reaction)

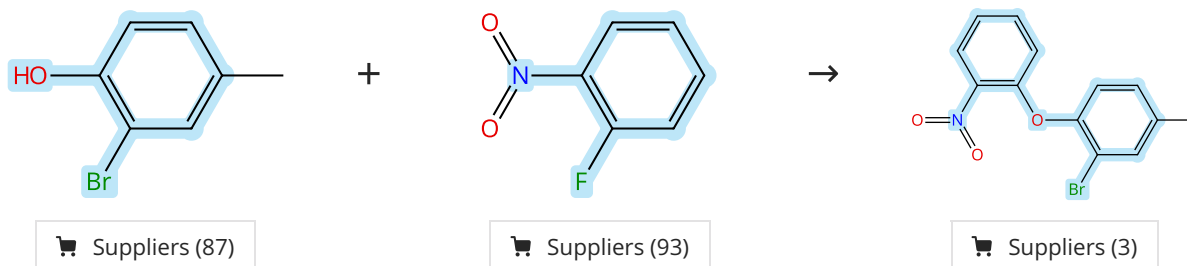
Steps: 1 Yield: 100%



31-170-CAS-22912733 Steps: 1 Yield: 100%	Total synthesis of polymorphatin A, a macrocyclic bisbibenzyl with boat configured arenes By: Almalki, Faisal A.; et al Tetrahedron (2020), 76(42), 131521.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 18 h, reflux	
Experimental Protocols	

Scheme 35 (1 Reaction)

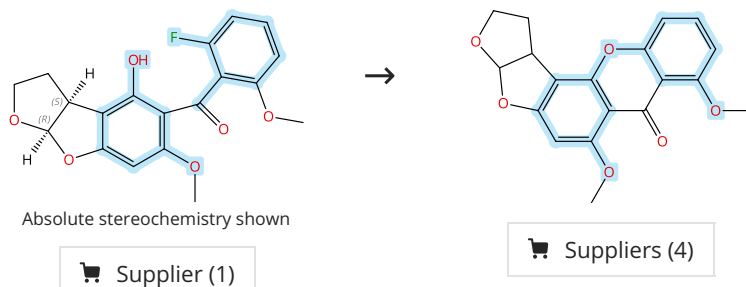
Steps: 1 Yield: 100%



31-170-CAS-11427809 Steps: 1 Yield: 100%	Synthesis and Photochemical Investigations of Tetrasubstituted Alkenes as Molecular Switches-The Effect of Substituents By: Tietze, Lutz F.; et al Chemistry - A European Journal (2011), 17(30), 8452-8461, S8452/1-S8452/65.
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 24 h, rt → 95 °C	
Experimental Protocols	

Scheme 36 (1 Reaction)

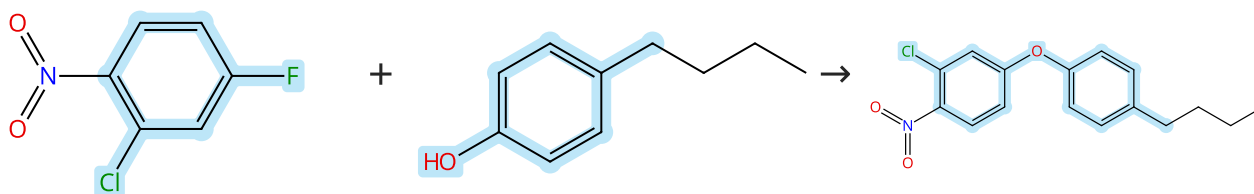
Steps: 1 Yield: 100%



31-614-CAS-29098175	Steps: 1 Yield: 100%	A complex induced proximity effect in the anionic Fries rearrangement of o-iodophenyl benzoates: synthesis of dihydro-O-methylsterigmatocystin and other xanthenes By: Horne, Stephen; et al Journal of Organic Chemistry (1990), 55(15), 4520-2.
1.1 Reagents: Potassium hydroxide Solvents: Methanol		

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



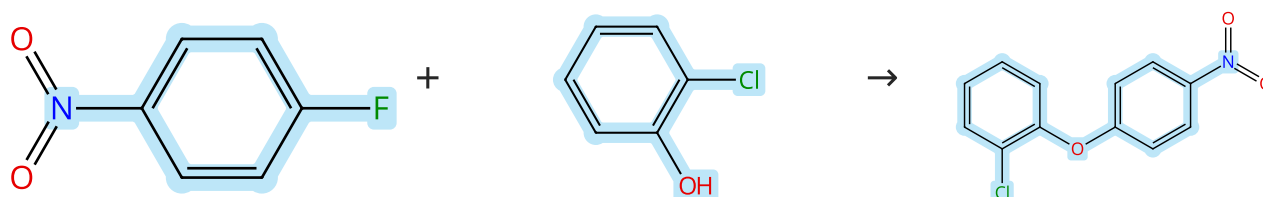
Suppliers (74)

Suppliers (71)

31-170-CAS-19212507	Steps: 1 Yield: 100%	Synthesis of N-Substituted Iminosugar Derivatives and Evaluation of Their Immunosuppressive Activities By: Lv, Zhuo; et al ChemMedChem (2018), 13(4), 338-351.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 70 °C; 70 °C → rt		
1.2 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (100)

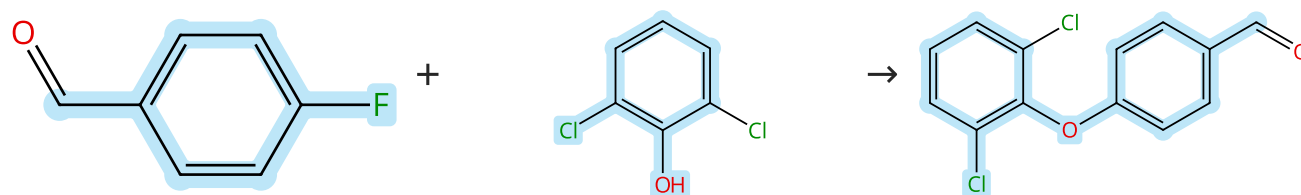
Suppliers (92)

Suppliers (32)

31-170-CAS-11695557	Steps: 1 Yield: 100%	CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions By: Keipour, Hoda; et al Canadian Journal of Chemistry (2016), 94(1), 95-104.
1.1 Catalysts: Cesium fluoride Solvents: Dimethyl sulfoxide; 0.15 h, 110 °C		
Experimental Protocols		

Scheme 39 (1 Reaction)

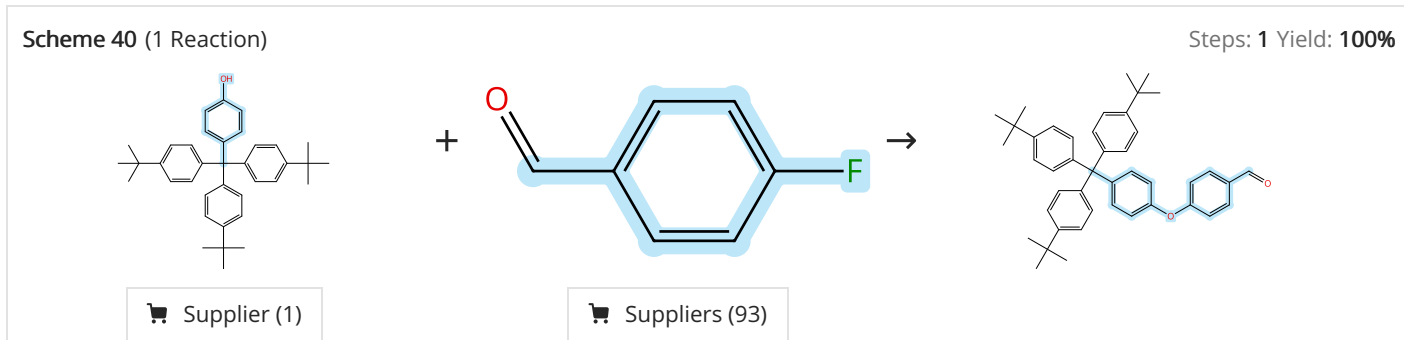
Steps: 1 Yield: 100%



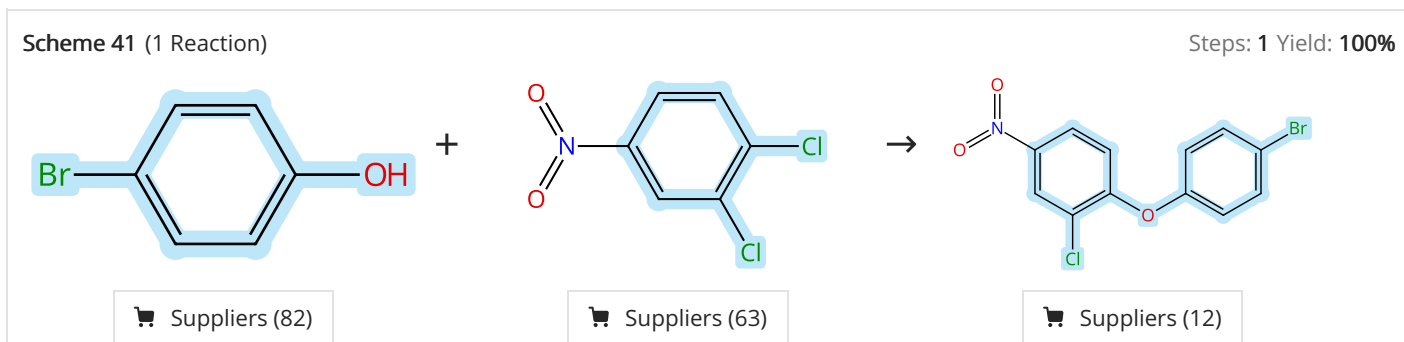
Suppliers (93)

Suppliers (86)

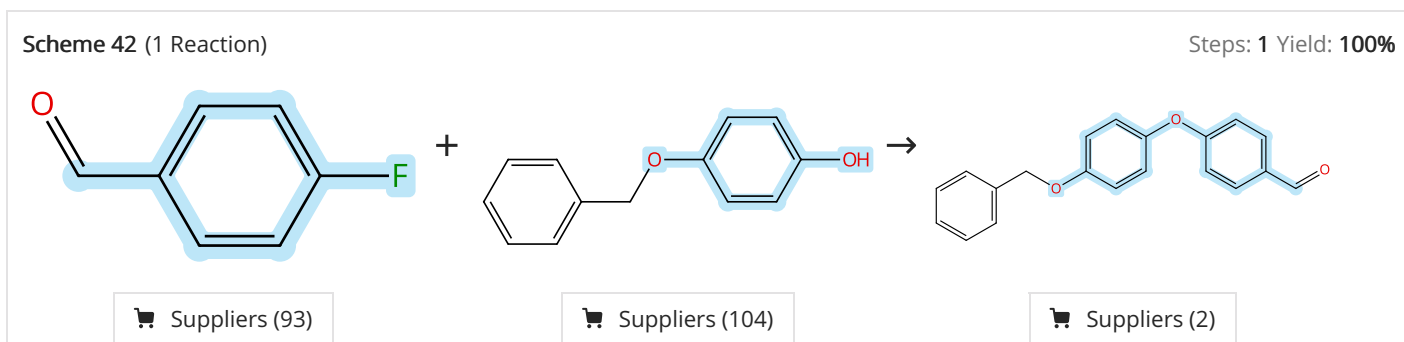
31-614-CAS-34652764 Steps: 1 Yield: 100%	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt	By: Chen, Tao; et al
1.2 Reagents: Water; rt Experimental Protocols	Letters in Drug Design & Discovery (2022), 19(4), 263-268.



31-170-CAS-6858330 Steps: 1 Yield: 100%	Rotaxanes of a macrocyclic ferrocenophane with dialkylammonium axle components
1.1 Reagents: Cesium fluoride Solvents: <i>N</i> -Methyl-2-pyrrolidone; 24 h, 100 °C	By: Suzuki, Yuji; et al
Experimental Protocols	Dalton Transactions (2009), (44), 9881-9891.



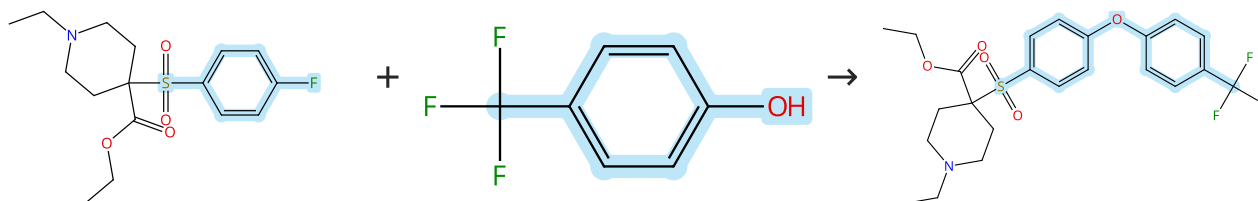
31-170-CAS-1832427 Steps: 1 Yield: 100%	Translation of microwave methodology to continuous flow for the efficient synthesis of diaryl ethers via a base-mediated S_NAr reaction
1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Solvents: Acetonitrile; 1 min, 25 bar, 195 °C	By: Wiles, Charlotte; et al
	Beilstein Journal of Organic Chemistry (2011), 7, 1360-1371.



31-170-CAS-18612294	Steps: 1 Yield: 100%	Switching subtype-selectivity: Fragment replacement strategy affords novel class of peroxisome proliferator-activated receptor α/δ (PPARα/δ) dual agonists By: Shioi, Ryuta; et al Bioorganic & Medicinal Chemistry Letters (2017), 27(14), 3131-3134.
1.1 Reagents: Potassium carbonate Solvents: Dimethylacetamide; 7 h, 80 °C		
Experimental Protocols		

Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%

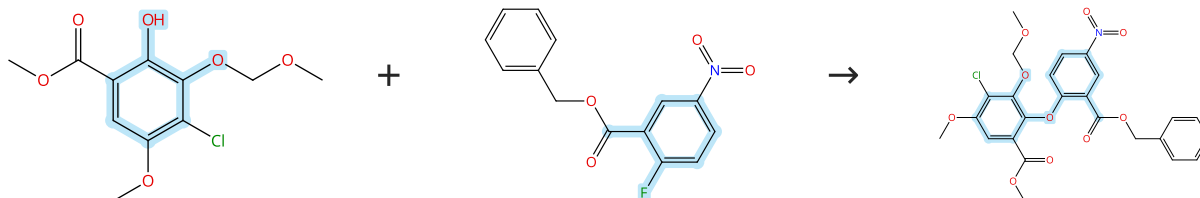


Suppliers (96)

31-170-CAS-4682785	Steps: 1 Yield: 100%	Orally Active MMP-1 Sparing α-Tetrahydropyranyl and α-Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease By: Becker, Daniel P.; et al Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 25 h, 90 °C		
Experimental Protocols		

Scheme 44 (1 Reaction)

Steps: 1 Yield: 100%

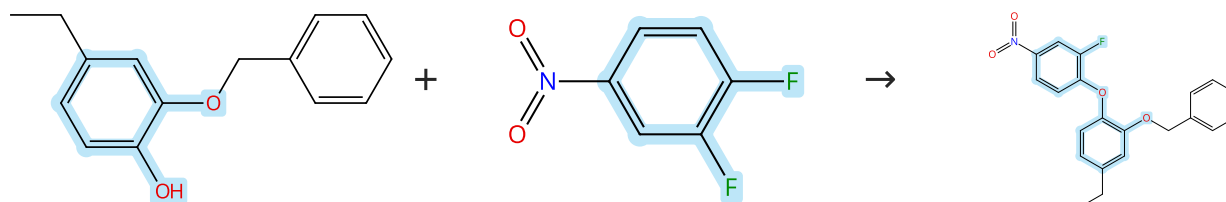


Suppliers (7)

31-170-CAS-11289575	Steps: 1 Yield: 100%	Unexpected Diels-Alder/Carbonyl-ene Cascade toward the Biomimetic Synthesis of Chloropupukeanin By: Suzuki, Takahiro; et al Organic Letters (2013), 15(7), 1748-1751.
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 5 min, 0 °C		
1.2 0 °C; overnight, rt		
1.3 Reagents: Water		
Experimental Protocols		

Scheme 45 (1 Reaction)

Steps: 1 Yield: 100%



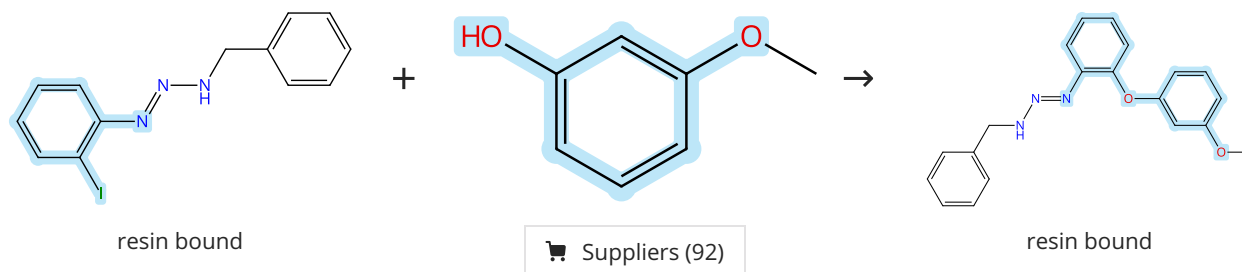
Suppliers (5)

Suppliers (88)

31-170-CAS-9737420	Steps: 1 Yield: 100%	From Triclosan toward the Clinic: Discovery of Nonbiocidal, Potent FabI Inhibitors for the Treatment of Resistant Bacteria
1.1 Reagents: Potassium hydroxide Solvents: Acetonitrile; rt → 80 °C; 2 h, 80 °C		By: Gerusz, Vincent; et al
Experimental Protocols		Journal of Medicinal Chemistry (2012), 55(22), 9914-9928.

Scheme 46 (1 Reaction)

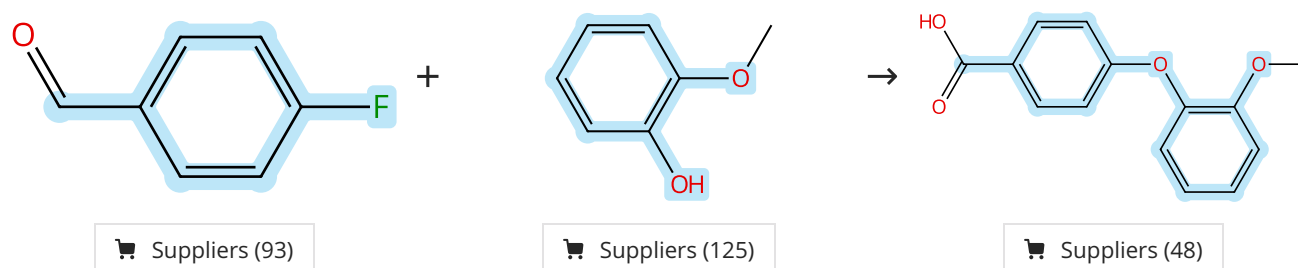
Steps: 1 Yield: 100%



31-170-CAS-1671593	Steps: 1 Yield: 100%	Solid phase synthesis of selectively deuterated arenes
1.1 Reagents: Sodium carbonate, Bromo[1,1'-thiobis[methane]] copper Solvents: Acetonitrile, Pyridine; 48 h, 75 °C		By: Vanderheiden, Sylvia; et al
Experimental Protocols		Chemical Communications (Cambridge, United Kingdom) (2011), 47(32), 9063-9065.

Scheme 47 (1 Reaction)

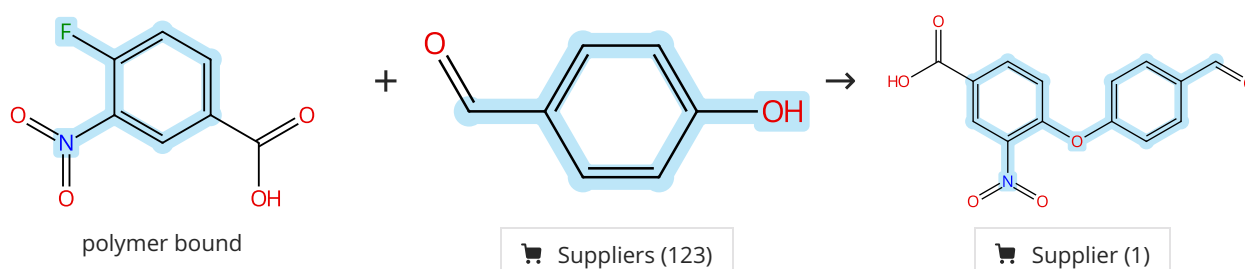
Steps: 1 Yield: 100%



31-170-CAS-10733623	Steps: 1 Yield: 100%	Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 100 °C		By: Nakamura, Tsuyoshi; et al
1.2 Reagents: 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate Solvents: <i>tert</i> -Butanol, Tetrahydrofuran, Water		Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.

Scheme 48 (1 Reaction)

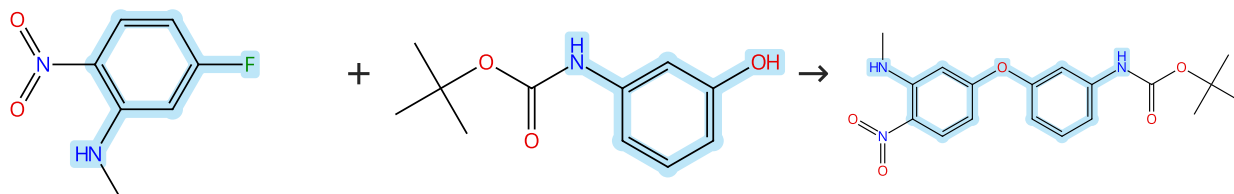
Steps: 1 Yield: 100%



31-170-CAS-8588497 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide 1.2 Reagents: Trifluoroacetic acid Solvents: Dichloromethane Experimental Protocols	Solid phase synthesis of functionalized biaryl ethers: Versatile scaffolds for combinatorial chemistry By: Wijkmans, Jac C. H. M.; et al Molecular Diversity (1998), 3(2), 117-120.
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Scheme 49 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (58)

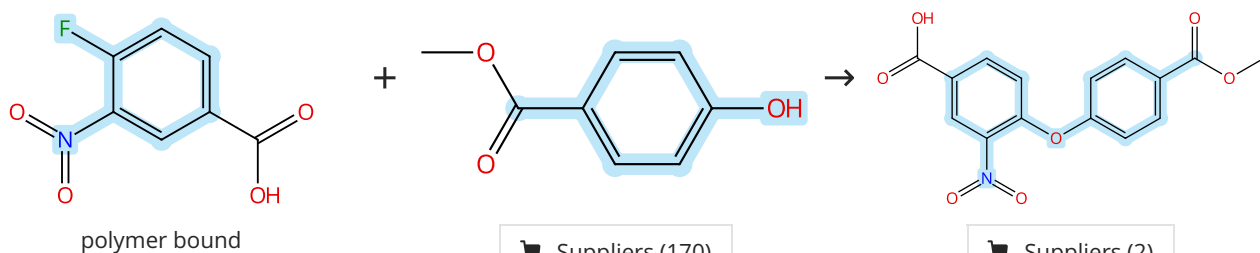
Suppliers (70)

Supplier (1)

31-170-CAS-3599229 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 14 h, 100 °C Experimental Protocols	Design and Synthesis of Novel DFG-Out RAF/Vascular Endothelial Growth Factor Receptor 2 (VEGFR2) Inhibitors. 1. Exploration of [5,6]-Fused Bicyclic Scaffolds By: Okaniwa, Masanori; et al Journal of Medicinal Chemistry (2012), 55(7), 3452-3478.
---	--

Scheme 50 (1 Reaction)

Steps: 1 Yield: 100%



polymer bound

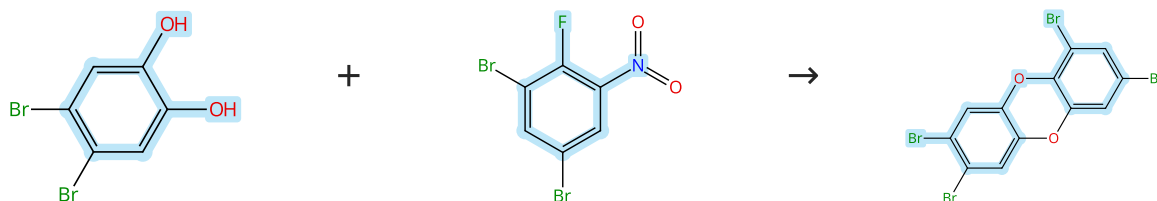
Suppliers (170)

Suppliers (2)

31-170-CAS-15303196 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide 1.2 Reagents: Trifluoroacetic acid Solvents: Dichloromethane Experimental Protocols	Solid phase synthesis of functionalized biaryl ethers: Versatile scaffolds for combinatorial chemistry By: Wijkmans, Jac C. H. M.; et al Molecular Diversity (1998), 3(2), 117-120.
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Scheme 51 (1 Reaction)

Steps: 1 Yield: 100%

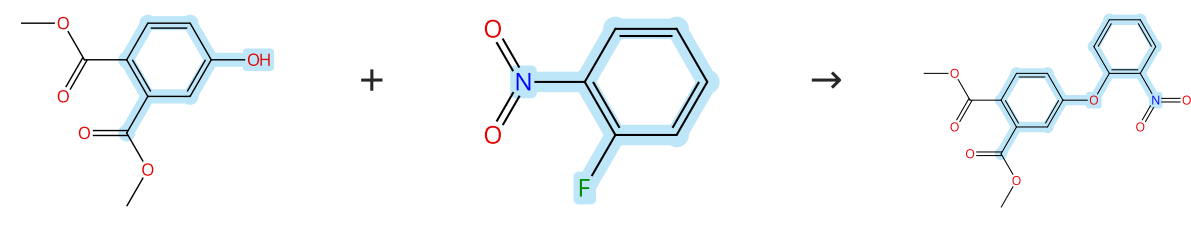


Suppliers (51)

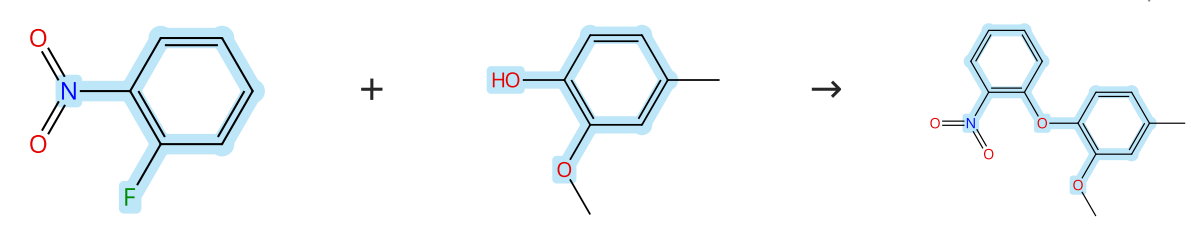
Suppliers (27)

Suppliers (6)

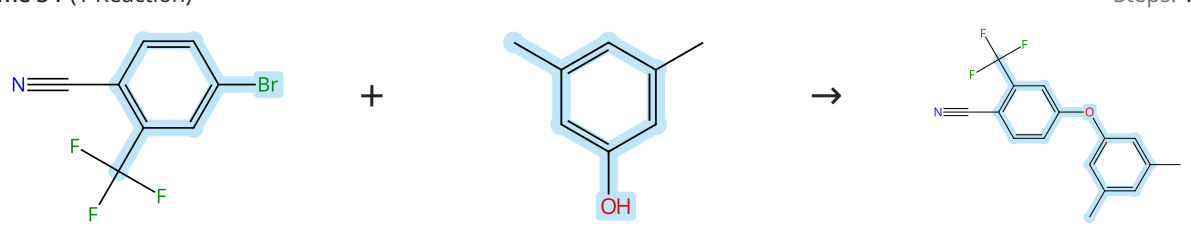
31-170-CAS-20583870 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 10 min, rt 1.2 6 h, 145 - 150 °C Experimental Protocols	Facile synthesis of bromo- and mixed bromo/ chloro dibenzo-p-dioxins and [¹⁴C]-labeled 1,3,7,8-tetrabromodibenzo-p-dioxin By: Singh, Anuradha; et al Chemosphere (2020), 239, 124626.
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Scheme 52 (1 Reaction)  🛒 Suppliers (66) 🛒 Suppliers (93) 🛒 Supplier (1)	Steps: 1 Yield: 100%
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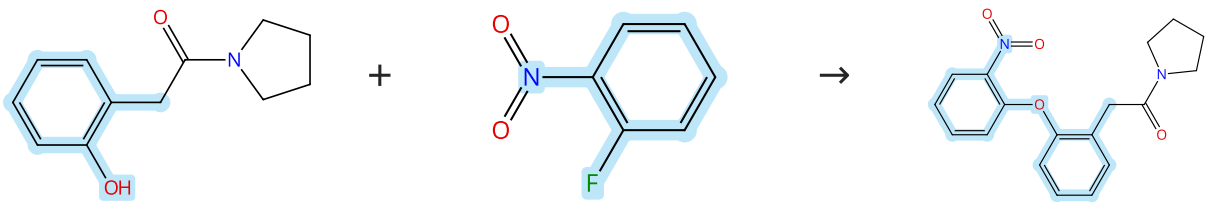
31-170-CAS-9763795 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 1 h, 100 °C; 100 °C → rt Experimental Protocols	Identification, Synthesis, and Characterization of New Glycogen Phosphorylase Inhibitors Binding to the Allosteric A MP Site By: Kristiansen, Marit; et al Journal of Medicinal Chemistry (2004), 47(14), 3537-3545.
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Scheme 53 (1 Reaction)  🛒 Suppliers (93) 🛒 Suppliers (103) 🛒 Supplier (1)	Steps: 1 Yield: 100%
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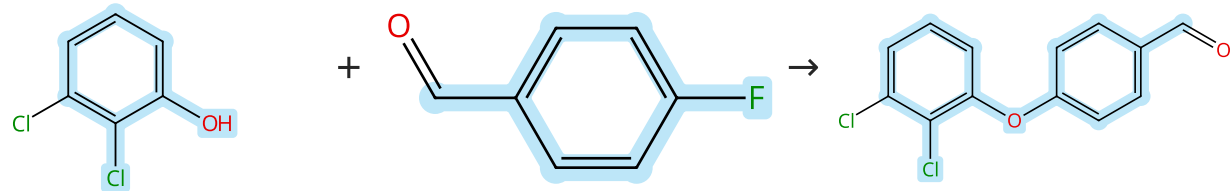
31-614-CAS-42944372 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Catalysts: Tetrabutylammonium bromide Solvents: 2-Methyltetrahydrofuran, Water; 10 min, 95 °C Experimental Protocols	S_NAr Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media By: Wong, Madison J.; et al ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.
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Scheme 54 (1 Reaction)  🛒 Suppliers (86) 🛒 Suppliers (89) 🛒 Suppliers (2)	Steps: 1 Yield: 100%
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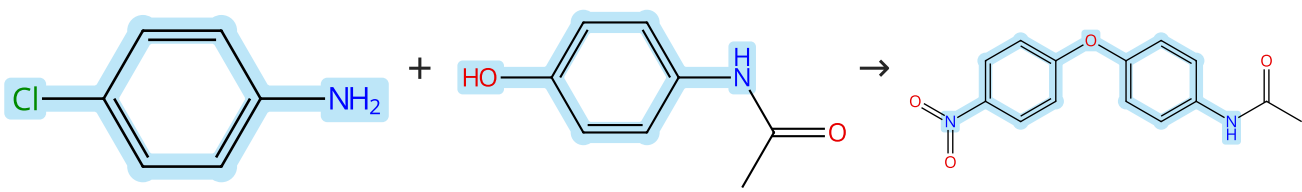
31-614-CAS-43350517 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: Cuprous iodide, 1<i>H</i>-imidazolium, 3-[4-(trifluoromethyl)-2-pyridinyl]-1-(2,4,6-trimethylphenyl)-, bromide (1:1) Solvents: Acetonitrile; 20 h, 110 °C Experimental Protocols	Bidentate NHC-Containing Ligands for Copper Catalyzed Synthesis of Functionalised Diaryl Ethers By: Richards, Ethan M.; et al Advanced Synthesis & Catalysis (2025), 367(6), e202400909.
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Scheme 55 (1 Reaction) Steps: 1 Yield: 100%   Suppliers (6)  Suppliers (93)  Supplier (1)		
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31-170-CAS-15220494 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 30 min, rt 1.2 24 h, 25 °C 1.3 Reagents: Hydrochloric acid Solvents: Water; acidified Experimental Protocols	Developing the Scope of O→C Aryl Migrations: Exploring Amide Substrates as Potential Precursors for Asymmetric Reactions By: Ameen, Dana; et al European Journal of Organic Chemistry (2014), 2014(9), 1925-1934.
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Scheme 56 (1 Reaction) Steps: 1 Yield: 100%   Suppliers (59)  Suppliers (93)  Suppliers (7)		
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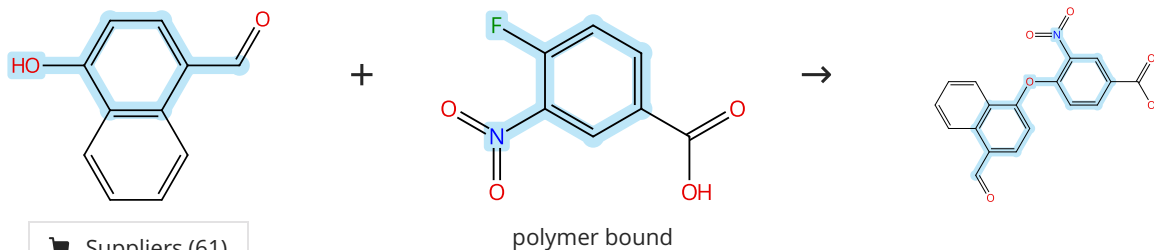
31-614-CAS-34652759 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt 1.2 Reagents: Water; rt Experimental Protocols	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide By: Chen, Tao; et al Letters in Drug Design & Discovery (2022), 19(4), 263-268.
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Scheme 57 (1 Reaction) Steps: 1 Yield: 100%   Suppliers (98)  Suppliers (143)  Suppliers (13)		
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31-170-CAS-11004638	Steps: 1 Yield: 100%	Novel synthesis of anthelmintic drug 4-isothiocyanato-4'-nitrodiphenyl ether and its analogs
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 5 h, 153 °C		By: Chaskar, A. C.; et al
1.2 Solvents: Water		Synthetic Communications (2009), 39(6), 992-1001.
Experimental Protocols		

Scheme 58 (1 Reaction)

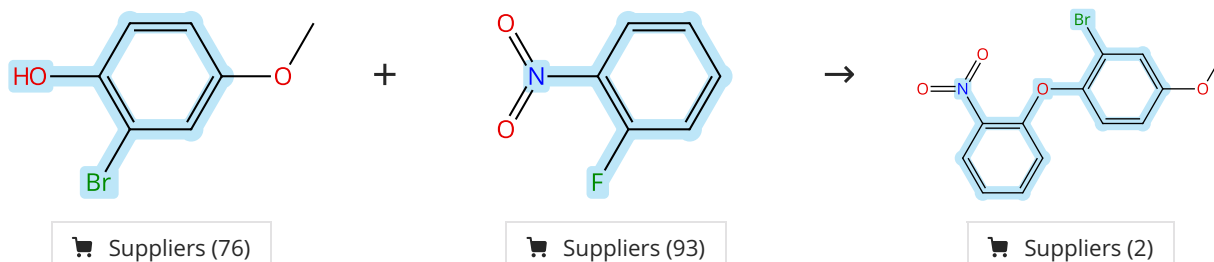
Steps: 1 Yield: 100%



31-170-CAS-14977427	Steps: 1 Yield: 100%	Solid phase synthesis of functionalized biaryl ethers: Versatile scaffolds for combinatorial chemistry
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide		By: Wijkman, Jac C. H. M.; et al
1.2 Reagents: Trifluoroacetic acid Solvents: Dichloromethane		Molecular Diversity (1998), 3(2), 117-120.
Experimental Protocols		

Scheme 59 (2 Reactions)

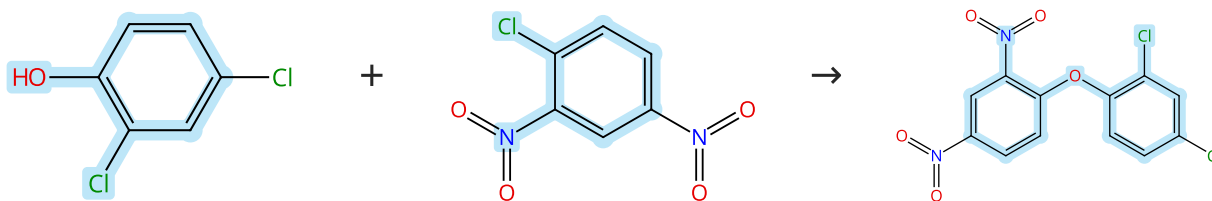
Steps: 1 Yield: 99-100%



31-170-CAS-450479	Steps: 1 Yield: 100%	Synthesis and Photochemical Investigations of Tetrasubstituted Alkenes as Molecular Switches-The Effect of Substituents
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 24 h, rt → 95 °C		By: Tietze, Lutz F.; et al
Experimental Protocols		Chemistry - A European Journal (2011), 17(30), 8452-8461, S8452/1-S8452/65.
31-614-CAS-31558912	Steps: 1 Yield: 99%	Synthesis and Bond Activation Chemistry of Palladium (II) Pincer Complexes with a Weakly Coordinating Side Arm
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 24 h, 100 °C		By: Prakasham, A. P.; et al
Experimental Protocols		Organometallics (2022), 41(5), 634-641.

Scheme 60 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (102)

Suppliers (75)

Suppliers (4)

31-614-CAS-25139582

Steps: 1 Yield: 100%

Chemical validation of a druggable site on Hsp27/H SPB1 using in silico solvent mapping and biophysical methods

By: Makley, Leah N.; et al

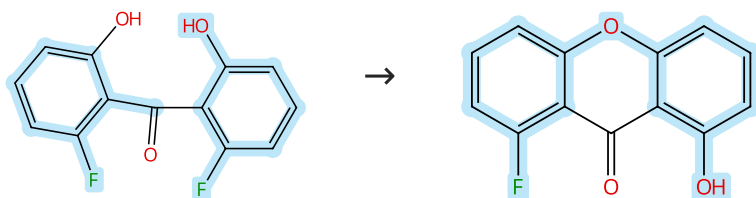
Bioorganic & Medicinal Chemistry (2021), 34, 115990.

- 1.1 Reagents: Cesium carbonate
Solvents: Dimethylformamide; rt
- 1.2 Reagents: Hydrochloric acid
Solvents: Water; acidified

Experimental Protocols

Scheme 61 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-30679890

Steps: 1 Yield: 100%

An Efficient Two-Step Protocol for the Isoprenylation of Xanthone at the C2 Position Starting from 1-Fluoroxanthone Derivative

By: Fujimoto, Yuuki; et al

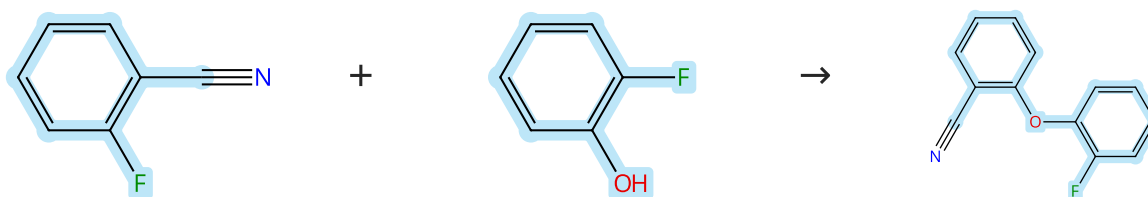
Synlett (2020), 31(14), 1423-1429.

- 1.1 Reagents: Cesium carbonate
Solvents: Acetone; 25 °C; 1 h, 25 °C
- 1.2 Reagents: Hydrochloric acid
Solvents: Water; 25 °C

Experimental Protocols

Scheme 62 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (101)

Suppliers (24)

31-614-CAS-32528351

Steps: 1 Yield: 100%

Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement

By: Jinno, Shunki; et al

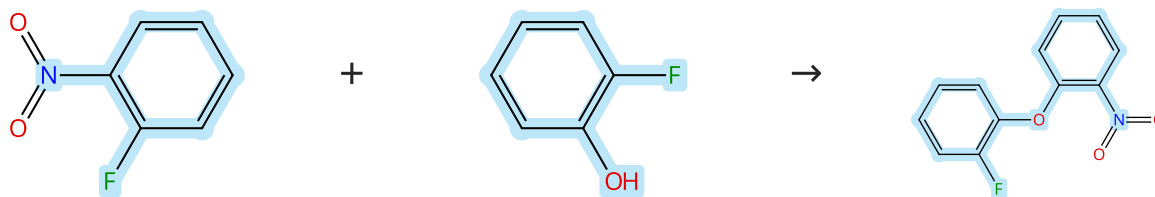
Organic Letters (2022), 24(23), 4140-4144.

- 1.1 Reagents: Potassium hydroxide
Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt
- 1.2 Reagents: Water; rt

Experimental Protocols

Scheme 63 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (101)

Suppliers (43)

31-170-CAS-13500550

Steps: 1 Yield: 100%

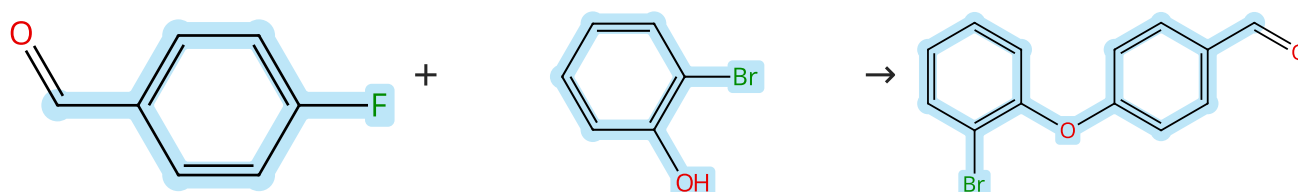
A Fast Way to Fluorescence: A Fourfold Domino Reaction to Condensed Polycyclic Compounds

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; 14 h, 95 °C

By: Tietze, Lutz F.; et al
 Chemistry - A European Journal (2014), 20(39), 12553-12558.

Scheme 64 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (82)

Suppliers (27)

31-614-CAS-34652755

Steps: 1 Yield: 100%

Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt

By: Chen, Tao; et al

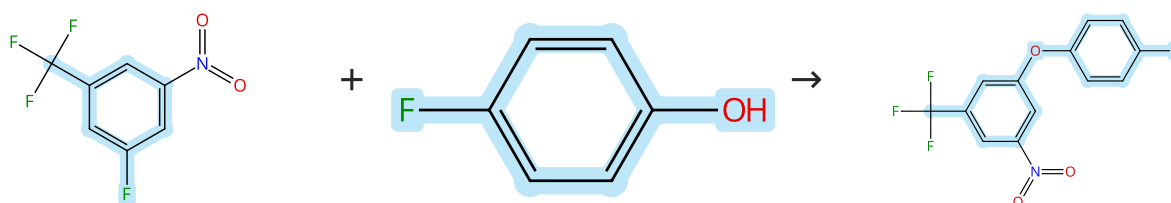
- 1.2 **Reagents:** Water; rt

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

Experimental Protocols

Scheme 65 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

Suppliers (86)

31-170-CAS-4868686

Steps: 1 Yield: 100%

Effect of the nucleophile structure on the relative mobility of the nitro group and fluorine atom in reactions of 3,5-dinitro- and 3-fluoro-5-nitrobenzotrifluorides with phenols in the presence of potassium carbonate

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide

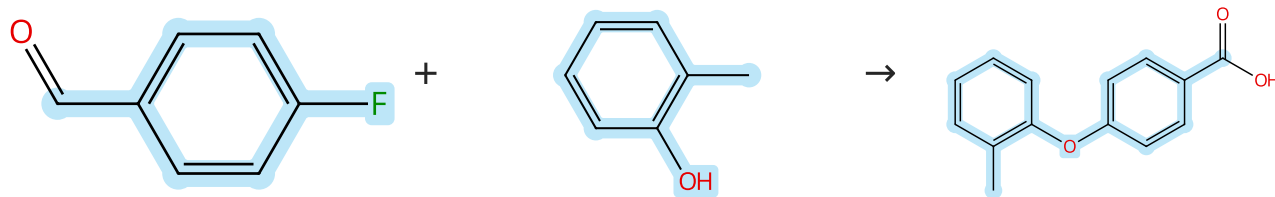
By: Khalfina, I. A.; et al

Experimental Protocols

Russian Journal of Organic Chemistry (Translation of Zhurnal Organicheskoi Khimii) (2002), 38(1), 47-58.

Scheme 66 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (83)

Suppliers (42)

31-170-CAS-65736

Steps: 1 Yield: 100%

Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist

By: Nakamura, Tsuyoshi; et al

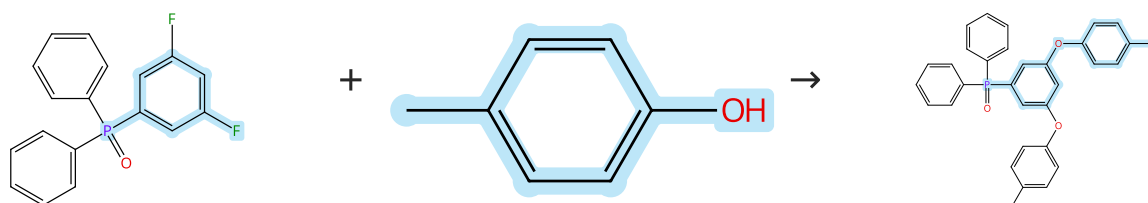
Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 100 °C

1.2 **Reagents:** 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate
Solvents: *tert*-Butanol, Tetrahydrofuran, Water

Scheme 67 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (89)

31-170-CAS-14841348

Steps: 1 Yield: 100%

Linear poly(arylene ether)s with pendant diphenyl phosphoryl groups via nucleophilic aromatic substitution reactions of 3,5-difluorotriphenylphosphine oxide

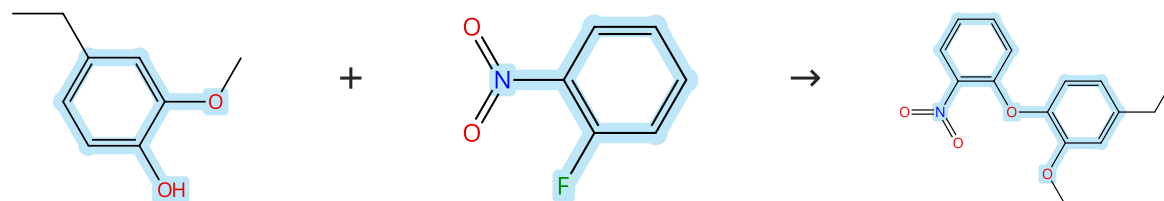
By: Tienda, Kevin; et al

Polymer Preprints (American Chemical Society, Division of Polymer Chemistry) (2010), 51(1), 666-667.

1.1 **Reagents:** Potassium carbonate
Solvents: *N*-Methyl-2-pyrrolidone; 6 h, 150 °C; 16 h, 150 °C → 185 °C

Scheme 68 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (81)

Suppliers (93)

Supplier (1)

31-170-CAS-7308831

Steps: 1 Yield: 100%

From Triclosan toward the Clinic: Discovery of Nonbiocidal, Potent FabI Inhibitors for the Treatment of Resistant Bacteria

By: Gerusz, Vincent; et al

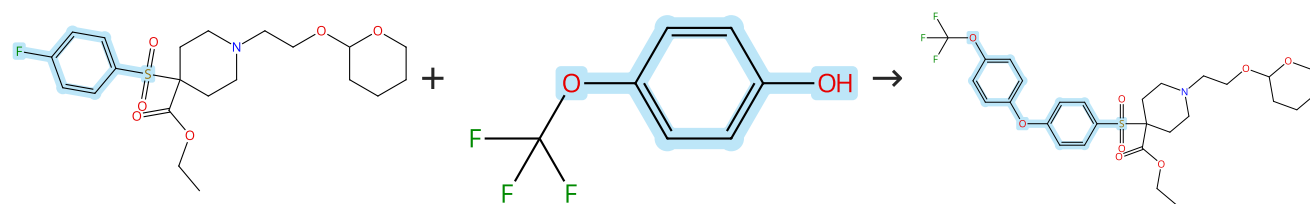
Journal of Medicinal Chemistry (2012), 55(22), 9914-9928.

1.1 **Reagents:** Potassium hydroxide
Solvents: Acetonitrile; rt → 80 °C; 2 h, 80 °C

Experimental Protocols

Scheme 69 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

31-170-CAS-12154556

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 20 h, 110 °C

Experimental Protocols

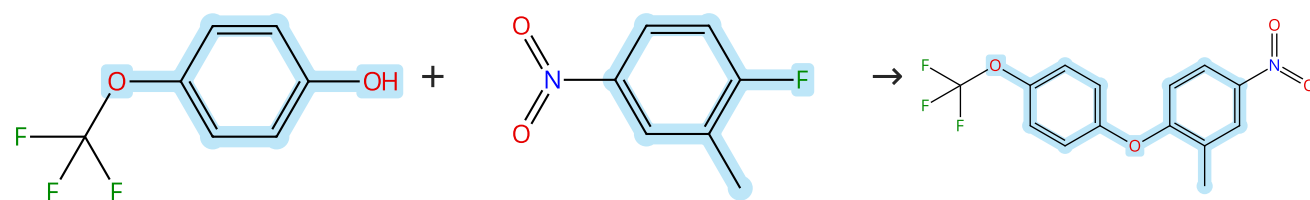
Orally Active MMP-1 Sparing α -Tetrahydropyranyl and α -Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease

By: Becker, Daniel P.; et al

Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.

Scheme 70 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (84)

Supplier (1)

31-614-CAS-41227168

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 22 h, 110 °C; 110 °C → rt
1.2 Solvents: Ethyl acetate, Water; rt

Experimental Protocols

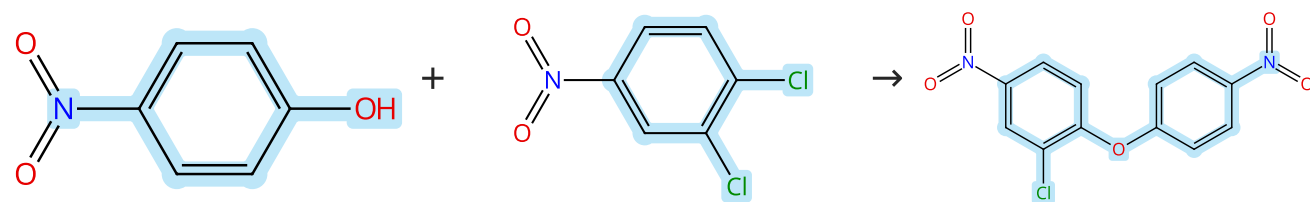
MPoMA protects against lung epithelial cell injury via p65 degradation

By: Lee, Soheun; et al

Biomedicine & Pharmacotherapy (2024), 175, 116674.

Scheme 71 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (133)

Suppliers (63)

Suppliers (4)

31-170-CAS-13090706

Steps: 1 Yield: 100%

1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene
Solvents: Acetonitrile; 10 min, 25 bar, 195 °C

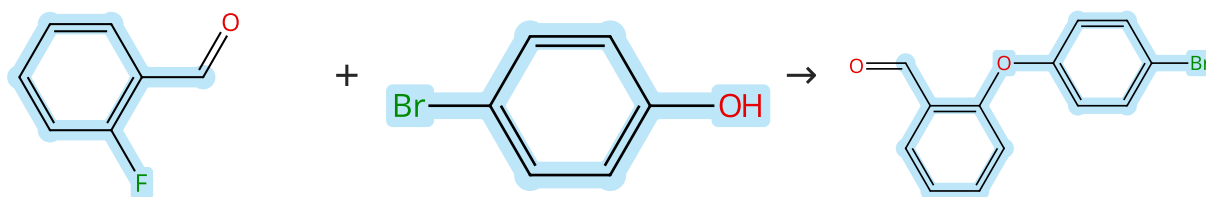
Translation of microwave methodology to continuous flow for the efficient synthesis of diaryl ethers via a base-mediated S_N Ar reaction

By: Wiles, Charlotte; et al

Beilstein Journal of Organic Chemistry (2011), 7, 1360-1371.

Scheme 72 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (82)

Suppliers (30)

31-614-CAS-36277495

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

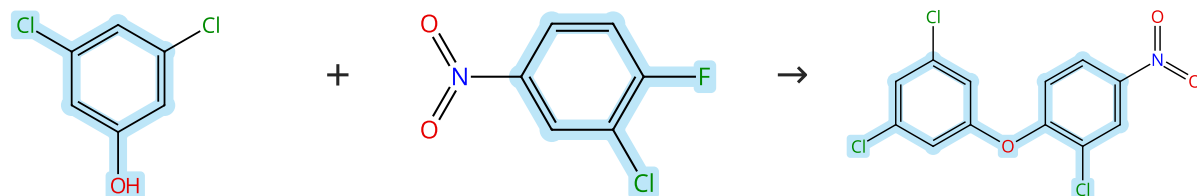
By: Can Uskup, Hacer; et al

Experimental Protocols

ACS Omega (2023), 8(15), 14047-14052.

Scheme 73 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (92)

Suppliers (86)

31-614-CAS-31980142

Steps: 1 Yield: 100%

Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

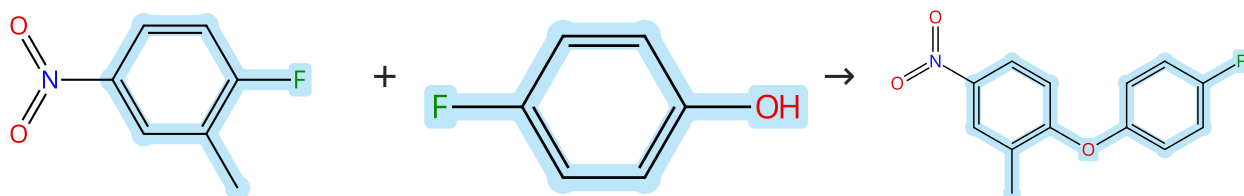
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt

By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

Scheme 74 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (86)

31-614-CAS-41227151

Steps: 1 Yield: 100%

MPoMA protects against lung epithelial cell injury via p65 degradation

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 19 h, 110 °C; 110 °C → rt

By: Lee, Soheun; et al

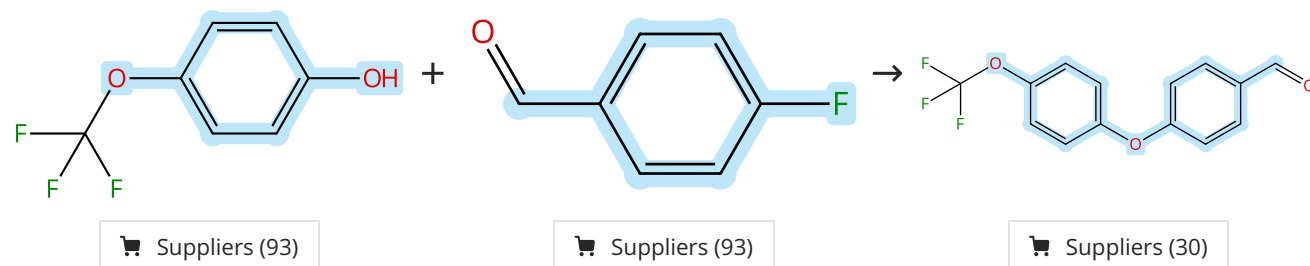
1.2 **Solvents:** Ethyl acetate, Water; rt

Biomedicine & Pharmacotherapy (2024), 175, 116674.

Experimental Protocols

Scheme 75 (1 Reaction)

Steps: 1 Yield: 100%



31-170-CAS-8625551

Steps: 1 Yield: 100%

1.1 Catalysts: Potassium carbonate
Solvents: Dimethylformamide; 6 h, 140 - 150 °C

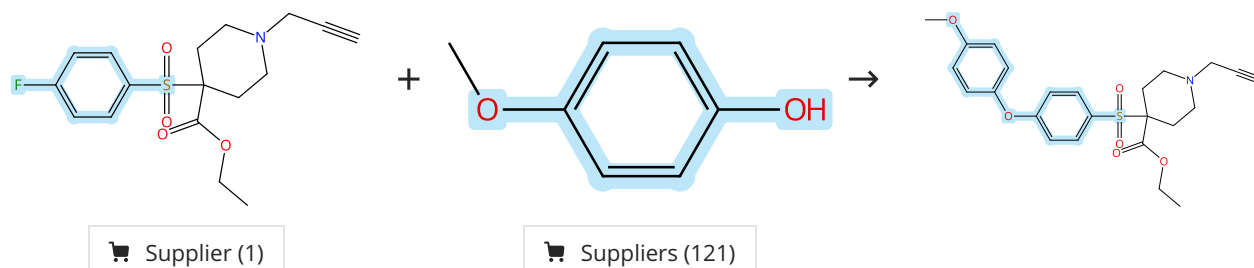
Synthesis and Structure-Activity Relationships for Extended Side Chain Analogues of the Antitubercular Drug (6S)-2-Nitro-6-[[4-(trifluoromethoxy)benzyl]oxy]-6,7-dihydro-5H-imidazo[2,1-b][1,3]oxazine (PA-824)

By: Palmer, Brian D.; et al

Journal of Medicinal Chemistry (2015), 58(7), 3036-3059.

Scheme 76 (1 Reaction)

Steps: 1 Yield: 100%



31-170-CAS-1937097

Steps: 1 Yield: 100%

1.1 Reagents: Cesium carbonate
Solvents: Dimethylformamide; 24 h, 95 °C

Experimental Protocols

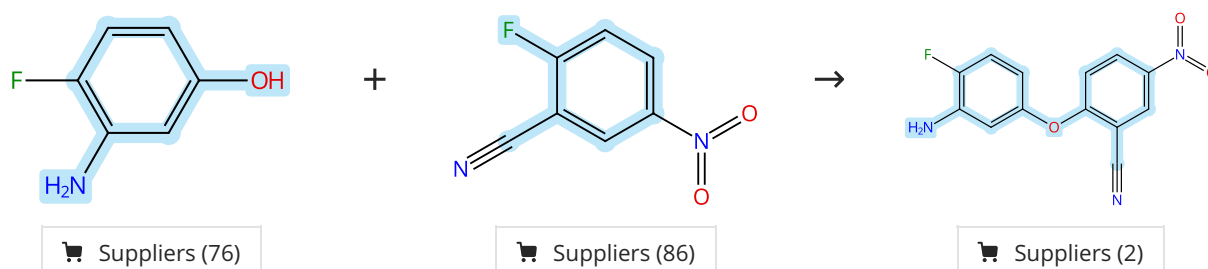
Orally Active MMP-1 Sparing α -Tetrahydropyranyl and α -Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease

By: Becker, Daniel P.; et al

Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.

Scheme 77 (1 Reaction)

Steps: 1 Yield: 100%



31-170-CAS-4572243

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; rt; 18 h, 80 °C

Experimental Protocols

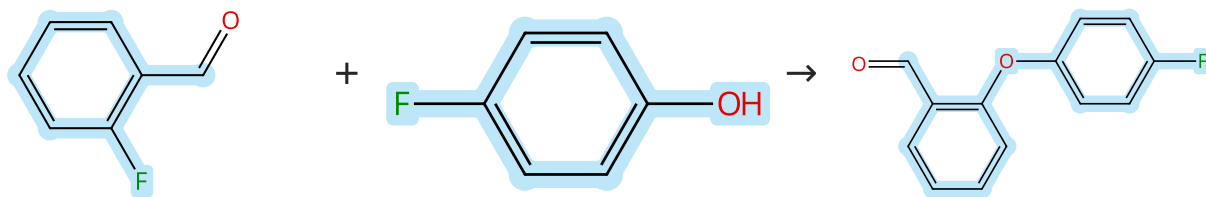
Discovery of a Selective Kinase Inhibitor (TAK-632) Targeting Pan-RAF Inhibition: Design, Synthesis, and Biological Evaluation of C-7-Substituted 1,3-Benzothiazole Derivatives

By: Okaniwa, Masanori; et al

Journal of Medicinal Chemistry (2013), 56(16), 6478-6494.

Scheme 78 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (86)

Suppliers (44)

31-614-CAS-36277497

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

By: Can Uskup, Hacer; et al

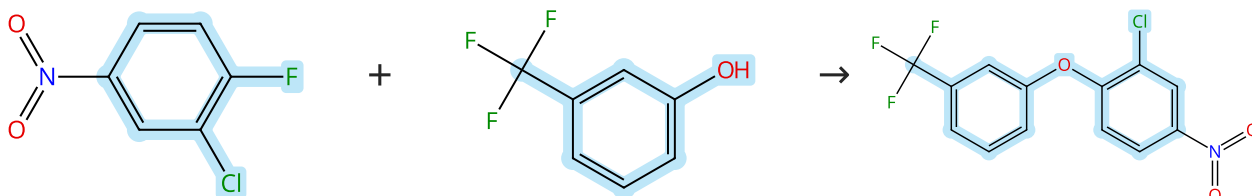
ACS Omega (2023), 8(15), 14047-14052.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Scheme 79 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (89)

Suppliers (44)

31-170-CAS-19152389

Steps: 1 Yield: 100%

Surmounting the resistance against EGFR inhibitors through the development of thieno[2,3-d]pyrimidine-based dual EGF R/HER2 inhibitors

By: Milik, Sandra N.; et al

European Journal of Medicinal Chemistry (2018), 155, 316-336.

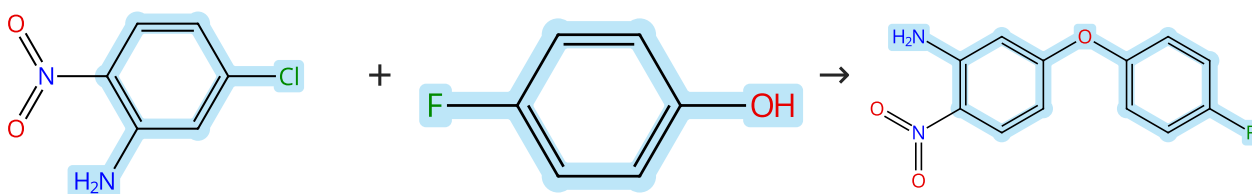
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 50 °C; cooled

1.2 **Reagents:** Water; cooled

Experimental Protocols

Scheme 80 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (97)

Suppliers (86)

Suppliers (4)

31-170-CAS-4125207

Steps: 1 Yield: 100%

Discovery of a novel class of potent and orally bioavailable sphingosine 1-phosphate receptor 1 antagonists

By: Ibrahim, Mohamed A.; et al

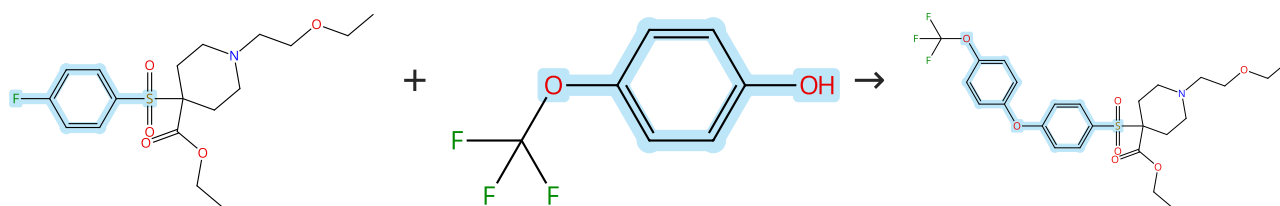
Journal of Medicinal Chemistry (2012), 55(3), 1368-1381.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 120 °C

Experimental Protocols

Scheme 81 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

31-170-CAS-9744122

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 25 h, 90 °C

Experimental Protocols

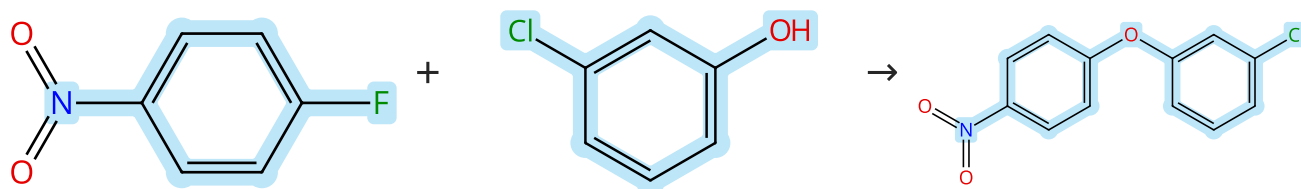
Orally Active MMP-1 Sparing α -Tetrahydropyranyl and α -Piperidiny Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease

By: Becker, Daniel P.; et al

Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.

Scheme 82 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (100)

Suppliers (85)

Suppliers (36)

31-170-CAS-2245250

Steps: 1 Yield: 100%

1.1 Catalysts: Cesium fluoride
Solvents: Dimethyl sulfoxide; 0.15 h, 110 °C

Experimental Protocols

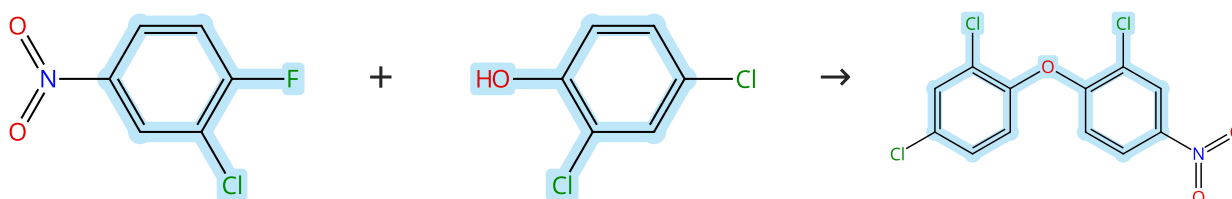
CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions

By: Keipour, Hoda; et al

Canadian Journal of Chemistry (2016), 94(1), 95-104.

Scheme 83 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (102)

Suppliers (20)

31-614-CAS-31980141

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; rt

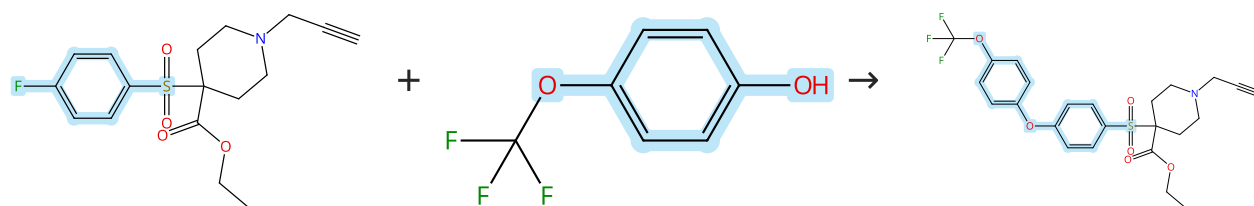
Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

Scheme 84 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (93)

31-614-CAS-40909850

Steps: 1 Yield: 100%

Carborane-Containing Matrix Metalloprotease (MMP) Ligands as Candidates for Boron Neutron-Capture Therapy (BNCT)

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 19 h, 90 °C

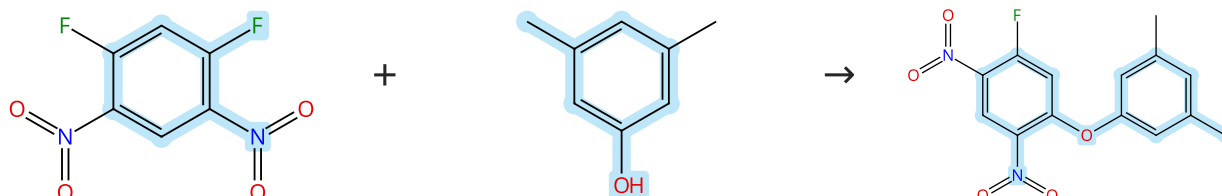
By: Lutz, Marlon R. Jr.; et al

Experimental Protocols

ChemMedChem (2020), 15(20), 1897-1908.

Scheme 85 (2 Reactions)

Steps: 1 Yield: 98-100%



Suppliers (64)

Suppliers (89)

31-170-CAS-1607781

Steps: 1 Yield: 100%

Dynamic Covalent Chemistry of Disulfides Offers a Highly Efficient Synthesis of Diverse Benzofused Nitrogen-Sulfur Heterocycles in One Pot

1.1 **Reagents:** Potassium carbonate
Solvents: Acetone; 5 h, rt

By: Zhu, Ning; et al

Experimental Protocols

Journal of Combinatorial Chemistry (2010), 12(4), 531-540.

31-170-CAS-2316859

Steps: 1 Yield: 98%

Solution-Phase Parallel Synthesis of Diverse 1,5-Benzodiazepin-2-ones

1.1 **Reagents:** Potassium carbonate
Solvents: Acetone; 5 h, rt

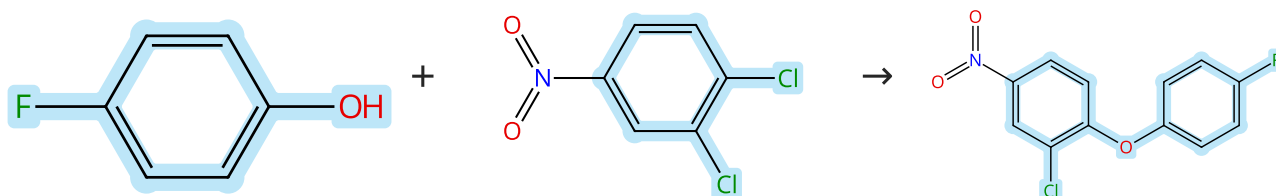
By: Zhao, Hai-Yan; et al

Experimental Protocols

Journal of Combinatorial Chemistry (2007), 9(6), 1164-1176.

Scheme 86 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

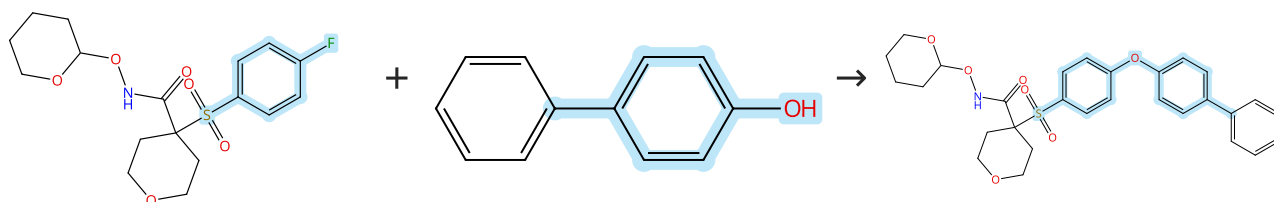
Suppliers (63)

Suppliers (13)

31-170-CAS-3972090 Steps: 1 Yield: 100%	Translation of microwave methodology to continuous flow for the efficient synthesis of diaryl ethers via a base-mediated S_NAr reaction By: Wiles, Charlotte; et al Beilstein Journal of Organic Chemistry (2011), 7, 1360-1371.
1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Solvents: Acetonitrile; 1 min, 25 bar, 195 °C	

Scheme 87 (1 Reaction)

Steps: 1 Yield: 100%



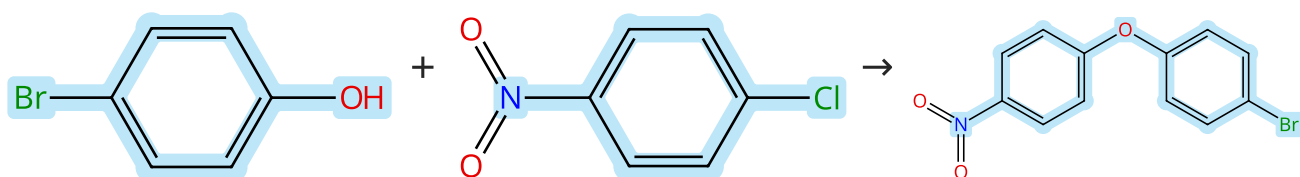
Supplier (1)

Suppliers (105)

31-170-CAS-15143646 Steps: 1 Yield: 100%	Orally Active MMP-1 Sparing α-Tetrahydropyranyl and α-Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease By: Becker, Daniel P.; et al Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.
1.1 Reagents: Cesium carbonate Solvents: Dimethylacetamide; 5 h, 95 °C	
Experimental Protocols	

Scheme 88 (2 Reactions)

Steps: 1 Yield: 98-100%



Suppliers (82)

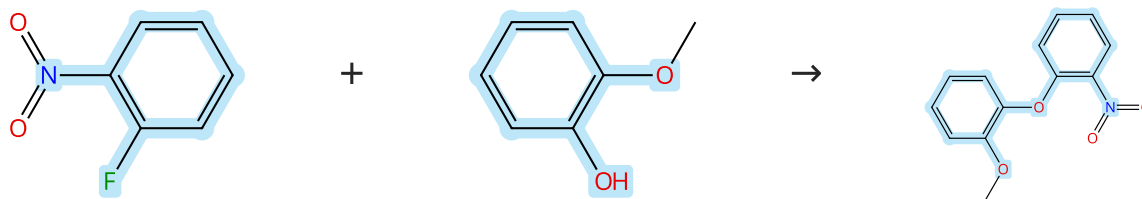
Suppliers (65)

Suppliers (54)

31-170-CAS-17492721 Steps: 1 Yield: 100%	The effect of O-nucleophiles structure on the features of halogen substitution in heterogeneous systems By: Orlov, V. Yu.; et al Russian Journal of General Chemistry (2017), 87(3), 386-388.
1.1 Reagents: Potassium carbonate, Iron oxide (Fe ₂ O ₃) Solvents: Dimethylformamide; 270 min, 130 °C Experimental Protocols	
31-170-CAS-10765026 Steps: 1 Yield: 98%	CuI Nanoparticles for C-N and C-O Cross Coupling of Heterocyclic Amines and Phenols with Chlorobenzenes By: Sreedhar, B.; et al Journal of Organic Chemistry (2009), 74(20), 7951-7954.
1.1 Reagents: Potassium carbonate Catalysts: Cuprous iodide Solvents: Dimethylformamide; 30 min, rt; 2 h, 110 °C; 110 °C → rt Experimental Protocols	

Scheme 89 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (125)

Suppliers (10)

31-170-CAS-15203573

Steps: 1 Yield: 100%

A Fast Way to Fluorescence: A Fourfold Domino Reaction to Condensed Polycyclic Compounds

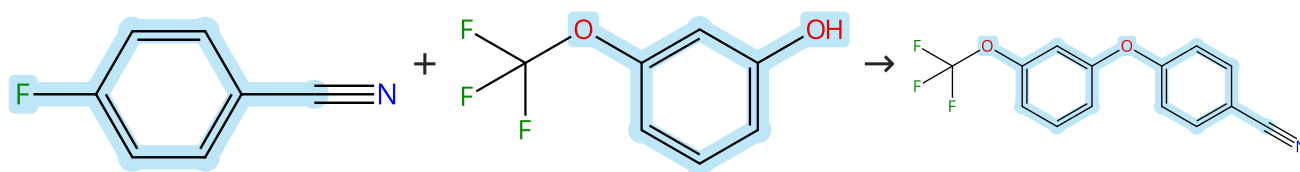
1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; 20 h, 95 °C

By: Tietze, Lutz F.; et al

Chemistry - A European Journal (2014), 20(39), 12553-12558.

Scheme 90 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (92)

Suppliers (74)

31-170-CAS-10854798

Steps: 1 Yield: 100%

Flavones as isosteres of 4(1H)-quinolones: Discovery of ligand efficient and dual stage antimalarial lead compounds

1.1 Reagents: Sodium carbonate
Solvents: Dimethylformamide; 24 h, reflux

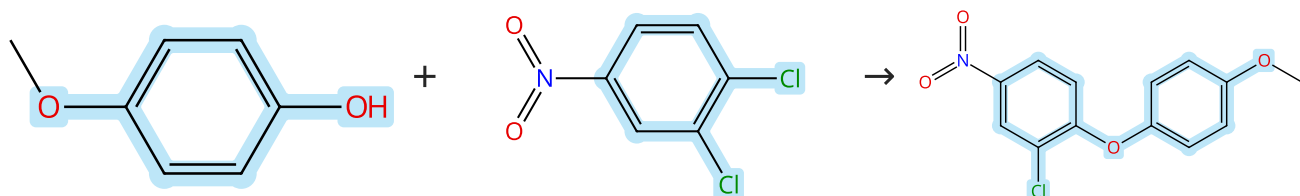
By: Rodrigues, Tiago; et al

Experimental Protocols

European Journal of Medicinal Chemistry (2013), 69, 872-880.

Scheme 91 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (121)

Suppliers (63)

Suppliers (14)

31-170-CAS-2042985

Steps: 1 Yield: 100%

Translation of microwave methodology to continuous flow for the efficient synthesis of diaryl ethers via a base-mediated S_NAr reaction

1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene
Solvents: Acetonitrile; 60 s, 25 bar, 195 °C

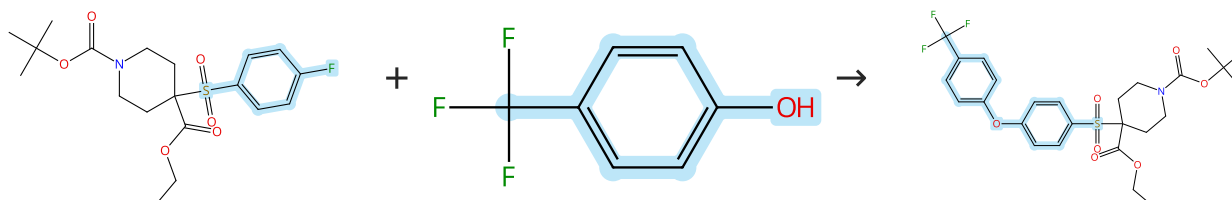
By: Wiles, Charlotte; et al

Beilstein Journal of Organic Chemistry (2011), 7, 1360-1371.

31-170-CAS-10967327 Steps: 1 Yield: 100%	Translation of microwave methodology to continuous flow for the efficient synthesis of diaryl ethers via a base-mediated S_NAr reaction By: Wiles, Charlotte; et al Beilstein Journal of Organic Chemistry (2011), 7, 1360-1371.
1.1 Reagents: Potassium carbonate Solvents: Acetonitrile; 1 min, 25 bar, 195 °C	

Scheme 92 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

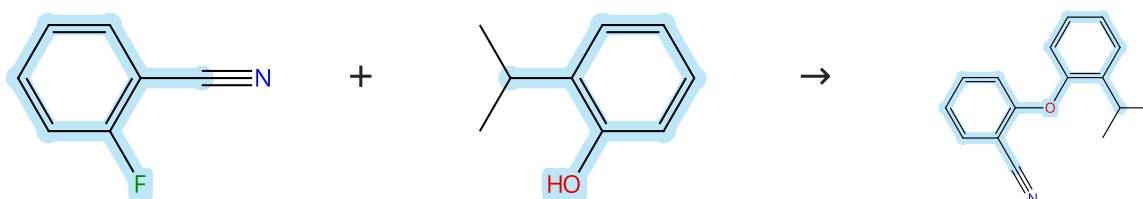
Suppliers (96)

Supplier (1)

31-170-CAS-7096406 Steps: 1 Yield: 100%	Orally Active MMP-1 Sparing α-Tetrahydropyranyl and α-Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease By: Becker, Daniel P.; et al Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 20 h, 90 °C; 90 °C → rt	
Experimental Protocols	

Scheme 93 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

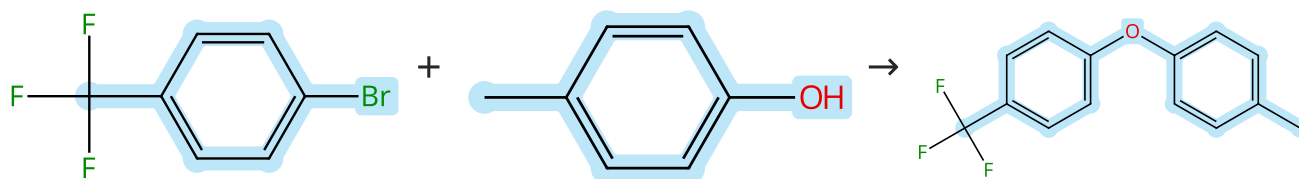
Suppliers (76)

Suppliers (22)

31-614-CAS-32528352 Steps: 1 Yield: 100%	Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement By: Jinno, Shunki; et al Organic Letters (2022), 24(23), 4140-4144.
1.1 Reagents: Potassium hydroxide Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt	
1.2 Reagents: Water; rt Experimental Protocols	

Scheme 94 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (79)

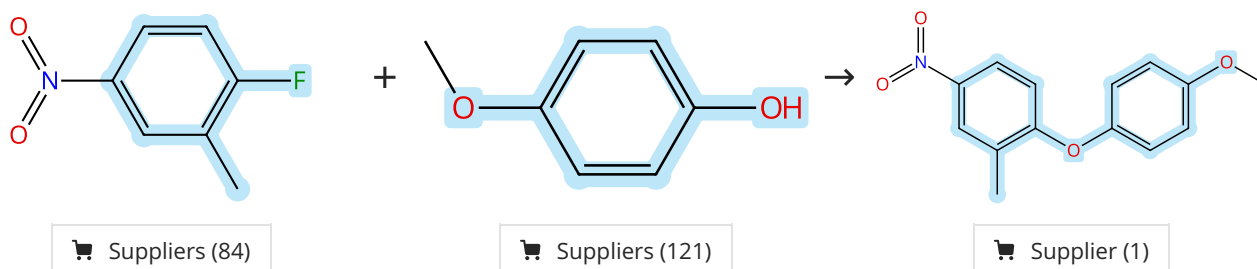
Suppliers (89)

Suppliers (6)

31-170-CAS-12949463 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: Potassium iodide, Cuprous iodide, <i>N,N'</i>-[2,2'-Bipyridine]-4,4'-diylbis[<i>N</i>'-[3-(trihydroxysilyl)propyl]urea] (Sol-Gel immobilized) Solvents: Dimethylformamide; rt; rt → 120 °C; 22 h, 120 °C; 120 °C → rt Experimental Protocols	Efficient and versatile sol-gel immobilized copper catalyst for Ullmann arylation of phenols By: Benyahya, Sofia; et al Advanced Synthesis & Catalysis (2008), 350(14+15), 2205-2208.
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Scheme 95 (1 Reaction)

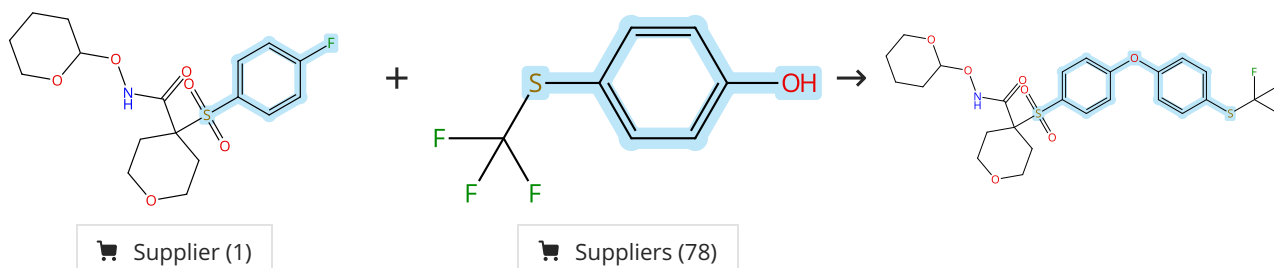
Steps: 1 Yield: 100%



31-614-CAS-41227155 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide, Ethyl acetate; 22 h, 110 °C; 110 °C → rt 1.2 Solvents: Water; rt Experimental Protocols	MPoMA protects against lung epithelial cell injury via p65 degradation By: Lee, Soheun; et al Biomedicine & Pharmacotherapy (2024), 175, 116674.
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Scheme 96 (1 Reaction)

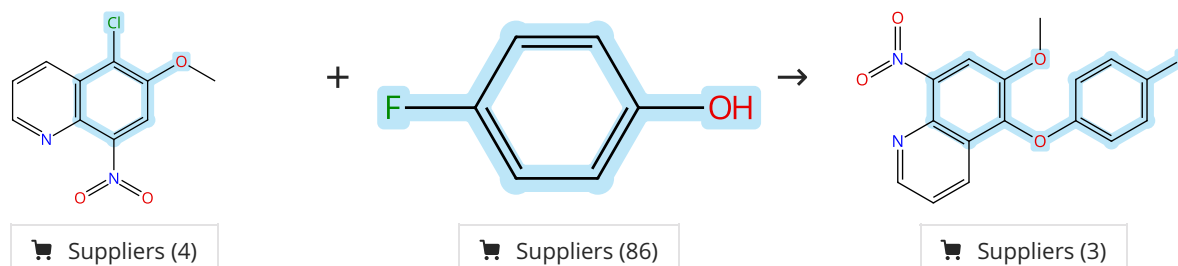
Steps: 1 Yield: 100%



31-170-CAS-2263699 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: Potassium fluoride Solvents: Dimethylacetamide; 12 h, 95 °C Experimental Protocols	Orally Active MMP-1 Sparing α-Tetrahydropyranyl and α-Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease By: Becker, Daniel P.; et al Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.
--	---

Scheme 97 (1 Reaction)

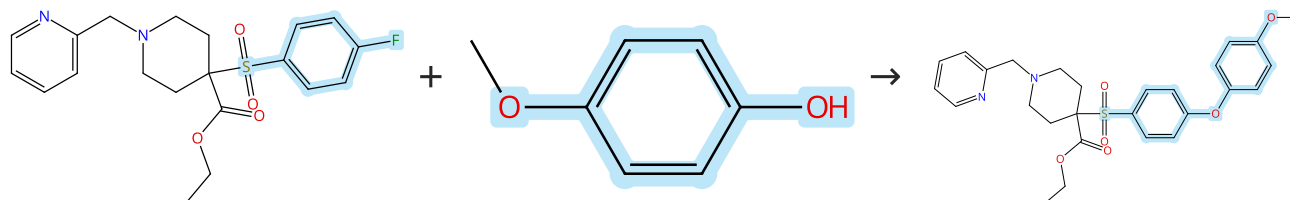
Steps: 1 Yield: 100%



31-614-CAS-24716243	5-Phenoxy Primaquine Analogs and the Tetraoxane Hybrid as Antimalarial Agents
1.1 Reagents: Lithium hydroxide Solvents: Dimethyl sulfoxide; rt; 4 h, 100 °C	By: Jansongsaeng, Somruedee; et al
1.2 Reagents: Water Experimental Protocols	Molecules (2021), 26(13), 3991.

Scheme 98 (1 Reaction)

Steps: 1 Yield: 100%

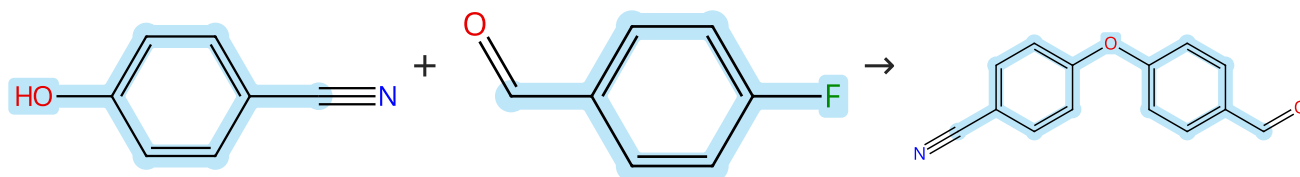


Suppliers (121)

31-170-CAS-12556729	Orally Active MMP-1 Sparing α-Tetrahydropyranyl and α-Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 75 - 80 °C; 2 d, rt	By: Becker, Daniel P.; et al
Experimental Protocols	Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.

Scheme 99 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

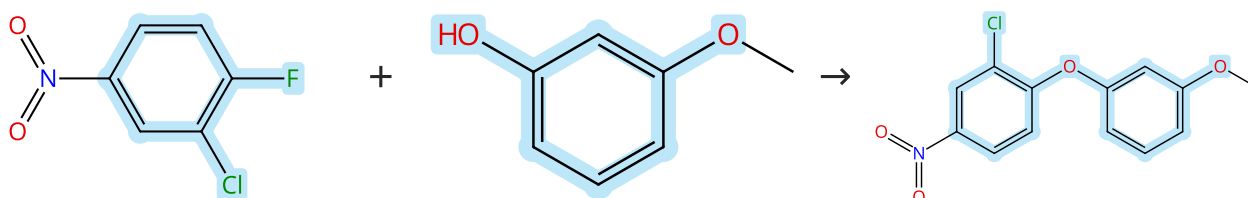
Suppliers (93)

Suppliers (68)

31-614-CAS-34652762	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt $\rightarrow$ 100 °C; 100 °C $\rightarrow$ rt	By: Chen, Tao; et al
1.2 Reagents: Water; rt Experimental Protocols	Letters in Drug Design & Discovery (2022), 19(4), 263-268.

Scheme 100 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

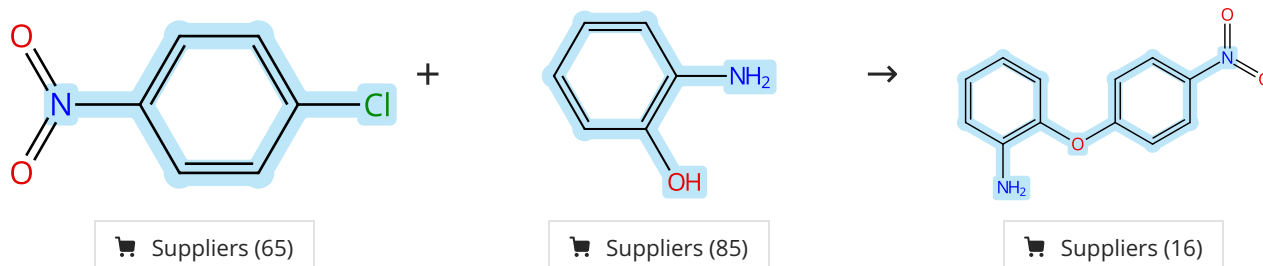
Suppliers (92)

Suppliers (10)

31-170-CAS-23045842	Steps: 1 Yield: 100%	Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 17 h, rt		By: Fujii, Shinya; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

Scheme 101 (1 Reaction)

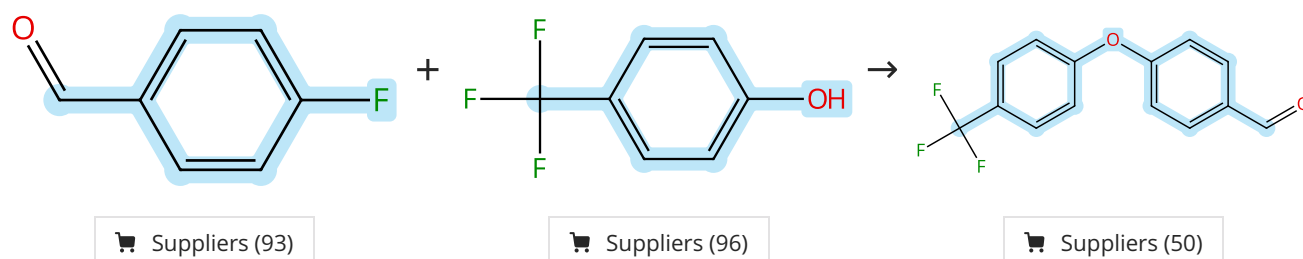
Steps: 1 Yield: 100%



31-170-CAS-17492718	Steps: 1 Yield: 100%	The effect of O-nucleophiles structure on the features of halogen substitution in heterogeneous systems
1.1 Reagents: Potassium carbonate, Iron oxide (Fe ₂ O ₃) Solvents: Dimethylformamide; 270 min, 130 °C		By: Orlov, V. Yu.; et al
Experimental Protocols		Russian Journal of General Chemistry (2017), 87(3), 386-388.

Scheme 102 (2 Reactions)

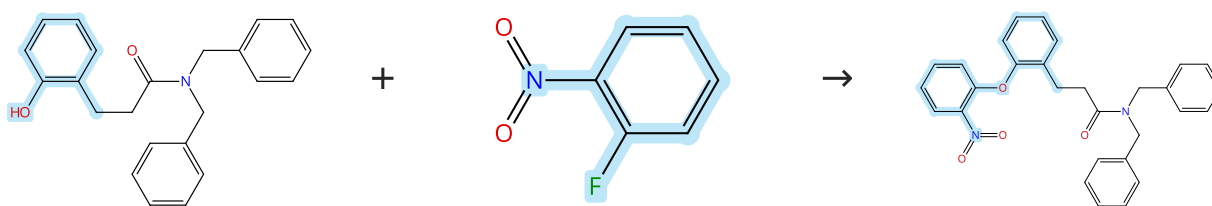
Steps: 1 Yield: 100%



31-614-CAS-34652760	Steps: 1 Yield: 100%	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt		By: Chen, Tao; et al
1.2 Reagents: Water; rt		Letters in Drug Design & Discovery (2022), 19(4), 263-268.
Experimental Protocols		
31-170-CAS-12891905	Steps: 1 Yield: 100%	Synthesis and Structure-Activity Relationships for Extended Side Chain Analogues of the Antitubercular Drug (6S)-2-Nitro-6-[[4-(trifluoromethoxy)benzyl]oxy]-6,7-dihydro-5H-imidazo[2,1-b][1,3]oxazine (PA-824)
1.1 Catalysts: Potassium carbonate Solvents: Dimethylformamide; 6 h, 140 - 150 °C		By: Palmer, Brian D.; et al
		Journal of Medicinal Chemistry (2015), 58(7), 3036-3059.

Scheme 103 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (93)

31-170-CAS-10668230

Steps: 1 Yield: 100%

Developing the Scope of O→C Aryl Migrations: Exploring Amide Substrates as Potential Precursors for Asymmetric Reactions

By: Ameen, Dana; et al

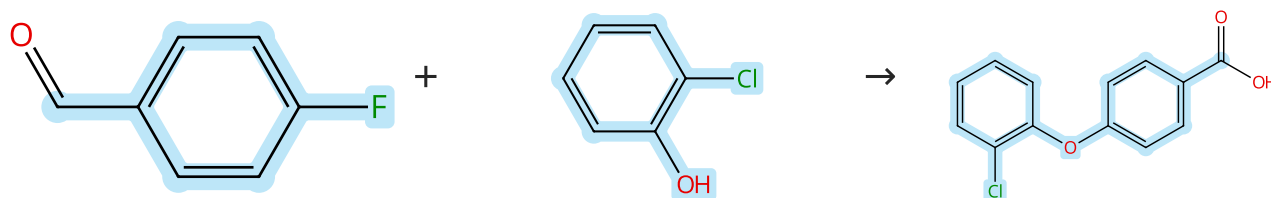
European Journal of Organic Chemistry (2014), 2014(9), 1925-1934.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; 30 min, rt
- 1.2 24 h, 60 °C
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Water; acidified

Experimental Protocols

Scheme 104 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (92)

Suppliers (45)

31-170-CAS-8604591

Steps: 1 Yield: 100%

Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist

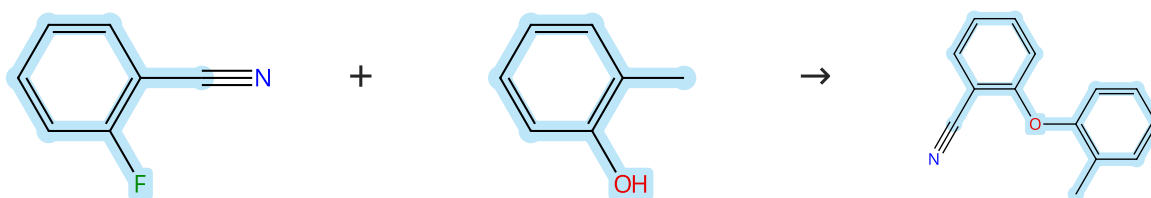
By: Nakamura, Tsuyoshi; et al

Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 100 °C
- 1.2 **Reagents:** 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate
Solvents: *tert*-Butanol, Tetrahydrofuran, Water

Scheme 105 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (83)

Suppliers (25)

31-614-CAS-32528349

Steps: 1 Yield: 100%

Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement

By: Jinno, Shunki; et al

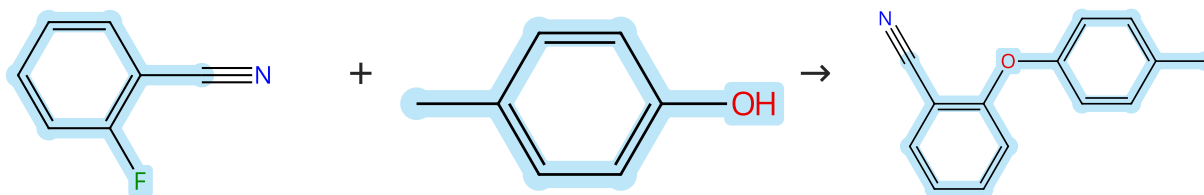
Organic Letters (2022), 24(23), 4140-4144.

- 1.1 **Reagents:** Potassium hydroxide
Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt
- 1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 106 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (89)

Suppliers (20)

31-614-CAS-38942833

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; 15 h, 90 °C; 90 °C → rt

1.2 Reagents: Water; rt

Experimental Protocols

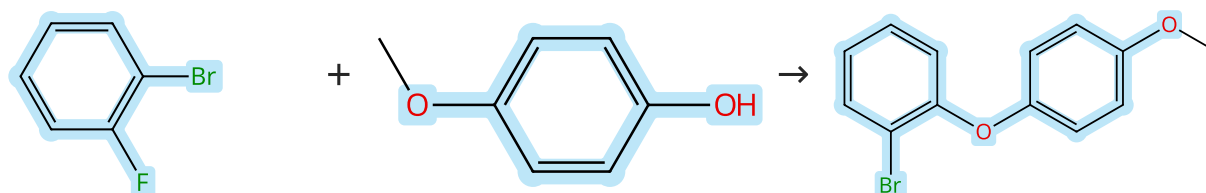
Update of the Imine-Anion-Mediated Smiles Rearrangement: Application to Migration of Electron-Neutral/Rich Aromatic Groups

By: Jinno, Shunki; et al

Synlett (2024), 35(13), 1565-1568.

Scheme 107 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (85)

Suppliers (121)

Suppliers (3)

31-170-CAS-8151247

Steps: 1 Yield: 100%

1.1 Reagents: (Diethylamino)trimethylsilane
Catalysts: *N*''-(1,1-Dimethylethyl)-*N,N,N*'-tris[tris(dimethylamino)phosphoranylidene]phosphorimidic triamide
Solvents: Dimethylformamide; 48 h, 100 °C

Experimental Protocols

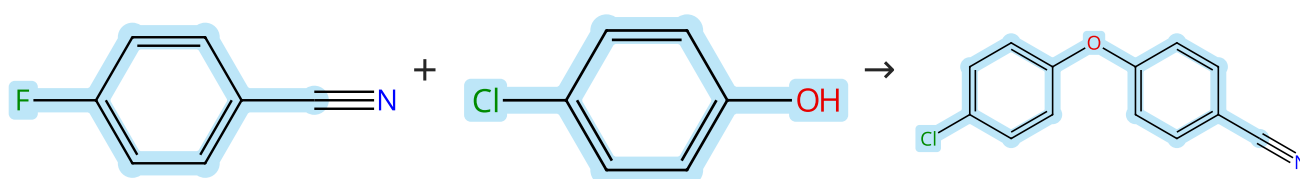
Synthesis of dictyomedins using phosphazene base catalyzed diaryl ether formation

By: Ebisawa, Masaru; et al

Tetrahedron Letters (2007), 48(50), 8918-8921.

Scheme 108 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (92)

Suppliers (90)

Suppliers (62)

31-170-CAS-4471682

Steps: 1 Yield: 100%

1.1 Reagents: Sodium carbonate
Solvents: Dimethylformamide; 24 h, reflux

Experimental Protocols

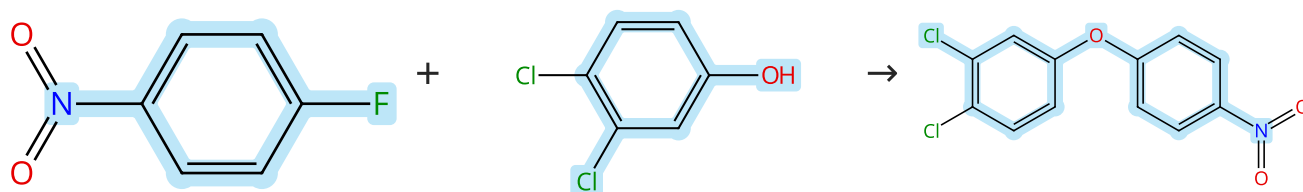
Flavones as isosteres of 4(1H)-quinolones: Discovery of ligand efficient and dual stage antimalarial lead compounds

By: Rodrigues, Tiago; et al

European Journal of Medicinal Chemistry (2013), 69, 872-880.

Scheme 109 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (100)

Suppliers (66)

Suppliers (11)

31-614-CAS-37615374

Steps: 1 Yield: 100%

Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors

By: Zhang, Haoming; et al

ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.

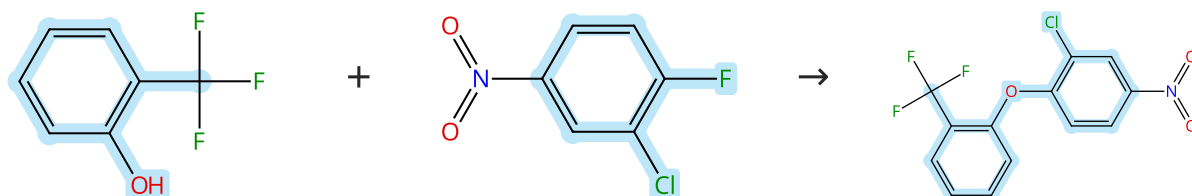
1.1 Reagents: Cesium carbonate

Solvents: Dimethylformamide; 6 h, 50 °C

Experimental Protocols

Scheme 110 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

Suppliers (86)

31-170-CAS-23040623

Steps: 1 Yield: 100%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

By: Fujii, Shinya; et al

Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

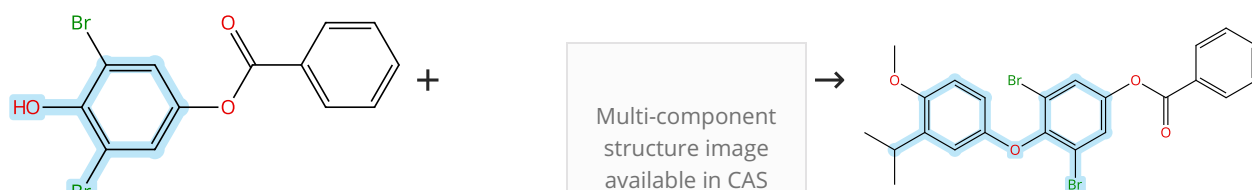
1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 17 h, rt

Experimental Protocols

Scheme 111 (1 Reaction)

Steps: 1 Yield: 100%



Multi-component structure image available in CAS SciFinder

Supplier (1)

Suppliers (19)

31-170-CAS-7194071

Steps: 1 Yield: 100%

Synthesis and Biological Evaluation of a Series of Liver-Selective Phosphonic Acid Thyroid Hormone Receptor Agonists and Their Prodrugs

By: Boyer, Serge H.; et al

Journal of Medicinal Chemistry (2008), 51(22), 7075-7093.

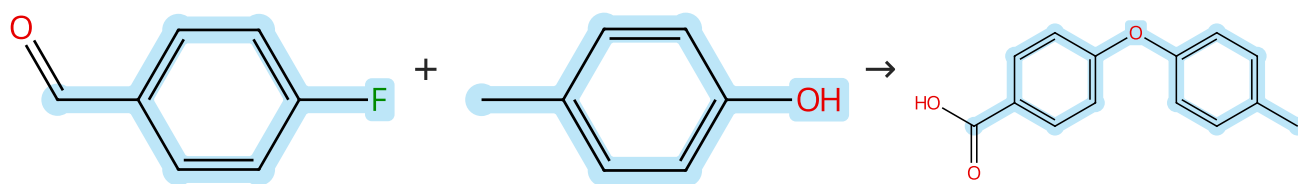
1.1 Reagents: Triethylamine, Copper

Solvents: Dichloromethane; 0 °C; 24 h, rt

Experimental Protocols

Scheme 112 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (89)

Suppliers (67)

31-170-CAS-11320466

Steps: 1 Yield: 100%

Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist

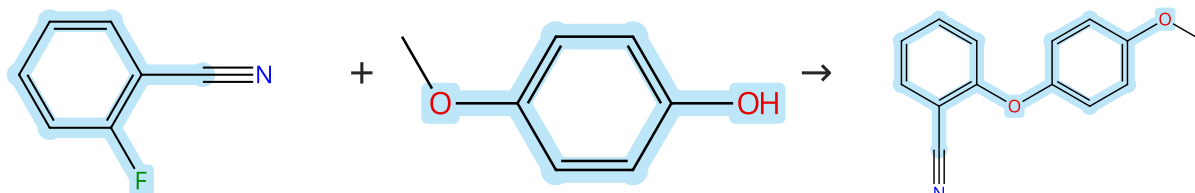
By: Nakamura, Tsuyoshi; et al

Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 100 °C
- 1.2 **Reagents:** 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate
Solvents: *tert*-Butanol, Tetrahydrofuran, Water

Scheme 113 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (121)

Suppliers (38)

31-170-CAS-12136641

Steps: 1 Yield: 100%

Synthesis of dictyomedins using phosphazene base catalyzed diaryl ether formation

By: Ebisawa, Masaru; et al

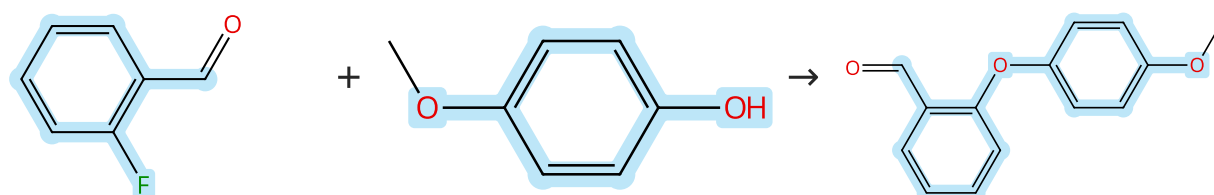
Tetrahedron Letters (2007), 48(50), 8918-8921.

- 1.1 **Reagents:** (Diethylamino)trimethylsilane
Catalysts: *N*''-(1,1-Dimethylethyl)-*N,N,N*'-tris[tris(dimethylamino)phosphoranylidene]phosphorimidic triamide
Solvents: Dimethylformamide; 2 h, rt

Experimental Protocols

Scheme 114 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (121)

Suppliers (56)

31-614-CAS-36277496

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

By: Can Uskup, Hacer; et al

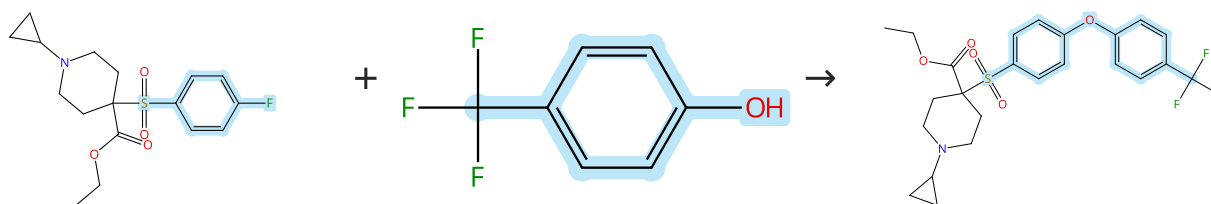
ACS Omega (2023), 8(15), 14047-14052.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Scheme 115 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (96)

31-170-CAS-8423442

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 28 h, 90 °C; 90 °C → rt

Experimental Protocols

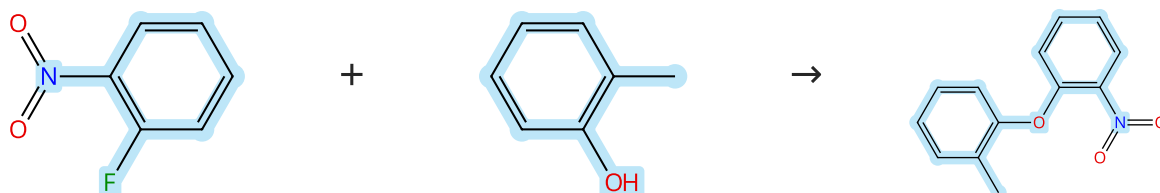
Orally Active MMP-1 Sparing α -Tetrahydropyranyl and α -PiperidinyI Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease

By: Becker, Daniel P.; et al

Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.

Scheme 116 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (83)

Suppliers (40)

31-170-CAS-2758563

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; 20 h, 95 °C

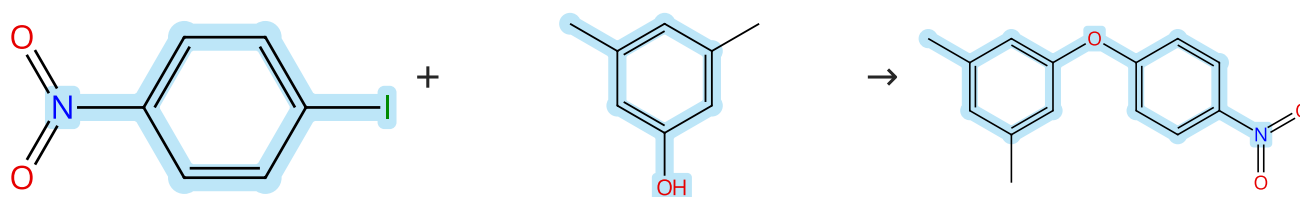
A Fast Way to Fluorescence: A Fourfold Domino Reaction to Condensed Polycyclic Compounds

By: Tietze, Lutz F.; et al

Chemistry - A European Journal (2014), 20(39), 12553-12558.

Scheme 117 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

Suppliers (89)

Suppliers (19)

31-170-CAS-4725031

Steps: 1 Yield: 100%

1.1 Reagents: Cesium carbonate
Solvents: Dimethylformamide; 18 h, 90 °C

Experimental Protocols

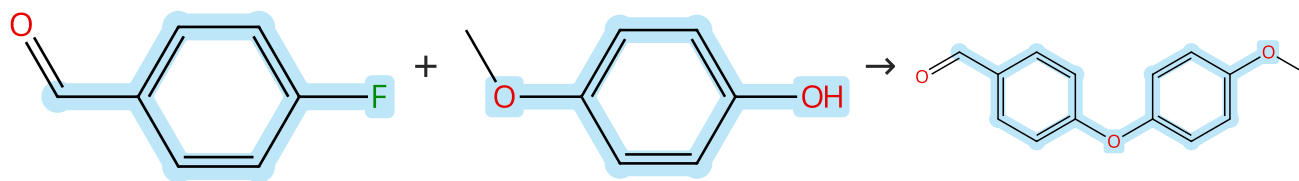
On the frontier between nucleophilic aromatic substitution and catalysis

By: Pichette Drapeau, Martin; et al

Chemistry - A European Journal (2014), 20(18), 5231-5236.

Scheme 118 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (121)

Suppliers (60)

31-614-CAS-41211204

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

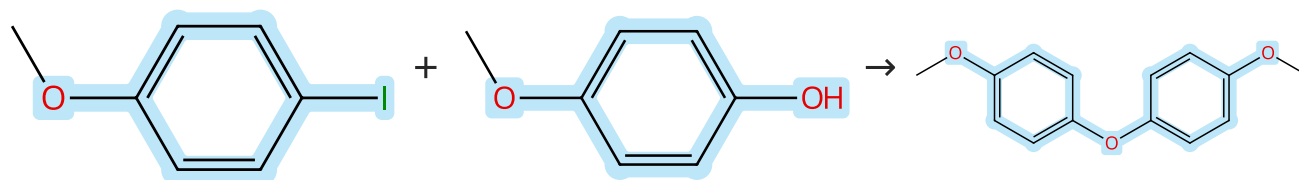
Synthesis and Characterization of New Bulky α -Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

Scheme 119 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (89)

Suppliers (121)

Suppliers (13)

31-170-CAS-7545168

Steps: 1 Yield: 100%

1.1 Reagents: Cesium carbonate
Catalysts: Copper
Solvents: Dimethylacetamide; 24 h, 140 °C

Experimental Protocols

A Cu Nanoparticle Embedded in Electrospondoped Carbon Nanofibers as Efficient Catalysts for Ullmann O-Arylation of Aryl Halides with Various Phenols

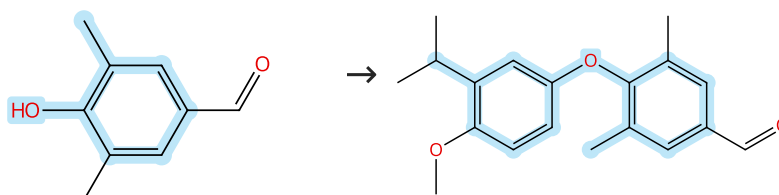
By: Zhang, Chenglin; et al

Catalysis Letters (2015), 145(9), 1764-1770.

Scheme 120 (1 Reaction)

Steps: 1 Yield: 100%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (19)

Suppliers (90)

Suppliers (3)

31-170-CAS-625431

Steps: 1 Yield: 100%

1.1 Reagents: Triethylamine, Copper
Solvents: Dichloromethane; 0 °C; 72 h, rt

Experimental Protocols

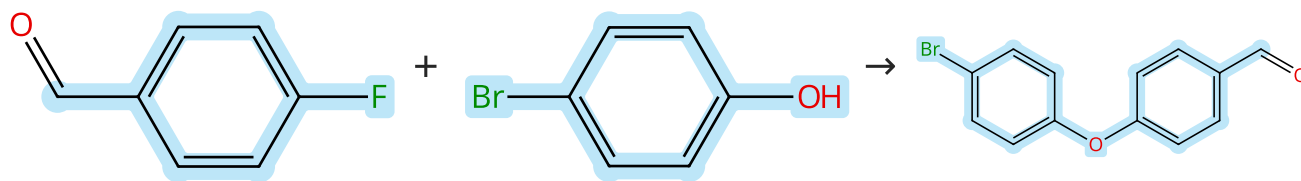
Synthesis and Biological Evaluation of a Series of Liver-Selective Phosphonic Acid Thyroid Hormone Receptor Agonists and Their Prodrugs

By: Boyer, Serge H.; et al

Journal of Medicinal Chemistry (2008), 51(22), 7075-7093.

Scheme 121 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (82)

Suppliers (71)

31-614-CAS-41211201

Steps: 1 Yield: 100%

Synthesis and Characterization of New Bulky α -Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

31-614-CAS-34652753

Steps: 1 Yield: 100%

Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt $\rightarrow$ 100 °C; 100 °C $\rightarrow$ rt

1.2 Reagents: Water; rt

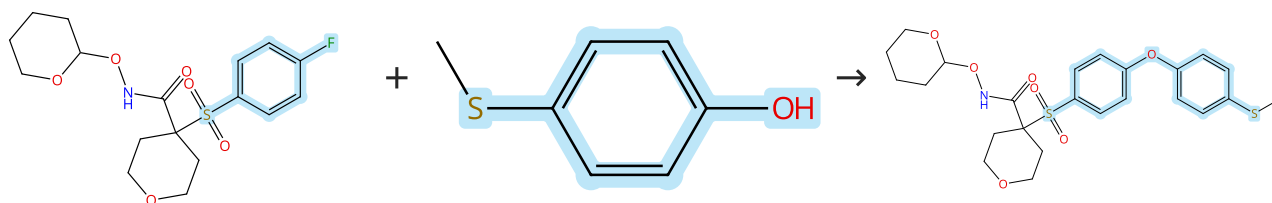
Experimental Protocols

By: Chen, Tao; et al

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

Scheme 122 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (62)

31-170-CAS-7815367

Steps: 1 Yield: 100%

Orally Active MMP-1 Sparing α -Tetrahydropyranyl and α -PiperidinyI Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease

1.1 Reagents: Cesium carbonate
Solvents: Dimethylacetamide; 24 h, 95 °C

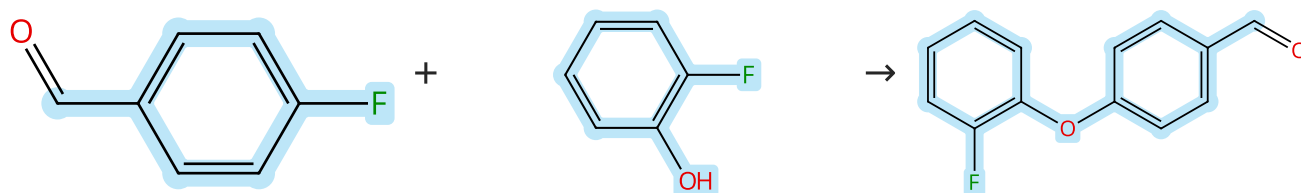
Experimental Protocols

By: Becker, Daniel P.; et al

Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.

Scheme 123 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (101)

Suppliers (52)

31-614-CAS-34652754	Steps: 1 Yield: 100%	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt → 100 °C; 100 °C → rt		By: Chen, Tao; et al
1.2 Reagents: Water; rt		Letters in Drug Design & Discovery (2022), 19(4), 263-268.
Experimental Protocols		

Scheme 124 (5 Reactions)		Steps: 1 Yield: 98-100%
Suppliers (83)	Suppliers (81)	Suppliers (22)

31-614-CAS-32016366	Steps: 1 Yield: 100%	Structure-guided discovery of novel potent and efficacious proteolysis targeting chimera (PROTAC) degrader of BRD4
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, rt		By: Xiang, Wang; et al
Experimental Protocols		Bioorganic Chemistry (2021), 115, 105238.

31-170-CAS-17836065	Steps: 1 Yield: 100%	Discovery of N-(4-(2,4-Difluorophenoxy)-3-(6-methyl-7-oxo-6,7-dihydro-1H-pyrrolo[2,3-c]pyridin-4-yl)phenyl)ethanesulfonamide (ABBV-075/Mivebresib), a Potent and Orally Available Bromodomain and Extraterminal Domain (BET) Family Bromodomain Inhibitor
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 1 h, rt → 110 °C		By: McDaniel, Keith F.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2017), 60(20), 8369-8384.

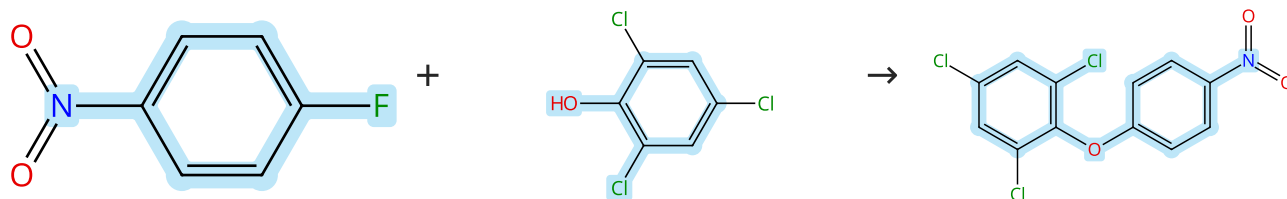
31-170-CAS-17085585	Steps: 1 Yield: 100%	Fragment-Based, Structure-Enabled Discovery of Novel Pyridones and Pyridone Macrocycles as Potent Bromodomain and Extra-Terminal Domain (BET) Family Bromodomain Inhibitors
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; rt → 110 °C; 60 min, 110 °C; cooled		By: Wang, Le; et al
1.2 Reagents: Sodium chloride Solvents: Water; rt		Journal of Medicinal Chemistry (2017), 60(9), 3828-3850.
Experimental Protocols		

31-614-CAS-38998627	Steps: 1 Yield: 98%	Novel Triazolopyridine-Based BRD4 Inhibitors as Potent HIV-1 Latency Reversing Agents
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; rt → 110 °C; 2 h, 110 °C		By: Wang, Yan-Kai; et al
Experimental Protocols		ACS Medicinal Chemistry Letters (2024), 15(1), 60-68.

31-614-CAS-39886652	Steps: 1 Yield: 98%	Structure-Based Discovery and Optimization of Furo[3,2-c]pyridin-4(5H)-one Derivatives as Potent and Second Bromodomain (BD2)-Selective Bromodomain and Extra Terminal Domain (BET) Inhibitors
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 4 h, rt → 80 °C		By: Li, Junhua; et al
Experimental Protocols		Journal of Medicinal Chemistry (2022), 65(7), 5760-5799.

Scheme 125 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (100)

Suppliers (87)

Suppliers (30)

31-170-CAS-6117971

Steps: 1 Yield: 100%

1.1 **Catalysts:** Cesium fluoride
Solvents: Dimethyl sulfoxide; 3 h, 110 °C

Experimental Protocols

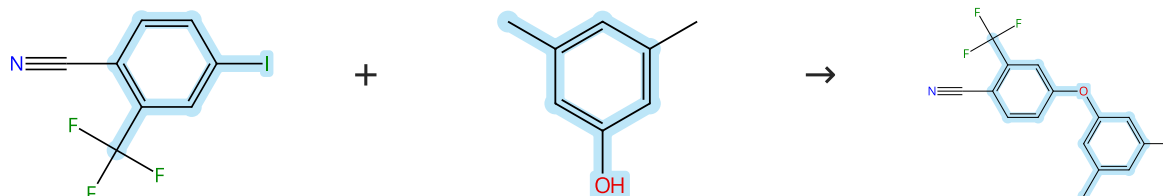
CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions

By: Keipour, Hoda; et al

Canadian Journal of Chemistry (2016), 94(1), 95-104.

Scheme 126 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

Suppliers (89)

Suppliers (2)

31-614-CAS-43350516

Steps: 1 Yield: 100%

1.1 **Reagents:** Cesium carbonate
Catalysts: Cuprous iodide, 1*H*-imidazolium, 3-[4-(trifluoromethyl)-2-pyridinyl]-1-(2,4,6-trimethylphenyl)-, bromide (1:1)
Solvents: Acetonitrile; 20 h, 110 °C

Experimental Protocols

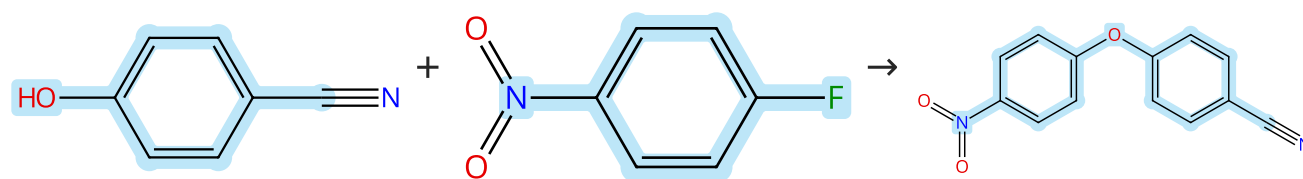
Bidentate NHC-Containing Ligands for Copper Catalyzed Synthesis of Functionalised Diaryl Ethers

By: Richards, Ethan M.; et al

Advanced Synthesis & Catalysis (2025), 367(6), e202400909.

Scheme 127 (2 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (90)

Suppliers (100)

Suppliers (50)

31-170-CAS-10989327

Steps: 1 Yield: 100%

1.1 **Catalysts:** Cesium fluoride
Solvents: Dimethyl sulfoxide; 0.5 h, 110 °C

Experimental Protocols

CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions

By: Keipour, Hoda; et al

Canadian Journal of Chemistry (2016), 94(1), 95-104.

31-170-CAS-19244823

Steps: 1 Yield: 99%

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 100 °C

Experimental Protocols

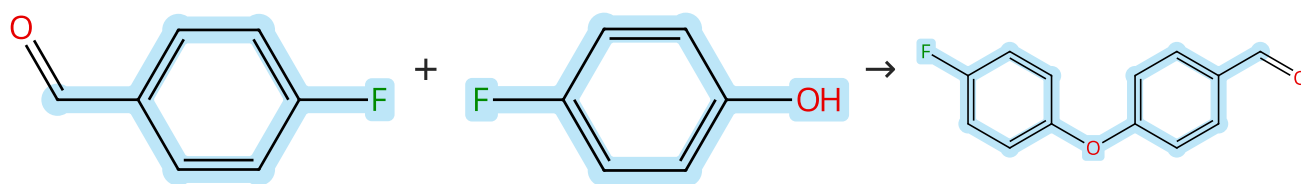
Synthesis and evaluation of aminobenzothiazoles as blockers of N- and T-type calcium channels

By: Sairaman, Anjali; et al

Bioorganic & Medicinal Chemistry (2018), 26(11), 3046-3059.

Scheme 128 (3 Reactions)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (86)

Suppliers (83)

31-614-CAS-41211198

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Synthesis and Characterization of New Bulky α -Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

31-614-CAS-34652756

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt $\rightarrow$ 100 °C; 100 °C $\rightarrow$ rt

1.2 Reagents: Water; rt

Experimental Protocols

Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

By: Chen, Tao; et al

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

31-170-CAS-10763003

Steps: 1 Yield: 100%

1.1 Catalysts: Potassium carbonate
Solvents: Dimethylformamide; 6 h, 140 - 150 °C

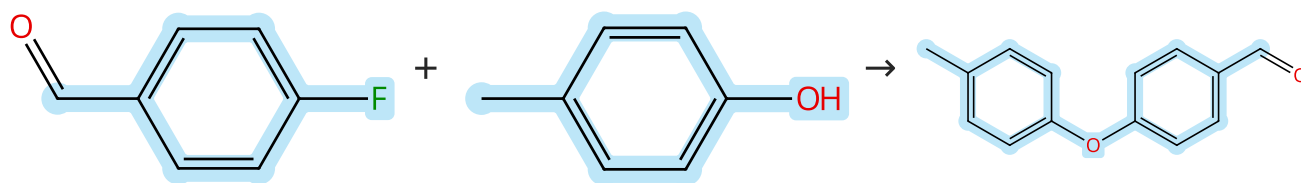
Synthesis and Structure-Activity Relationships for Extended Side Chain Analogues of the Antitubercular Drug (6S)-2-Nitro-6-[[4-(trifluoromethoxy)benzyl]oxy]-6,7-dihydro-5H-imidazo[2,1-b][1,3]oxazine (PA-824)

By: Palmer, Brian D.; et al

Journal of Medicinal Chemistry (2015), 58(7), 3036-3059.

Scheme 129 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (89)

Suppliers (55)

31-614-CAS-41211196

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Synthesis and Characterization of New Bulky α -Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

31-614-CAS-34652758

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 16 h, rt $\rightarrow$ 100 °C; 100 °C $\rightarrow$ rt

1.2 Reagents: Water; rt

Experimental Protocols

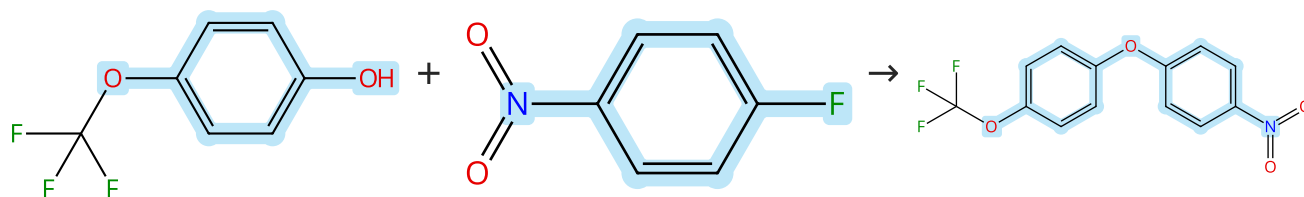
Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide

By: Chen, Tao; et al

Letters in Drug Design & Discovery (2022), 19(4), 263-268.

Scheme 130 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (100)

Suppliers (39)

31-170-CAS-6961561

Steps: 1 Yield: 100%

1.1 **Catalysts:** Potassium *tert*-butoxide
Solvents: Dimethyl sulfoxide; 2 h, 101 °C

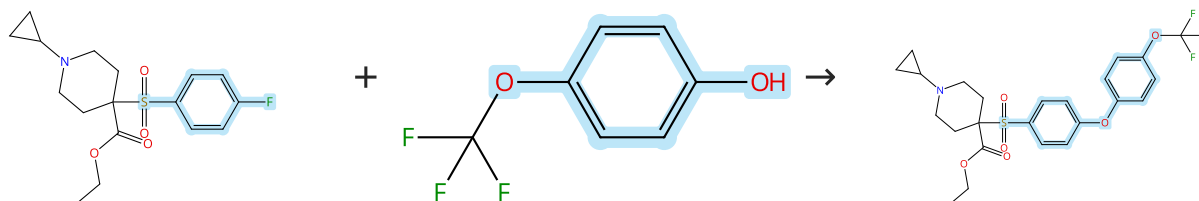
Synthesis and Structure-Activity Relationships for Extended Side Chain Analogues of the Antitubercular Drug (6S)-2-Nitro-6-[[4-(trifluoromethoxy)benzyl]oxy]-6,7-dihydro-5H-imidazo[2,1-b][1,3]oxazine (PA-824)

By: Palmer, Brian D.; et al

Journal of Medicinal Chemistry (2015), 58(7), 3036-3059.

Scheme 131 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

31-170-CAS-5815863

Steps: 1 Yield: 100%

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 40 h, 90 °C

Experimental Protocols

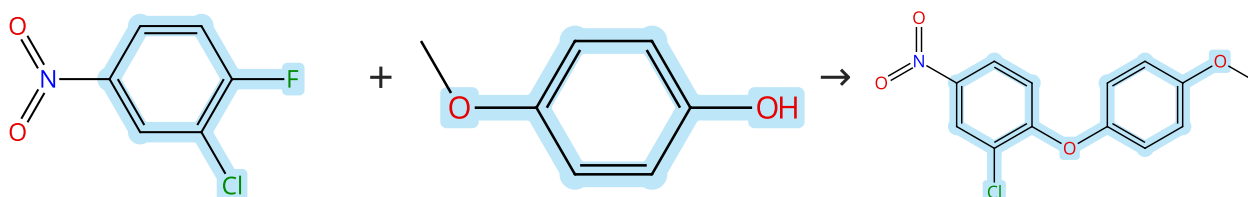
Orally Active MMP-1 Sparing α -Tetrahydropyranyl and α -Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease

By: Becker, Daniel P.; et al

Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.

Scheme 132 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (121)

Suppliers (14)

31-614-CAS-31980144

Steps: 1 Yield: 100%

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt

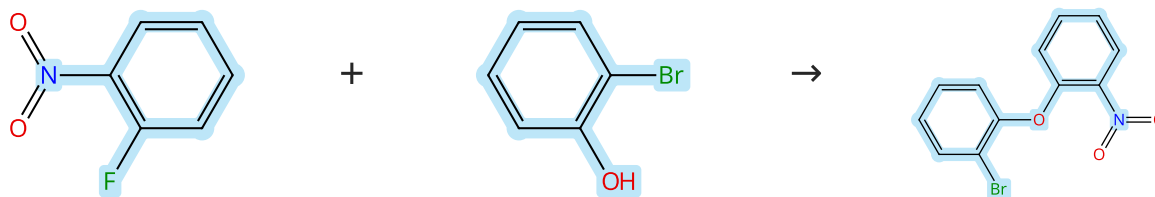
Structural Development of Salicylanilide-Based SPAK Inhibitors as Candidate Antihypertensive Agents

By: Fujii, Shinya; et al

ChemMedChem (2021), 16(18), 2817-2822.

Scheme 133 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (82)

Suppliers (12)

31-170-CAS-16101941

Steps: 1 Yield: 100%

An efficient domino Sonogashira/double carbopalladation/C-H-activation reaction leading to fluorescent polycyclic aromatic hydrocarbons

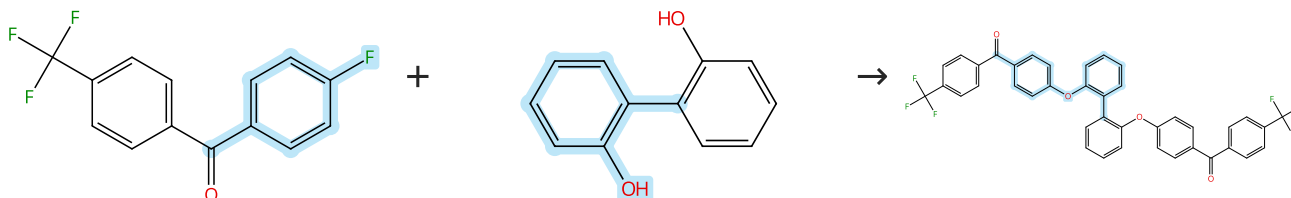
By: Tietze, Lutz F.; et al

Heterocycles (2015), 90(2), 919-927.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; 23 h, 95 °C

Scheme 134 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (34)

Suppliers (80)

31-170-CAS-13782172

Steps: 1 Yield: 100%

Synthesis of novel trifluoromethyl-bearing bifunctional acyl-acceptant arenes: 2,2'-bis(trifluoromethylated aryloxy) biphenyls

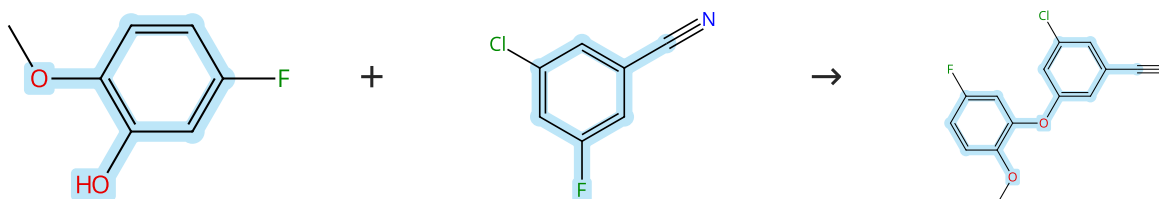
By: Okamoto, Akiko; et al

Journal of Oleo Science (2007), 56(9), 479-491.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylacetamide; 24 h, 150 °C

Scheme 135 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (65)

Suppliers (62)

Supplier (1)

31-170-CAS-7180196

Steps: 1 Yield: 100%

Synthetic chemistry-led creation of a difluorinated biaryl ether non-nucleoside reverse transcriptase inhibitor

By: Jones, Lyn H.; et al

Organic & Biomolecular Chemistry (2007), 5(21), 3431-3433.

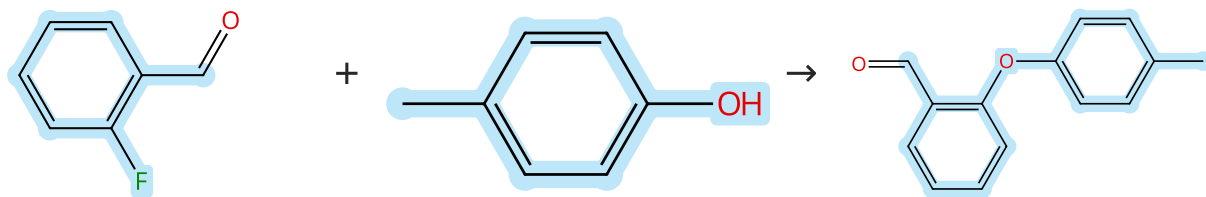
1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 10 min, rt

1.2 rt → 80 °C; 18 h, 80 °C

Experimental Protocols

Scheme 136 (2 Reactions)

Steps: 1 Yield: 98-100%



Suppliers (83)

Suppliers (89)

Suppliers (51)

31-614-CAS-36277492

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

By: Can Uskup, Hacer; et al

Experimental Protocols

ACS Omega (2023), 8(15), 14047-14052.

31-170-CAS-4968549

Steps: 1 Yield: 98%

Cross Dehydrogenative Coupling via Base-Promoted Homolytic Aromatic Substitution (BHAS): Synthesis of Fluorenones and Xanthenes

1.1 Reagents: Potassium carbonate
Solvents: Dimethylacetamide; 4 h, 170 °C

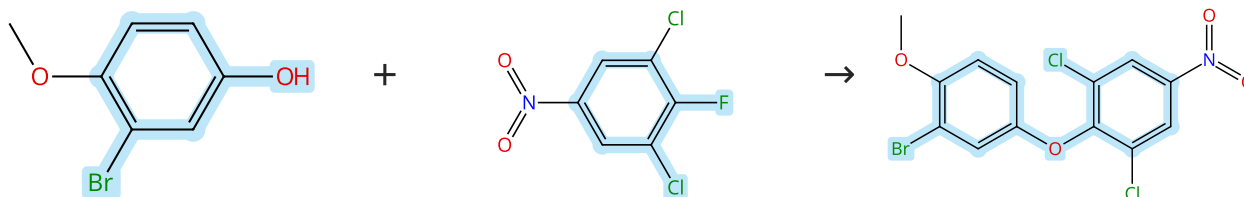
By: Wertz, Sebastian; et al

Experimental Protocols

Organic Letters (2013), 15(4), 928-931.

Scheme 137 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (62)

Suppliers (62)

31-170-CAS-15595472

Steps: 1 Yield: 100%

Rational Design of Potent and Selective Inhibitors of an Epoxide Hydrolase Virulence Factor from Pseudomonas aeruginosa

1.1 Reagents: Sodium hydroxide
Solvents: Dimethylformamide, Water; 0 °C; 0 °C → rt; 1 - 24 h, rt

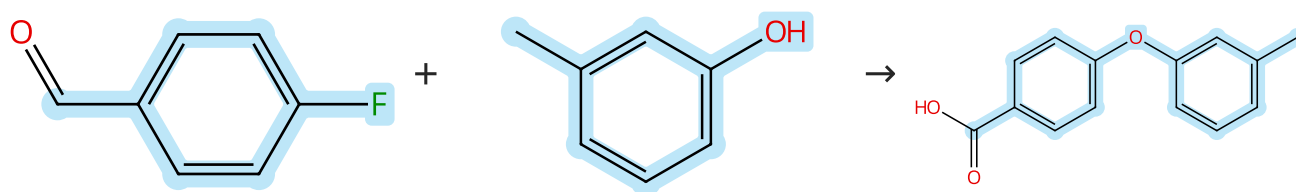
By: Kitamura, Seiya; et al

Experimental Protocols

Journal of Medicinal Chemistry (2016), 59(10), 4790-4799.

Scheme 138 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

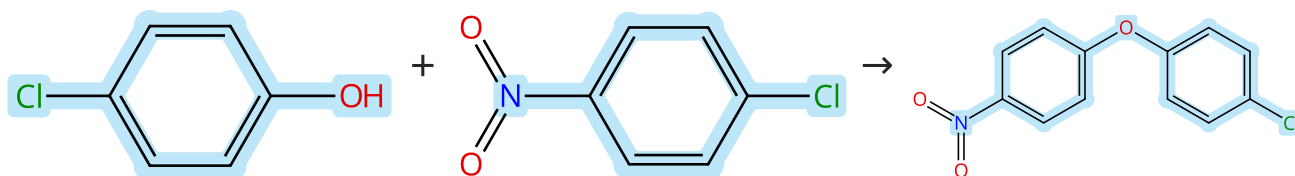
Suppliers (91)

Suppliers (42)

31-170-CAS-13470162 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 100 °C 1.2 Reagents: 2-Methyl-2-butene, Sodium chlorite, Monopotassium phosphate Solvents: <i>tert</i>-Butanol, Tetrahydrofuran, Water	Discovery of CS-2100, a potent, orally active and S1 P₃-sparing S1P₁ agonist By: Nakamura, Tsuyoshi; et al Bioorganic & Medicinal Chemistry Letters (2012), 22(4), 1788-1792.
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Scheme 139 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

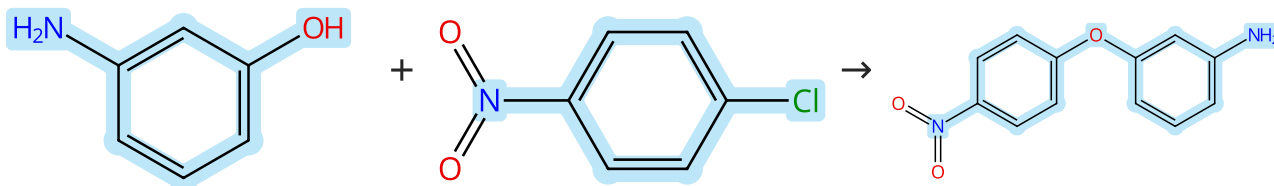
Suppliers (65)

Suppliers (45)

31-170-CAS-17492720 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate, Iron oxide (Fe₂O₃) Solvents: Dimethylformamide; 270 min, 130 °C Experimental Protocols	The effect of O-nucleophiles structure on the features of halogen substitution in heterogeneous systems By: Orlov, V. Yu.; et al Russian Journal of General Chemistry (2017), 87(3), 386-388.
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Scheme 140 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (81)

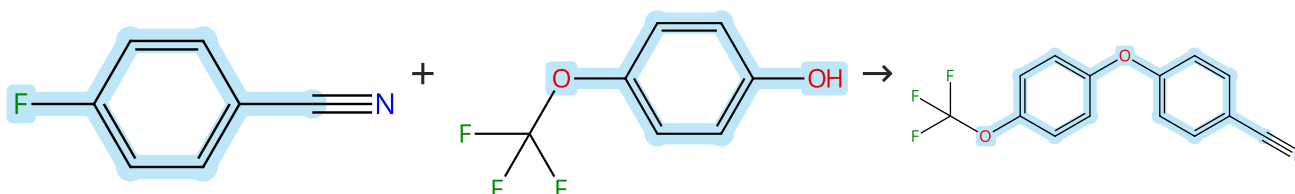
Suppliers (65)

Suppliers (22)

31-170-CAS-17492722 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate, Iron oxide (Fe₂O₃) Solvents: Dimethylformamide; 270 min, 130 °C Experimental Protocols	The effect of O-nucleophiles structure on the features of halogen substitution in heterogeneous systems By: Orlov, V. Yu.; et al Russian Journal of General Chemistry (2017), 87(3), 386-388.
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Scheme 141 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (92)

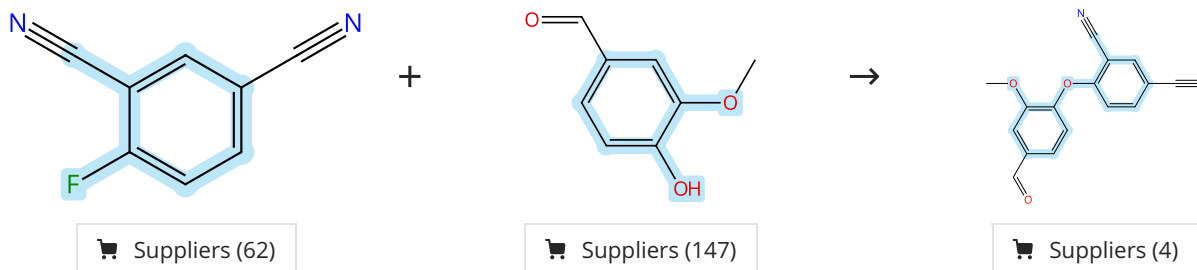
Suppliers (93)

Suppliers (8)

31-170-CAS-6604768	Steps: 1 Yield: 100%	Flavones as isosteres of 4(1H)-quinolones: Discovery of ligand efficient and dual stage antimalarial lead compounds
1.1 Reagents: Sodium carbonate Solvents: Dimethylformamide; 24 h, reflux		By: Rodrigues, Tiago; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2013), 69, 872-880.

Scheme 142 (1 Reaction)

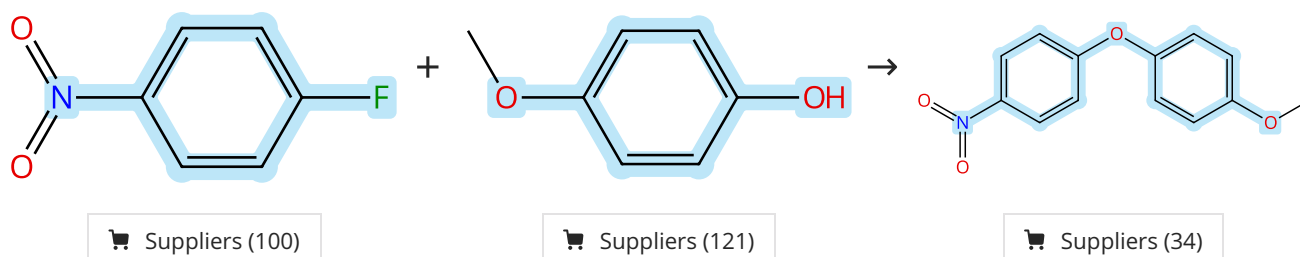
Steps: 1 Yield: 100%



31-170-CAS-11134450	Steps: 1 Yield: 100%	Identification of Diaryl Ether-Based Ligands for Estrogen-Related Receptor α as Potential Antidiabetic Agents
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 5 h, 95 °C		By: Patch, Raymond J.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2011), 54(3), 788-808.

Scheme 143 (3 Reactions)

Steps: 1 Yield: 100%



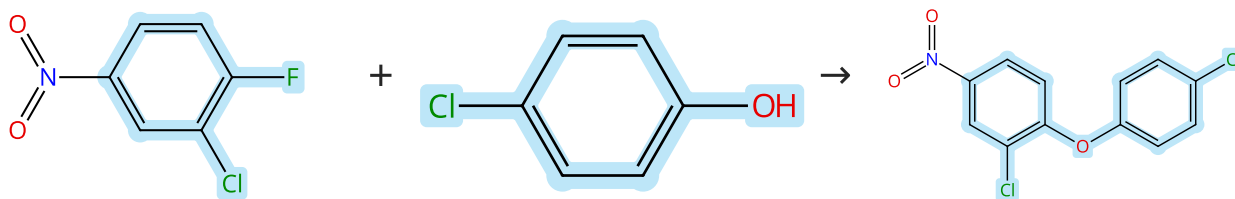
31-614-CAS-42944338	Steps: 1 Yield: 100%	S_NAr Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media
1.1 Reagents: Potassium hydroxide Catalysts: Tetrabutylammonium bromide Solvents: 2-Methyltetrahydrofuran, Water; 10 min, 95 °C		By: Wong, Madison J.; et al
Experimental Protocols		ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.

31-170-CAS-11342634	Steps: 1 Yield: 100%	CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions
1.1 Catalysts: Cesium fluoride Solvents: Dimethyl sulfoxide; 0.5 h, 110 °C		By: Keipour, Hoda; et al
Experimental Protocols		Canadian Journal of Chemistry (2016), 94(1), 95-104.

31-170-CAS-10006171	Steps: 1 Yield: 100%	Synthesis of dictyomedins using phosphazene base catalyzed diaryl ether formation
1.1 Reagents: (Diethylamino)trimethylsilane Catalysts: N'' -(1,1-Dimethylethyl)- N,N,N' -tris[tris(dimethylamino)phosphoranylidene]phosphorimidic triamide Solvents: Dimethylformamide; 2 h, rt		By: Ebisawa, Masaru; et al
Experimental Protocols		Tetrahedron Letters (2007), 48(50), 8918-8921.

Scheme 144 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (90)

Suppliers (47)

31-170-CAS-23041155

Steps: 1 Yield: 100%

Structural development of N-(4-phenoxyphenyl)benzamide derivatives as novel SPAK inhibitors blocking WNK kinase signaling

By: Fujii, Shinya; et al

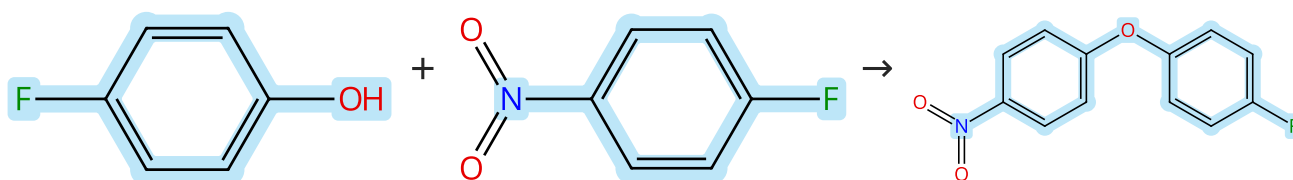
Bioorganic & Medicinal Chemistry Letters (2020), 30(17), 127408.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 17 h, rt

Experimental Protocols

Scheme 145 (2 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (86)

Suppliers (100)

Suppliers (42)

31-170-CAS-11419961

Steps: 1 Yield: 100%

CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions

By: Keipour, Hoda; et al

Canadian Journal of Chemistry (2016), 94(1), 95-104.

1.1 **Catalysts:** Cesium fluoride
Solvents: Dimethyl sulfoxide; 0.15 h, 110 °C

Experimental Protocols

31-614-CAS-37615360

Steps: 1 Yield: 99%

Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors

By: Zhang, Haoming; et al

ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.

1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 6 h, 50 °C

Experimental Protocols

Scheme 146 (3 Reactions)

Steps: 1 Yield: 98-100%



Suppliers (93)

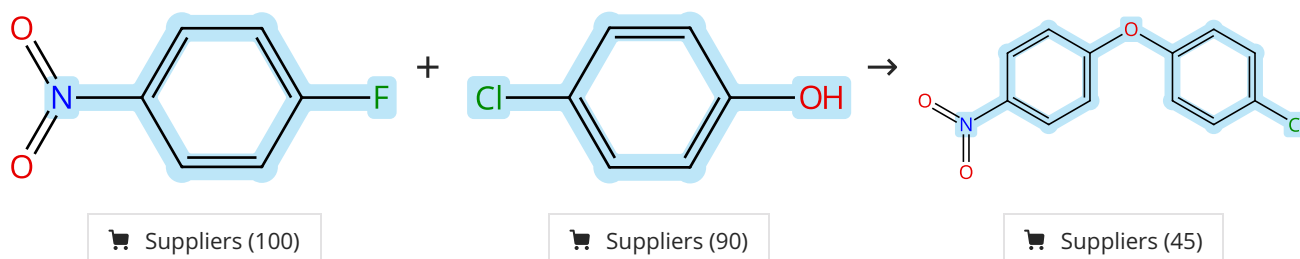
Suppliers (90)

Suppliers (63)

31-614-CAS-41211200 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C Experimental Protocols	Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation By: Hasdemir, Belma; et al ChemistrySelect (2024), 9(26), e202401028.
31-614-CAS-34652751 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, rt $\rightarrow$ 100 °C; 100 °C $\rightarrow$ rt 1.2 Reagents: Water; rt Experimental Protocols	Discovery of novel cytochrome bc1 complex inhibitor based on natural product neopeltolide By: Chen, Tao; et al Letters in Drug Design & Discovery (2022), 19(4), 263-268.
31-170-CAS-2944344 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethylacetamide; 1 h, 60 °C; cooled Experimental Protocols	Benzimidazole inhibitors of the protein kinase C HK2: Clarification of the binding mode by flexible side chain docking and protein-ligand crystallography By: Matijssen, Cornelis; et al Bioorganic & Medicinal Chemistry (2012), 20(22), 6630-6639.

Scheme 147 (1 Reaction)

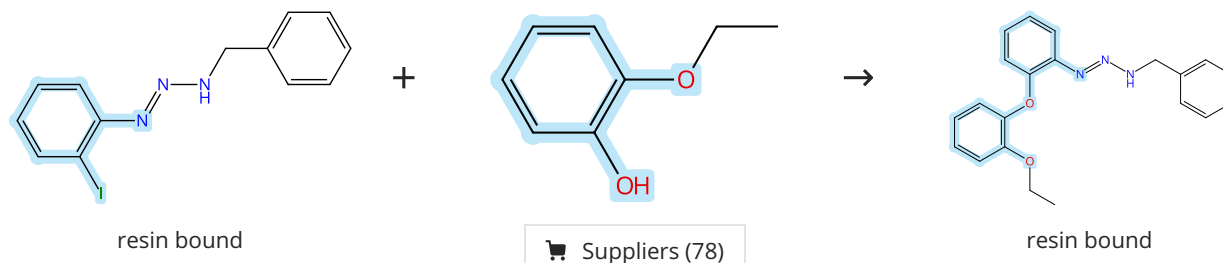
Steps: 1 Yield: 100%



31-170-CAS-4302515 Steps: 1 Yield: 100% 1.1 Catalysts: Cesium fluoride Solvents: Dimethyl sulfoxide; 0.15 h, 110 °C Experimental Protocols	CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions By: Keipour, Hoda; et al Canadian Journal of Chemistry (2016), 94(1), 95-104.
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Scheme 148 (1 Reaction)

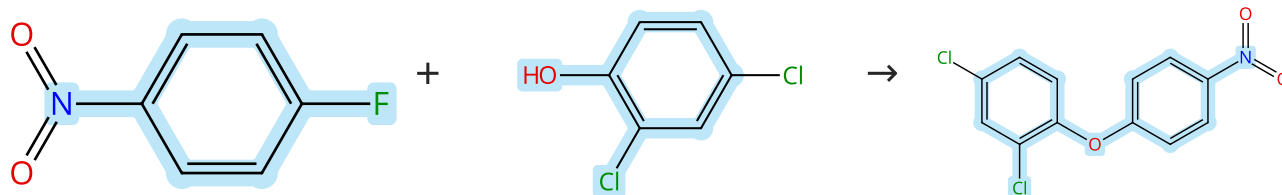
Steps: 1 Yield: 100%



31-170-CAS-12936402 Steps: 1 Yield: 100% 1.1 Reagents: Sodium carbonate, Bromo[1,1'-thiobis[methane]] copper Solvents: Acetonitrile, Pyridine; 48 h, 75 °C Experimental Protocols	Solid phase synthesis of selectively deuterated arenes By: Vanderheiden, Sylvia; et al Chemical Communications (Cambridge, United Kingdom) (2011), 47(32), 9063-9065.
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Scheme 149 (2 Reactions)

Steps: 1 Yield: 98-100%



Suppliers (100)

Suppliers (102)

Suppliers (64)

31-170-CAS-14973560

Steps: 1 Yield: 100%

1.1 Catalysts: Cesium fluoride
Solvents: Dimethyl sulfoxide; 0.15 h, 110 °C

Experimental Protocols

CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions

By: Keipour, Hoda; et al

Canadian Journal of Chemistry (2016), 94(1), 95-104.

31-614-CAS-37615377

Steps: 1 Yield: 98%

1.1 Reagents: Cesium carbonate
Solvents: Dimethylformamide; 6 h, 50 °C

Experimental Protocols

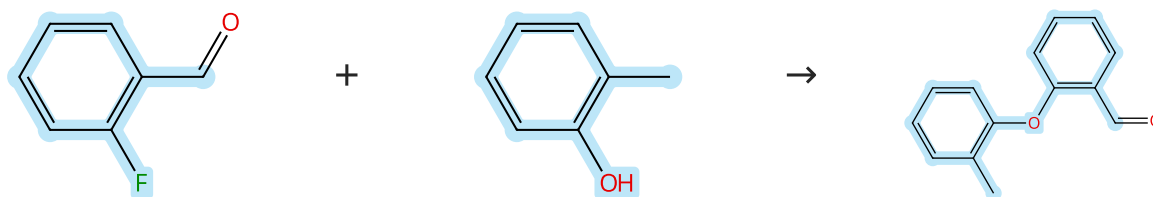
Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors

By: Zhang, Haoming; et al

ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.

Scheme 150 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (83)

Suppliers (20)

31-614-CAS-36277498

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

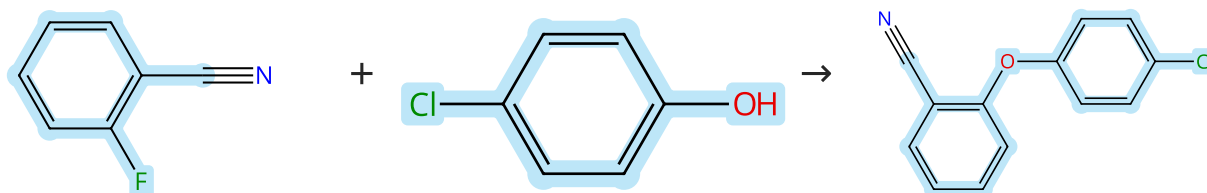
Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

By: Can Uskup, Hacer; et al

ACS Omega (2023), 8(15), 14047-14052.

Scheme 151 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (90)

Suppliers (17)

31-614-CAS-32528350

Steps: 1 Yield: 100%

1.1 Reagents: Potassium hydroxide
Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt

1.2 Reagents: Water; rt

Experimental Protocols

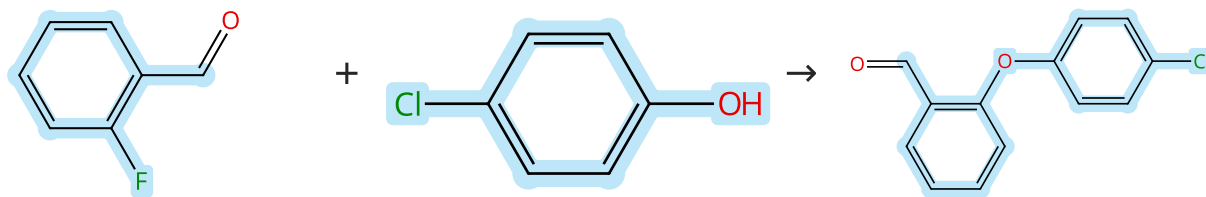
Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement

By: Jinno, Shunki; et al

Organic Letters (2022), 24(23), 4140-4144.

Scheme 152 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (90)

Suppliers (61)

31-614-CAS-36277490

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

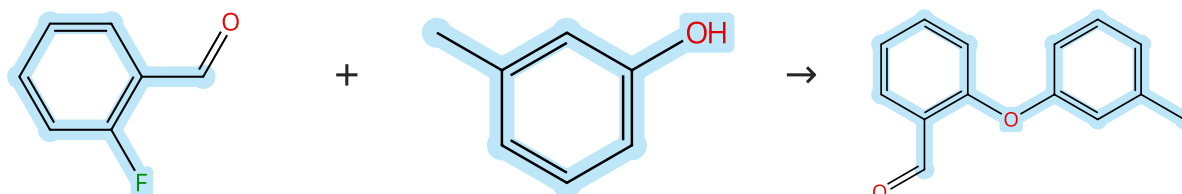
By: Can Uskup, Hacer; et al

Experimental Protocols

ACS Omega (2023), 8(15), 14047-14052.

Scheme 153 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (91)

Suppliers (14)

31-614-CAS-36277479

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

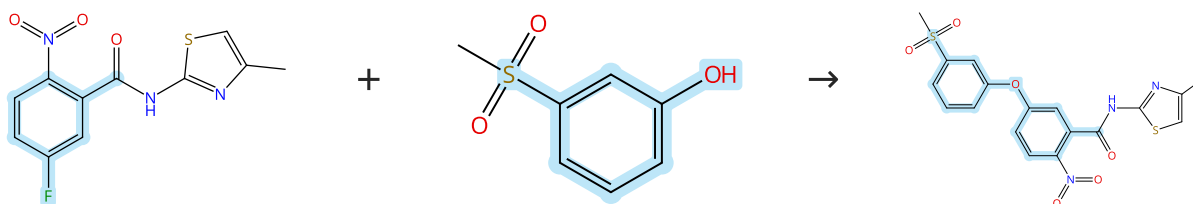
By: Can Uskup, Hacer; et al

Experimental Protocols

ACS Omega (2023), 8(15), 14047-14052.

Scheme 154 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (3)

Suppliers (61)

31-170-CAS-7133577

Steps: 1 Yield: 99%

Discovery of potent and orally active 3-alkoxy-5-phenoxy-N-thiazolyl benzamides as novel allosteric glucokinase activators

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; overnight, 100 °C

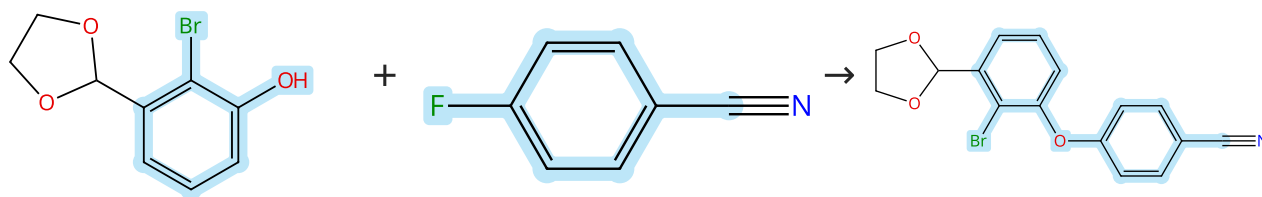
By: Iino, Tomoharu; et al

Experimental Protocols

Bioorganic & Medicinal Chemistry (2009), 17(7), 2733-2743.

Scheme 155 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (24)

Suppliers (92)

31-170-CAS-5180358

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; overnight, 100 °C

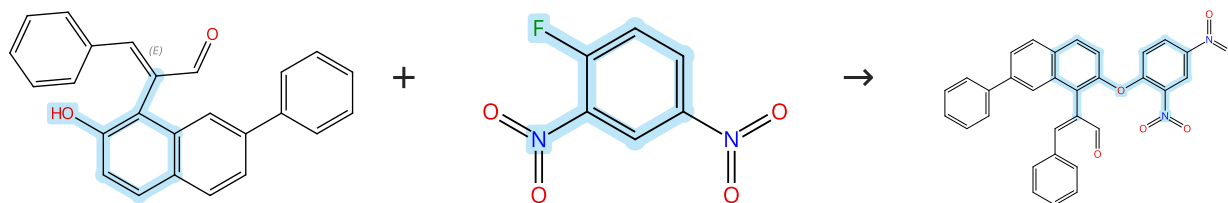
Discovery and structure-activity study of a novel benzoxa borole anti-inflammatory agent (AN2728) for the potential topical treatment of psoriasis and atopic dermatitis

By: Akama, Tsutomu; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(8), 2129-2132.

Scheme 156 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560087

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

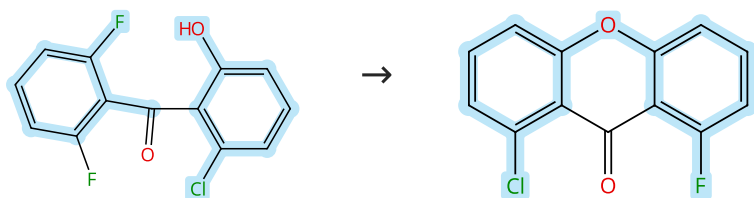
By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Experimental Protocols

Scheme 157 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-28802093

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 25 °C; 30 min, 25 °C
 1.2 **Reagents:** Hydrochloric acid
Solvents: Water; 25 °C

An Efficient Two-Step Protocol for the Isoprenylation of Xanthone at the C2 Position Starting from 1-Fluoroxanthone Derivative

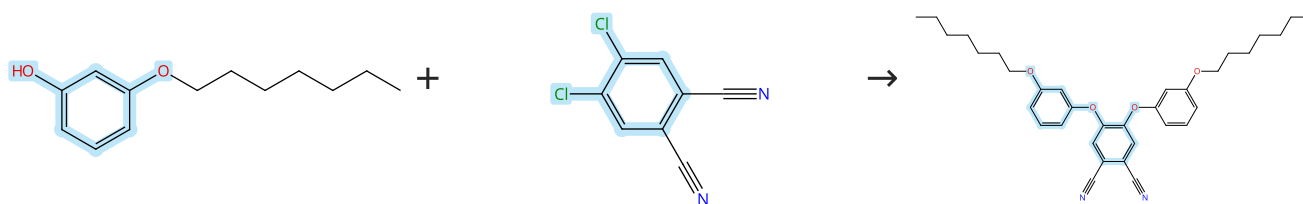
By: Fujimoto, Yuuki; et al

Synlett (2020), 31(14), 1423-1429.

Experimental Protocols

Scheme 158 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (7)

Suppliers (72)

31-170-CAS-18371575

Steps: 1 Yield: 99%

Phthalocyanine-based discotic liquid crystals switching from a molten alkyl chain type to a flying-seed-like type

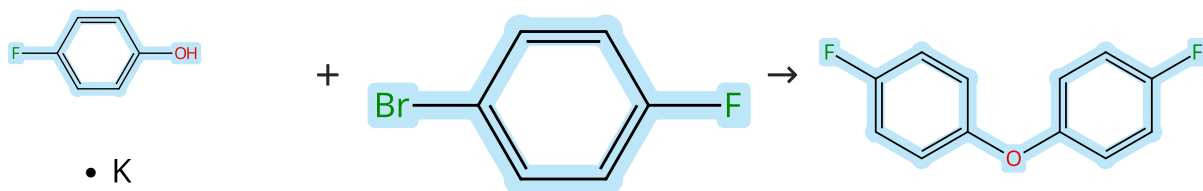
By: Nakamura, Hiromu; et al

Journal of Materials Chemistry C: Materials for Optical and Electronic Devices (2017), 5(29), 7297-7306.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 10 min, 90 °C
- 1.2 110 min, 100 °C

Scheme 159 (1 Reaction)

Steps: 1 Yield: 99%



• K

Suppliers (103)

Suppliers (63)

31-614-CAS-37728583

Steps: 1 Yield: 99%

Cross-coupling by a noncan onical mechanism involving the addition of aryl halide to Cu(II)

By: Delaney, Connor P.; et al

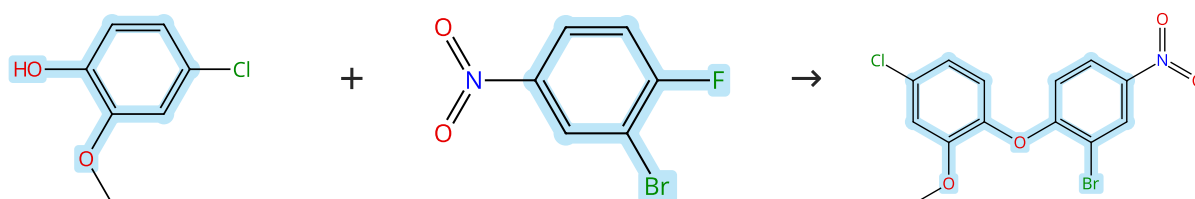
Science (Washington, DC, United States) (2023), 381(6662), 1079-1085.

- 1.1 **Reagents:** 1-Fluoronaphthalene
Catalysts: 2983730-57-8
Solvents: Dimethyl sulfoxide; rt → 100 °C; 5 min, 100 °C

Experimental Protocols

Scheme 160 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

Suppliers (83)

31-614-CAS-35312180

Steps: 1 Yield: 99%

Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors

By: Xu, Kai-Yan; et al

Bioorganic & Medicinal Chemistry (2023), 78, 117152.

- 1.1 **Reagents:** Cesium carbonate
Solvents: Dimethyl sulfoxide; 2 h, 110 °C

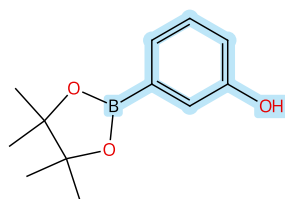
Experimental Protocols

Scheme 161 (1 Reaction)

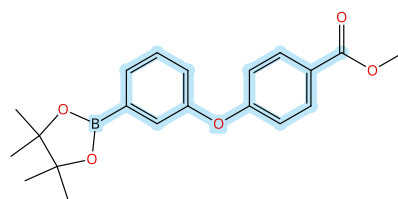
Steps: 1 Yield: 99%

Multi-component
structure image
available in CAS
SciFinder

+



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Suppliers (90)

31-614-CAS-31534025

Steps: 1 Yield: 99%

Ligand- and Counterion-Assisted Phenol O-Arylation with TM P-Iodonium(III) Acetates

By: Kikushima, Kotaro; et al

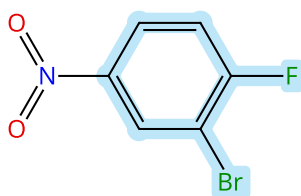
Organic Letters (2022), 24(10), 1924-1928.

- 1.1 **Reagents:** Sodium carbonate
Solvents: Dichloromethane, Water; 18 h, 40 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

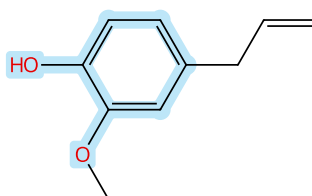
Experimental Protocols

Scheme 162 (1 Reaction)

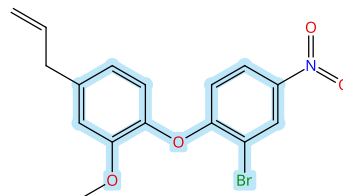
Steps: 1 Yield: 99%



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Suppliers (83)

Suppliers (120)

31-614-CAS-42944341

Steps: 1 Yield: 99%

S_NAr Reactions Using Continuous Plug Flow...in Aqueous Biphasic Media

By: Wong, Madison J.; et al

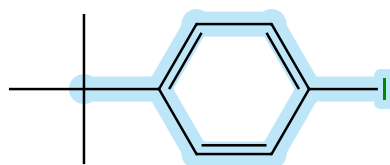
ACS Sustainable Chemistry & Engineering (2024), 12(52), 18725-18734.

- 1.1 **Reagents:** Potassium hydroxide
Catalysts: Tetrabutylammonium bromide
Solvents: 2-Methyltetrahydrofuran, Water; 10 min, 95 °C

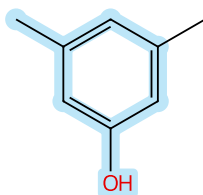
Experimental Protocols

Scheme 163 (1 Reaction)

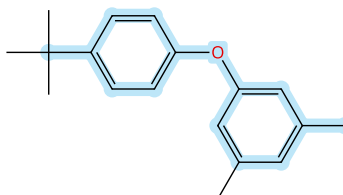
Steps: 1 Yield: 99%



+



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Suppliers (68)

Suppliers (89)

31-614-CAS-43350512

Steps: 1 Yield: 99%

Bidentate NHC-Containing Ligands for Copper Catalyzed Synthesis of Functionalised Diaryl Ethers

By: Richards, Ethan M.; et al

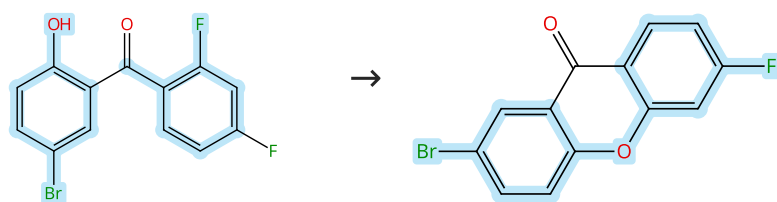
Advanced Synthesis & Catalysis (2025), 367(6), e202400909.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Cuprous iodide, 1*H*-imidazolium, 3-[4-(trifluoromethyl)-2-pyridinyl]-1-(2,4,6-trimethylphenyl)-, bromide (1:1)
Solvents: Acetonitrile; 20 h, 110 °C

Experimental Protocols

Scheme 164 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

31-614-CAS-27843525

Steps: 1 Yield: 99%

Fluorinated 9H-xanthene-9-carboxylic acid oxazol-2-ylamides as potent, orally available mGlu1 receptor enhancers

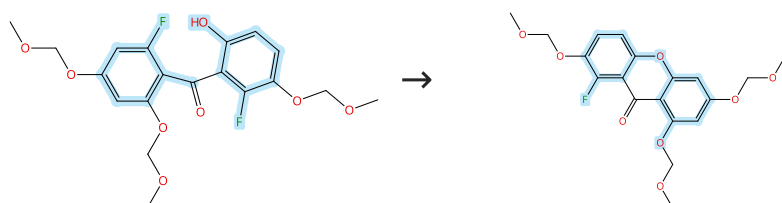
1.1 **Reagents:** Potassium carbonate
Solvents: Methyl ethyl ketone; reflux

By: Vieira, Eric; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(6), 1666-1669.

Scheme 165 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-29944010

Steps: 1 Yield: 99%

Anion-Accelerated Aromatic Oxy-Cope Rearrangement in Geranylation/Nerylation of Xanthone: Stereochemical Insights and Synthesis of Fuscaxanthone F

1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 25 °C; 1.5 h, 25 °C

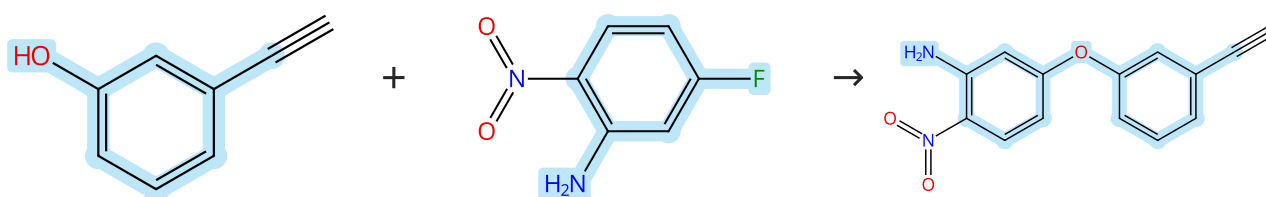
Experimental Protocols

By: Fujimoto, Yuuki; et al

Synlett (2020), 31(14), 1378-1383.

Scheme 166 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (70)

Suppliers (82)

31-170-CAS-7576251

Steps: 1 Yield: 99%

New benzimidazole-2-urea derivatives as tubulin inhibitors

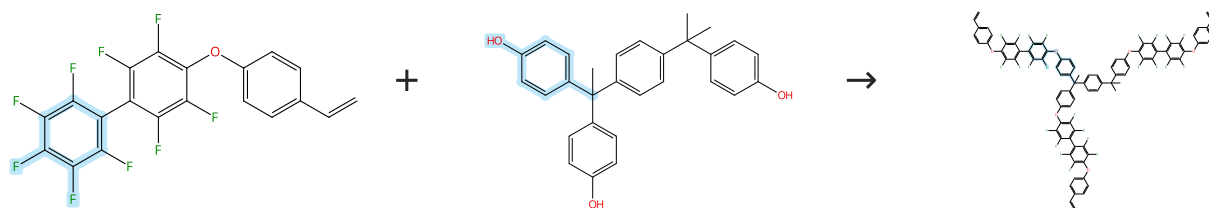
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 100 °C

By: Wang, Wenna; et al

Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 167 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (55)

31-614-CAS-28447126

Steps: 1 Yield: 99%

Cross-Linkable Molecular Glasses: Low Dielectric Constant Materials Patternable in Hydrofluoroethers

1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 16 h, 60 °C; 5 h, 60 °C

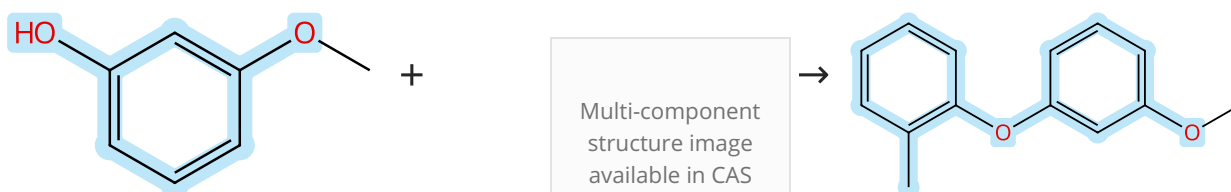
By: Murotani, Eisuke; et al

Experimental Protocols

ACS Applied Materials & Interfaces (2009), 1(10), 2363-2370.

Scheme 168 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (92)

Multi-component
structure image
available in CAS
SciFinder

Suppliers (16)

31-170-CAS-15461276

Steps: 1 Yield: 99%

Metal-free arylation of oxygen nucleophiles with diaryliodonium salts1.1 Reagents: Potassium *tert*-butoxide

Solvents: Tetrahydrofuran; 15 min, 0 °C

By: Jalalian, Nazli; et al

1.2 30 min, 40 °C

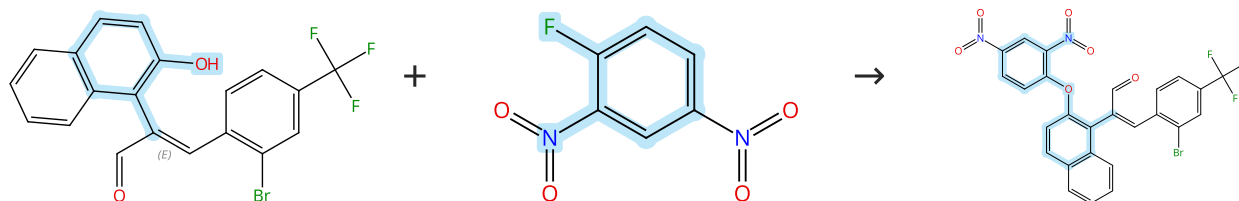
1.3 Reagents: Water; 0 °C

Chemistry - A European Journal (2012), 18(44), 14140-14149, S14140/1-S14140/63.

Experimental Protocols

Scheme 169 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560064

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

1.1 Reagents: Potassium carbonate

Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)

Solvents: Chloroform; 24 h, -10 °C

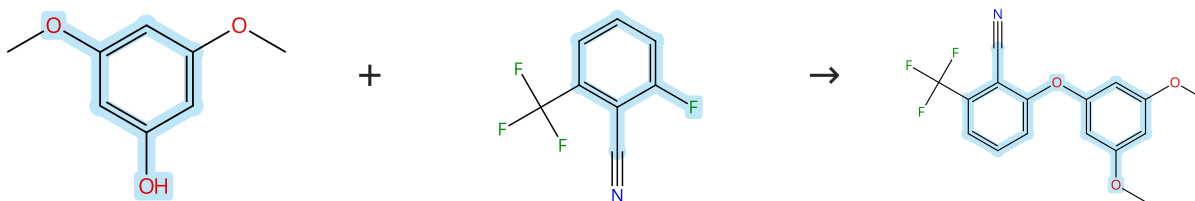
By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Experimental Protocols

Scheme 170 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (92)

Suppliers (71)

Supplier (1)

31-170-CAS-15131974

Steps: 1 Yield: 99%

Synthesis of phloroglucinol monoaryl ethers

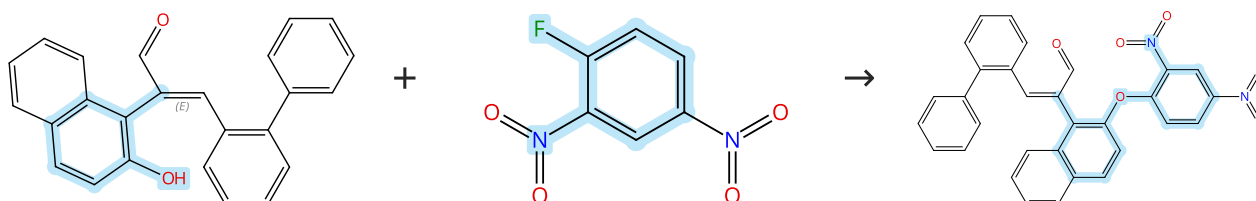
- 1.1 **Reagents:** Cesium carbonate
Solvents: *N*-Methyl-2-pyrrolidone; 30 min, 50 °C
 1.2 4 h, 65 °C; 65 °C → rt

By: Sherwood, Alexander M.; et al
 Synthesis (2012), 44(8), 1208-1212.

Experimental Protocols

Scheme 171 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560038

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

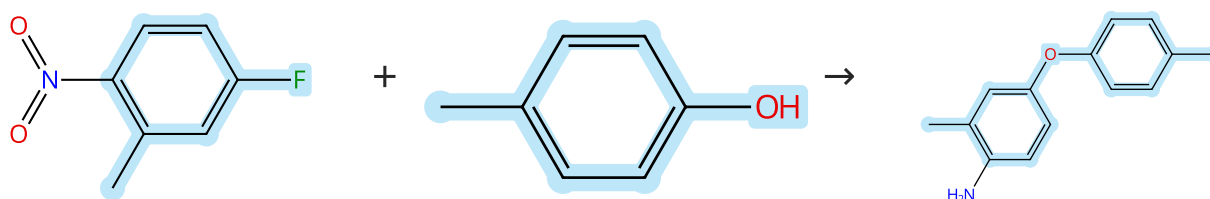
- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

By: Guo, Fengyuan; et al
 Nature Communications (2023), 14(1), 5050.

Experimental Protocols

Scheme 172 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

Suppliers (89)

Suppliers (15)

31-170-CAS-3468320

Steps: 1 Yield: 99%

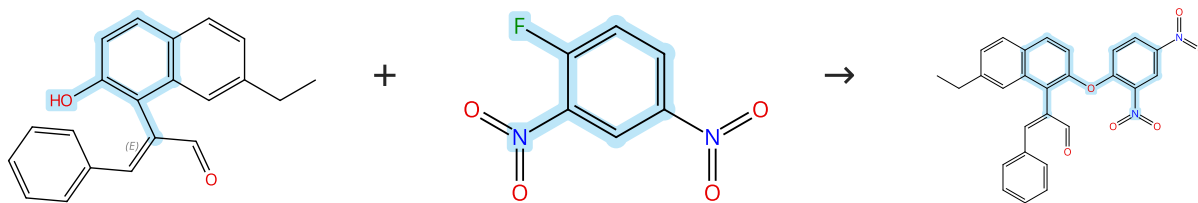
Discovery and structure-activity study of a novel benzoxa borole anti-inflammatory agent (AN2728) for the potential topical treatment of psoriasis and atopic dermatitis

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; overnight, 70 °C
 1.2 **Reagents:** Hydrochloric acid, Stannous chloride
Solvents: Ethanol; 2 h, 50 °C

By: Akama, Tsutomu; et al
 Bioorganic & Medicinal Chemistry Letters (2009), 19(8), 2129-2132.

Scheme 173 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560085

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

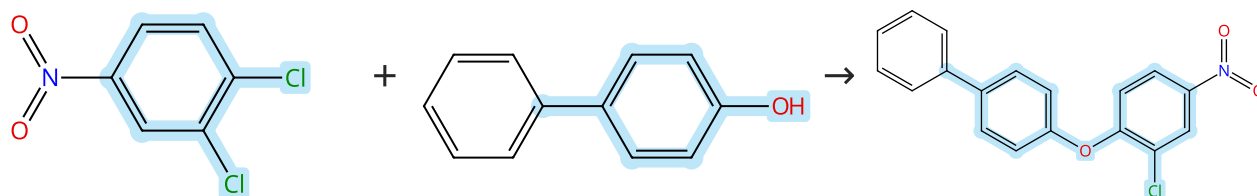
Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Scheme 174 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (63)

Suppliers (105)

Supplier (1)

31-170-CAS-12707911

Steps: 1 Yield: 99%

- 1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene
Solvents: Dimethylacetamide; rt → 150 °C; 10 min, 150 °C; 150 °C → 90 °C

Experimental Protocols

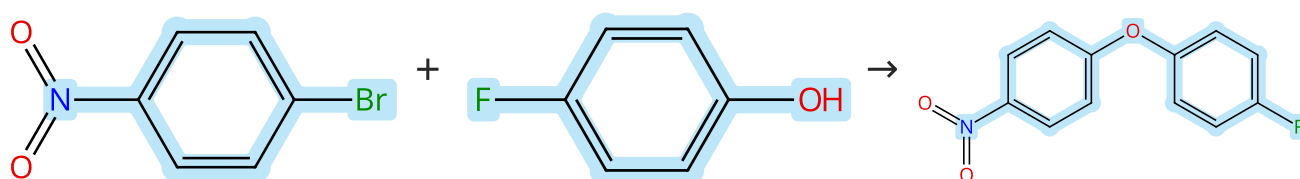
The application of stop-flow microwave technology to scaling-out S_NAr reactions using a soluble organic base

By: Marafie, Jameel A.; et al

Organic & Biomolecular Chemistry (2010), 8(9), 2219-2227.

Scheme 175 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (77)

Suppliers (86)

Suppliers (42)

31-170-CAS-10031163

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Cesium carbonate, 1,1'-Azobis(cyclohexane-1-carbonitrile)
Catalysts: Copper bromide (CuBr)
Solvents: Dimethylformamide; 90 min, 100 °C

Experimental Protocols

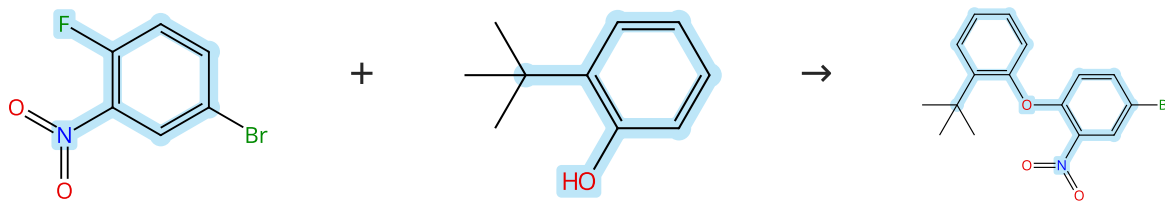
Microwave assisted synthesis of selected diaryl ethers under Cu(I)-catalysis

By: Navarro, Lorena; et al

Tetrahedron Letters (2015), 56(14), 1812-1815.

Scheme 176 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

Suppliers (54)

Suppliers (5)

31-170-CAS-16856064

Steps: 1 Yield: 99%

Identification and optimization of a novel series of indole amine 2,3-dioxygenase inhibitors

By: Markwalder, Jay A.; et al

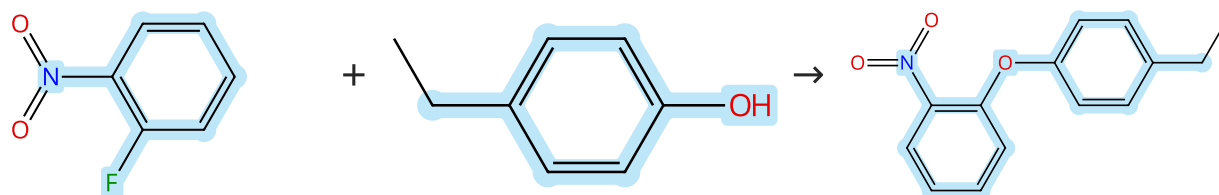
Bioorganic & Medicinal Chemistry Letters (2017), 27(3), 582-585.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 4 h, 100 °C

Experimental Protocols

Scheme 177 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (93)

Suppliers (100)

Supplier (1)

31-170-CAS-15901087

Steps: 1 Yield: 99%

Distance-Dependent Attractive and Repulsive Interactions of Bulky Alkyl Groups

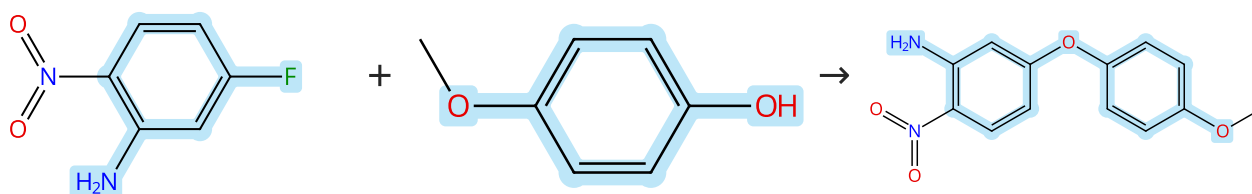
By: Hwang, Jungwun; et al

Angewandte Chemie, International Edition (2016), 55(28), 8086-8089.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt; 3 h, 90 °C

Scheme 178 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

Suppliers (121)

Suppliers (4)

31-170-CAS-11171030

Steps: 1 Yield: 99%

New benzimidazole-2-urea derivatives as tubulin inhibitors

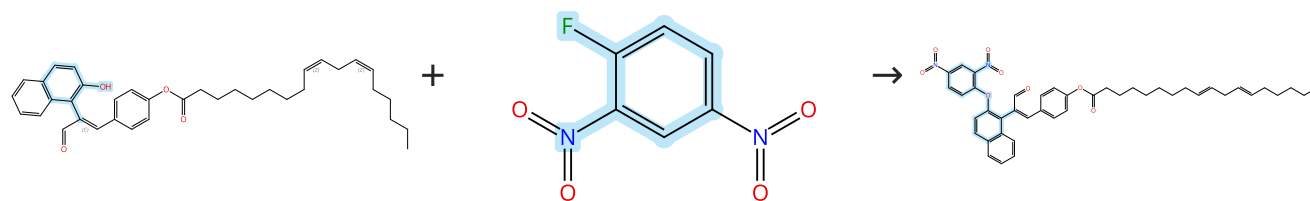
By: Wang, Wenna; et al

Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 100 °C

Scheme 179 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560110

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

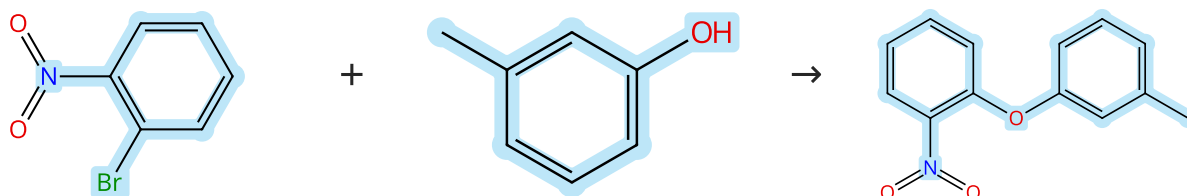
Nature Communications (2023), 14(1), 5050.

1.1 **Reagents:** Potassium carbonate
Catalysts: 2412758-02-0
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

Scheme 180 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (72)

Suppliers (91)

Suppliers (12)

31-170-CAS-1381182

Steps: 1 Yield: 99%

Betti Base as an Efficient Ligand for Copper-Catalyzed Ullmann Coupling of Phenol with Aryl Halides

By: Yang, Li; et al

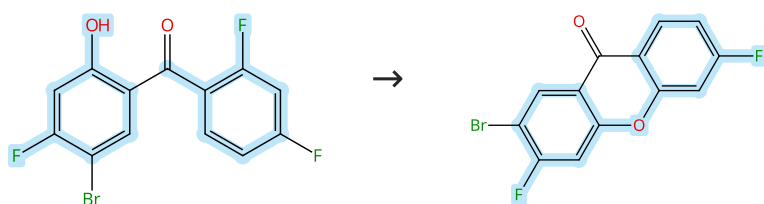
Synthetic Communications (2014), 44(17), 2468-2477.

1.1 **Reagents:** Potassium carbonate
Catalysts: Cuprous iodide, 1-[(Dimethylamino)-2-pyridinylmethyl]-2-naphthalenol
Solvents: Dimethyl sulfoxide; 24 h, 120 °C

Experimental Protocols

Scheme 181 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-30248127

Steps: 1 Yield: 99%

Fluorinated 9H-xanthene-9-carboxylic acid oxazol-2-ylamides as potent, orally available mGlu1 receptor enhancers

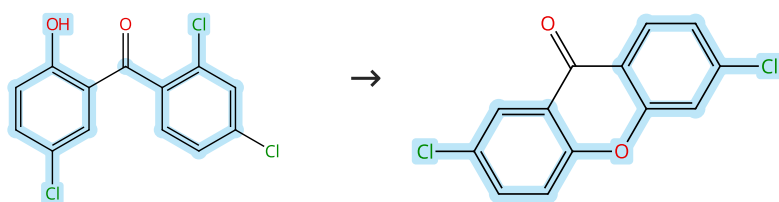
By: Vieira, Eric; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(6), 1666-1669.

1.1 **Reagents:** Potassium carbonate
Solvents: Methyl ethyl ketone; reflux

Scheme 182 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (3)

Suppliers (13)

31-614-CAS-25494759

Steps: 1 Yield: 99%

Microwave-assisted transition-metal-free intramolecular Ullmann-type O-arylation in water for the synthesis of xanthenes and azaxanthenes

1.1 **Reagents:** Tetrabutylammonium hydroxide
Solvents: Water; 4 min, 120 °C

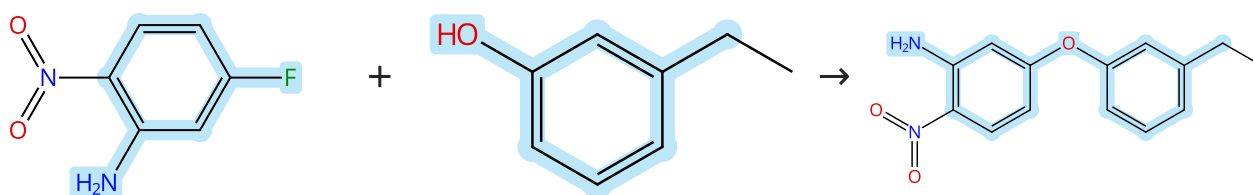
Experimental Protocols

By: Zhang, Xiaojin; et al

Tetrahedron Letters (2014), 55(35), 4883-4887.

Scheme 183 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

Suppliers (74)

Supplier (1)

31-170-CAS-10830418

Steps: 1 Yield: 99%

New benzimidazole-2-urea derivatives as tubulin inhibitors

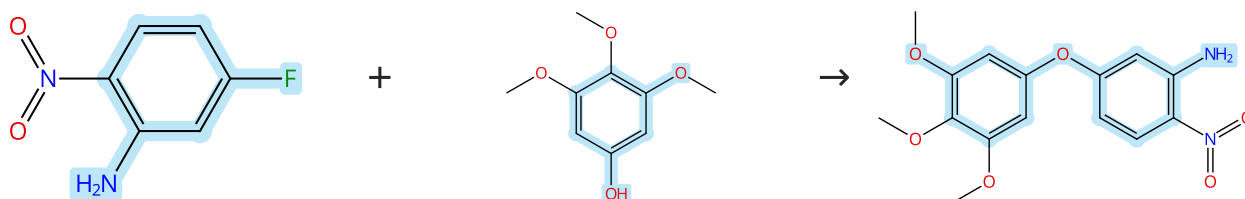
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 100 °C

By: Wang, Wenna; et al

Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 184 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

Suppliers (77)

31-170-CAS-2038321

Steps: 1 Yield: 99%

New benzimidazole-2-urea derivatives as tubulin inhibitors

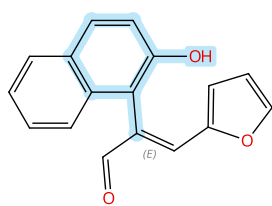
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 100 °C

By: Wang, Wenna; et al

Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

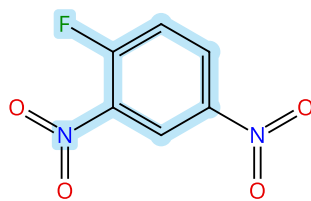
Scheme 185 (1 Reaction)

Steps: 1 Yield: 99%

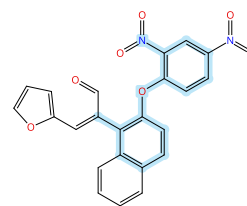


Double bond geometry shown

+



→



Suppliers (75)

31-614-CAS-37560072

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

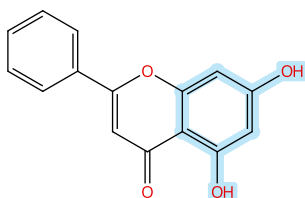
Nature Communications (2023), 14(1), 5050.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

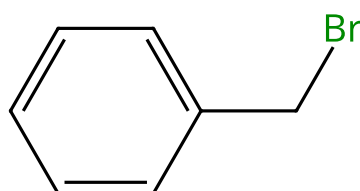
Scheme 186 (1 Reaction)

Steps: 1 Yield: 99%



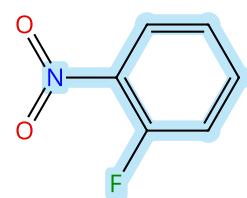
Suppliers (114)

+

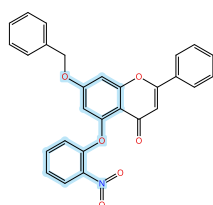


Suppliers (67)

+



→



Suppliers (93)

31-008-CAS-19628578

Steps: 1 Yield: 99%

Directional modification of chrysin for exerting apoptosis and enhancing significantly anti-cancer effects of 10-hydroxy camptothecin

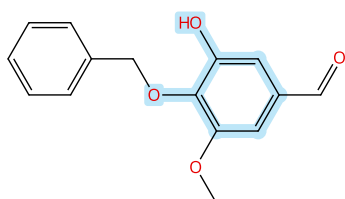
By: Tang, Qin; et al

Biomedicine & Pharmacotherapy (2016), 82, 693-703.

- 1.1 **Reagents:** Potassium carbonate
Solvents: *N*-Methyl-2-pyrrolidone; 75 °C
 1.2 **Reagents:** Potassium carbonate; 85 °C

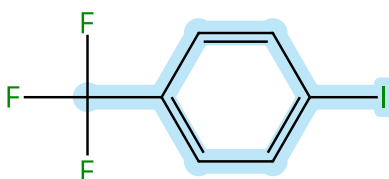
Scheme 187 (1 Reaction)

Steps: 1 Yield: 99%



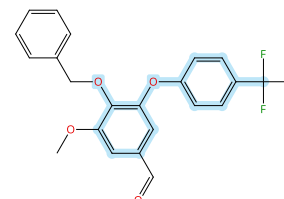
Suppliers (17)

+



Suppliers (82)

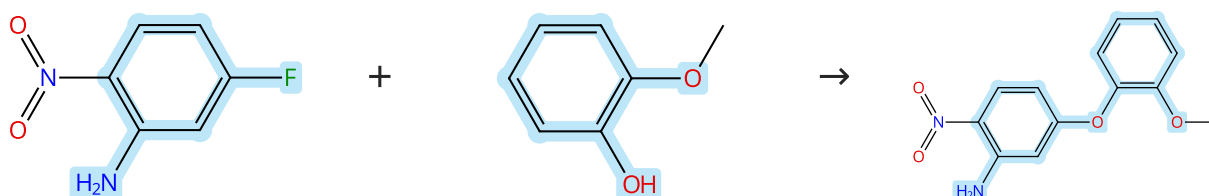
→



31-170-CAS-4754055	Steps: 1 Yield: 99%	Unexpected copper mediated benzyl O→O migration during an Ullmann ether coupling
1.1 Reagents: Tripotassium phosphate Catalysts: Picolinic acid, Cuprous iodide Solvents: Dimethyl sulfoxide; 24 h, 100 °C		By: Vanucci-Bacque, Corinne; et al
Experimental Protocols		Tetrahedron Letters (2014), 55(2), 528-530.

Scheme 188 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

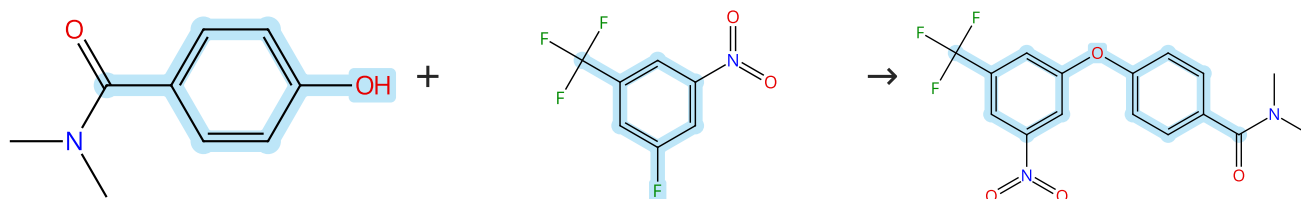
Suppliers (125)

Suppliers (2)

31-170-CAS-6921356	Steps: 1 Yield: 99%	New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C		By: Wang, Wenna; et al
		Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 189 (1 Reaction)

Steps: 1 Yield: 99%



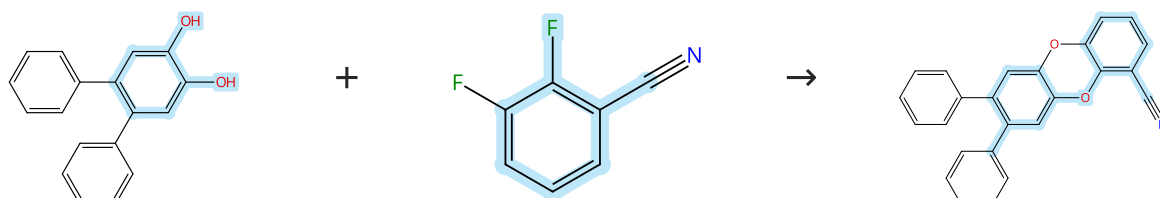
Suppliers (63)

Suppliers (70)

31-170-CAS-20939824	Steps: 1 Yield: 99%	Design, synthesis, and in vitro bioactivity evaluation of fluorine-containing analogues for sphingosine-1-phosphate 2 receptor
1.1 Reagents: Monopotassium phosphate Solvents: Dimethylacetamide; 12 h, 100 °C		By: Luo, Zonghua; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2019), 27(16), 3619-3631.

Scheme 190 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

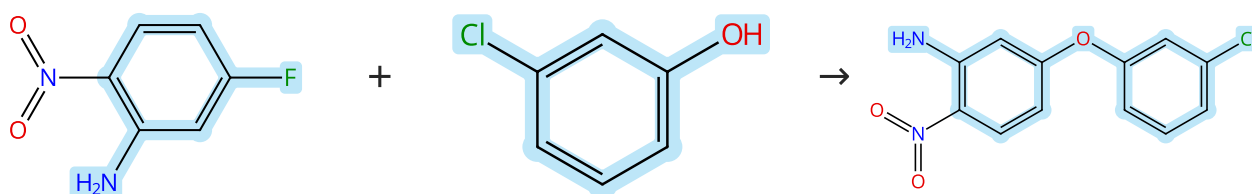
Suppliers (76)

Suppliers (6)

31-170-CAS-9813173	Steps: 1 Yield: 99% Synthesis of cyanodibenzo[1,4]dioxines and their derivatives by cyano-activated fluoro displacement reactions
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide, Toluene	By: Eastmond, Geoffrey C.; et al New Journal of Chemistry (2001), 25(3), 379-384.

Scheme 191 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

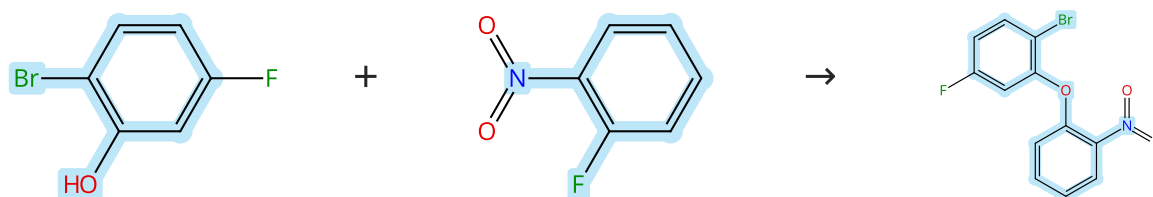
Suppliers (85)

Suppliers (3)

31-170-CAS-10889031	Steps: 1 Yield: 99% New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C	By: Wang, Wenna; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 192 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (88)

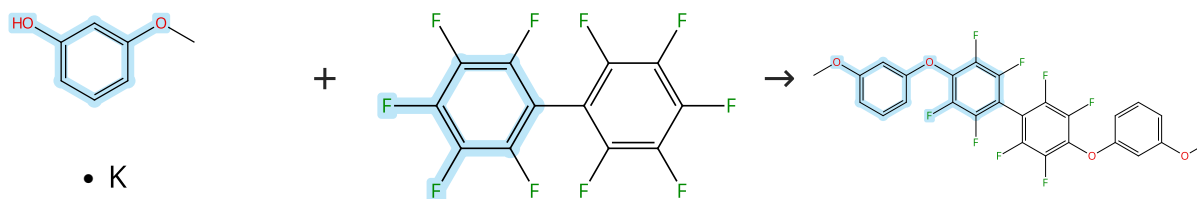
Suppliers (93)

Suppliers (6)

31-170-CAS-511764	Steps: 1 Yield: 99% Synthesis and Photochemical Investigations of Tetrasubstituted Alkenes as Molecular Switches-The Effect of Substituents
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 22 h, rt → 95 °C	By: Tietze, Lutz F.; et al
Experimental Protocols	Chemistry - A European Journal (2011), 17(30), 8452-8461, S8452/1-S8452/65.

Scheme 193 (1 Reaction)

Steps: 1 Yield: 99%



• K

Supplier (1)

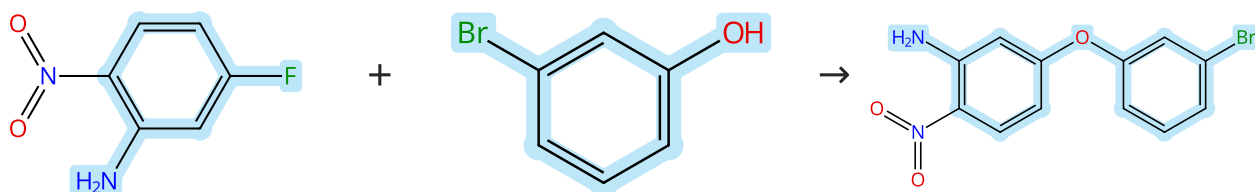
Suppliers (68)

Suppliers (5)

31-170-CAS-9152958	Steps: 1 Yield: 99%	Aromatic fluorinated bisphenols
No Data Available		By: Kameneva, T. M.; et al Dopovidi Akademii Nauk Ukrain'skoi RSR, Seriya B: Geologichni, Khimichni ta Biologichni Nauki (1983), (1), 22-4.

Scheme 194 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

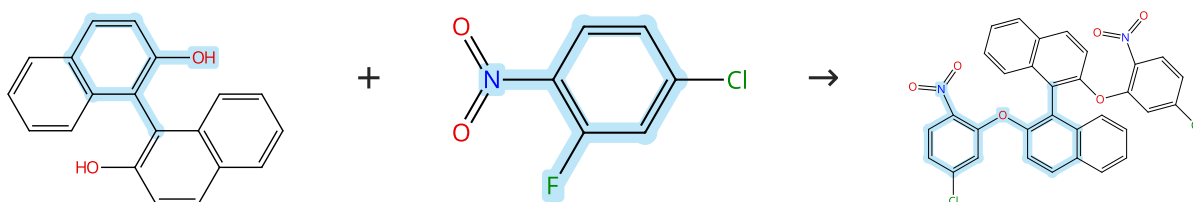
Suppliers (74)

Suppliers (4)

31-170-CAS-6583899	Steps: 1 Yield: 99%	New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C		By: Wang, Wenna; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 195 (1 Reaction)

Steps: 1 Yield: 99%



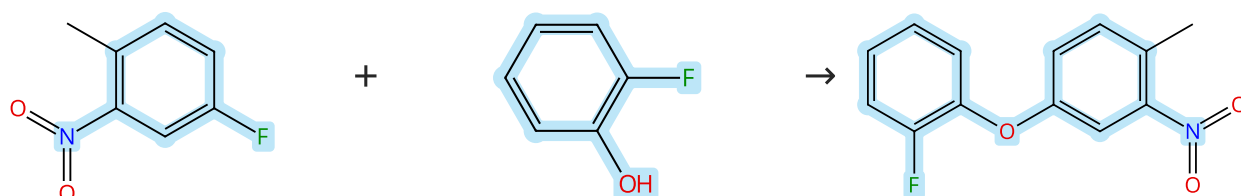
Suppliers (85)

Suppliers (79)

31-170-CAS-63234	Steps: 1 Yield: 99%	Enantioselective alkylation to aldimines catalyzed by chiral 2,2'-bis[(2-aminoaryl)oxy]-1,1'-binaphthyl-copper(I) complexes
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, 80 °C		By: Hatano, Manabu; et al Tetrahedron Letters (2008), 49(2), 379-382.
Experimental Protocols		

Scheme 196 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

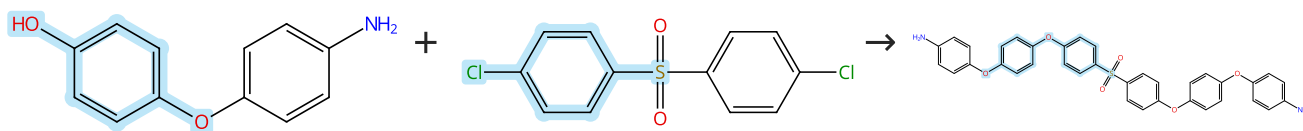
Suppliers (101)

Suppliers (6)

31-170-CAS-5120430	Synthesis of heteroaryl-fused pyrazoles as p38 kinase inhibitors
Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: <i>N</i> -Methyl-2-pyrrolidone; 48 h, 160 °C	By: Abbot, Sarah C.; et al Heterocycles (2009), 78(11), 2811-2826.

Scheme 197 (1 Reaction)

Steps: 1 Yield: 99%



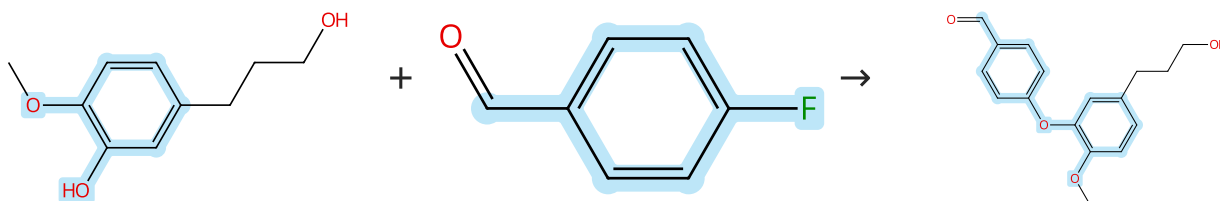
Suppliers (27)

Suppliers (84)

31-170-CAS-23277429	Novel type of highly soluble and thermally stable poly(sulfone ether imide)s
Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Toluene, <i>N</i> -Methyl-2-pyrrolidone; rt → 140 °C Experimental Protocols	By: Abbasi, Farideh; et al Designed Monomers and Polymers (2015), 18(8), 789-798.

Scheme 198 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (27)

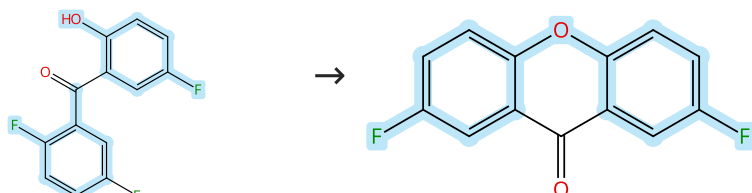
Suppliers (93)

Supplier (1)

31-170-CAS-1810991	Synthesis of isomeric corniculatolides
Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 2 h, 120 °C; 120 °C → rt 1.2 Reagents: Water; cooled Experimental Protocols	By: Raut, Gajanan N.; et al Tetrahedron Letters (2012), 53(47), 6343-6346.

Scheme 199 (1 Reaction)

Steps: 1 Yield: 99%

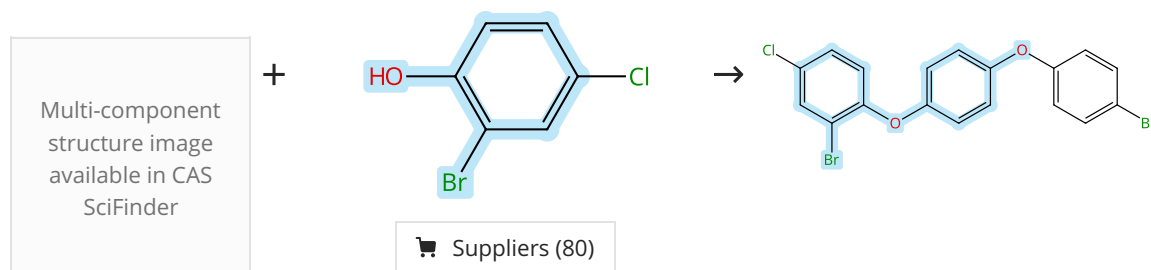


Suppliers (7)

31-614-CAS-27033782	Steps: 1 Yield: 99%	Fluorinated 9H-xanthene-9-carboxylic acid oxazol-2-ylamides as potent, orally available mGlu1 receptor enhancers By: Vieira, Eric; et al Bioorganic & Medicinal Chemistry Letters (2009), 19(6), 1666-1669.
1.1 Reagents: Potassium carbonate Solvents: Methyl ethyl ketone; reflux		

Scheme 200 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-41866199	Steps: 1 Yield: 99%	Synthesis of Diaryl Ethers via Hypervalent Iodine-Mediated C-H Functionalization By: Hu, Chenghu; et al Organic Letters (2024), 26(37), 7864-7868.
1.1 Reagents: Cesium carbonate Solvents: Toluene; 8 h, 80 °C		
Experimental Protocols		

Scheme 201 (1 Reaction)

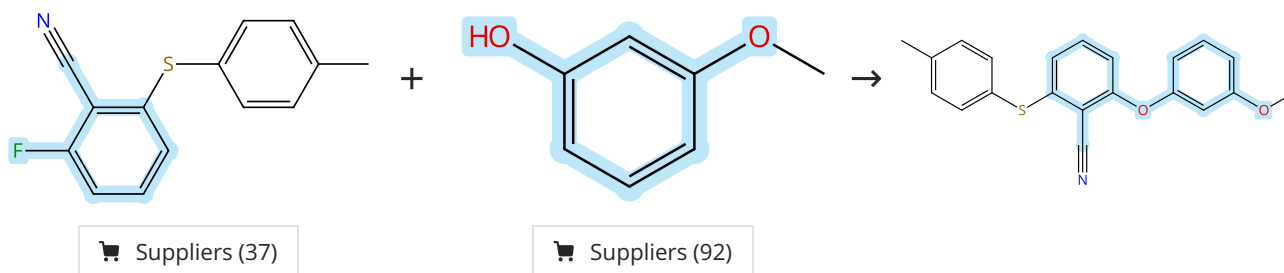
Steps: 1 Yield: 99%



31-170-CAS-18666890	Steps: 1 Yield: 99%	Discovery of AAT-008, a novel, potent, and selective prostaglandin EP4 receptor antagonist By: Okumura, Yoshiyuki; et al Bioorganic & Medicinal Chemistry Letters (2017), 27(5), 1186-1192.
1.1 Reagents: Potassium carbonate Solvents: Toluene; reflux		

Scheme 202 (1 Reaction)

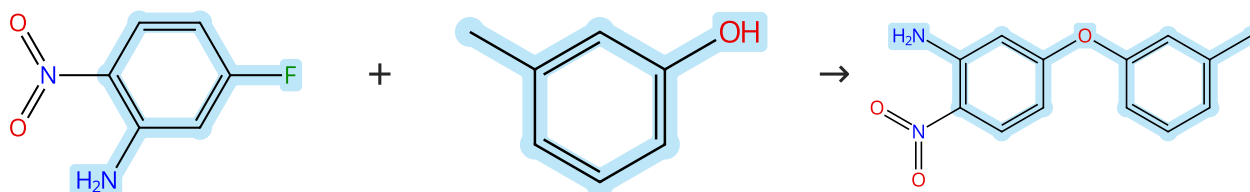
Steps: 1 Yield: 99%



31-170-CAS-1976544 Steps: 1 Yield: 99% 1.1 Catalysts: Alumina, Potassium fluoride, 18-Crown-6 Solvents: Acetonitrile	Synthesis of diaryl ethers, diaryl thioethers, and diaryl amines mediated by potassium fluoride-alumina and 18-crown-6: Expansion of scope and utility By: Sawyer, J. Scott; et al Journal of Organic Chemistry (1998), 63(18), 6338-6343.
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Scheme 203 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

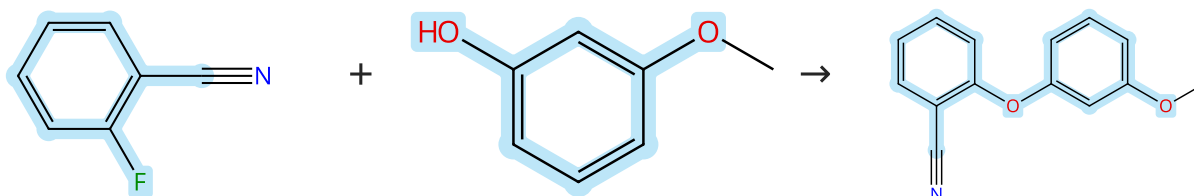
Suppliers (91)

Suppliers (7)

31-170-CAS-8719983 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C	New benzimidazole-2-urea derivatives as tubulin inhibitors By: Wang, Wenna; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.
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Scheme 204 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (87)

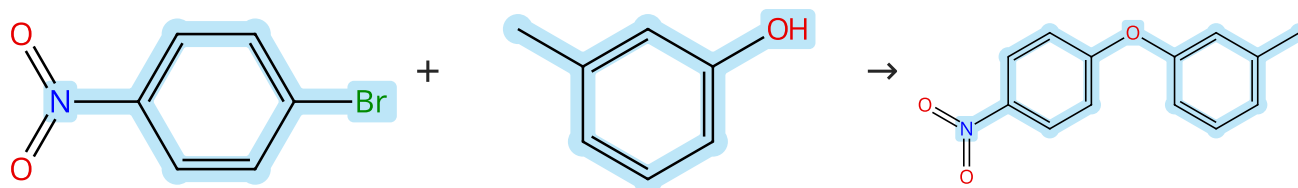
Suppliers (92)

Suppliers (8)

31-170-CAS-11104509 Steps: 1 Yield: 99% 1.1 Catalysts: Alumina, Potassium fluoride, 18-Crown-6 Solvents: Acetonitrile	Synthesis of diaryl ethers, diaryl thioethers, and diaryl amines mediated by potassium fluoride-alumina and 18-crown-6: Expansion of scope and utility By: Sawyer, J. Scott; et al Journal of Organic Chemistry (1998), 63(18), 6338-6343.
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Scheme 205 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (77)

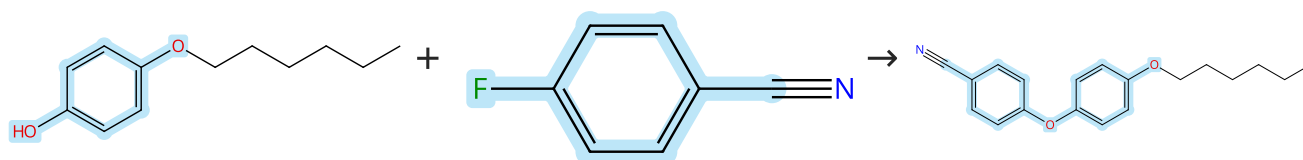
Suppliers (91)

Suppliers (39)

31-170-CAS-9910824 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Cuprous iodide, 1-[(Dimethylamino)-2-pyridinylmethyl]-2-naphthalenol Solvents: Dimethyl sulfoxide; 24 h, 120 °C Experimental Protocols	Betti Base as an Efficient Ligand for Copper-Catalyzed Ullmann Coupling of Phenol with Aryl Halides By: Yang, Li; et al Synthetic Communications (2014), 44(17), 2468-2477.
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Scheme 206 (1 Reaction)

Steps: 1 Yield: 99%



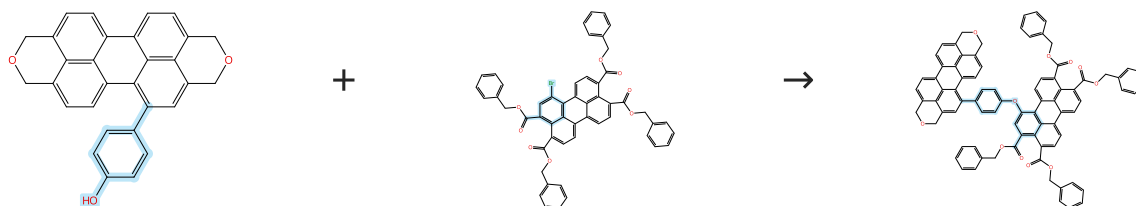
Suppliers (74)

Suppliers (92)

31-170-CAS-16314172 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 - 9 h, 150 °C Experimental Protocols	Discovery of N-(1-(3-(4-phenoxyphenyl)-1,2,4-oxadiazol-5-yl)ethyl)acetamides as novel acetyl-CoA carboxylase 2 (ACC2) inhibitors with peroxisome proliferator-activated receptor α/δ (PPARα/δ) dual agonistic activity By: Okazaki, Shogo; et al Bioorganic & Medicinal Chemistry (2016), 24(21), 5258-5269.
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Scheme 207 (1 Reaction)

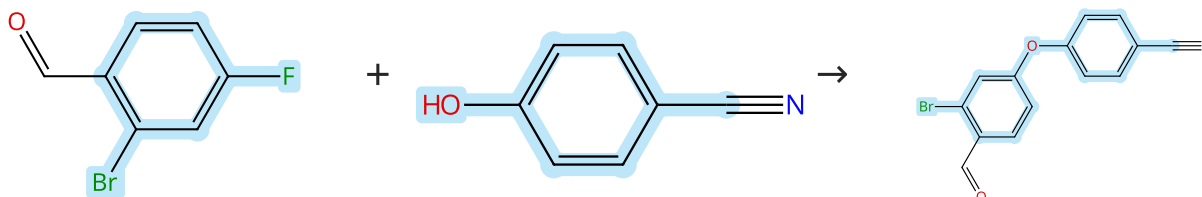
Steps: 1 Yield: 99%



31-170-CAS-18359075 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: 18-Crown-6 Solvents: Dimethylformamide; rt; rt $\rightarrow$ 50 °C; 5 h, 50 °C; 50 °C $\rightarrow$ rt 1.2 Solvents: Water; rt Experimental Protocols	Controllable Monobromination of Perylene Ring System: Synthesis of Bay-Functionalized Perylene Dyes By: Takahashi, Masaki; et al Journal of Organic Chemistry (2018), 83(2), 624-631.
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Scheme 208 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (84)

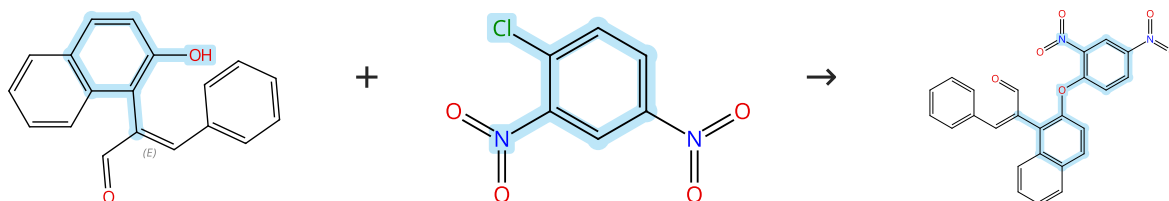
Suppliers (90)

Suppliers (8)

31-170-CAS-3046580	Steps: 1 Yield: 99%	Discovery and structure-activity study of a novel benzoxa borole anti-inflammatory agent (AN2728) for the potential topical treatment of psoriasis and atopic dermatitis By: Akama, Tsutomu; et al Bioorganic & Medicinal Chemistry Letters (2009), 19(8), 2129-2132.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 100 °C		

Scheme 209 (1 Reaction)

Steps: 1 Yield: 99%



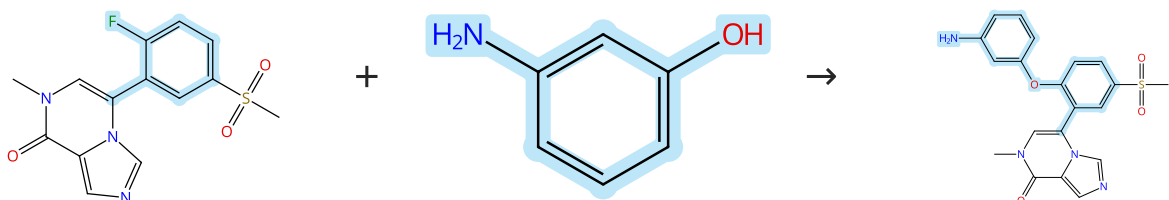
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560233	Steps: 1 Yield: 99%	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][(2 <i>S</i> ,3 <i>R</i>)-2-[[[(2 <i>R</i> ,3 <i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C		
Experimental Protocols		

Scheme 210 (1 Reaction)

Steps: 1 Yield: 99%



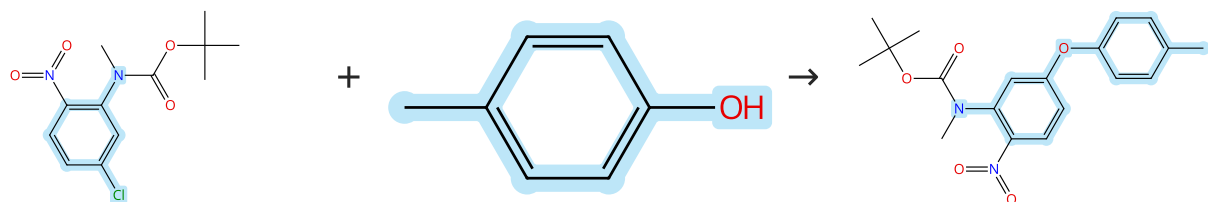
Suppliers (81)

Supplier (1)

31-170-CAS-17139443	Steps: 1 Yield: 99%	Design, synthesis and biological evaluation of 7-methylimidazo[1,5-a]pyrazin-8(7H)-one derivatives as BRD4 inhibitors By: Zhao, Leilei; et al Bioorganic & Medicinal Chemistry (2017), 25(8), 2482-2490.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 12 h, 80 °C		
Experimental Protocols		

Scheme 211 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (8)

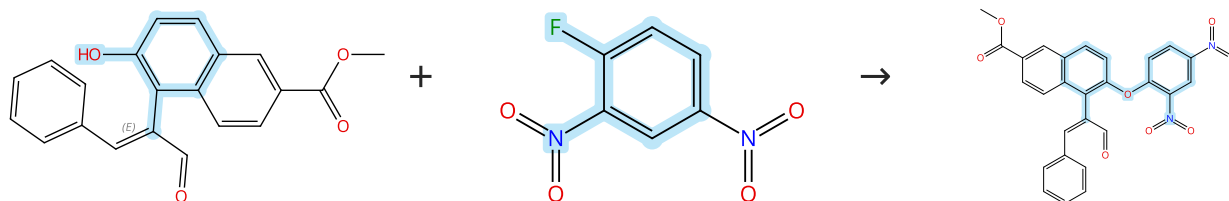
Suppliers (89)

Supplier (1)

31-170-CAS-19673616	Steps: 1 Yield: 99%	Discovery of DS-6930, a potent selective PPAR γ modulator. Part I: Lead identification
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; rt; 10 h, 80 °C		By: Shinozuka, Tsuyoshi; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2018), 26(18), 5079-5098.

Scheme 212 (1 Reaction)

Steps: 1 Yield: 99%



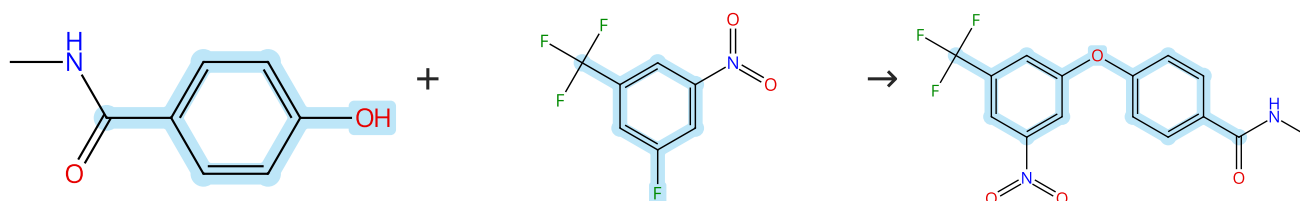
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560103	Steps: 1 Yield: 99%	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution
1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2 <i>S</i> ,3 <i>R</i>)-2-[[[(2 <i>R</i> ,3 <i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C		By: Guo, Fengyuan; et al
Experimental Protocols		Nature Communications (2023), 14(1), 5050.

Scheme 213 (1 Reaction)

Steps: 1 Yield: 99%



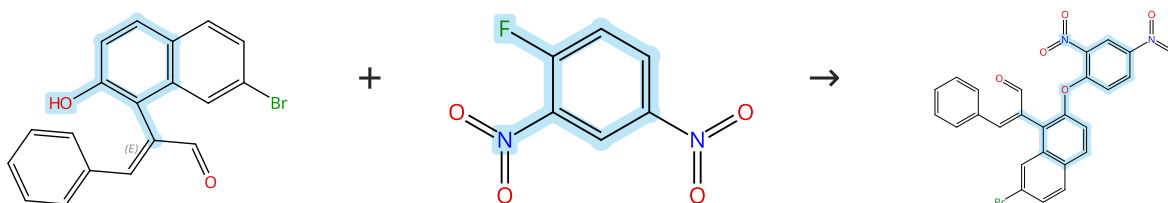
Suppliers (58)

Suppliers (70)

31-170-CAS-20939823	Steps: 1 Yield: 99%	Design, synthesis, and in vitro bioactivity evaluation of fluorine-containing analogues for sphingosine-1-phosphate 2 receptor
1.1 Reagents: Monopotassium phosphate Solvents: Dimethylacetamide; 12 h, 100 °C		By: Luo, Zonghua; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2019), 27(16), 3619-3631.

Scheme 214 (1 Reaction)

Steps: 1 Yield: 99%



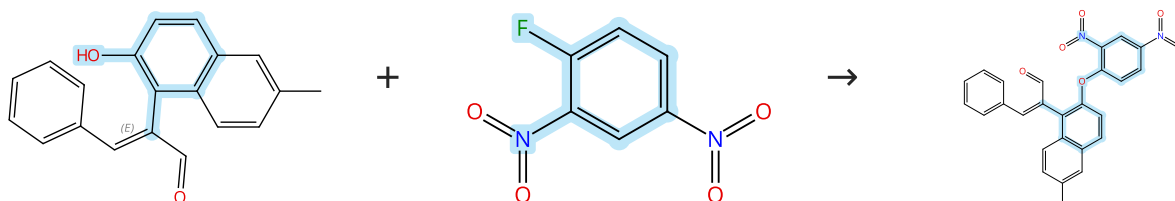
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560095 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 215 (1 Reaction)

Steps: 1 Yield: 99%



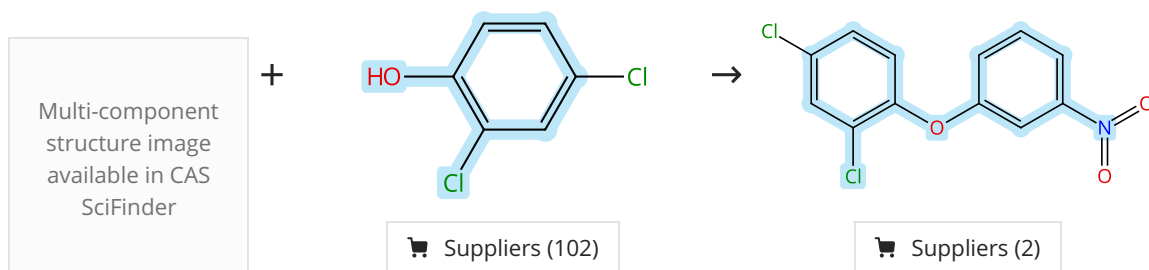
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560097 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 216 (1 Reaction)

Steps: 1 Yield: 99%

Multi-component
structure image
available in CAS
SciFinder

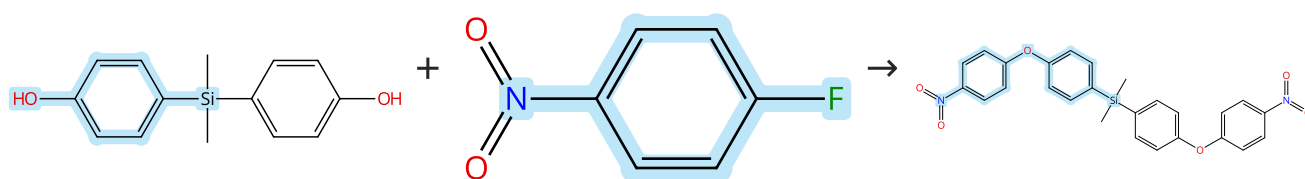
Suppliers (102)

Suppliers (2)

31-614-CAS-41866225 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Solvents: Toluene; 10 h, 80 °C Experimental Protocols	Synthesis of Diaryl Ethers via Hypervalent Iodine-Mediated C-H Functionalization By: Hu, Chenghu; et al Organic Letters (2024), 26(37), 7864-7868.
---	--

Scheme 217 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (3)

Suppliers (100)

31-170-CAS-17207705	Steps: 1 Yield: 99%	Silarylene-containing oligo(ether-amide)s based on bis(4-(4-amino phenoxy)phenyl)dimethylsilane. Effect of the dicarboxylic acid structure on some properties
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 60 °C		By: Terraza, Claudio A.; et al
Experimental Protocols		RSC Advances (2015), 5(36), 28515-28526.

Scheme 218 (1 Reaction)

Steps: 1 Yield: 99%

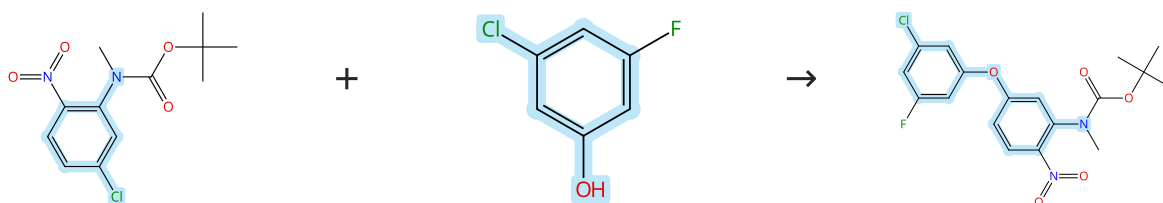


Supplier (1)

31-614-CAS-28633964	Steps: 1 Yield: 99%	Fluorinated 9H-xanthene-9-carboxylic acid oxazol-2-ylamides as potent, orally available mGlu1 receptor enhancers
1.1 Reagents: Potassium carbonate Solvents: Methyl ethyl ketone; reflux		By: Vieira, Eric; et al
		Bioorganic & Medicinal Chemistry Letters (2009), 19(6), 1666-1669.

Scheme 219 (1 Reaction)

Steps: 1 Yield: 99%



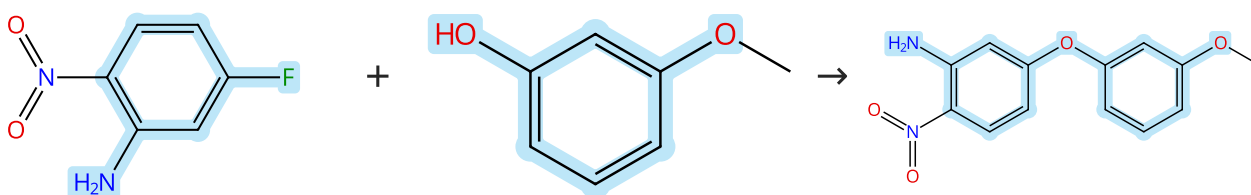
Suppliers (8)

Suppliers (67)

31-170-CAS-19673638	Steps: 1 Yield: 99%	Discovery of DS-6930, a potent selective PPARγ modulator. Part I: Lead identification
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; rt; 10 h, 80 °C		By: Shinozuka, Tsuyoshi; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2018), 26(18), 5079-5098.

Scheme 220 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

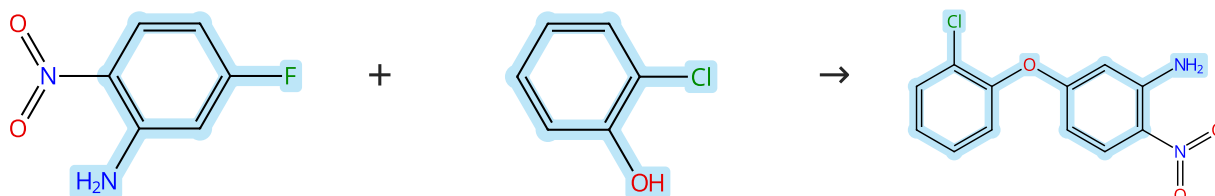
Suppliers (92)

Suppliers (4)

31-170-CAS-9042723	Steps: 1 Yield: 99%	New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C	By: Wang, Wenna; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.	

Scheme 221 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

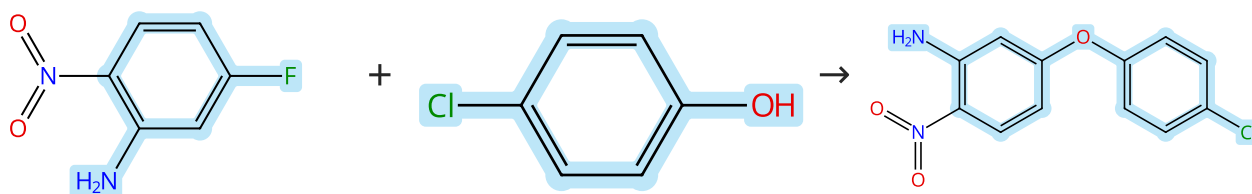
Suppliers (92)

Suppliers (5)

31-170-CAS-8765385	Steps: 1 Yield: 99%	New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C	By: Wang, Wenna; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.	

Scheme 222 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

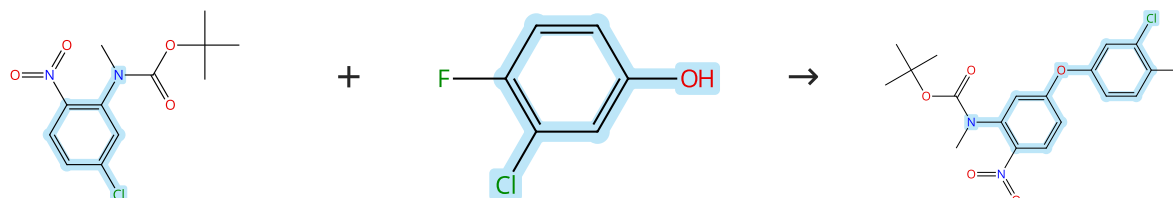
Suppliers (90)

Suppliers (5)

31-170-CAS-4791827	Steps: 1 Yield: 99%	New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C	By: Wang, Wenna; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.	

Scheme 223 (1 Reaction)

Steps: 1 Yield: 99%



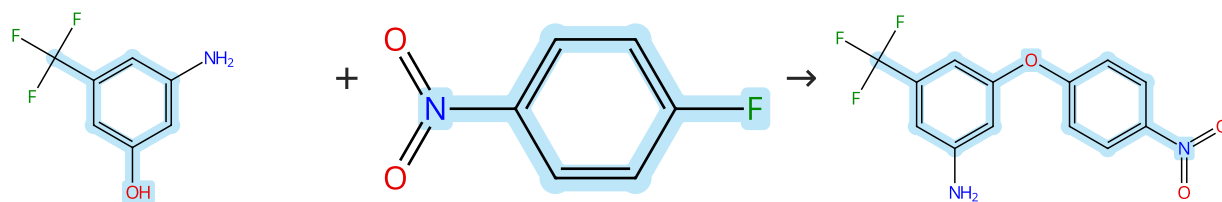
Suppliers (8)

Suppliers (82)

31-170-CAS-19673639	Steps: 1 Yield: 99%	Discovery of DS-6930, a potent selective PPARγ modulator. Part I: Lead identification
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; rt; 10 h, 80 °C	By: Shinozuka, Tsuyoshi; et al	
Experimental Protocols	Bioorganic & Medicinal Chemistry (2018), 26(18), 5079-5098.	

Scheme 224 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (50)

Suppliers (100)

Suppliers (3)

31-170-CAS-23272226

Steps: 1 Yield: 99%

Discovery of Novel Small Molecule Dual Inhibitors Targeting Toll-Like Receptors 7 and 8

By: Padilla-Salinas, Rosaura; et al

Journal of Medicinal Chemistry (2019), 62(22), 10221-10244.

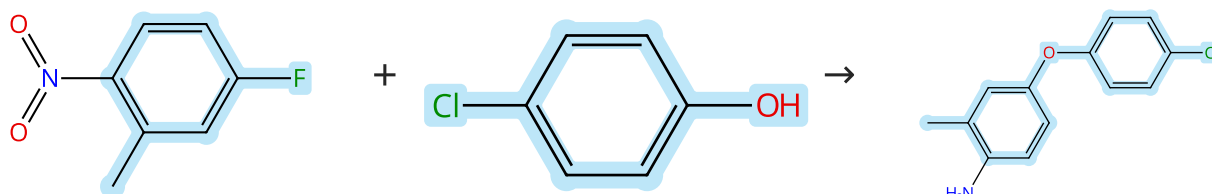
1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 24 - 76 h, 90 °C

Experimental Protocols

Scheme 225 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

Suppliers (90)

Suppliers (27)

31-170-CAS-1323898

Steps: 1 Yield: 99%

Discovery and structure-activity study of a novel benzoxa borole anti-inflammatory agent (AN2728) for the potential topical treatment of psoriasis and atopic dermatitis

By: Akama, Tsutomu; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(8), 2129-2132.

1.1 Reagents: Potassium carbonate

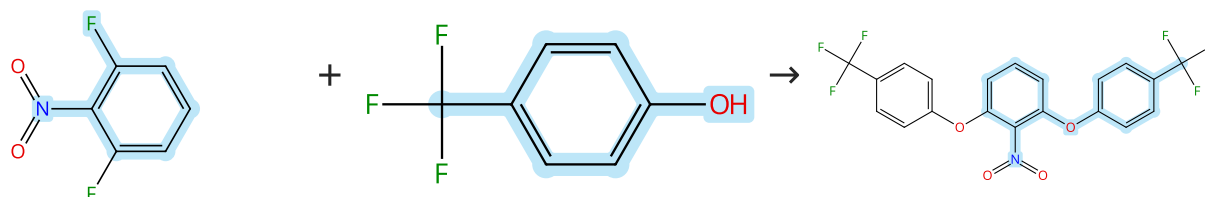
Solvents: Dimethylformamide; overnight, 70 °C

1.2 Reagents: Hydrochloric acid, Stannous chloride

Solvents: Ethanol; 2 h, 50 °C

Scheme 226 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (96)

31-614-CAS-40274057

Steps: 1 Yield: 99%

A Strategy for Improving the Efficiency of Boronic Acid Catalysis in the Synthesis of Amides

By: Almetwali, Fatima; et al

European Journal of Organic Chemistry (2023), 26(45), e202300720.

1.1 Reagents: Potassium carbonate

Solvents: Dimethyl sulfoxide; 4 h, 100 °C; 100 °C → rt

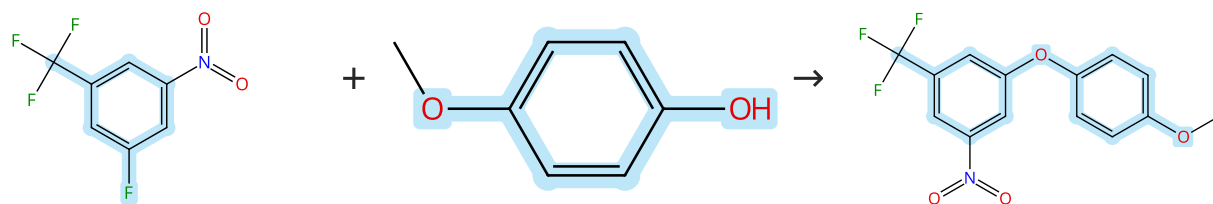
1.2 Reagents: Sodium hydroxide

Solvents: Water; rt

Experimental Protocols

Scheme 227 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (70)

Suppliers (121)

31-170-CAS-20939821

Steps: 1 Yield: 99%

Design, synthesis, and in vitro bioactivity evaluation of fluorine-containing analogues for sphingosine-1-phosphate 2 receptor

1.1 **Reagents:** Monopotassium phosphate
Solvents: Dimethylacetamide; 12 h, 100 °C

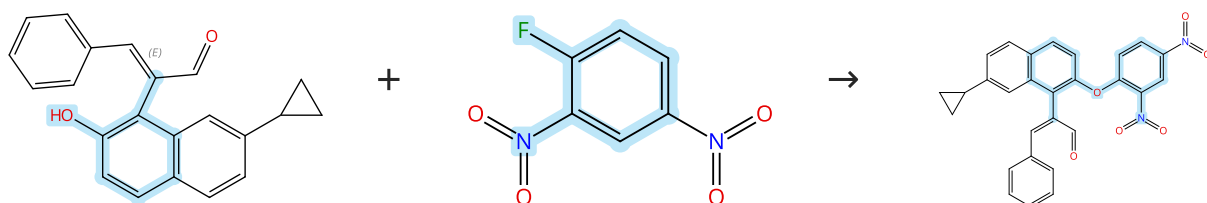
Experimental Protocols

By: Luo, Zonghua; et al

Bioorganic & Medicinal Chemistry (2019), 27(16), 3619-3631.

Scheme 228 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560084

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

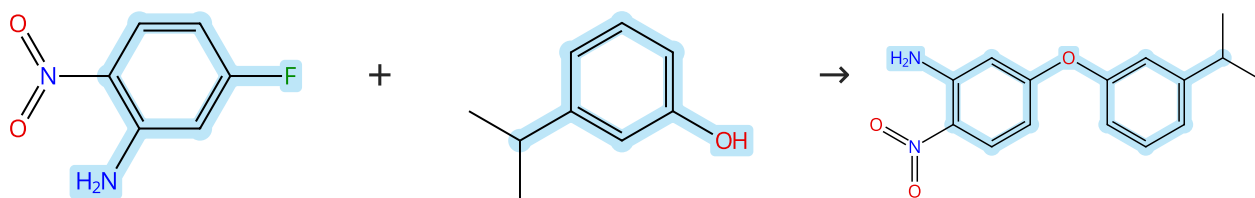
By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Experimental Protocols

Scheme 229 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

Suppliers (51)

Supplier (1)

31-170-CAS-2320467

Steps: 1 Yield: 99%

New benzimidazole-2-urea derivatives as tubulin inhibitors

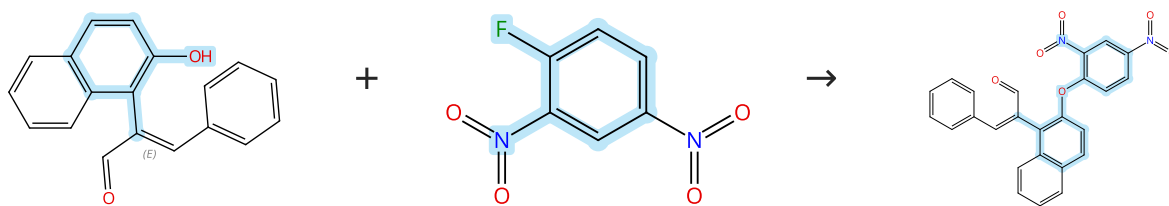
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 100 °C

By: Wang, Wenna; et al

Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 230 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560035

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

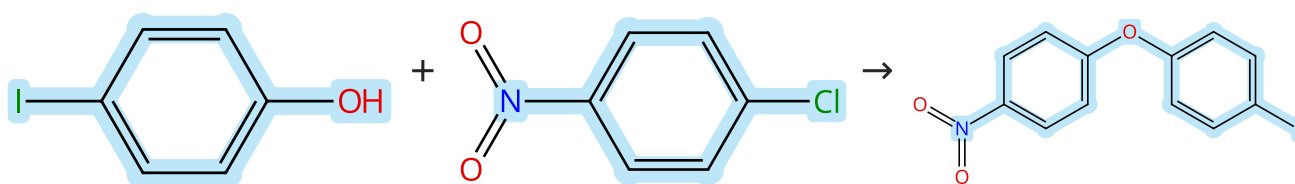
Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Scheme 231 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (86)

Suppliers (65)

Suppliers (43)

31-170-CAS-1653451

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Cuprous iodide
Solvents: Dimethylformamide; 30 min, rt; 2 h, 110 °C; 110 °C → rt

Experimental Protocols

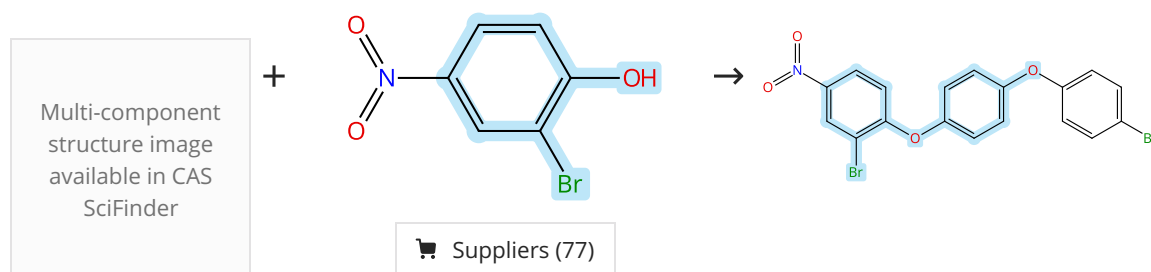
CuI Nanoparticles for C-N and C-O Cross Coupling of Heterocyclic Amines and Phenols with Chlorobenzenes

By: Sreedhar, B.; et al

Journal of Organic Chemistry (2009), 74(20), 7951-7954.

Scheme 232 (1 Reaction)

Steps: 1 Yield: 99%



Multi-component
structure image
available in CAS
SciFinder

Suppliers (77)

31-614-CAS-41866186

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Cesium carbonate
Solvents: Toluene; 8 h, 80 °C

Experimental Protocols

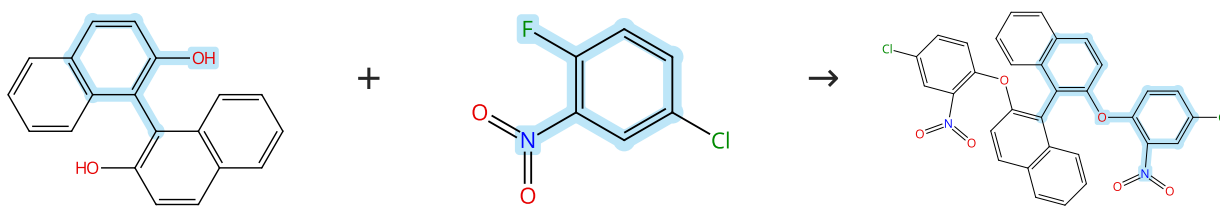
Synthesis of Diaryl Ethers via Hypervalent Iodine-Mediated C-H Functionalization

By: Hu, Chenghu; et al

Organic Letters (2024), 26(37), 7864-7868.

Scheme 233 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

Suppliers (81)

31-170-CAS-2208708

Steps: 1 Yield: 99%

Enantioselective alkynylation to aldimines catalyzed by chiral 2,2'-bis[(2-aminoaryl)oxy]-1,1'-binaphthyl-copper(I) complexes

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 16 h, 80 °C

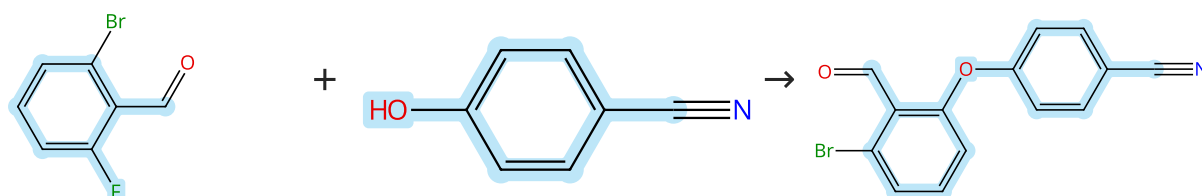
Experimental Protocols

By: Hatano, Manabu; et al

Tetrahedron Letters (2008), 49(2), 379-382.

Scheme 234 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (84)

Suppliers (90)

Suppliers (9)

31-170-CAS-948982

Steps: 1 Yield: 99%

Discovery and structure-activity study of a novel benzoxa borole anti-inflammatory agent (AN2728) for the potential topical treatment of psoriasis and atopic dermatitis

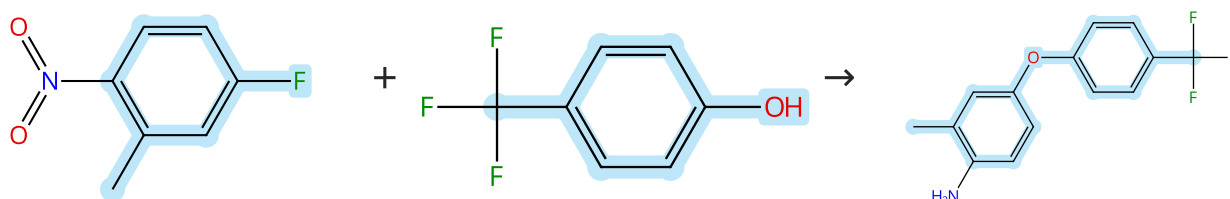
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; overnight, 100 °C

By: Akama, Tsutomu; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(8), 2129-2132.

Scheme 235 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

Suppliers (96)

Suppliers (29)

31-170-CAS-7721219

Steps: 1 Yield: 99%

Discovery and structure-activity study of a novel benzoxa borole anti-inflammatory agent (AN2728) for the potential topical treatment of psoriasis and atopic dermatitis

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; overnight, 70 °C

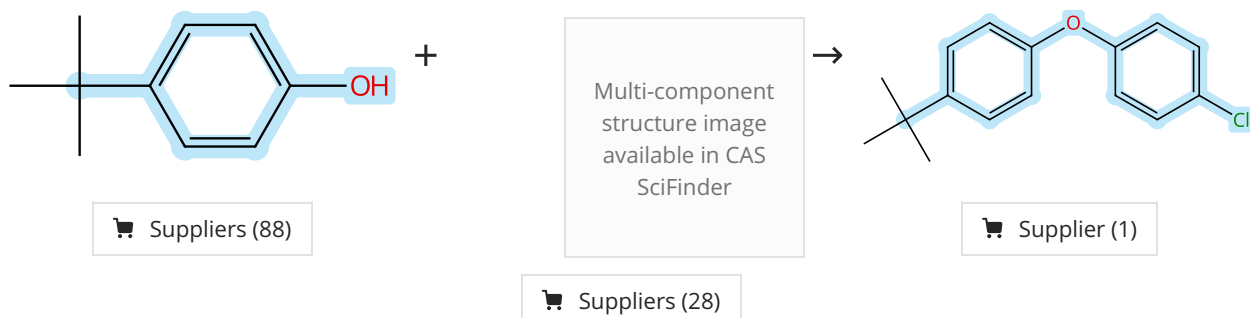
1.2 **Reagents:** Hydrochloric acid, Stannous chloride
Solvents: Ethanol; 2 h, 50 °C

By: Akama, Tsutomu; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(8), 2129-2132.

Scheme 236 (2 Reactions)

Steps: 1 Yield: 99%



31-170-CAS-2916862

Steps: 1 Yield: 99%

Metal-free arylation of oxygen nucleophiles with diaryliodonium salts

By: Jalalian, Nazli; et al

Chemistry - A European Journal (2012), 18(44), 14140-14149, S14140/1-S14140/63.

- 1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; 15 min, 0 °C
- 1.2 40 min, 40 °C
- 1.3 **Reagents:** Water; 0 °C

Experimental Protocols

31-170-CAS-12032922

Steps: 1 Yield: 99%

Room temperature, metal-free synthesis of diaryl ethers with use of diaryliodonium salts

By: Jalalian, Nazli; et al

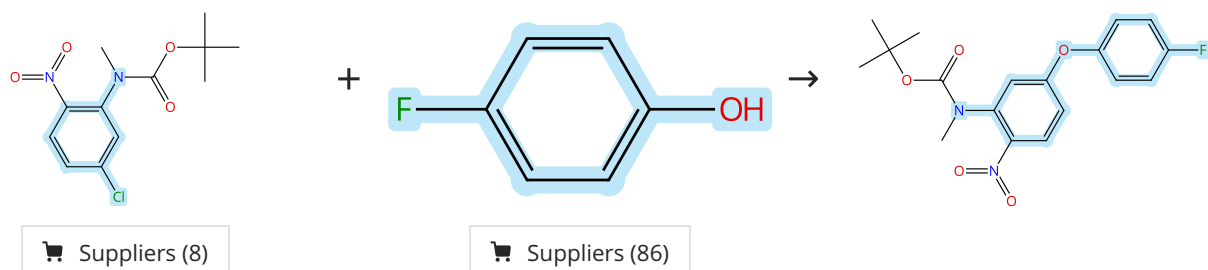
Organic Letters (2011), 13(6), 1552-1555.

- 1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; 15 min, 0 °C
- 1.2 0 °C; 40 min, 40 °C

Experimental Protocols

Scheme 237 (1 Reaction)

Steps: 1 Yield: 99%



31-170-CAS-19673613

Steps: 1 Yield: 99%

Discovery of DS-6930, a potent selective PPAR γ modulator. Part I: Lead identification

By: Shinozuka, Tsuyoshi; et al

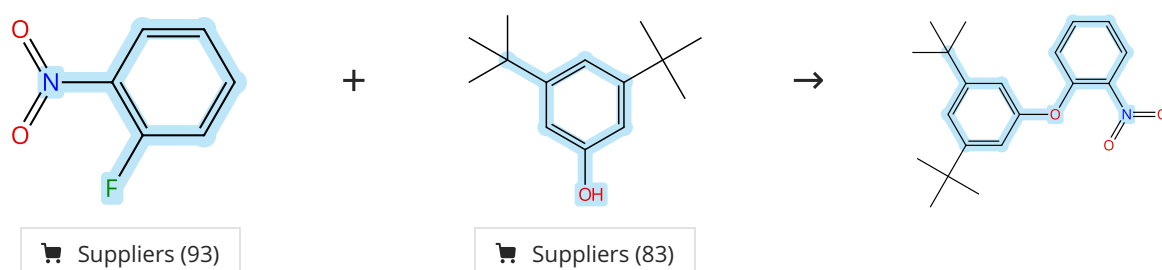
Bioorganic & Medicinal Chemistry (2018), 26(18), 5079-5098.

- 1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide; rt; 10 h, 80 °C

Experimental Protocols

Scheme 238 (1 Reaction)

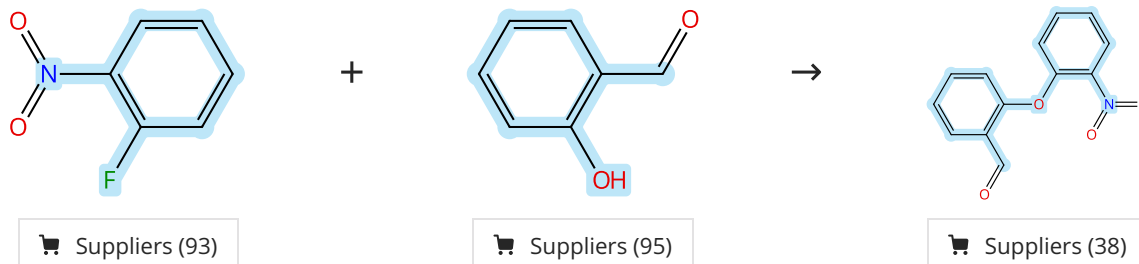
Steps: 1 Yield: 99%



31-170-CAS-22969183	Steps: 1 Yield: 99%	Synthesis of Cycloheptatrienes, Oxepines, Thiepinines, and Silepinines: A Comparison between Broensted Acid and Au-Catalysis
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt → 95 °C; 24 h, 95 °C		By: Sprenger, Kristin; et al
Experimental Protocols		European Journal of Organic Chemistry (2020), 2020(39), 6245-6254.

Scheme 239 (1 Reaction)

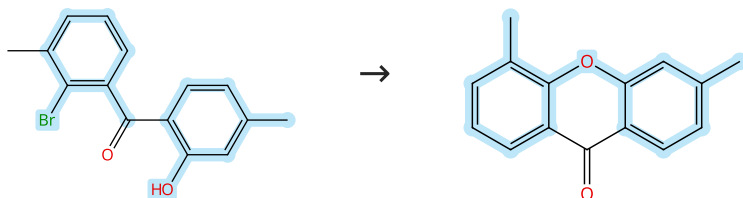
Steps: 1 Yield: 99%



31-170-CAS-17072062	Steps: 1 Yield: 99%	NHC-Organocatalyzed C_{Ar}-O Bond Cleavage: Mild Access to 2-Hydroxybenzophenones
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 100 °C		By: Janssen-Mueller, Daniel; et al
		Angewandte Chemie, International Edition (2017), 56(22), 6276-6279.

Scheme 240 (1 Reaction)

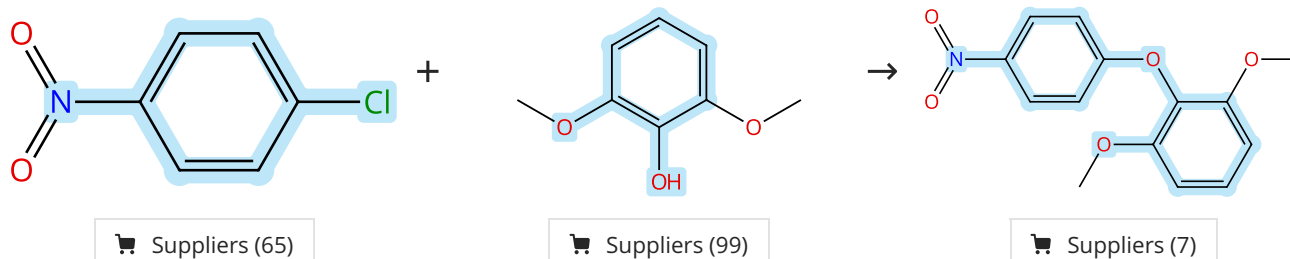
Steps: 1 Yield: 99%



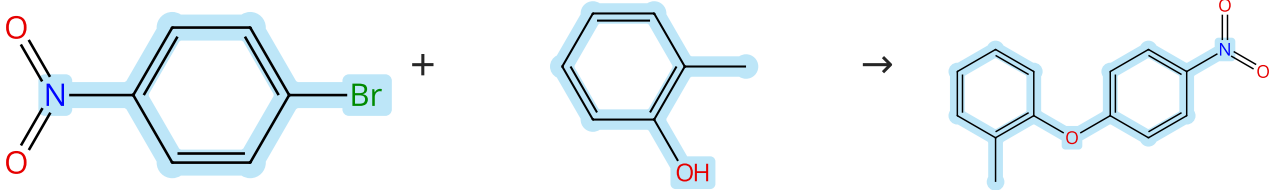
31-614-CAS-25023334	Steps: 1 Yield: 99%	Microwave-assisted transition-metal-free intramolecular Ullmann-type O-arylation in water for the synthesis of xanthenes and azaxanthenes
1.1 Reagents: Tetrabutylammonium hydroxide Solvents: Water; 8 min, 120 °C		By: Zhang, Xiaojin; et al
Experimental Protocols		Tetrahedron Letters (2014), 55(35), 4883-4887.

Scheme 241 (1 Reaction)

Steps: 1 Yield: 99%

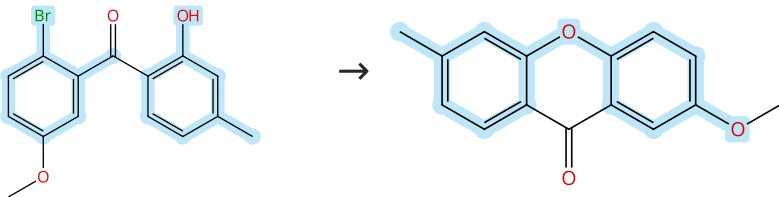


31-170-CAS-13314388	Steps: 1 Yield: 99%	Recyclable mesoporous ceria supported gold nanoparticles based catalyst for O-arylation of phenols
1.1 Reagents: Potassium hydroxide Catalysts: Ceria, Gold (mesoporous ceria supported) Solvents: Dimethyl sulfoxide; 2 h, 110 °C		By: Emayavaramban, P.; et al
1.2 Reagents: Water; cooled		Journal of Nanoscience and Nanotechnology (2015), 15(12), 9358-9368.
Experimental Protocols		

Scheme 242 (2 Reactions)		Steps: 1 Yield: 99%
		
 Suppliers (77)	 Suppliers (83)	 Suppliers (10)

31-170-CAS-1681012	Steps: 1 Yield: 99%	CuO Nanoparticles on Ionic Polymer for Coupling Reactions of Aryl Bromides and Chlorides with Phenols
1.1 Reagents: Tripotassium phosphate Catalysts: Copper oxide (CuO), Benzenemethanaminium, <i>N</i> -[bis(dimethylamino)methylene]-4-ethenyl- <i>N</i> -[(4-ethenyl phenyl)methyl]-, chloride (1:1), polymer with diethenylbenzene Solvents: 1,4-Dioxane; 16 h, 120 °C		By: Hu, Tongjie; et al
Experimental Protocols		ChemCatChem (2011), 3(4), 661-665.

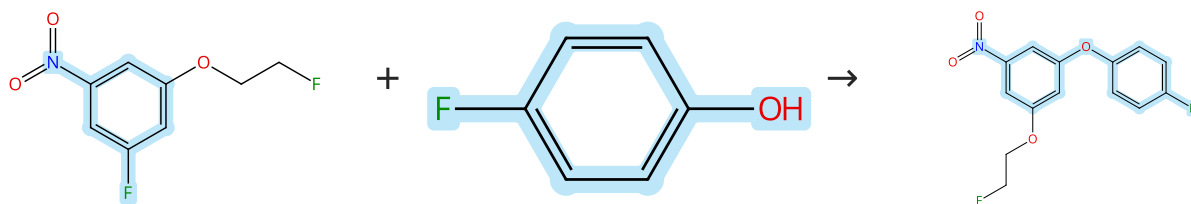
31-170-CAS-15072106	Steps: 1 Yield: 99%	Bio-inspired copper catalysts for the formation of diaryl ethers
1.1 Reagents: Potassium carbonate, Butylimidazole Catalysts: Cuprous chloride Solvents: Toluene; 16 h, 120 °C		By: Schareina, Thomas; et al
Experimental Protocols		Tetrahedron Letters (2008), 49(11), 1851-1855.

Scheme 243 (1 Reaction)		Steps: 1 Yield: 99%
		

31-614-CAS-26427750	Steps: 1 Yield: 99%	Microwave-assisted transition-metal-free intramolecular Ullmann-type O-arylation in water for the synthesis of xanthenes and azaxanthenes
1.1 Reagents: Tetrabutylammonium hydroxide Solvents: Water; 8 min, 120 °C		By: Zhang, Xiaojin; et al
Experimental Protocols		Tetrahedron Letters (2014), 55(35), 4883-4887.

Scheme 244 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

Suppliers (86)

31-170-CAS-20939805

Steps: 1 Yield: 99%

1.1 Solvents: Dimethylacetamide; 12 h, 100 °C

Experimental Protocols

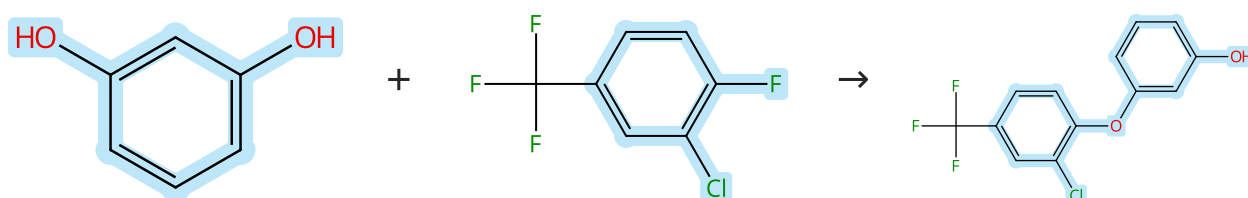
Design, synthesis, and in vitro bioactivity evaluation of fluorine-containing analogues for sphingosine-1-phosphate 2 receptor

By: Luo, Zonghua; et al

Bioorganic & Medicinal Chemistry (2019), 27(16), 3619-3631.

Scheme 245 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (109)

Suppliers (68)

Suppliers (6)

31-614-CAS-32517468

Steps: 1 Yield: 99%

1.1 Reagents: Potassium hydroxide
Solvents: Dimethyl sulfoxide; 20 - 30 °C

1.2 3 h, 50 °C; 24 h, 130 °C; 130 °C → 50 °C

1.3 Reagents: Hydrochloric acid
Solvents: Water

Experimental Protocols

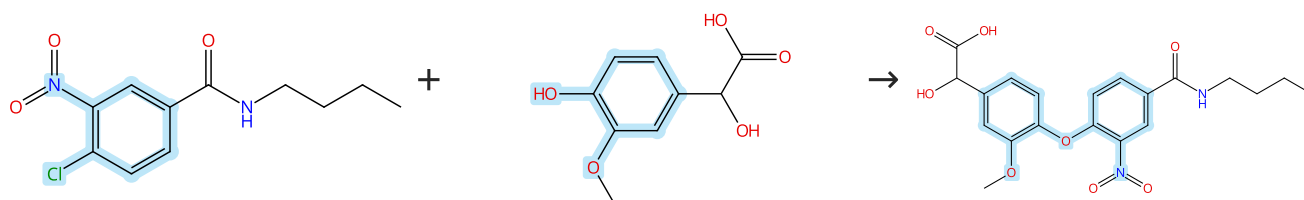
Highly Enantioselective Rhodium-Catalyzed Transfer Hydrogenation of Tetrasubstituted Olefins: Application toward the Synthesis of GPR40 Agonist MK-2305

By: Zhong, Yong-Li; et al

Organic Letters (2022), 24(17), 3254-3258.

Scheme 246 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (28)

Suppliers (100)

31-170-CAS-558518

Steps: 1 Yield: 99%

1.1 Reagents: Cesium carbonate
Solvents: Dimethyl sulfoxide; rt; rt → 60 °C; 4 h, 60 °C; 60 °C → rt1.2 Reagents: Hydrochloric acid
Solvents: Water; pH 3

Experimental Protocols

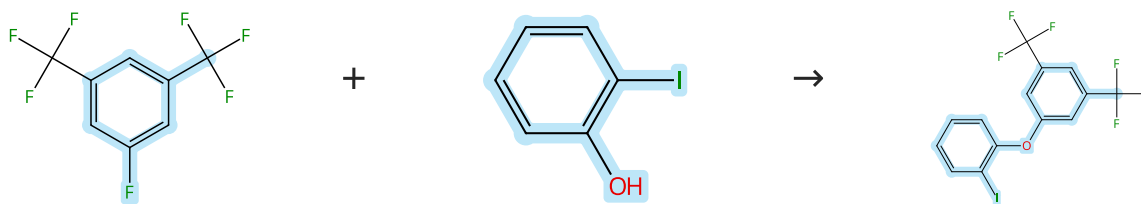
Pd-Catalyzed Deoxygenation of Mandelate Esters

By: Milne, Jacqueline E.; et al

Journal of Organic Chemistry (2009), 74(1), 445-447.

Scheme 247 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (45)

Suppliers (85)

31-614-CAS-40274042

Steps: 1 Yield: 99%

A Strategy for Improving the Efficiency of Boronic Acid Catalysis in the Synthesis of Amides

By: Almetwali, Fatima; et al

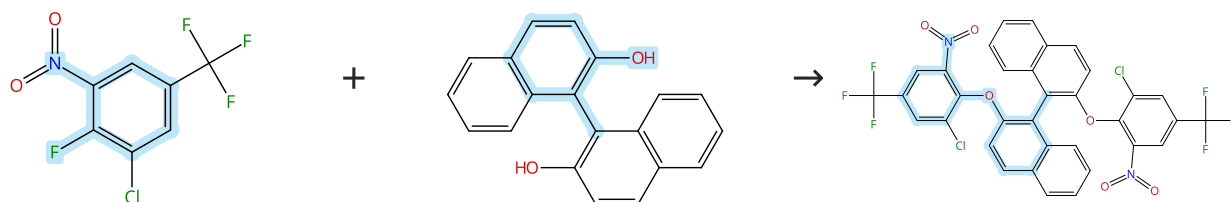
European Journal of Organic Chemistry (2023), 26(45), e202300720.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; 100 °C; 100 °C → 25 °C
- 1.2 **Reagents:** Sodium hydroxide
Solvents: Water; 25 °C

Experimental Protocols

Scheme 248 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (48)

Suppliers (85)

31-170-CAS-6471559

Steps: 1 Yield: 99%

Enantioselective alkylation to aldimines catalyzed by chiral 2,2'-bis[(2-aminoaryl)oxy]-1,1'-binaphthyl-copper(I) complexes

By: Hatano, Manabu; et al

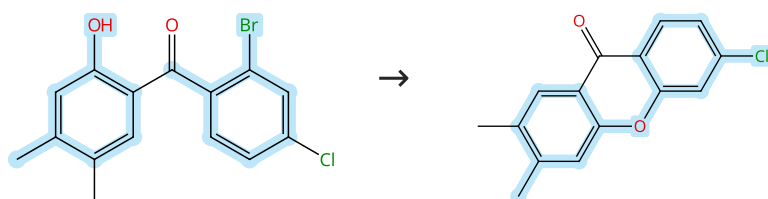
Tetrahedron Letters (2008), 49(2), 379-382.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 16 h, 80 °C

Experimental Protocols

Scheme 249 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

Suppliers (2)

31-614-CAS-26451208

Steps: 1 Yield: 99%

A convenient approach to the xanthone scaffold by an aqueous aromatic substitution of bromo- and iodoarenes

By: Barbero, Nekane; et al

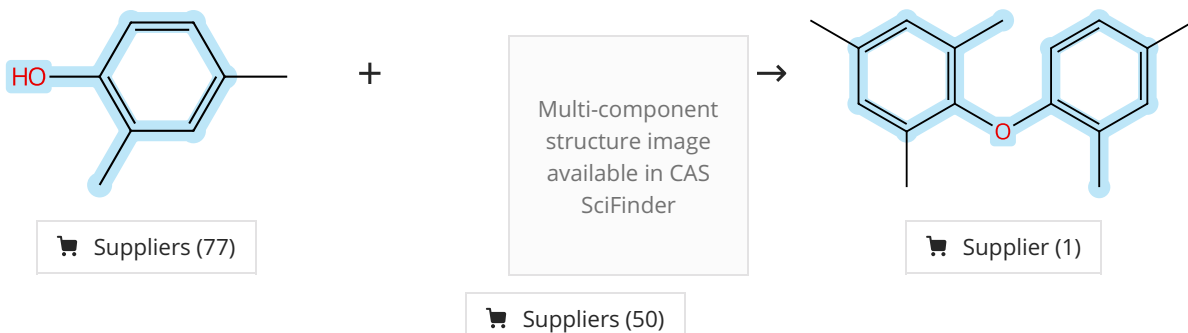
Tetrahedron (2009), 65(29-30), 5729-5732.

- 1.1 **Reagents:** Potassium hydroxide
Solvents: Water; 15 h, 100 °C

Experimental Protocols

Scheme 250 (2 Reactions)

Steps: 1 Yield: 99%



31-170-CAS-14163409

Steps: 1 Yield: 99%

Metal-free arylation of oxygen nucleophiles with diaryliodonium salts

By: Jalalian, Nazli; et al

Chemistry - A European Journal (2012), 18(44), 14140-14149, S14140/1-S14140/63.

- 1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; 15 min, 0 °C
- 1.2 15 min, 40 °C
- 1.3 **Reagents:** Water; 0 °C

Experimental Protocols

31-170-CAS-9902171

Steps: 1 Yield: 99%

Room temperature, metal-free synthesis of diaryl ethers with use of diaryliodonium salts

By: Jalalian, Nazli; et al

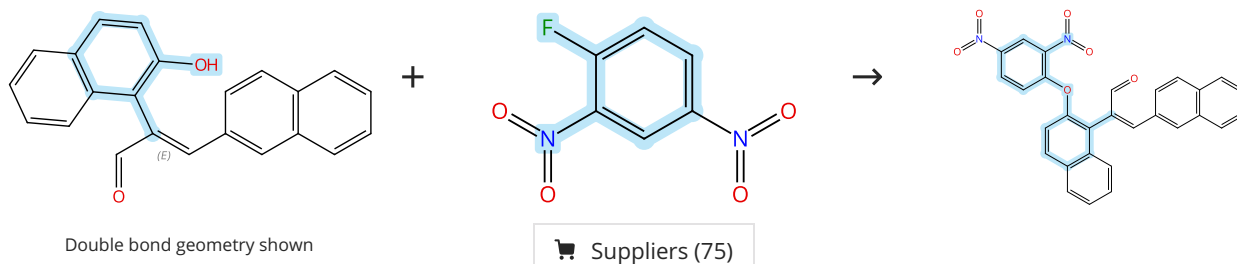
Organic Letters (2011), 13(6), 1552-1555.

- 1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; 15 min, 0 °C
- 1.2 0 °C; 15 min, 40 °C

Experimental Protocols

Scheme 251 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-37560065

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

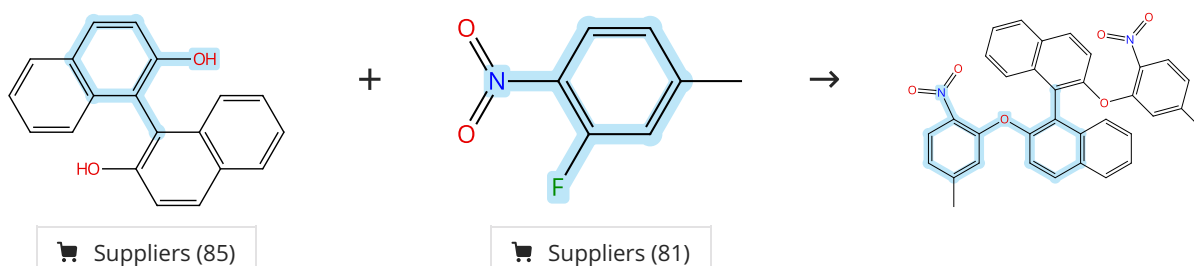
Nature Communications (2023), 14(1), 5050.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

Scheme 252 (1 Reaction)

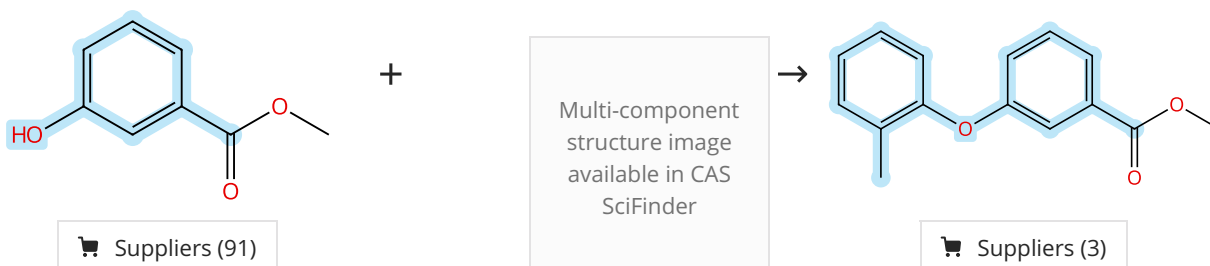
Steps: 1 Yield: 99%



31-170-CAS-13468993	Steps: 1 Yield: 99%	Enantioselective alkynylation to aldimines catalyzed by chiral 2,2'-bis[(2-aminoaryl)oxy]-1,1'-binaphthyl-copper(I) complexes
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, 80 °C		By: Hatano, Manabu; et al
Experimental Protocols		Tetrahedron Letters (2008), 49(2), 379-382.

Scheme 253 (1 Reaction)

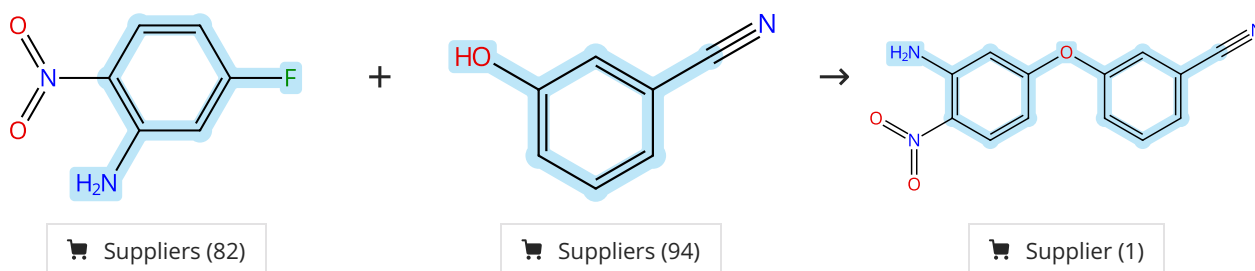
Steps: 1 Yield: 99%



31-170-CAS-4946599	Steps: 1 Yield: 99%	Chelation-Assisted Nickel-Catalyzed Oxidative Annulation via Double C-H Activation/Alkyne Insertion Reaction
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Tetrahydrofuran; 15 min, 0 °C		By: Misal Castro, Luis C.; et al
1.2 0 °C → rt; 3 h, rt		Chemistry - A European Journal (2016), 22(4), 1362-1367.
Experimental Protocols		

Scheme 254 (1 Reaction)

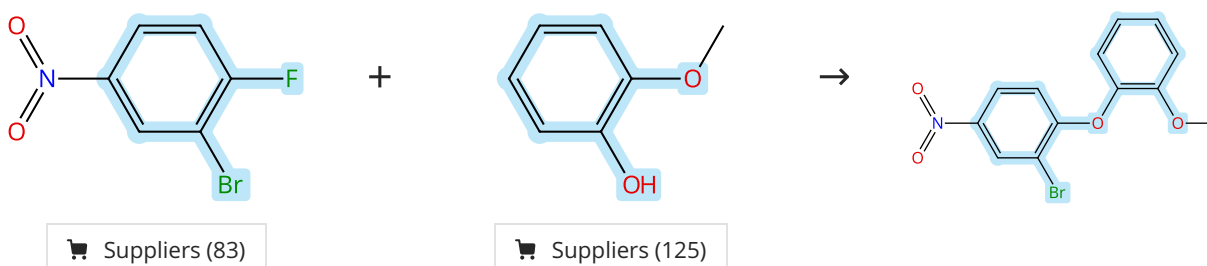
Steps: 1 Yield: 99%



31-170-CAS-6314722	Steps: 1 Yield: 99%	New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C		By: Wang, Wenna; et al
		Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 255 (1 Reaction)

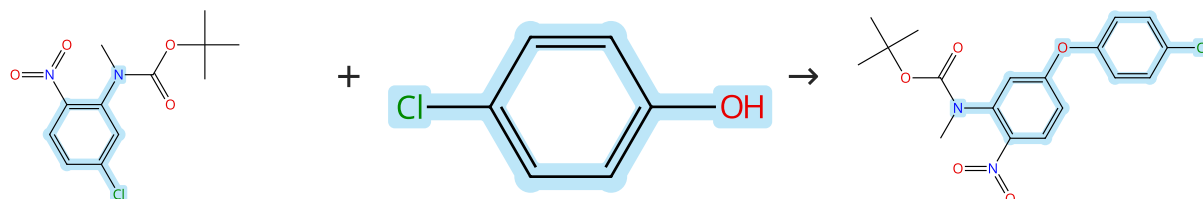
Steps: 1 Yield: 99%



31-614-CAS-35312185	Steps: 1 Yield: 99%	Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 2 h, 110 °C		By: Xu, Kai-Yan; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2023), 78, 117152.

Scheme 256 (1 Reaction)

Steps: 1 Yield: 99%



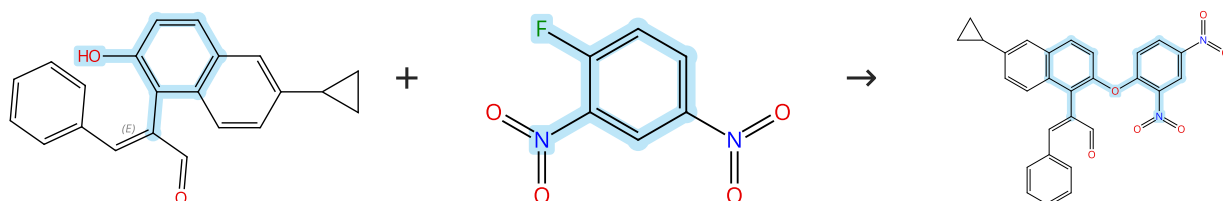
Suppliers (8)

Suppliers (90)

31-170-CAS-19673618	Steps: 1 Yield: 99%	Discovery of DS-6930, a potent selective PPAR γ modulator. Part I: Lead identification
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; rt; 10 h, 80 °C		By: Shinozuka, Tsuyoshi; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2018), 26(18), 5079-5098.

Scheme 257 (1 Reaction)

Steps: 1 Yield: 99%



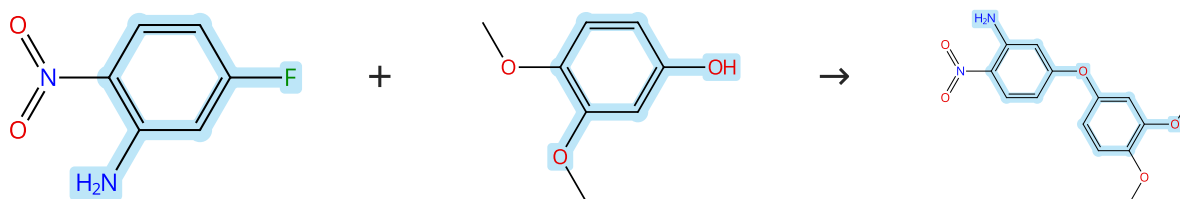
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560092	Steps: 1 Yield: 99%	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S _N Ar reaction-guided dynamic kinetic resolution
1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2 <i>S</i> ,3 <i>R</i>)-2-[[[(1 <i>R</i> ,3 <i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C		By: Guo, Fengyuan; et al
Experimental Protocols		Nature Communications (2023), 14(1), 5050.

Scheme 258 (1 Reaction)

Steps: 1 Yield: 99%



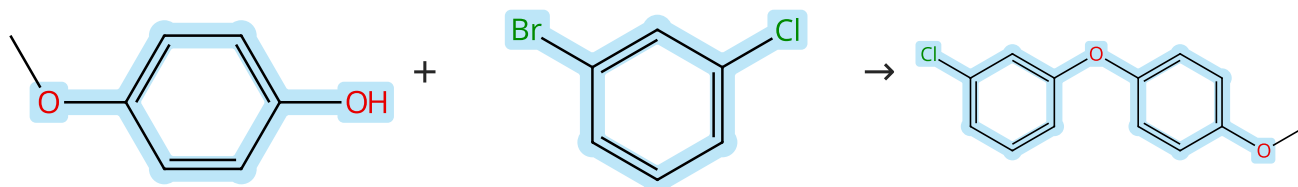
Suppliers (82)

Suppliers (103)

31-170-CAS-13302980	Steps: 1 Yield: 99%	New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C		By: Wang, Wenna; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 259 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (121)

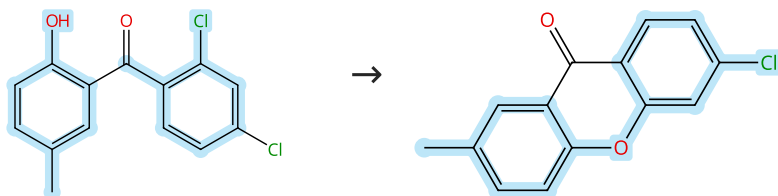
Suppliers (83)

Suppliers (7)

31-614-CAS-43130456	Steps: 1 Yield: 99%	Migratory Aryl Cross-Coupling
1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, [3,6-Dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]bis(1,1-dimethylethyl)phosphine Solvents: Toluene; 13 h, 100 °C 1.2 Reagents: Ammonium chloride Solvents: Water		By: Sekiguchi, Yoshiya; et al Journal of the American Chemical Society (2025), 147(1), 1224-1230.
Experimental Protocols		

Scheme 260 (1 Reaction)

Steps: 1 Yield: 99%



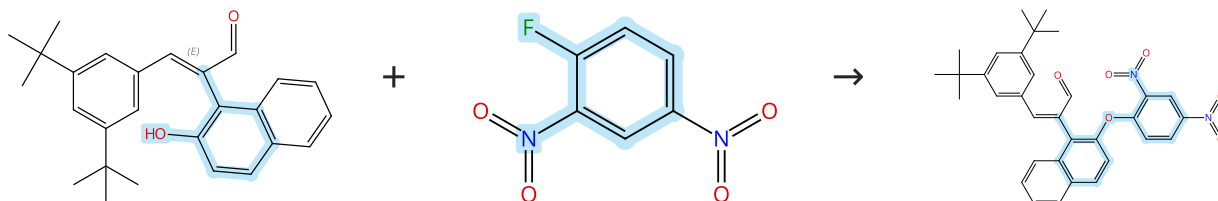
Suppliers (3)

Suppliers (4)

31-614-CAS-28954034	Steps: 1 Yield: 99%	Microwave-assisted transition-metal-free intramolecular Ullmann-type O-arylation in water for the synthesis of xanthenes and azaxanthenes
1.1 Reagents: Tetrabutylammonium hydroxide Solvents: Water; 4 min, 120 °C		By: Zhang, Xiaojin; et al Tetrahedron Letters (2014), 55(35), 4883-4887.
Experimental Protocols		

Scheme 261 (1 Reaction)

Steps: 1 Yield: 99%



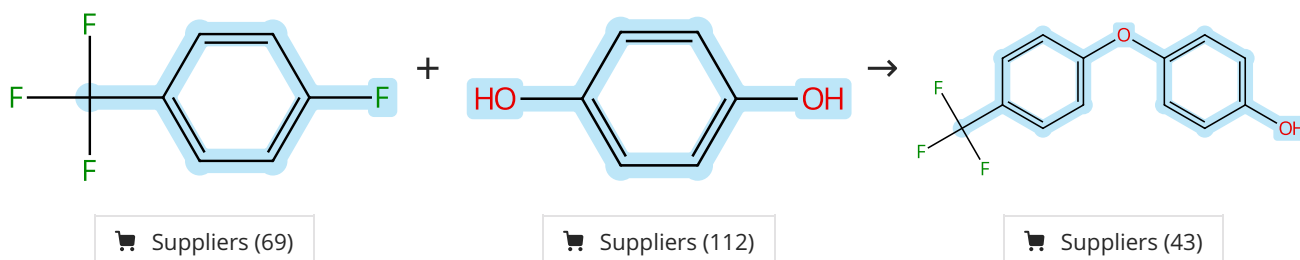
Double bond geometry shown

Suppliers (75)

31-614-CAS-37560070 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 262 (1 Reaction)

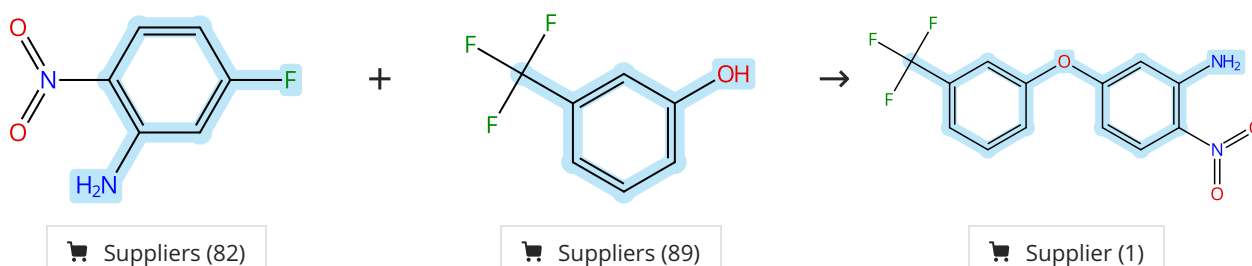
Steps: 1 Yield: 99%



31-170-CAS-8152044 Steps: 1 Yield: 99% 1.1 Reagents: Potassium <i>tert</i>-butoxide, 18-Crown-6 Solvents: Dimethyl sulfoxide; 24 h, 100 °C	Selective displacement of aryl fluorides with hydroquinone: synthesis of 4-phenoxyphenols By: Marcune, Benjamin F.; et al Tetrahedron Letters (2005), 46(45), 7823-7826.
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Scheme 263 (1 Reaction)

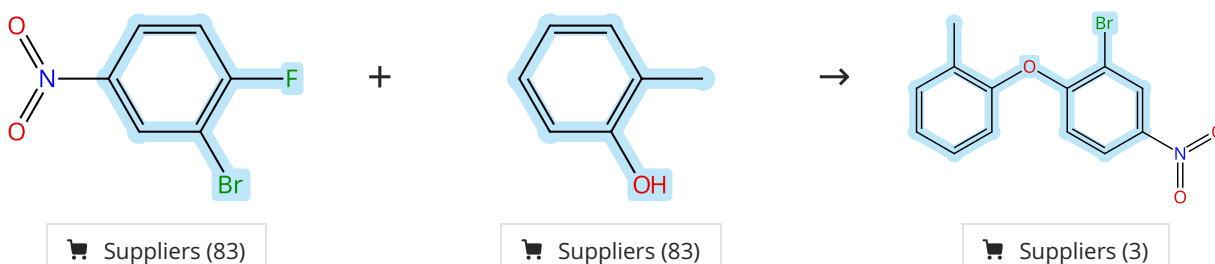
Steps: 1 Yield: 99%



31-170-CAS-8435917 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C	New benzimidazole-2-urea derivatives as tubulin inhibitors By: Wang, Wenna; et al Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.
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Scheme 264 (1 Reaction)

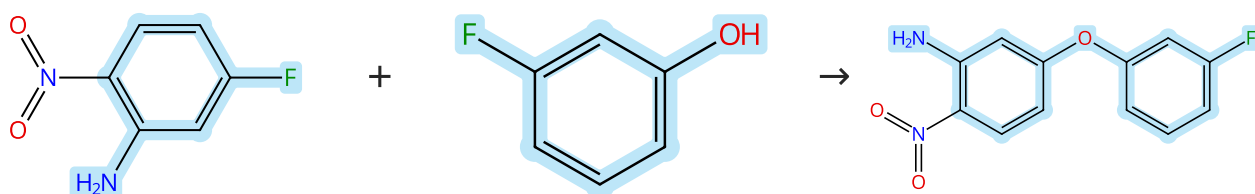
Steps: 1 Yield: 99%



31-614-CAS-35312184	Steps: 1 Yield: 99%	Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 2 h, 110 °C		By: Xu, Kai-Yan; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2023), 78, 117152.

Scheme 265 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

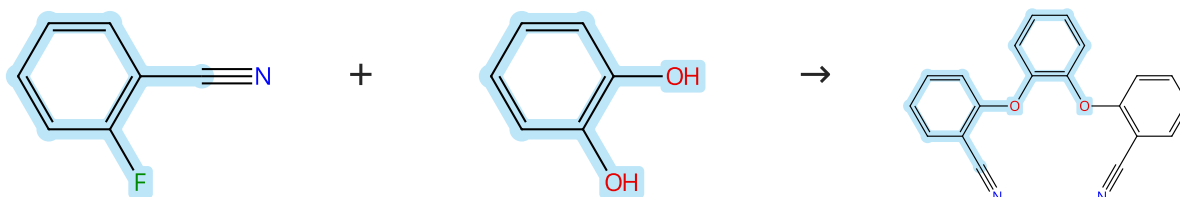
Suppliers (78)

Supplier (1)

31-170-CAS-4451217	Steps: 1 Yield: 99%	New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C		By: Wang, Wenna; et al
		Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 266 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (87)

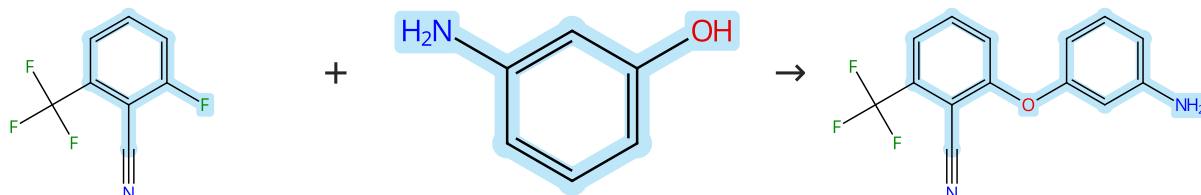
Suppliers (106)

Suppliers (2)

31-170-CAS-10430322	Steps: 1 Yield: 99%	Synthesis of Di- and Trixanthones that Display High Stability and a Visual Fluorescence Response to Strong Acid
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 12 h, 80 °C		By: Wang, Baolin; et al
		Chemistry - An Asian Journal (2014), 9(11), 3307-3312.

Scheme 267 (1 Reaction)

Steps: 1 Yield: 99%



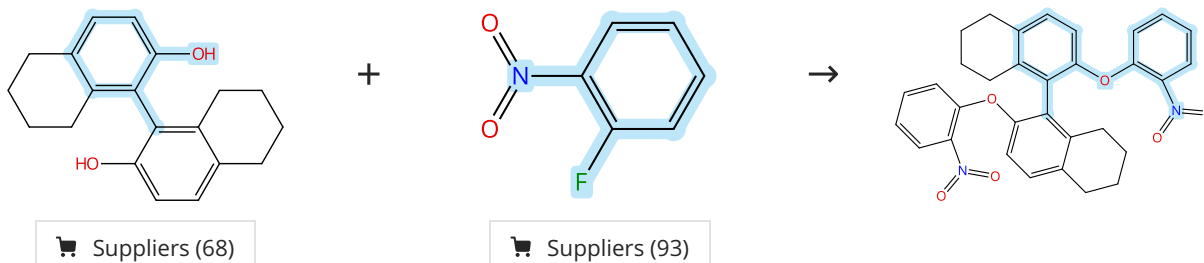
Suppliers (71)

Suppliers (81)

31-614-CAS-34902390	Steps: 1 Yield: 99%	Structure-Activity Relationship Development Efforts towards Peripherally Selective Analogs of the Cannabinoid Receptor Partial Agonist BAY 59-3074 By: Amato, George; et al Molecules (2022), 27(17), 5672.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 15 min, rt; 4 h, 80 °C		
Experimental Protocols		

Scheme 268 (1 Reaction)

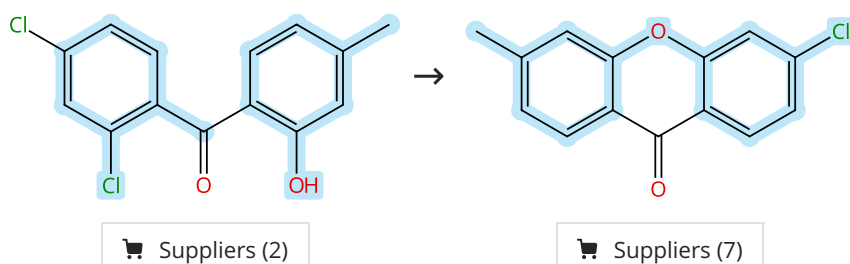
Steps: 1 Yield: 99%



31-170-CAS-11320682	Steps: 1 Yield: 99%	Enantioselective alkynylation to aldimines catalyzed by chiral 2,2'-bis[(2-aminoaryl)oxy]-1,1'-binaphthyl-copper(I) complexes By: Hatano, Manabu; et al Tetrahedron Letters (2008), 49(2), 379-382.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, 80 °C		
Experimental Protocols		

Scheme 269 (1 Reaction)

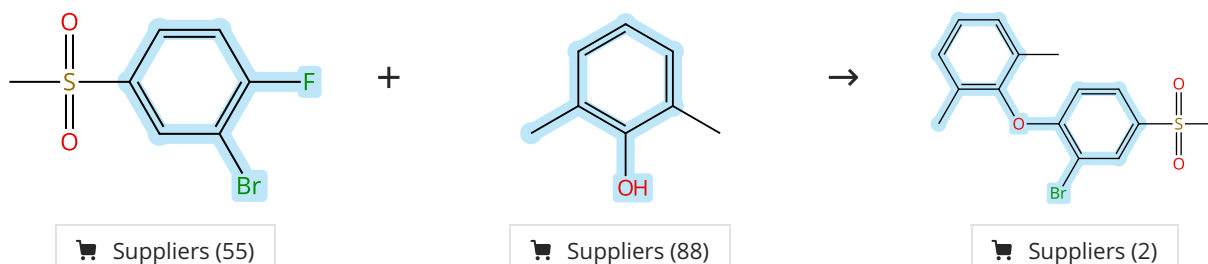
Steps: 1 Yield: 99%



31-614-CAS-29753964	Steps: 1 Yield: 99%	Microwave-assisted transition-metal-free intramolecular Ullmann-type O-arylation in water for the synthesis of xanthenes and azaxanthenes By: Zhang, Xiaojin; et al Tetrahedron Letters (2014), 55(35), 4883-4887.
1.1 Reagents: Tetrabutylammonium hydroxide Solvents: Water; 4 min, 120 °C		
Experimental Protocols		

Scheme 270 (1 Reaction)

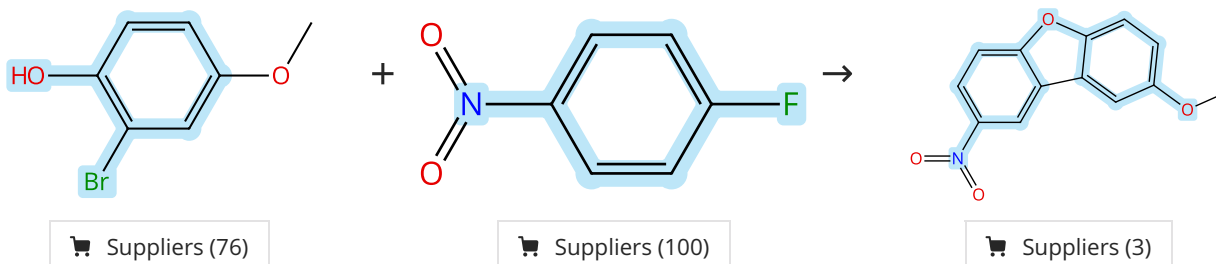
Steps: 1 Yield: 99%



31-614-CAS-24358140	Steps: 1 Yield: 99%	Discovery of N-Ethyl-4-[2-(4-fluoro-2,6-dimethyl-phenoxy)-5-(1-hydroxy-1-methyl-ethyl)phenyl]-6-methyl-7-oxo-1H-pyrrolo [2,3-c]pyridine-2-carboxamide (ABBV-744), a BET Bromodomain Inhibitor with Selectivity for the Second Bromodomain By: Sheppard, George S.; et al Journal of Medicinal Chemistry (2020), 63(10), 5585-5623.
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 18 h, 110 °C		
Experimental Protocols		

Scheme 271 (1 Reaction)

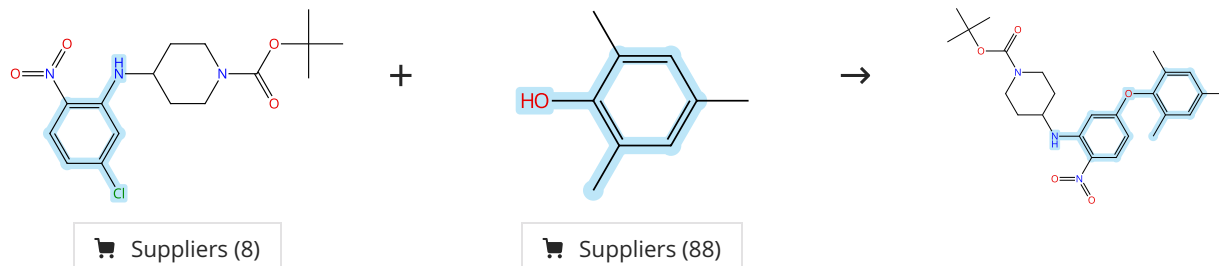
Steps: 1 Yield: 99%



31-614-CAS-28853728	Steps: 1 Yield: 99%	Synthesis of dibenzofurans directly from aryl halides and ortho-bromophenols via one-pot consecutive S_NAr and intramolecular palladium-catalyzed aryl-aryl coupling reactions By: Xu, Hui; et al Chemical & Pharmaceutical Bulletin (2008), 56(10), 1496-1498.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 0.5 h, 90 °C		
1.2 Catalysts: Triphenylphosphine, Palladium diacetate; 0.5 h, 90 °C		

Scheme 272 (1 Reaction)

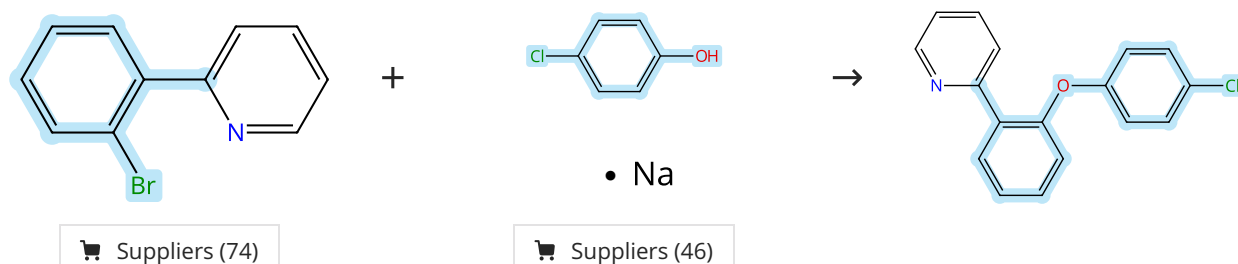
Steps: 1 Yield: 99%



31-170-CAS-9779594	Steps: 1 Yield: 99%	Design, synthesis and preliminary SAR studies of novel N-arylmethyl substituted piperidine-linked aniline derivatives as potent HIV-1 NNRTIs By: Zhang, Lingzi; et al Bioorganic & Medicinal Chemistry (2014), 22(1), 633-642.
1.1 Reagents: Potassium carbonate Catalysts: Tetrabutylammonium bromide, Potassium iodide Solvents: Dimethylformamide; 12 h, 120 - 130 °C		
Experimental Protocols		

Scheme 273 (1 Reaction)

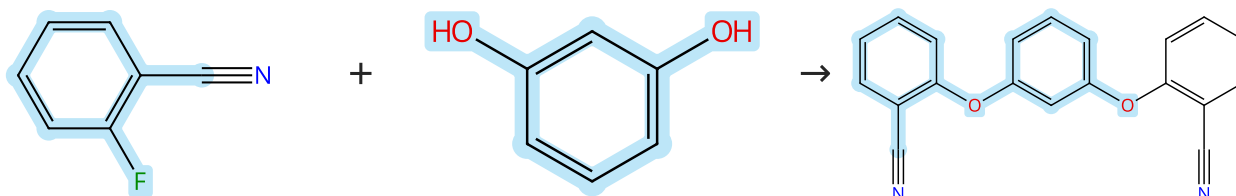
Steps: 1 Yield: 99%



31-170-CAS-17208712 Steps: 1 Yield: 99% 1.1 Catalysts: Gold(1+), (acetonitrile)[1,3-bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2 <i>H</i> -imidazol-2-ylidene]-, (<i>OC</i> -6-11)-hexafluoroantimonate(1-) (1:1) Solvents: Acetonitrile; 6 h, 110 °C	Au(III)-aryl intermediates in oxidant-free C-N and C-O cross-coupling catalysis By: Serra, Jordi; et al Chemical Science (2017), 8(2), 946-952.
--	--

Scheme 274 (1 Reaction)

Steps: 1 Yield: 99%



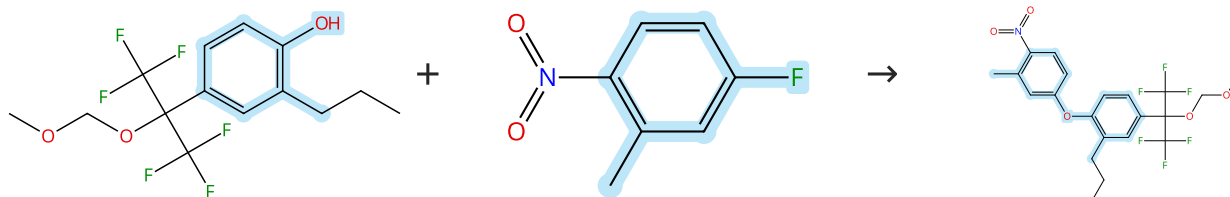
Suppliers (87)

Suppliers (109)

31-170-CAS-3054758 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 12 h, 80 °C	Synthesis of Di- and Trixan thones that Display High Stability and a Visual Fluorescence Response to Strong Acid By: Wang, Baolin; et al Chemistry - An Asian Journal (2014), 9(11), 3307-3312.
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Scheme 275 (1 Reaction)

Steps: 1 Yield: 99%



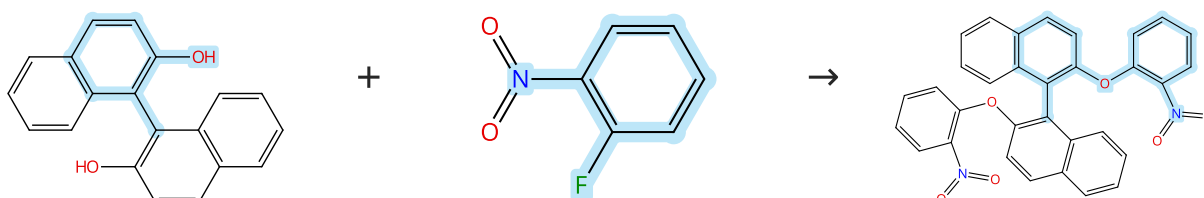
Supplier (1)

Suppliers (85)

31-170-CAS-16062982 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; rt; 2 h, 80 °C Experimental Protocols	Discovery of a 2-hydroxyacetophenone derivative as an outstanding linker to enhance potency and β-selectivity of liver X receptor agonist By: Koura, Minoru; et al Bioorganic & Medicinal Chemistry (2016), 24(16), 3436-3446.
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Scheme 276 (2 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (85)

Suppliers (93)

31-170-CAS-7087572 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 16 h, 80 °C Experimental Protocols	Enantioselective alkynylation to aldimines catalyzed by chiral 2,2'-bis[(2-aminoaryl)oxy]-1,1'-binaphthyl-copper(I) complexes By: Hatano, Manabu; et al Tetrahedron Letters (2008), 49(2), 379-382.
31-170-CAS-3921413 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 6 h, 80 °C Experimental Protocols	Racemic and chiral expanded salen-type complexes derived from biphenol and binaphthol: Salbip and salbin By: Grill, Joseph M.; et al Journal of Organometallic Chemistry (2005), 690(12), 3009-3017.

Scheme 277 (1 Reaction)

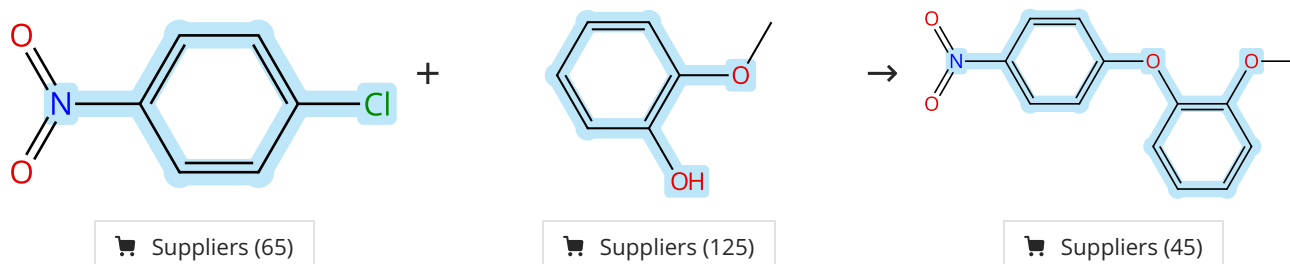
Steps: 1 Yield: 99%



31-170-CAS-6786095 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate Catalysts: Copper(1+), bis(hydroboron)hexakis[μ-(1-methyl-1<i>H</i>-imidazol-2-yl-κC²:κN³)]tri-, tetrafluoroborate(1-) (1:1) Solvents: Dimethyl sulfoxide; 24 h, 100 °C Experimental Protocols	Efficient catalysis of Ullmann-type arylation reactions by a novel trinuclear copper(I) complex with a chelating tricarbene ligand By: Tubaro, Cristina; et al Tetrahedron (2008), 64(19), 4187-4195.
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Scheme 278 (1 Reaction)

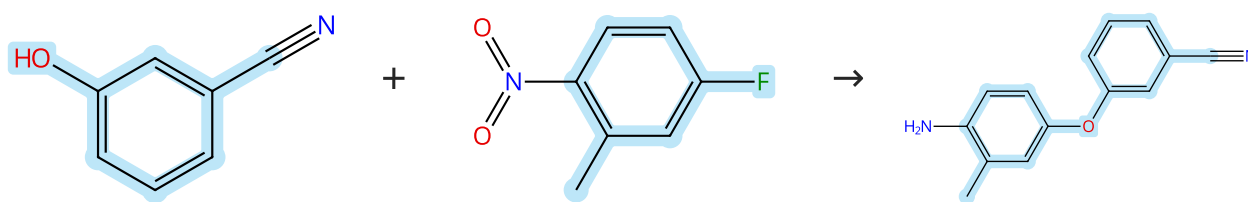
Steps: 1 Yield: 99%



31-170-CAS-2122426 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 30 - 40 s, rt → reflux; 5 min, reflux; reflux → rt 1.2 Solvents: Water; 30 min, < rt Experimental Protocols	Synthesis of diaryl ethers, diaryl sulfides, heteroaryl ethers and heteroaryl sulfides under microwave dielectric heating By: Li, Feng; et al Synthesis (2005), (8), 1305-1313.
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Scheme 279 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (94)

Suppliers (85)

Suppliers (5)

31-170-CAS-12592991

Steps: 1 Yield: 99%

Discovery and structure-activity study of a novel benzoxa borole anti-inflammatory agent (AN2728) for the potential topical treatment of psoriasis and atopic dermatitis

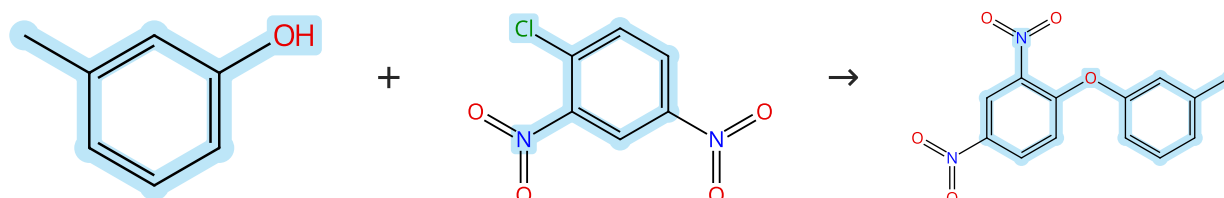
By: Akama, Tsutomu; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(8), 2129-2132.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; overnight, 70 °C
- 1.2 **Reagents:** Hydrochloric acid, Stannous chloride
Solvents: Ethanol; 2 h, 50 °C

Scheme 280 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (91)

Suppliers (75)

Suppliers (4)

31-170-CAS-7830915

Steps: 1 Yield: 99%

Betti Base as an Efficient Ligand for Copper-Catalyzed Ullmann Coupling of Phenol with Aryl Halides

By: Yang, Li; et al

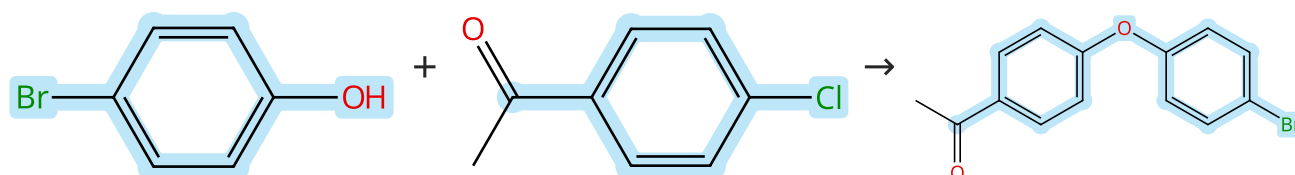
Synthetic Communications (2014), 44(17), 2468-2477.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Cuprous iodide, 1-[(Dimethylamino)-2-pyridinylmethyl]-2-naphthalenol
Solvents: Dimethyl sulfoxide; 24 h, 120 °C

Experimental Protocols

Scheme 281 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

Suppliers (68)

Suppliers (61)

31-170-CAS-15032170

Steps: 1 Yield: 99%

CuI Nanoparticles for C-N and C-O Cross Coupling of Heterocyclic Amines and Phenols with Chlorobenzene

By: Sreedhar, B.; et al

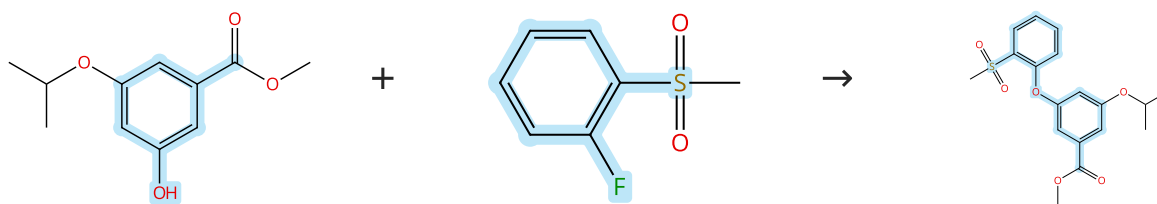
Journal of Organic Chemistry (2009), 74(20), 7951-7954.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Cuprous iodide
Solvents: Dimethylformamide; 30 min, rt; 4 h, 110 °C; 110 °C → rt

Experimental Protocols

Scheme 282 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (49)

Suppliers (49)

31-170-CAS-14997766

Steps: 1 Yield: 99%

Discovery of potent and orally active 3-alkoxy-5-phenoxy-N-thiazolyl benzamides as novel allosteric glucokinase activators

By: Iino, Tomoharu; et al

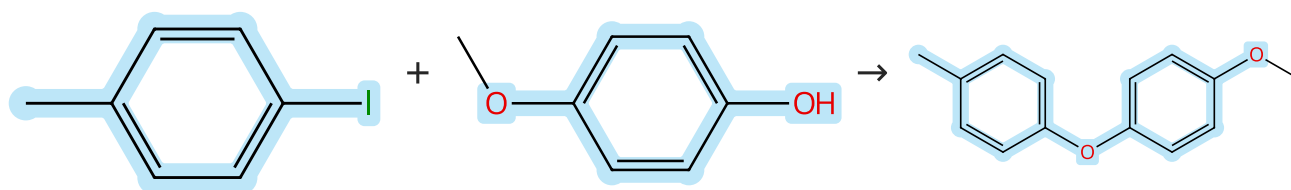
Bioorganic & Medicinal Chemistry (2009), 17(7), 2733-2743.

1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 2 h, 130 °C

Experimental Protocols

Scheme 283 (3 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (82)

Suppliers (121)

Suppliers (17)

31-170-CAS-14780556

Steps: 1 Yield: 99%

Ligand-free Ullmann-type C-heteroatom couplings under practical conditions

By: Gueell, Imma; et al

European Journal of Organic Chemistry (2014), 2014(15), 3188-3195.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Cuprous iodide
Solvents: Dimethyl sulfoxide; 24 h, 110 °C

1.2 **Solvents:** Ethyl acetate

Experimental Protocols

31-170-CAS-13815434

Steps: 1 Yield: 99%

Facile synthesis of 9-azajulolidine and its application to post-Ullmann reactions

By: Wong, Ken-Tsung; et al

Tetrahedron Letters (2007), 48(29), 5051-5054.

1.1 **Reagents:** Cesium carbonate, 5,6,9,10-Tetrahydro-4*H*,8*H*-pyrido[3,2-*i*][1,6]naphthyridine
Catalysts: Cuprous iodide
Solvents: Toluene; reflux

Experimental Protocols

31-170-CAS-2019156

Steps: 1 Yield: 98%

Maghemite-Copper Nanocomposites: Applications for Ligand-Free Cross-Coupling (C-O, C-S, and C-N) Reactions

By: Sharma, Rakesh K.; et al

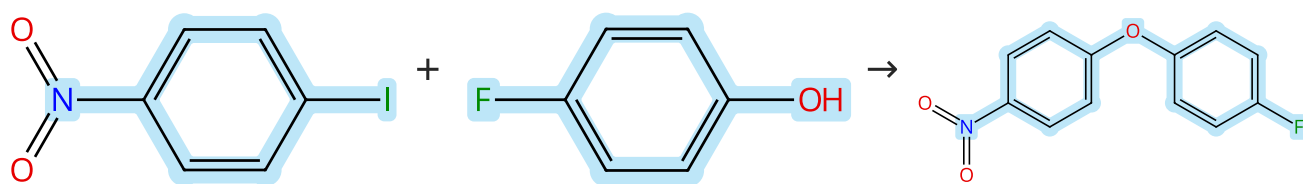
ChemCatChem (2015), 7(21), 3495-3502.

1.1 **Reagents:** Cesium carbonate
Catalysts: Copper
Solvents: Dimethylformamide; 24 h, 130 °C

Experimental Protocols

Scheme 284 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (70)

Suppliers (86)

Suppliers (42)

31-170-CAS-284649

Steps: 1 Yield: 99%

On the frontier between nucleophilic aromatic substitution and catalysis

1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylformamide; 18 h, 90 °C

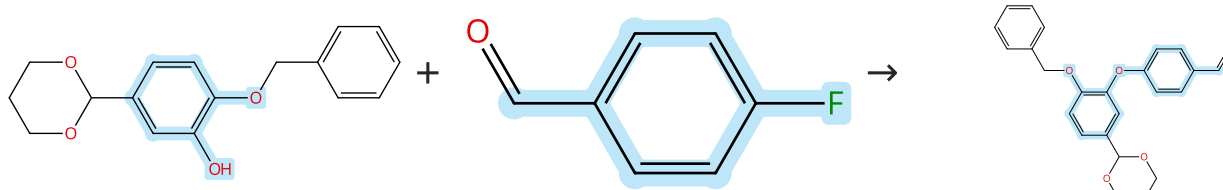
By: Pichette Drapeau, Martin; et al

Experimental Protocols

Chemistry - A European Journal (2014), 20(18), 5231-5236.

Scheme 285 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (93)

31-170-CAS-9412391

Steps: 1 Yield: 99%

Syntheses of marchantins C, O and P as promising highly bioactive compounds

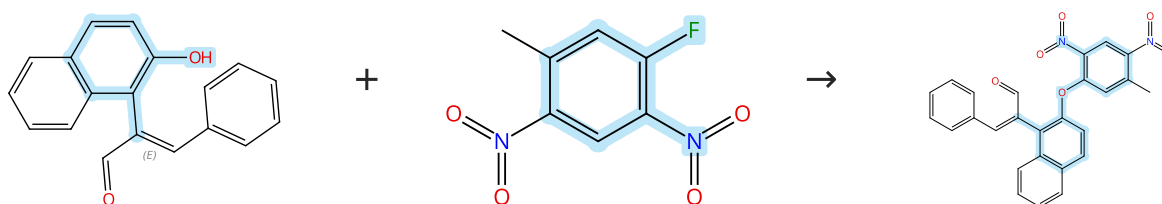
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 48 h, rt → 165 °C

By: Speicher, Andreas; et al

Natural Product Communications (2011), 6(3), 393-402.

Scheme 286 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (55)

31-614-CAS-37560115

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

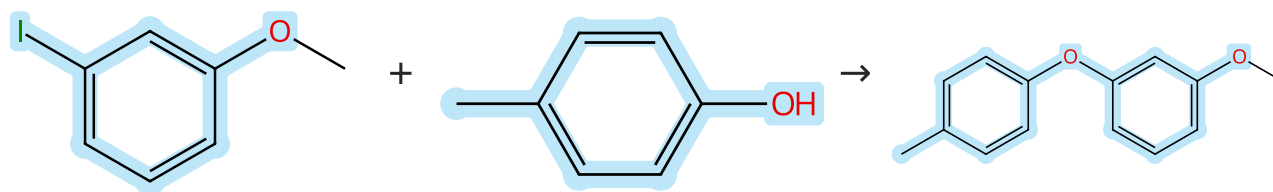
By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Experimental Protocols

Scheme 287 (3 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (76)

Suppliers (89)

Suppliers (8)

31-170-CAS-7466364

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Cesium carbonate
Catalysts: *rel*-(11*R*,12*R*)-9,10-Dihydro-9,10-ethanoanthracene-11,12-dimethanol, Cuprous iodide
Solvents: Acetonitrile; rt; 22 h, 82 °C

Experimental Protocols

General, Mild, and Intermolecular Ullmann-Type Synthesis of Diaryl and Alkyl Aryl Ethers Catalyzed by Diol-Copper(I) Complex

By: Naidu, Ajay B.; et al

Journal of Organic Chemistry (2009), 74(10), 3675-3679.

31-170-CAS-14649553

Steps: 1 Yield: 98%

- 1.1 **Reagents:** Cesium carbonate
Catalysts: β -Alanine (chloromethylated polystyrene-bound, copper complexes), Tetrabutylammonium bromide, Copper (complexes with chloromethylated polystyrene-bound β -alanine), Polystyrene (chloromethylated, reaction products with β -alanine, copper complexes)
Solvents: Dimethyl sulfoxide; 14 h, 120 °C

A reusable polymer supported copper catalyst for the C-N and C-O bond cross-coupling reaction of aryl halides as well as arylboronic acids

By: Islam, Sk. Manirul; et al

Journal of Organometallic Chemistry (2012), 696(26), 4264-4274.

31-170-CAS-12342641

Steps: 1 Yield: 98%

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Dihexyl ether, Cupric acetate, 2-Propanaminium, *N,N,N*-tris(1,1-dimethylethyl)-2-methyl-, bromide (1:1)
Solvents: *N*-Methyl-2-pyrrolidone; 12 h, 120 °C

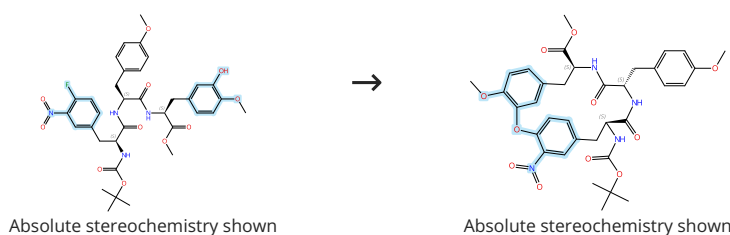
A recyclable heterogeneous copper(II) Schiff base catalyst for the O-arylation reaction of phenols with aryl halides

By: Islam, Manirul; et al

Journal of Chemical Research (2010), 34(3), 170-174.

Scheme 288 (1 Reaction)

Steps: 1 Yield: 99%



Absolute stereochemistry shown

Absolute stereochemistry shown

31-170-CAS-5310429

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide

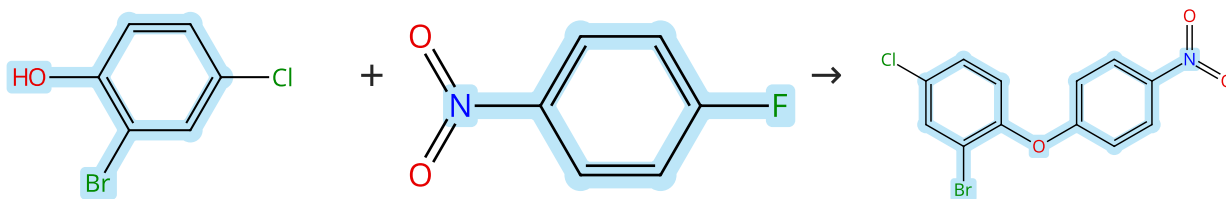
An efficient total synthesis of K-13, a non-competitive inhibitor of ACE I

By: Bigot, Antony; et al

Tetrahedron Letters (2000), 41(23), 4573-4577.

Scheme 289 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (80)

Suppliers (100)

Suppliers (6)

31-614-CAS-37615387

Steps: 1 Yield: 99%

Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors

By: Zhang, Haoming; et al

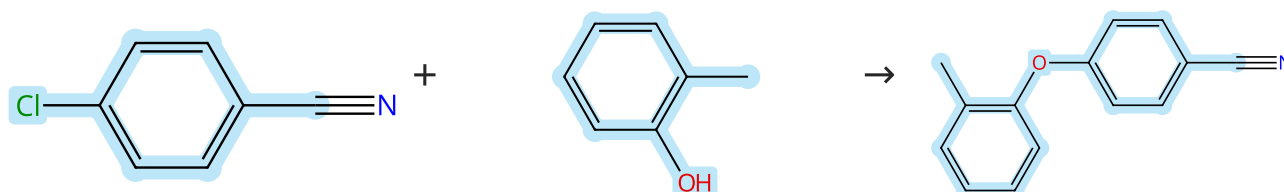
ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.

1.1 Reagents: Cesium carbonate
Solvents: Dimethylformamide; 6 h, 50 °C

Experimental Protocols

Scheme 290 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

Suppliers (83)

Suppliers (30)

31-170-CAS-10134838

Steps: 1 Yield: 99%

An efficient catalyst system for diaryl ether synthesis from aryl chlorides

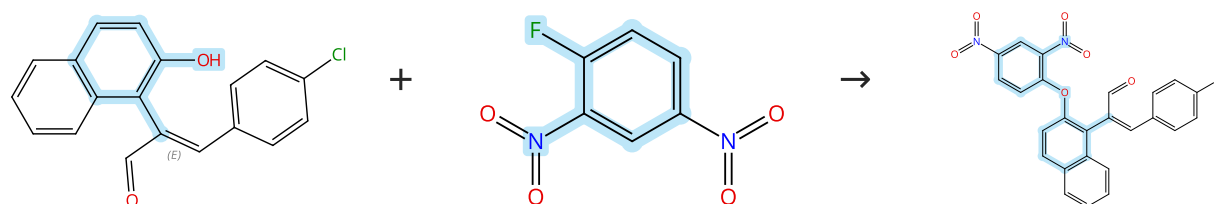
By: Harkal, Surendra; et al

Tetrahedron Letters (2005), 46(18), 3237-3240.

1.1 Reagents: Tripotassium phosphate
Catalysts: Palladium diacetate, 2-[Bis(1,1-dimethylethyl)phosphino]-1-phenyl-1*H*-indole
Solvents: Toluene; 20 h, 120 °C

Scheme 291 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (75)

31-614-CAS-37560057

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

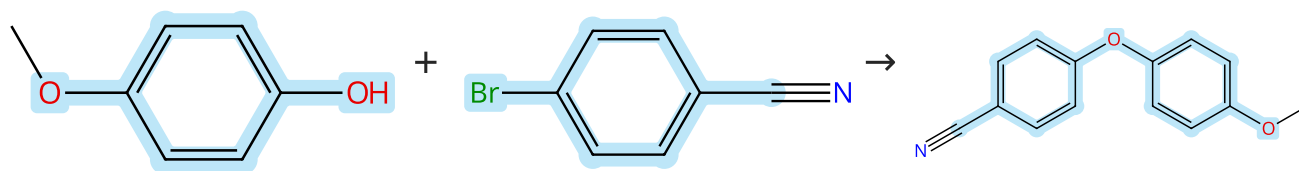
Nature Communications (2023), 14(1), 5050.

1.1 Reagents: Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

Scheme 292 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (121)

Suppliers (93)

Suppliers (38)

31-170-CAS-2015915

Steps: 1 Yield: 99%

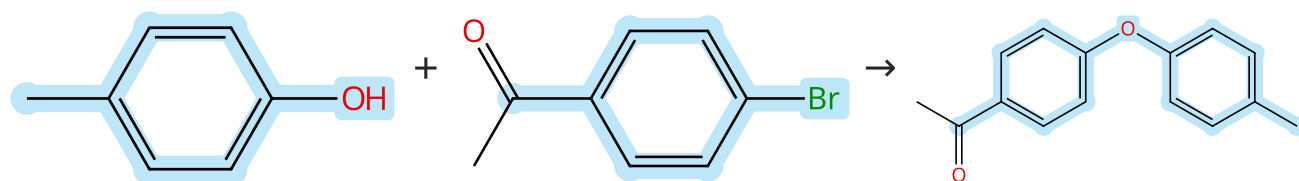
Copper promoted synthesis of diaryl ethers

- 1.1 Reagents: Sodium hydride
Solvents: Tetrahydrofuran
- 1.2 Reagents: Tripotassium phosphate
Catalysts: Malachite; cooled; 18 h, 115 °C

By: Ghosh, Rajshekhar; et al
New Journal of Chemistry (2004), 28(11), 1390-1393.

Scheme 293 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (89)

Suppliers (85)

Suppliers (38)

31-170-CAS-14082210

Steps: 1 Yield: 99%

Formation of Aryl-Nitrogen, Aryl-Oxygen, and Aryl-Carbon Bonds Using Well-Defined Copper(I)-Based Catalysts

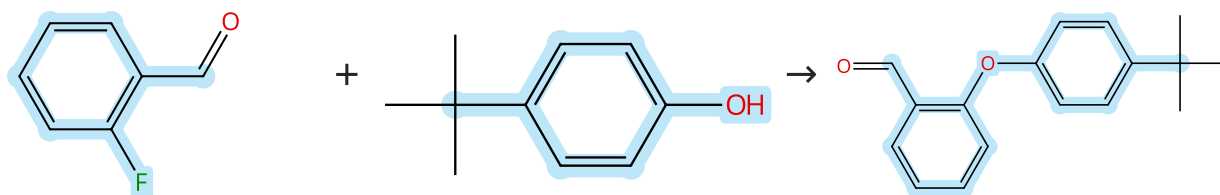
- 1.1 Reagents: Cesium carbonate
Catalysts: (7-4)-Bromo(2,9-dimethyl-1,10-phenanthroline- κN^1 , κN^{10})(triphenylphosphine)copper
Solvents: Toluene

By: Gujadhur, Rattan K.; et al
Organic Letters (2001), 3(26), 4315-4317.

Experimental Protocols

Scheme 294 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (88)

Suppliers (35)

31-170-CAS-7084788

Steps: 1 Yield: 99%

Cross Dehydrogenative Coupling via Base-Promoted Homolytic Aromatic Substitution (BHAS): Synthesis of Fluorenes and Xanthenes

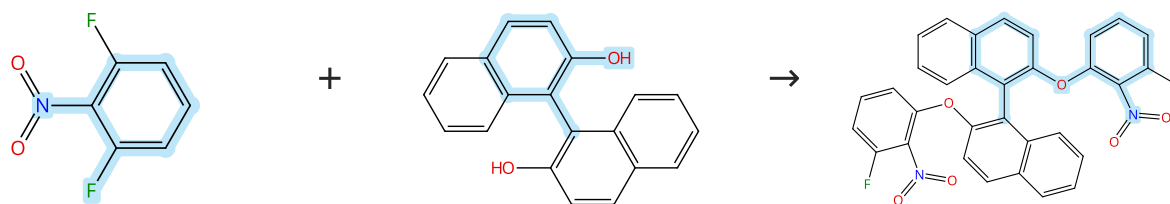
- 1.1 Reagents: Potassium carbonate
Solvents: Dimethylacetamide; 4 h, 170 °C

By: Wertz, Sebastian; et al
Organic Letters (2013), 15(4), 928-931.

Experimental Protocols

Scheme 295 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (85)

31-170-CAS-4330473

Steps: 1 Yield: 99%

Enantioselective alkynylation to aldimines catalyzed by chiral 2,2'-bis[(2-aminoaryl)oxy]-1,1'-binaphthyl-copper(I) complexes

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 16 h, 80 °C

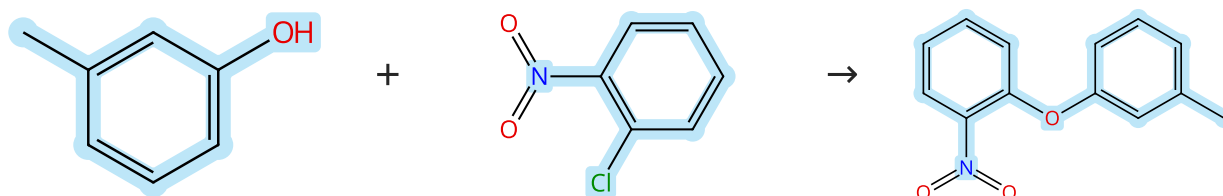
Experimental Protocols

By: Hatano, Manabu; et al

Tetrahedron Letters (2008), 49(2), 379-382.

Scheme 296 (3 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (91)

Suppliers (76)

Suppliers (12)

31-170-CAS-12419331

Steps: 1 Yield: 99%

Betti Base as an Efficient Ligand for Copper-Catalyzed Ullmann Coupling of Phenol with Aryl Halides

1.1 **Reagents:** Potassium carbonate
Catalysts: Cuprous iodide, 1-[(Dimethylamino)-2-pyridiny Imethyl]-2-naphthalenol
Solvents: Dimethyl sulfoxide; 24 h, 120 °C

Experimental Protocols

By: Yang, Li; et al

Synthetic Communications (2014), 44(17), 2468-2477.

31-170-CAS-5962414

Steps: 1 Yield: 98%

Design, synthesis and antifungal activity of novel furancar boxamide derivatives

1.1 **Reagents:** Potassium hydroxide; 15 min, 90 °C
 1.2 90 °C; 2 h, 150 °C

Experimental Protocols

By: Wen, Fang; et al

European Journal of Medicinal Chemistry (2016), 120, 244-251.

31-170-CAS-5625887

Steps: 1 Yield: 98%

Design, synthesis and antifungal/insecticidal evaluation of novel nicotinamide derivatives

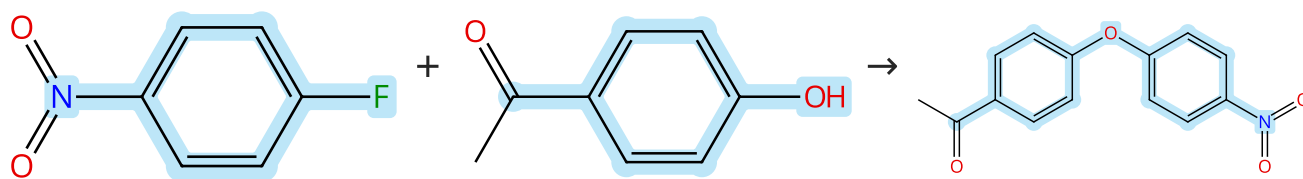
1.1 **Reagents:** Potassium hydroxide; 15 min, 90 °C
 1.2 90 °C; 2 h, 150 °C

By: Wen, Fang; et al

Pesticide Biochemistry and Physiology (2010), 98(2), 248-253.

Scheme 297 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (100)

Suppliers (124)

Suppliers (49)

31-170-CAS-1289887

Steps: 1 Yield: 99%

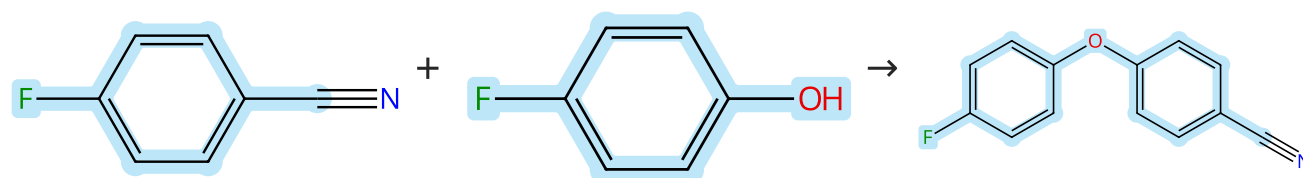
A homologous enolate Truce-Smiles rearrangement

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 120 °C
- 1.2 **Solvents:** Water

By: Mitchell, Lorna H.; et al
 Tetrahedron Letters (2004), 45(29), 5669-5671.

Scheme 298 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (92)

Suppliers (86)

Suppliers (72)

31-170-CAS-2124745

Steps: 1 Yield: 99%

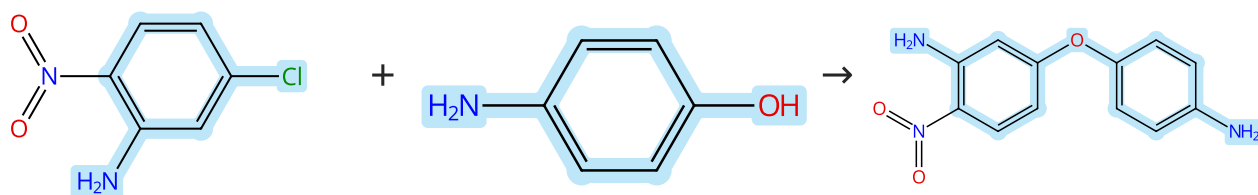
Inhibitors of Acyl-CoA:Cholesterol O-Acyltransferase. 3. Discovery of a Novel Series of N-Alkyl-N-[(fluorophenoxy)benzyl]-N'-arylureas with Weak Toxicological Effects on Adrenal Glands

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 6 h, 150 °C

By: Tanaka, Akira; et al
 Journal of Medicinal Chemistry (1998), 41(22), 4408-4420.

Scheme 299 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (97)

Suppliers (100)

Suppliers (45)

31-170-CAS-13932982

Steps: 1 Yield: 99%

Discovery of Novel Benzimidazoles as Potent Inhibitors of TIE-2 and VEGFR-2 Tyrosine Kinase Receptors

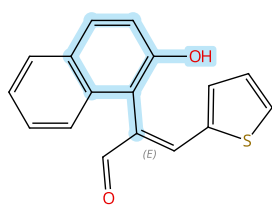
- 1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide; overnight, 90 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

By: Hasegawa, Masaichi; et al
 Journal of Medicinal Chemistry (2007), 50(18), 4453-4470.

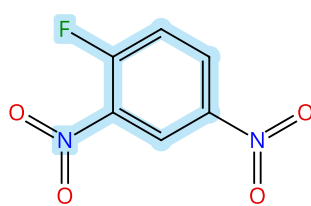
Experimental Protocols

Scheme 300 (1 Reaction)

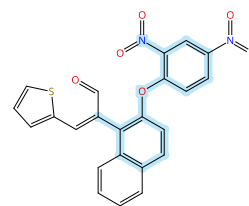
Steps: 1 Yield: 99%



+



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Double bond geometry shown

Suppliers (75)

31-614-CAS-37560076

Steps: 1 Yield: 99%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

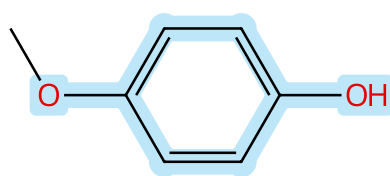
Nature Communications (2023), 14(1), 5050.

1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

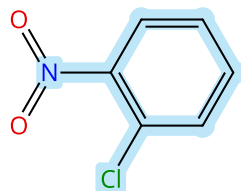
Experimental Protocols

Scheme 301 (3 Reactions)

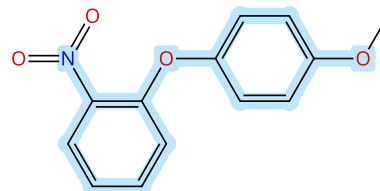
Steps: 1 Yield: 98-99%



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Suppliers (121)

Suppliers (76)

Suppliers (6)

31-170-CAS-10687657

Steps: 1 Yield: 99%

A facile and efficient synthesis of diaryl amines or ethers under microwave irradiation in the presence of KF/Al_2O_3 without solvent and their anti-fungal biological activities against six phytopathogens

By: Huang, Liang-Zhu; et al

International Journal of Molecular Sciences (2013), 14(9), 18850-18860.

1.1 **Catalysts:** Alumina, Potassium fluoride; 15 min, 150 °C

31-170-CAS-12599384

Steps: 1 Yield: 98%

Design, synthesis and antifungal activity of novel furancar boxamide derivatives

By: Wen, Fang; et al

European Journal of Medicinal Chemistry (2016), 120, 244-251.

1.1 **Reagents:** Potassium hydroxide; 15 min, 90 °C

1.2 90 °C; 2 h, 150 °C

Experimental Protocols

31-170-CAS-7758533

Steps: 1 Yield: 98%

Design, synthesis and antifungal/insecticidal evaluation of novel nicotinamide derivatives

By: Wen, Fang; et al

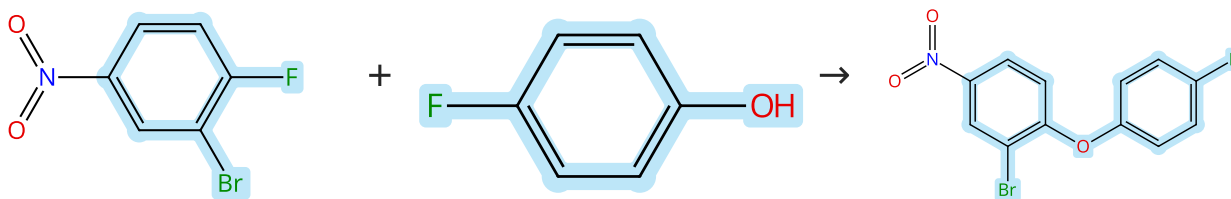
Pesticide Biochemistry and Physiology (2010), 98(2), 248-253.

1.1 **Reagents:** Potassium hydroxide; 15 min, 90 °C

1.2 90 °C; 2 h, 150 °C

Scheme 302 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (86)

Suppliers (28)

31-614-CAS-35312181

Steps: 1 Yield: 99%

Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors

By: Xu, Kai-Yan; et al

Bioorganic & Medicinal Chemistry (2023), 78, 117152.

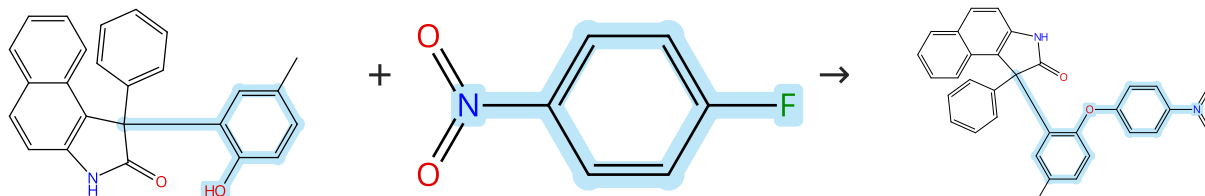
1.1 Reagents: Cesium carbonate

Solvents: Dimethyl sulfoxide; 2 h, 110 °C

Experimental Protocols

Scheme 303 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (100)

31-170-CAS-23191824

Steps: 1 Yield: 99%

I₂-Mediated Cross-Dehydrogenative Coupling and Amidation of 3-Aryl Benzofuranones with Aryl Amines for the Synthesis of 3,3-Diaryl Indolin-2-ones

By: Tang, Zhi; et al

Journal of Organic Chemistry (2021), 86(3), 2965-2973.

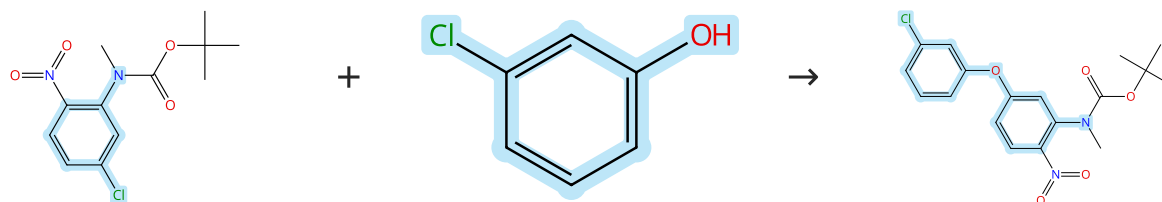
1.1 Reagents: Tempo

Solvents: Dimethyl sulfoxide; 4 h, 40 °C

Experimental Protocols

Scheme 304 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (8)

Suppliers (85)

31-170-CAS-19673617

Steps: 1 Yield: 99%

Discovery of DS-6930, a potent selective PPAR γ modulator. Part I: Lead identification

By: Shinozuka, Tsuyoshi; et al

Bioorganic & Medicinal Chemistry (2018), 26(18), 5079-5098.

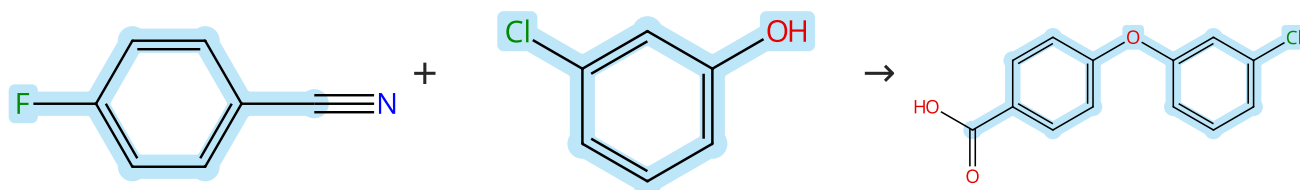
1.1 Reagents: Sodium hydride

Solvents: Dimethylformamide; rt; 10 h, 80 °C

Experimental Protocols

Scheme 305 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (92)

Suppliers (85)

Suppliers (53)

31-170-CAS-2381376

Steps: 1 Yield: 99%

Discovery of N-{4-[(3-Hydroxyphenyl)-3-methylpiperazin-1-yl]methyl-2-methylpropyl}-4-phenoxybenzamide Analogues as Selective Kappa Opioid Receptor Antagonists

By: Kormos, Chad M.; et al

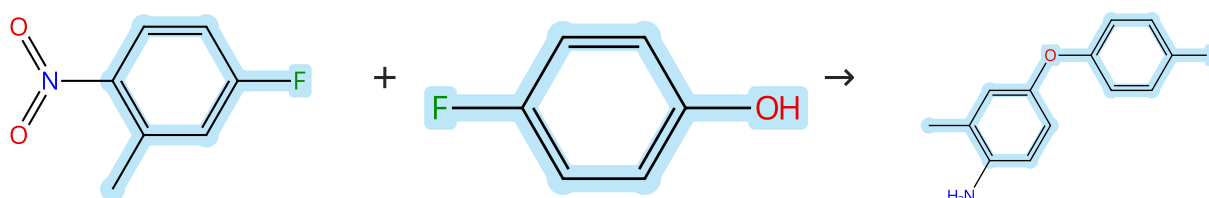
Journal of Medicinal Chemistry (2013), 56(11), 4551-4567.

- 1.1 Reagents: Potassium hydroxide
Solvents: Dimethylformamide; 20 min, rt → 175 °C
- 1.2 Reagents: Potassium hydroxide
Solvents: Water; 12 h, reflux
- 1.3 Reagents: Hydrogen ion
Solvents: Water; acidified

Experimental Protocols

Scheme 306 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

Suppliers (86)

Suppliers (41)

31-170-CAS-14724159

Steps: 1 Yield: 99%

Discovery and structure-activity study of a novel benzoxa borole anti-inflammatory agent (AN2728) for the potential topical treatment of psoriasis and atopic dermatitis

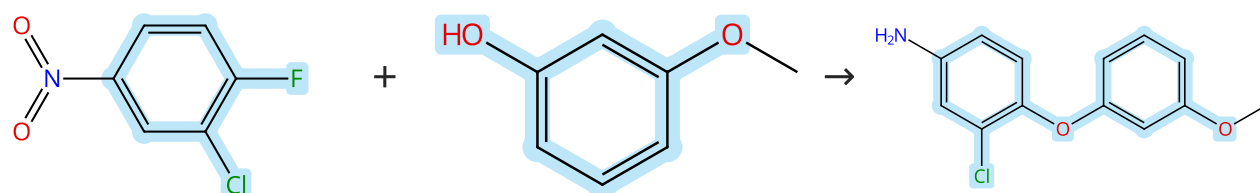
By: Akama, Tsutomu; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(8), 2129-2132.

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; overnight, 70 °C
- 1.2 Reagents: Hydrochloric acid, Stannous chloride
Solvents: Ethanol; 2 h, 50 °C

Scheme 307 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (86)

Suppliers (92)

Suppliers (32)

31-170-CAS-3568384

Steps: 1 Yield: 99%

Design and Synthesis of Novel Human Epidermal Growth Factor Receptor 2 (HER2)/Epidermal Growth Factor Receptor (EGFR) Dual Inhibitors Bearing a Pyrrolo[3,2-d]pyrimidine Scaffold

By: Ishikawa, Tomoyasu; et al

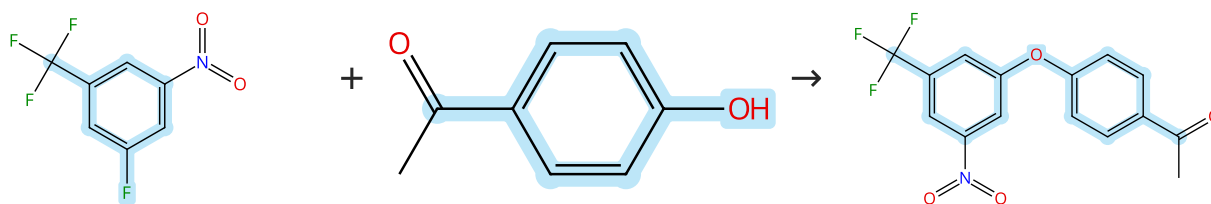
Journal of Medicinal Chemistry (2011), 54(23), 8030-8050.

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 80 °C
- 1.2 Reagents: Hydrogen
Catalysts: Platinum
Solvents: Ethyl acetate; 2 h, rt

Experimental Protocols

Scheme 308 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (70)

Suppliers (124)

Supplier (1)

31-170-CAS-13733022

Steps: 1 Yield: 99%

Effect of the nucleophile structure on the relative mobility of the nitro group and fluorine atom in reactions of 3,5-dinitro- and 3-fluoro-5-nitrobenzotrifluorides with phenols in the presence of potassium carbonate

By: Khalfina, I. A.; et al

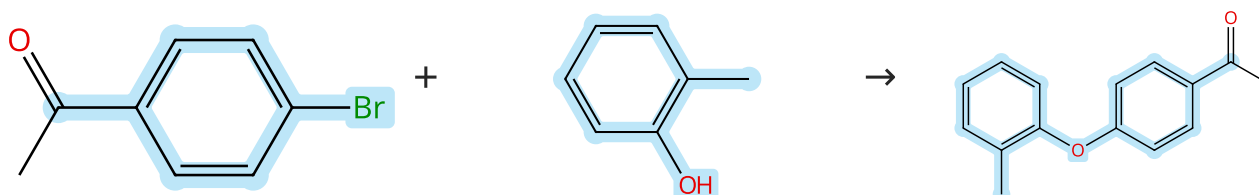
Russian Journal of Organic Chemistry (Translation of Zhurnal Organicheskoi Khimii) (2002), 38(1), 47-58.

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide

Experimental Protocols

Scheme 309 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

Suppliers (83)

Suppliers (7)

31-170-CAS-1389897

Steps: 1 Yield: 99%

CuO Nanoparticles on Ionic Polymer for Coupling Reactions of Aryl Bromides and Chlorides with Phenols

By: Hu, Tongjie; et al

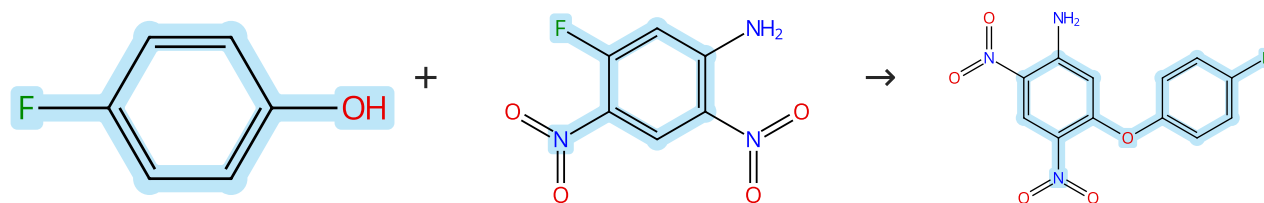
ChemCatChem (2011), 3(4), 661-665.

1.1 Reagents: Tripotassium phosphate
Catalysts: Copper oxide (CuO), Benzenemethanaminium, *N*-[bis(dimethylamino)methylene]-4-ethenyl-*N*-[(4-ethenyl phenyl)methyl]-, chloride (1:1), polymer with diethenylbenzene
Solvents: 1,4-Dioxane; 16 h, 120 °C

Experimental Protocols

Scheme 310 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (86)

Suppliers (64)

Suppliers (2)

31-170-CAS-4834098

Steps: 1 Yield: 99%

Design, synthesis and evaluation of novel 2,5,6-trisubstituted benzimidazoles targeting FtsZ as antitubercular agents

By: Park, Bora; et al

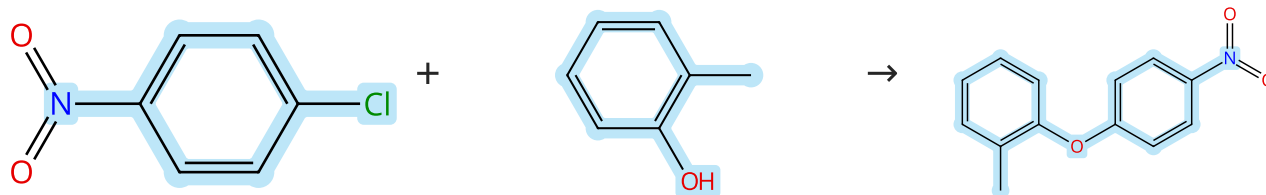
Bioorganic & Medicinal Chemistry (2014), 22(9), 2602-2612.

1.1 Reagents: Potassium carbonate
Solvents: Acetone; 16 h, rt

Experimental Protocols

Scheme 311 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (65)

Suppliers (83)

Suppliers (10)

31-170-CAS-3795400

Steps: 1 Yield: 99%

CuO Nanoparticles on Ionic Polymer for Coupling Reactions of Aryl Bromides and Chlorides with Phenols

By: Hu, Tongjie; et al

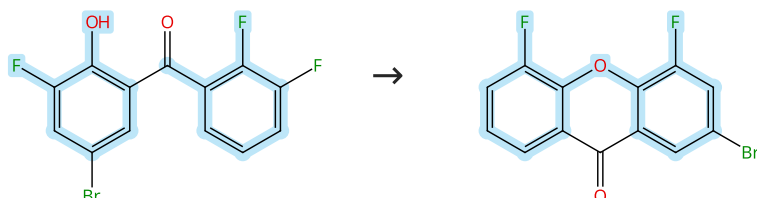
ChemCatChem (2011), 3(4), 661-665.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Copper oxide (CuO), Benzenemethanaminium, *N*-[bis(dimethylamino)methylene]-4-ethenyl-*N*-[(4-ethenyl phenyl)methyl]-, chloride (1:1), polymer with diethenylbenzene
Solvents: 1,4-Dioxane; 16 h, 135 °C

Experimental Protocols

Scheme 312 (1 Reaction)

Steps: 1 Yield: 99%



31-084-CAS-11465355

Steps: 1 Yield: 99%

Fluorinated 9H-xanthene-9-carboxylic acid oxazol-2-ylamides as potent, orally available mGlu1 receptor enhancers

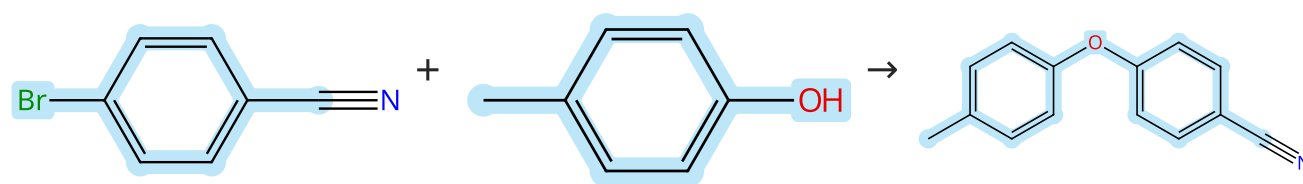
By: Vieira, Eric; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(6), 1666-1669.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Methyl ethyl ketone; reflux

Scheme 313 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (93)

Suppliers (89)

Suppliers (56)

31-170-CAS-12835062

Steps: 1 Yield: 99%

Immobilization of copper in organic-inorganic hybrid materials: a highly efficient and reusable catalyst for the Ullmann diaryl etherification

By: Miao, Tao; et al

Tetrahedron Letters (2007), 48(1), 95-99.

- 1.1 **Reagents:** Potassium fluoride
Catalysts: Copper (3-[(2-aminoethyl)amino]propyl)-functionalized silica-gel-immobilized)
Solvents: Dimethyl sulfoxide; 16 h, 130 °C

Experimental Protocols

31-170-CAS-13275084	Steps: 1 Yield: 99%	Copper promoted synthesis of diaryl ethers
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran		By: Ghosh, Rajshekhar; et al
1.2 Reagents: Tripotassium phosphate Catalysts: Malachite; cooled; 18 h, 115 °C		New Journal of Chemistry (2004), 28(11), 1390-1393.

Scheme 314 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (89)

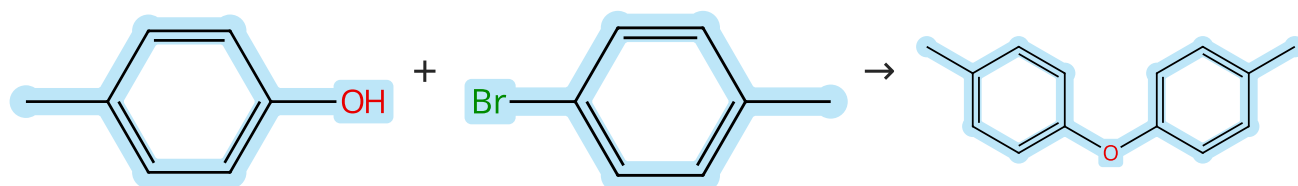
Suppliers (82)

Suppliers (32)

31-170-CAS-13018454	Steps: 1 Yield: 99%	Highly efficient heterogeneous catalyst for O-arylation of phenols with aryl halides using natural ferrous chamosite
1.1 Reagents: Potassium carbonate Catalysts: Chamosite ($\text{Al}_2\text{Fe}_5(\text{OH})_8\text{O}(\text{SiO}_3)_3$) Solvents: Dimethylformamide; 12 h, 110 °C		By: Arundhathi, R.; et al
		Applied Clay Science (2011), 51(1-2), 131-137.

Scheme 315 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (89)

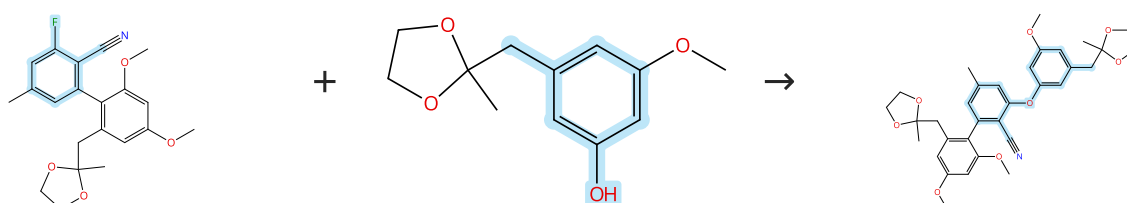
Suppliers (67)

Suppliers (55)

31-170-CAS-9824687	Steps: 1 Yield: 99%	Formation of Aryl-Nitrogen, Aryl-Oxygen, and Aryl-Carbon Bonds Using Well-Defined Copper(I)-Based Catalysts
1.1 Reagents: Cesium carbonate Catalysts: (7-4)-Bromo(2,9-dimethyl-1,10-phenanthroline- κN^1 , κN^{10})(triphenylphosphine)copper Solvents: Toluene		By: Gujadhur, Rattan K.; et al
Experimental Protocols		Organic Letters (2001), 3(26), 4315-4317.

Scheme 316 (1 Reaction)

Steps: 1 Yield: 99%

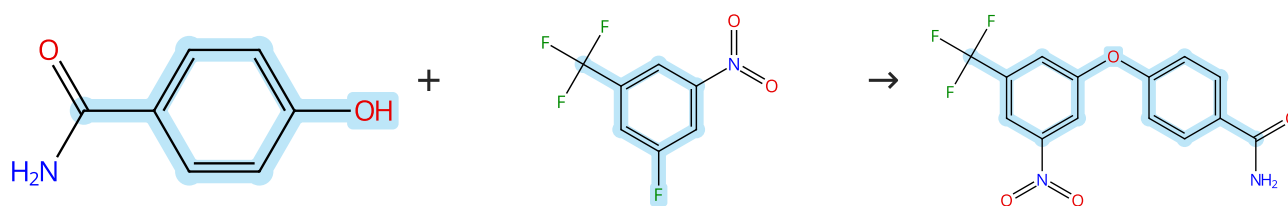


Supplier (1)

31-170-CAS-6207661	Steps: 1 Yield: 99%	Total Synthesis of A Novel Tetracyclic Alkaloid, Cassiarin F from the Flowers of Cassia siamea
1.1 Reagents: Cesium carbonate Solvents: Dimethyl sulfoxide; 12 h, 90 °C		By: Deguchi, Jun; et al
Experimental Protocols		Organic Letters (2011), 13(16), 4344-4347.

Scheme 317 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (76)

Suppliers (70)

31-170-CAS-20939822

Steps: 1 Yield: 99%

Design, synthesis, and in vitro bioactivity evaluation of fluorine-containing analogues for sphingosine-1-phosphate 2 receptor

1.1 Reagents: Monopotassium phosphate
Solvents: Dimethylacetamide; 12 h, 100 °C

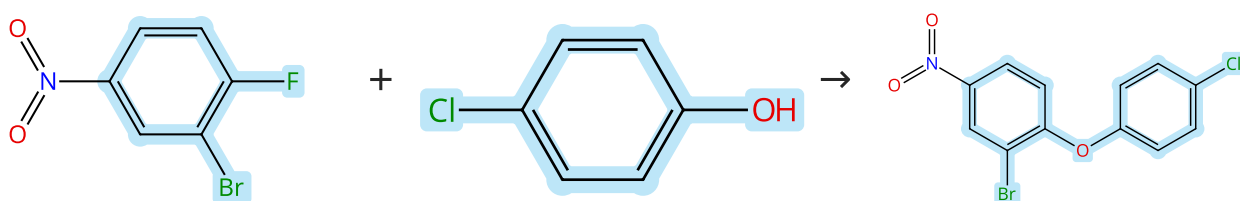
Experimental Protocols

By: Luo, Zonghua; et al

Bioorganic & Medicinal Chemistry (2019), 27(16), 3619-3631.

Scheme 318 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (90)

Suppliers (3)

31-614-CAS-35312187

Steps: 1 Yield: 99%

Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors

1.1 Reagents: Cesium carbonate
Solvents: Dimethyl sulfoxide; 2 h, 110 °C

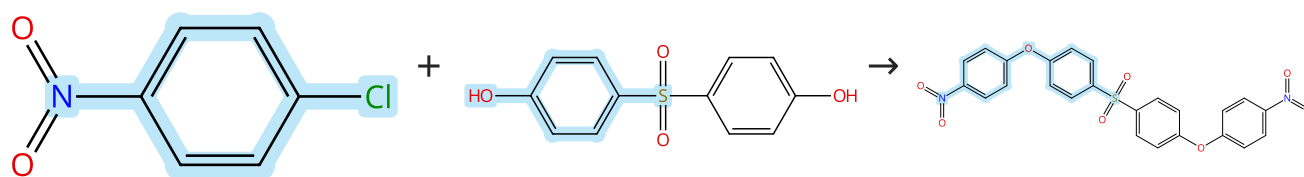
Experimental Protocols

By: Xu, Kai-Yan; et al

Bioorganic & Medicinal Chemistry (2023), 78, 117152.

Scheme 319 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (65)

Suppliers (97)

Suppliers (2)

31-170-CAS-15102316

Steps: 1 Yield: 99%

Synthesis and characterization of diethylenetriamine of polyimide

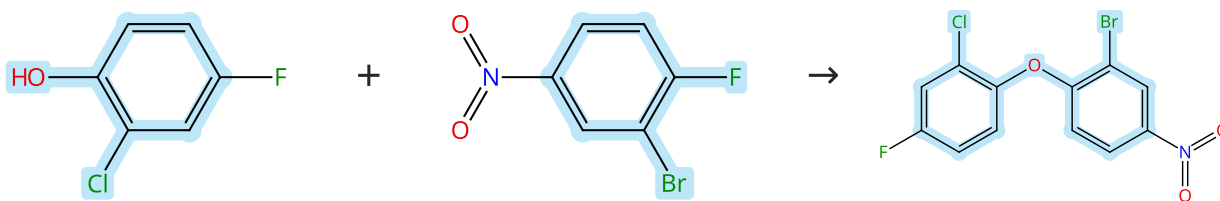
1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide, Toluene; 8 h, 134 - 146 °C

By: Zhu, Peng; et al

Huagong Xinxing Cailiao (2011), 39(9), 51-52.

Scheme 320 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (78)

Suppliers (83)

31-614-CAS-35312183

Steps: 1 Yield: 99%

Design, synthesis, and biological evaluation of quinoxaline derivatives as potent BRD4 inhibitors

1.1 Reagents: Cesium carbonate

Solvents: Dimethyl sulfoxide; 2 h, 110 °C

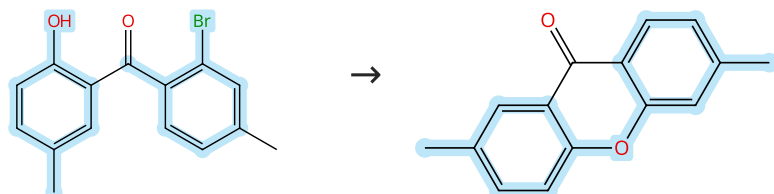
By: Xu, Kai-Yan; et al

Experimental Protocols

Bioorganic & Medicinal Chemistry (2023), 78, 117152.

Scheme 321 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

Suppliers (2)

31-614-CAS-30700408

Steps: 1 Yield: 99%

A convenient approach to the xanthone scaffold by an aqueous aromatic substitution of bromo- and iodoarenes

1.1 Reagents: Potassium hydroxide

Solvents: Water; 15 h, 100 °C

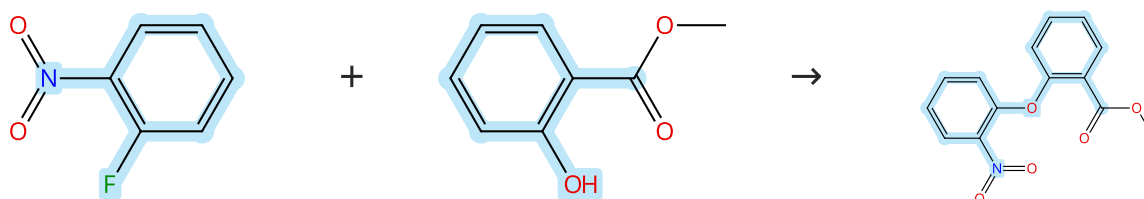
By: Barbero, Nekane; et al

Experimental Protocols

Tetrahedron (2009), 65(29-30), 5729-5732.

Scheme 322 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (93)

Suppliers (133)

Suppliers (3)

31-614-CAS-38294812

Steps: 1 Yield: 99%

Herringbone Helical Foldamers from Aromatic Ether Derived ε-Amino Acid Peptides

1.1 Reagents: Potassium carbonate

Solvents: DMSO-*d*₆; overnight, rt

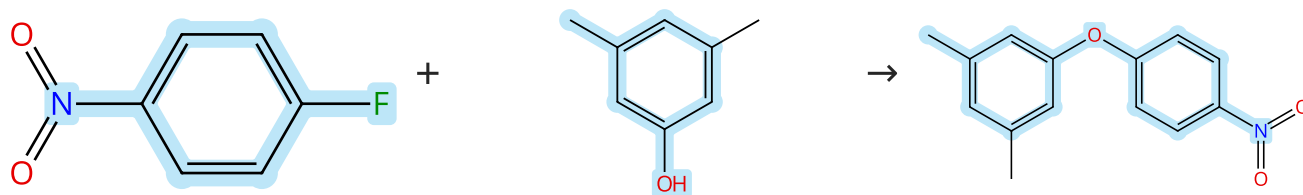
By: Yang, Yong; et al

Experimental Protocols

Chemistry - A European Journal (2023), 29(66), e202301832.

Scheme 323 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (100)

Suppliers (89)

Suppliers (19)

31-170-CAS-6608413

Steps: 1 Yield: 99%

On the frontier between nucleophilic aromatic substitution and catalysis

By: Pichette Drapeau, Martin; et al

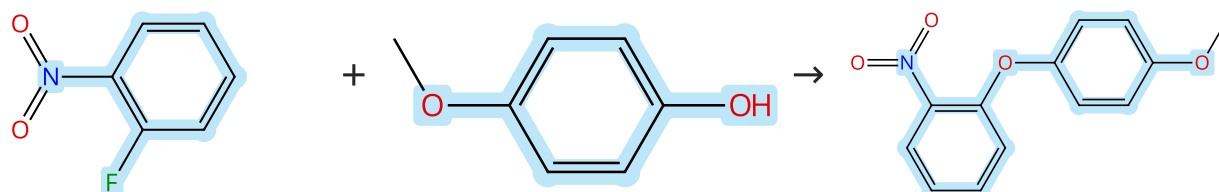
Chemistry - A European Journal (2014), 20(18), 5231-5236.

1.1 Reagents: Cesium carbonate
Solvents: Dimethylformamide; 6 h, 22 °C

Experimental Protocols

Scheme 324 (3 Reactions)

Steps: 1 Yield: 99%



Suppliers (93)

Suppliers (121)

Suppliers (6)

31-170-CAS-29848083

Steps: 1 Yield: 99%

Diphenyl ether derivatives occupy the expanded binding site of cyclohexanedione compounds at the colchicine site in tubulin by movement of the α T5 loop

By: Bueno, Oskia; et al

European Journal of Medicinal Chemistry (2019), 171, 195-208.

1.1 Reagents: Cesium carbonate
Solvents: Acetonitrile; rt; 3 h, 80 °C

31-170-CAS-8292724

Steps: 1 Yield: 99%

A facile and efficient synthesis of diaryl amines or ethers under microwave irradiation in the presence of $\text{KF}/\text{Al}_2\text{O}_3$ without solvent and their anti-fungal biological activities against six phytopathogens

By: Huang, Liang-Zhu; et al

International Journal of Molecular Sciences (2013), 14(9), 18850-18860.

1.1 Catalysts: Alumina, Potassium fluoride; 15 min, 150 °C

31-170-CAS-5394272

Steps: 1 Yield: 99%

Synthesis of tetrasubstituted alkenes through a palladium-catalyzed domino carbopalladation/C-H-activation reaction

By: Tietze, Lutz F.; et al

Chemistry - A European Journal (2012), 18(11), 3286-3291, S3286/1-S3286/50.

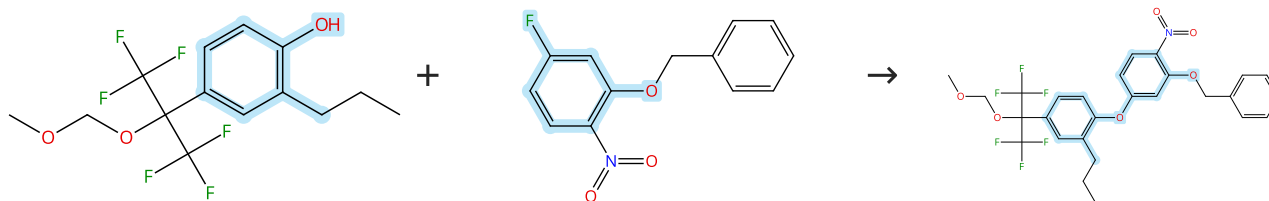
1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; 23 h, 100 °C; 100 °C → rt

1.2 Solvents: Water; rt

Experimental Protocols

Scheme 325 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

Suppliers (55)

31-170-CAS-16065720

Steps: 1 Yield: 99%

Discovery of a 2-hydroxyacetophenone derivative as an outstanding linker to enhance potency and β -selectivity of liver X receptor agonist

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt; 3 h, 80 °C

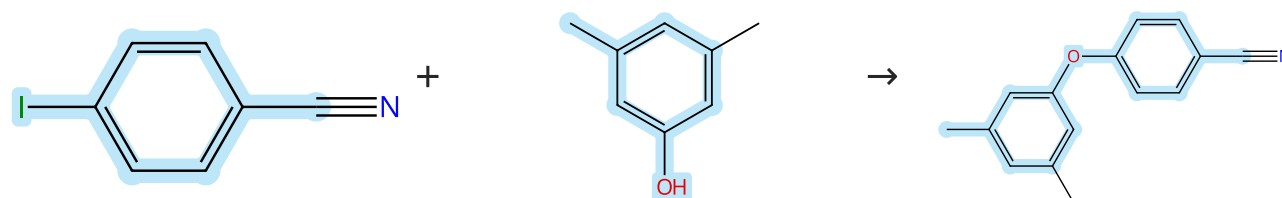
Experimental Protocols

By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry (2016), 24(16), 3436-3446.

Scheme 326 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (81)

Suppliers (89)

Suppliers (22)

31-170-CAS-15029268

Steps: 1 Yield: 99%

On the frontier between nucleophilic aromatic substitution and catalysis

1.1 **Reagents:** Cesium carbonate, 2,2,6,6-Tetramethyl-3,5-heptanedione

Catalysts: Cuprous iodide**Solvents:** Dimethylformamide; 24 h, 50 °C

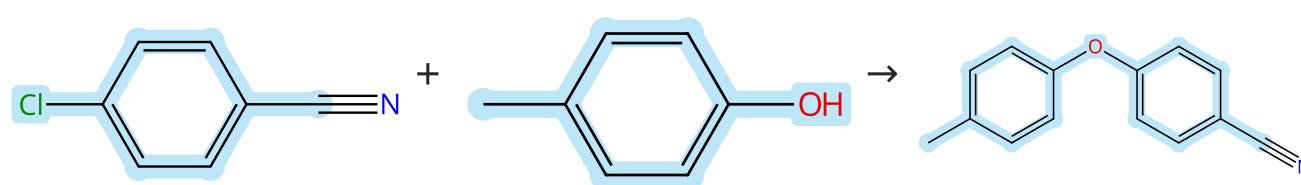
Experimental Protocols

By: Pichette Drapeau, Martin; et al

Chemistry - A European Journal (2014), 20(18), 5231-5236.

Scheme 327 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

Suppliers (89)

Suppliers (56)

31-170-CAS-23365413

Steps: 1 Yield: 99%

Oxalohydrazide Ligands for Copper-Catalyzed C-O Coupling Reactions with High Turnover Numbers

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Copper bromide (CuBr), *N*¹,*N*²-Bis(2,5-dimethyl-1*H*-pyrrol-1-yl)ethanediamide

Solvents: Dimethyl sulfoxide; 5 - 10 min, rt

1.2 24 h, 120 °C; 120 °C → rt

1.3 **Reagents:** Sodium hydroxide**Solvents:** Water; rt

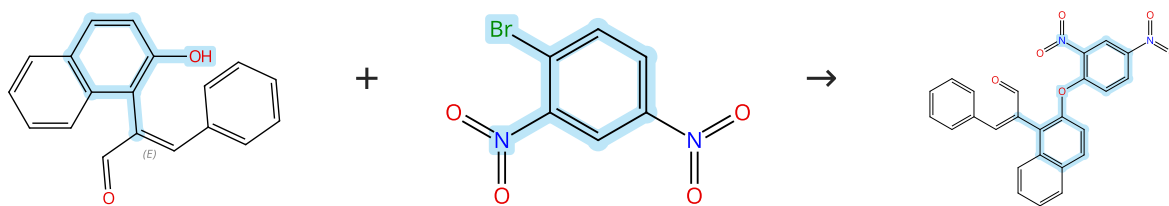
Experimental Protocols

By: Ray, Ritwika; et al

Angewandte Chemie, International Edition (2021), 60(15), 8203-8211.

Scheme 328 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (57)

31-614-CAS-37560235

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

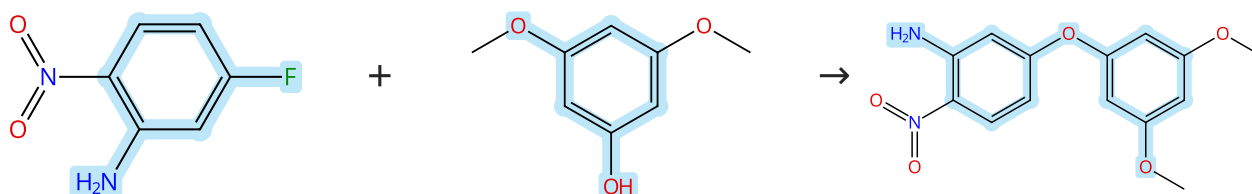
Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

Nature Communications (2023), 14(1), 5050.

Scheme 329 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (82)

Suppliers (92)

Supplier (1)

31-170-CAS-15433969

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 24 h, 100 °C

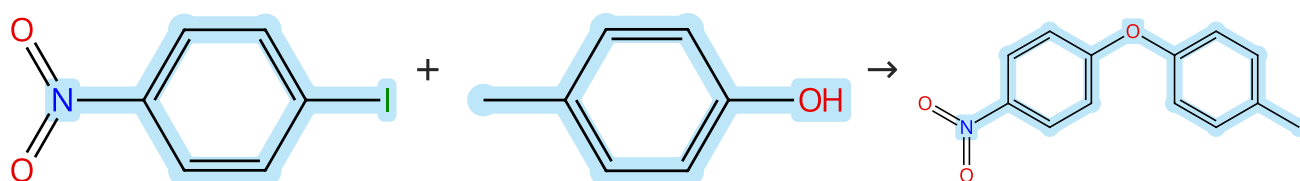
New benzimidazole-2-urea derivatives as tubulin inhibitors

By: Wang, Wenna; et al

Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 330 (2 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (70)

Suppliers (89)

Suppliers (37)

31-170-CAS-13323769

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Copper
Solvents: Dimethylformamide; 24 h, 130 °C

Experimental Protocols

Maghemite-Copper Nanocomposites: Applications for Ligand-Free Cross-Coupling (C-O, C-S, and C-N) Reactions

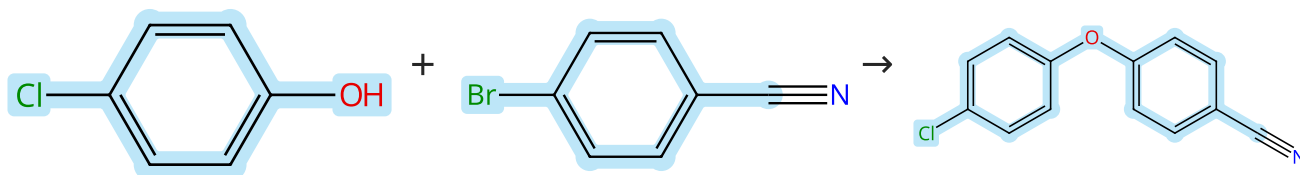
By: Sharma, Rakesh K.; et al

ChemCatChem (2015), 7(21), 3495-3502.

31-170-CAS-17776501	Steps: 1 Yield: 98%	A highly active and recyclable catalyst for the synthesis of indole and phenyl ether By: Qiao, Bo; et al RSC Advances (2015), 5(113), 93463-93469.
1.1 Reagents: Potassium fluoride Catalysts: Alumina, Copper Solvents: Dimethyl sulfoxide; 16 h, 135 °C		
Experimental Protocols		

Scheme 331 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

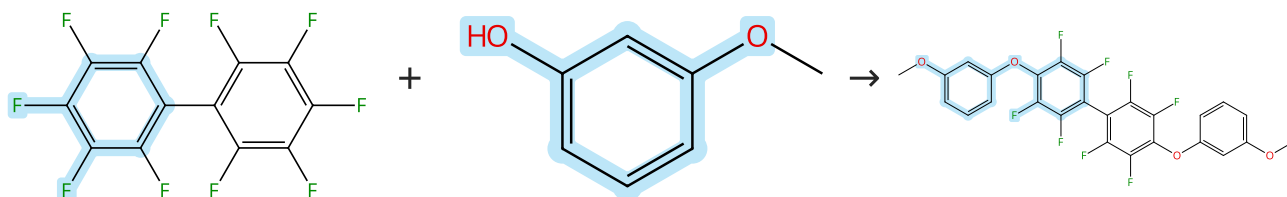
Suppliers (93)

Suppliers (62)

31-170-CAS-4122008	Steps: 1 Yield: 99%	Copper promoted synthesis of diaryl ethers By: Ghosh, Rajshekhar; et al New Journal of Chemistry (2004), 28(11), 1390-1393.
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran		
1.2 Reagents: Tripotassium phosphate Catalysts: Malachite; cooled; 18 h, 115 °C		

Scheme 332 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (68)

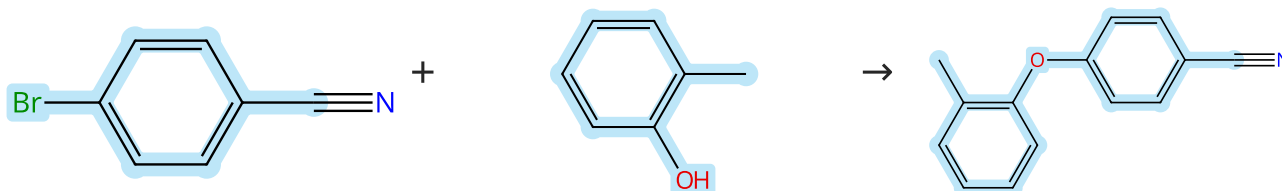
Suppliers (92)

Suppliers (5)

31-170-CAS-2752480	Steps: 1 Yield: 99%	Bisphenols from hexafluorobenzene and decafluorobiphenyl By: Kameneva, T. M.; et al Zhurnal Organicheskoi Khimii (1989), 25(3), 576-82.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide		

Scheme 333 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (93)

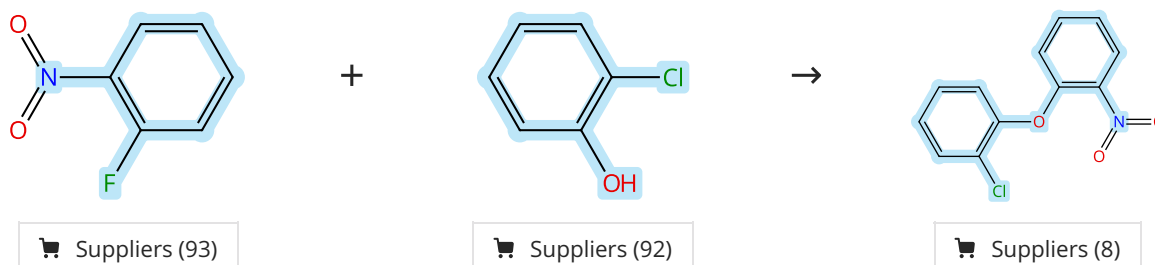
Suppliers (83)

Suppliers (30)

31-170-CAS-12933337 Steps: 1 Yield: 99% 1.1 Reagents: Tripotassium phosphate Catalysts: Copper oxide (CuO), Benzenemethanaminium, <i>N</i>-[bis(dimethylamino)methylene]-4-ethenyl-<i>N</i>-[(4-ethenyl phenyl)methyl]-, chloride (1:1), polymer with diethenylbenzene Solvents: 1,4-Dioxane; 16 h, 120 °C Experimental Protocols	CuO Nanoparticles on Ionic Polymer for Coupling Reactions of Aryl Bromides and Chlorides with Phenols By: Hu, Tongjie; et al ChemCatChem (2011), 3(4), 661-665.
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Scheme 334 (1 Reaction)

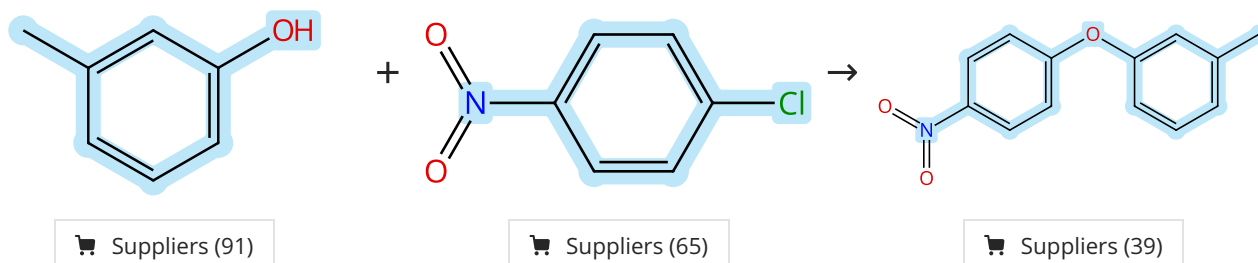
Steps: 1 Yield: 99%



31-170-CAS-22969850 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt → 95 °C; 24 h, 95 °C Experimental Protocols	Synthesis of Cycloheptatrienes, Oxepines, Thiepinines, and Silepinines: A Comparison between Brønsted Acid and Au-Catalysis By: Sprenger, Kristin; et al European Journal of Organic Chemistry (2020), 2020(39), 6245-6254.
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Scheme 335 (1 Reaction)

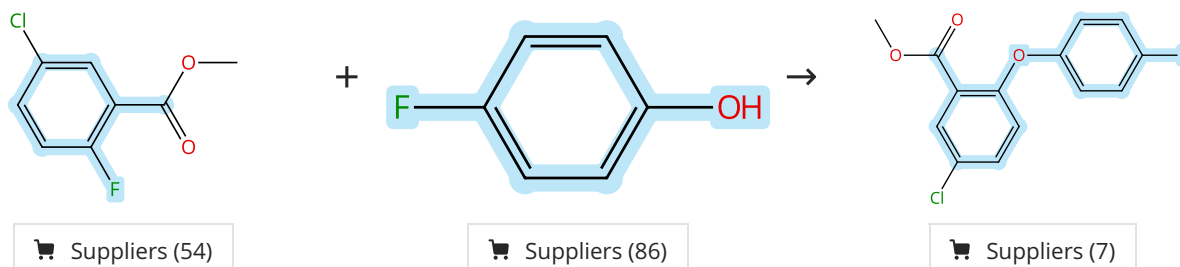
Steps: 1 Yield: 99%



31-170-CAS-12703073 Steps: 1 Yield: 99% 1.1 Reagents: Potassium carbonate Catalysts: Cuprous iodide, 1-[(Dimethylamino)-2-pyridinylmethyl]-2-naphthalenol Solvents: Dimethyl sulfoxide; 24 h, 120 °C Experimental Protocols	Betti Base as an Efficient Ligand for Copper-Catalyzed Ullmann Coupling of Phenol with Aryl Halides By: Yang, Li; et al Synthetic Communications (2014), 44(17), 2468-2477.
--	---

Scheme 336 (1 Reaction)

Steps: 1 Yield: 99%



31-170-CAS-18666923	Steps: 1 Yield: 99%	Discovery of AAT-008, a novel, potent, and selective prostaglandin EP4 receptor antagonist By: Okumura, Yoshiyuki; et al Bioorganic & Medicinal Chemistry Letters (2017), 27(5), 1186-1192.
1.1 Reagents: Potassium carbonate Solvents: Toluene; reflux		

Scheme 337 (2 Reactions)

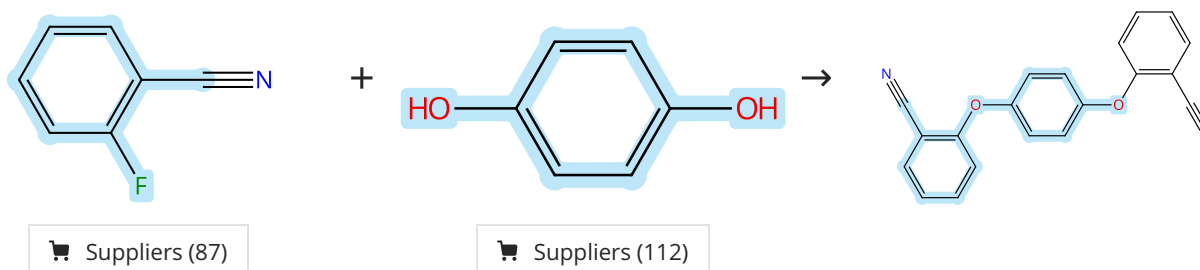
Steps: 1 Yield: 98-99%



31-170-CAS-4765002	Steps: 1 Yield: 99%	Maghemite-Copper Nanocomposites: Applications for Ligand-Free Cross-Coupling (C-O, C-S, and C-N) Reactions By: Sharma, Rakesh K.; et al ChemCatChem (2015), 7(21), 3495-3502.
1.1 Reagents: Cesium carbonate Catalysts: Copper Solvents: Dimethylformamide; 24 h, 130 °C		
Experimental Protocols		
31-170-CAS-872908	Steps: 1 Yield: 98%	Ligand-free coupling of phenols and alcohols with aryl halides by a recyclable heterogeneous copper catalyst By: Wang, Man; et al RSC Advances (2012), 2(13), 5528-5530.
1.1 Reagents: Cesium carbonate Catalysts: Cuprous iodide (reaction product with MOF-253), [[2,2'-Bipyridine]-5,5'-dicarboxylato(2-)-κO ⁵]hydroxyaluminum (reaction product with CuI) Solvents: Dimethyl sulfoxide; rt; 24 h, 80 °C		
Experimental Protocols		

Scheme 338 (1 Reaction)

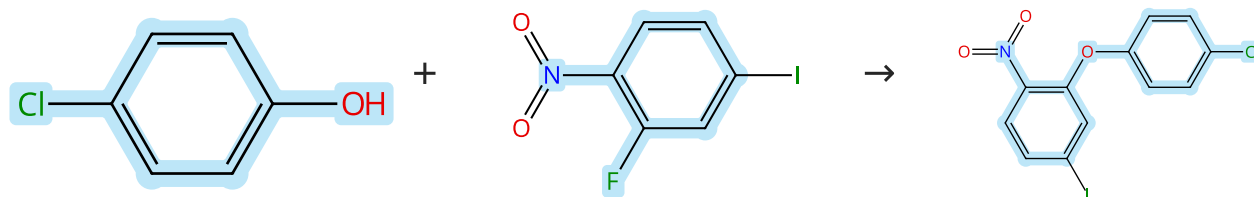
Steps: 1 Yield: 99%



31-170-CAS-11868743	Steps: 1 Yield: 99%	Synthesis of Di- and Triphenylones that Display High Stability and a Visual Fluorescence Response to Strong Acid By: Wang, Baolin; et al Chemistry - An Asian Journal (2014), 9(11), 3307-3312.
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 12 h, 80 °C		

Scheme 339 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

Suppliers (54)

31-614-CAS-38471053

Steps: 1 Yield: 99%

Exploring the Translational Gap of a Novel Class of Escherichia coli IspE Inhibitors

By: Ropponen, Henni-Karoliina; et al

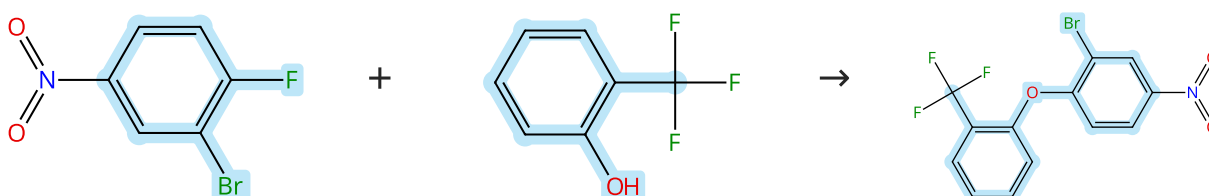
ChemMedChem (2023), 18(19), e202300346.

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; 20 min, rt
- 1.2 3 h, 90 °C

Experimental Protocols

Scheme 340 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (83)

Suppliers (80)

31-614-CAS-35312186

Steps: 1 Yield: 99%

Design, synthesis, and biological evaluation of quinoxalinone derivatives as potent BRD4 inhibitors

By: Xu, Kai-Yan; et al

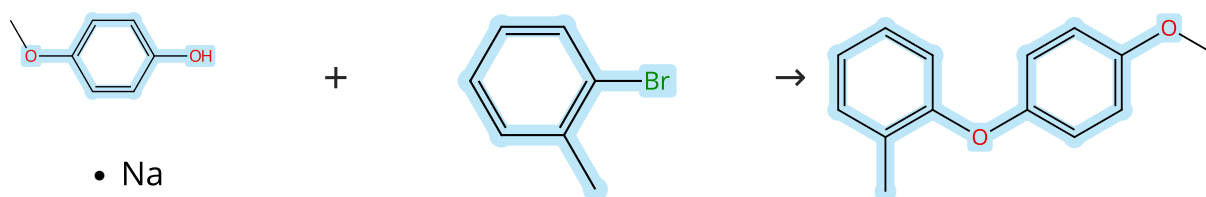
Bioorganic & Medicinal Chemistry (2023), 78, 117152.

- 1.1 Reagents: Cesium carbonate
Solvents: Dimethyl sulfoxide; 2 h, 110 °C

Experimental Protocols

Scheme 341 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (5)

Suppliers (60)

Suppliers (7)

31-170-CAS-13307319

Steps: 1 Yield: 99%

Air stable, sterically hindered ferrocenyl dialkylphosphines for palladium-catalyzed C-C, C-N, and C-O bond-forming cross-couplings

By: Kataoka, Noriyasu; et al

Journal of Organic Chemistry (2002), 67(16), 5553-5566.

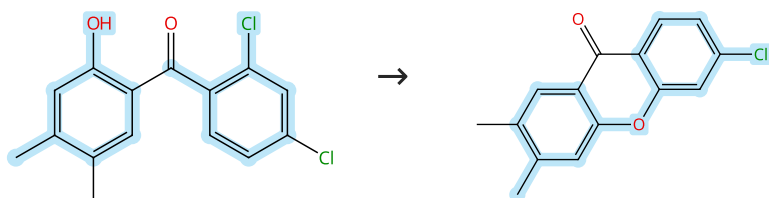
- 1.1 Catalysts: Bis(dibenzylideneacetone)palladium, 1'-[Bis(1,1-dimethylethyl)phosphino]-1,2,3,4,5-pentaphenylferrocene
Solvents: Toluene

Experimental Protocols

31-170-CAS-6255104	Steps: 1 Yield: 99%	Unusual in Situ Ligand Modification to Generate a Catalyst for Room Temperature Aromatic C-O Bond Formation
1.1 Catalysts: Bis(dibenzylideneacetone)palladium, 1'-[Bis(1,1-dimethylethyl)phosphino]-1,2,3,4,5-pentaphenylferrocene Solvents: Toluene		By: Shelby, Quinetta; et al
Experimental Protocols		Journal of the American Chemical Society (2000), 122(43), 10718-10719.

Scheme 342 (1 Reaction)

Steps: 1 Yield: 99%

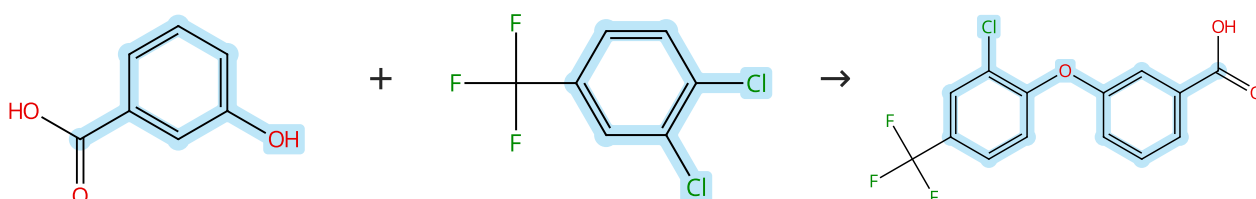


Suppliers (2)

31-614-CAS-30553855	Steps: 1 Yield: 99%	Microwave-assisted transition-metal-free intramolecular Ullmann-type O-arylation in water for the synthesis of xanthenes and azaxanthenes
1.1 Reagents: Tetrabutylammonium hydroxide Solvents: Water; 4 min, 120 °C		By: Zhang, Xiaojin; et al
Experimental Protocols		Tetrahedron Letters (2014), 55(35), 4883-4887.

Scheme 343 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (93)

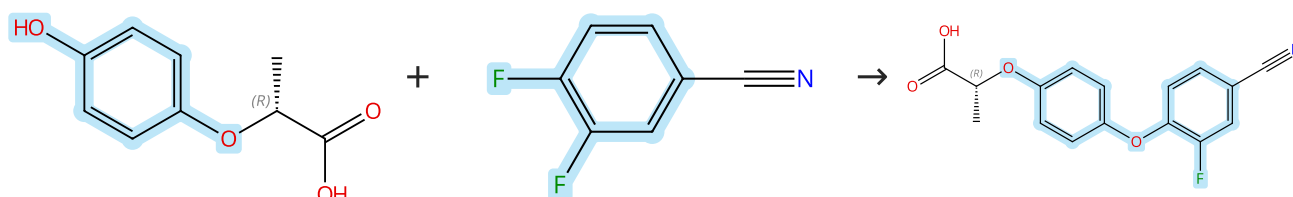
Suppliers (71)

Suppliers (43)

31-170-CAS-3930042	Steps: 1 Yield: 99%	The improvement on synthesis of 3-(2-chloro-α,α,α-trifluoro-<i>p</i>-tolyoxy)benzoic acid
1.1 Reagents: Chlorobenzene, Potassium hydroxide Solvents: Water; rt; reflux; cooled		By: Zhang, Hai-bin; et al
1.2 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 6 h, reflux		Xiandai Nongyao (2005), 4(5), 6-7.
1.3 Reagents: Hydrochloric acid Solvents: Water; acidified		

Scheme 344 (1 Reaction)

Steps: 1 Yield: 99%

Absolute stereochemistry shown,
Rotation (+)

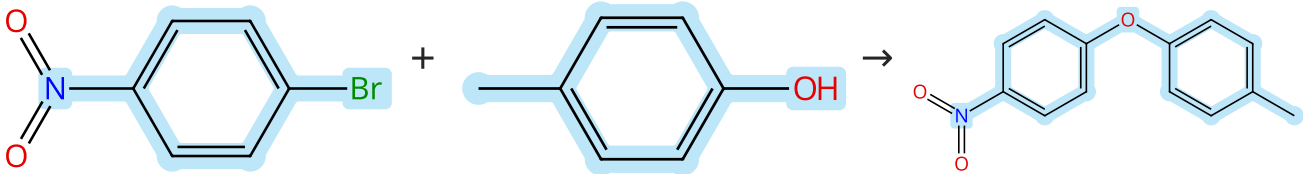
Suppliers (73)

Absolute stereochemistry shown,
Rotation (+)

Suppliers (55)

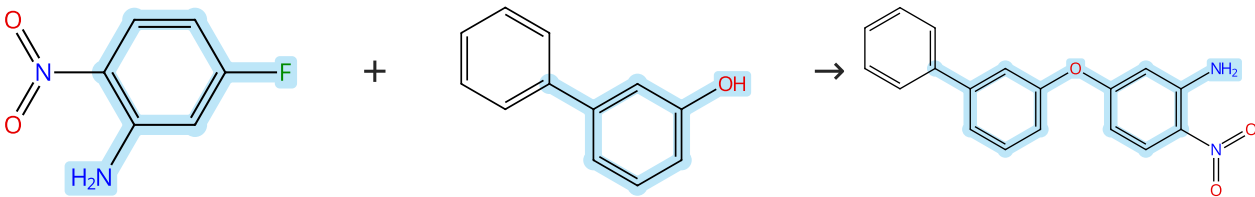
Suppliers (45)

31-170-CAS-24410323	Steps: 1 Yield: 99%	Synthesis and herbicidal activity of novel N-allyloxy/propargyloxy aryloxyphenoxy propionamide derivatives
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 1 h, 75 °C		By: Liu, Qixing; et al
1.2 75 °C; 6 h, 75 °C; cooled		Chemical Research in Chinese Universities (2016), 32(2), 188-194.
1.3 Reagents: Hydrochloric acid Solvents: Water; pH 4 - 5		
Experimental Protocols		

Scheme 345 (2 Reactions)		Steps: 1 Yield: 99%
		
 Suppliers (77)	 Suppliers (89)	 Suppliers (37)

31-170-CAS-15706344	Steps: 1 Yield: 99%	Ullmann Coupling Reaction of Nitro-Substituted Aryl Halides with Phenols under Mild Conditions: Micro-/Mesoporous Hierarchical LaAlPO-5 Zeolite Catalyst
1.1 Reagents: Potassium carbonate Catalysts: Lanthanum (complexes with zeolite AlPO-5), Aluminum phosphate (mesoporous, lanthanide-doped) Solvents: Dimethyl sulfoxide; 12 h, 80 °C		By: Ke, Qingping; et al
Experimental Protocols		ChemCatChem (2016), 8(8), 1557-1563.

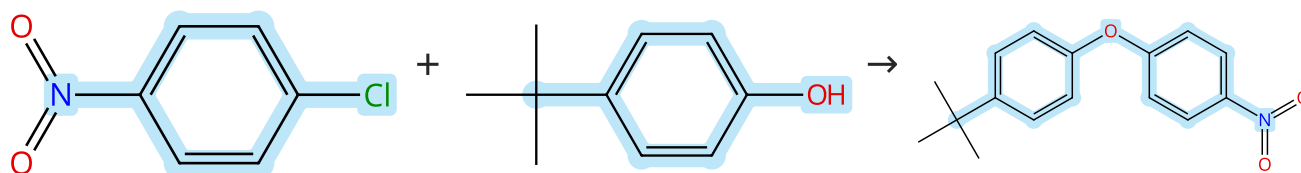
31-170-CAS-13319259	Steps: 1 Yield: 99%	Palladium nanoparticles supported on silica diphenylphosphinite as efficient catalyst for C-O and C-S arylation of aryl halides
1.1 Reagents: Sodium hydroxide, Tetrabutylammonium hydroxide Catalysts: Diphenylphosphine (silica-supported), Palladium chloride Solvents: Water; 6 h, 70 °C		By: Iranpoor, Nasser; et al
Experimental Protocols		Applied Organometallic Chemistry (2013), 27(9), 501-506.

Scheme 346 (1 Reaction)		Steps: 1 Yield: 99%
		
 Suppliers (82)	 Suppliers (73)	

31-170-CAS-4271042	Steps: 1 Yield: 99%	New benzimidazole-2-urea derivatives as tubulin inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C		By: Wang, Wenna; et al
		Bioorganic & Medicinal Chemistry Letters (2014), 24(17), 4250-4253.

Scheme 347 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (65)

Suppliers (88)

Suppliers (22)

31-170-CAS-10476591

Steps: 1 Yield: 99%

CuO Nanoparticles on Ionic Polymer for Coupling Reactions of Aryl Bromides and Chlorides with Phenols

By: Hu, Tongjie; et al

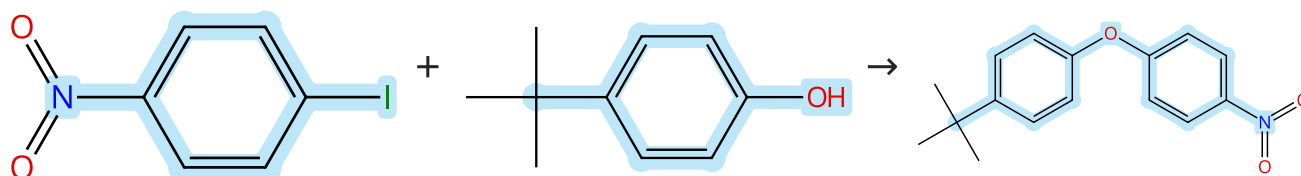
ChemCatChem (2011), 3(4), 661-665.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Copper oxide (CuO), Benzenemethanaminium, *N*-[bis(dimethylamino)methylene]-4-ethenyl-*N*-[(4-ethenyl phenyl)methyl]-, chloride (1:1), polymer with diethenylbenzene
Solvents: 1,4-Dioxane; 16 h, 135 °C

Experimental Protocols

Scheme 348 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (70)

Suppliers (88)

Suppliers (22)

31-170-CAS-6276946

Steps: 1 Yield: 99%

Maghemite-Copper Nanocomposites: Applications for Ligand-Free Cross-Coupling (C-O, C-S, and C-N) Reactions

By: Sharma, Rakesh K.; et al

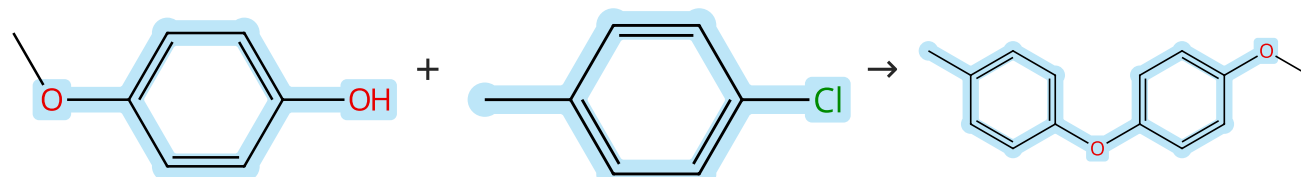
ChemCatChem (2015), 7(21), 3495-3502.

1.1 **Reagents:** Cesium carbonate
Catalysts: Copper
Solvents: Dimethylformamide; 24 h, 130 °C

Experimental Protocols

Scheme 349 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (121)

Suppliers (82)

Suppliers (17)

31-170-CAS-7627196

Steps: 1 Yield: 99%

Copper- or iron-catalyzed arylation of phenols from respectively aryl chlorides and aryl iodides

By: Xia, Ning; et al

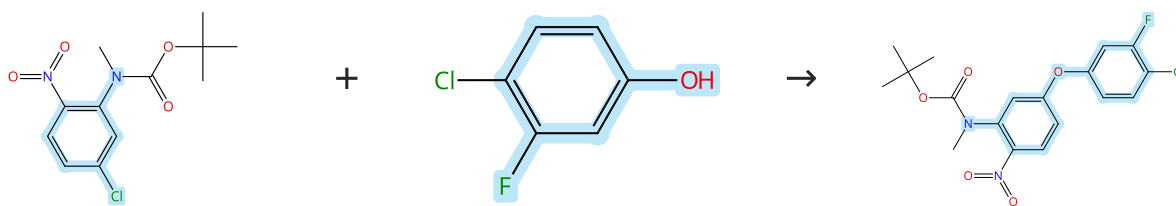
Chemistry - A European Journal (2008), 14(20), 6037-6039.

1.1 **Reagents:** Cesium carbonate
Catalysts: 2,2,6,6-Tetramethyl-3,5-heptanedione, Copper bromide
Solvents: Dimethylformamide; rt; rt → 135 °C; 36 h, 135 °C

Experimental Protocols

Scheme 350 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (8)

Suppliers (82)

31-170-CAS-19673637

Steps: 1 Yield: 99%

Discovery of DS-6930, a potent selective PPAR γ modulator.
Part I: Lead identification

1.1 **Reagents:** Sodium hydride
Solvents: Dimethylformamide; rt; 10 h, 80 °C

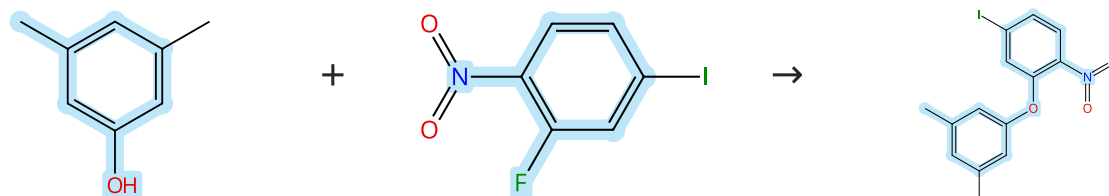
By: Shinozuka, Tsuyoshi; et al

Experimental Protocols

Bioorganic & Medicinal Chemistry (2018), 26(18), 5079-5098.

Scheme 351 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (89)

Suppliers (54)

31-614-CAS-38471052

Steps: 1 Yield: 99%

Exploring the Translational Gap of a Novel Class of Escherichia coli IspE Inhibitors

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; 20 min, rt
 1.2 3 h, 90 °C

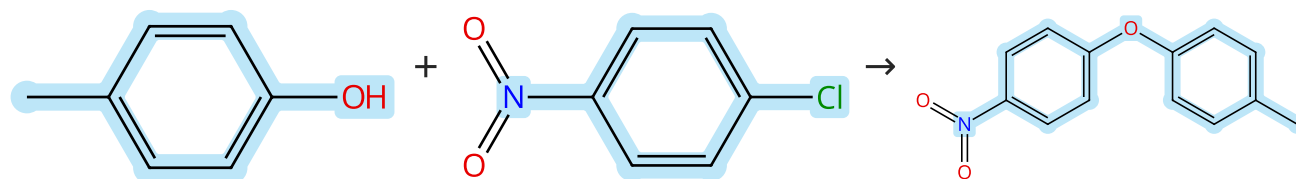
By: Ropponen, Henni-Karoliina; et al

ChemMedChem (2023), 18(19), e202300346.

Experimental Protocols

Scheme 352 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (89)

Suppliers (65)

Suppliers (37)

31-170-CAS-17492724

Steps: 1 Yield: 99%

The effect of O-nucleophiles structure on the features of halogen substitution in heterogeneous systems

1.1 **Reagents:** Potassium carbonate, Iron oxide (Fe₂O₃)
Solvents: Dimethylformamide; 270 min, 130 °C

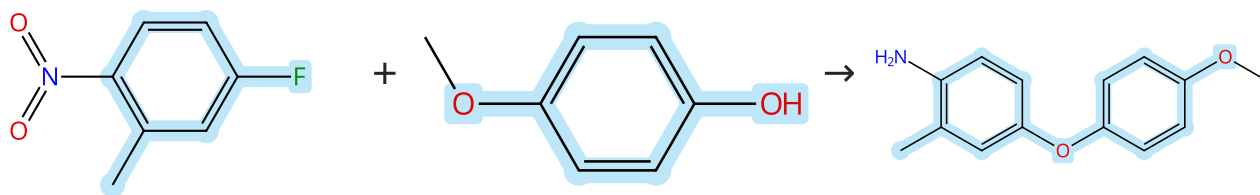
By: Orlov, V. Yu.; et al

Experimental Protocols

Russian Journal of General Chemistry (2017), 87(3), 386-388.

Scheme 353 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

Suppliers (121)

Suppliers (12)

31-170-CAS-5583902

Steps: 1 Yield: 99%

Discovery and structure-activity study of a novel benzoxa borole anti-inflammatory agent (AN2728) for the potential topical treatment of psoriasis and atopic dermatitis

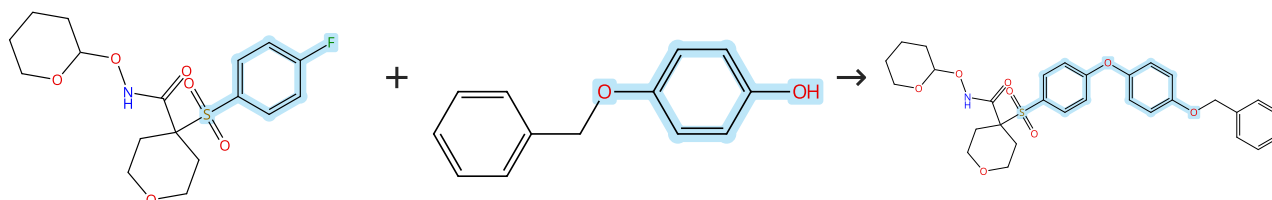
By: Akama, Tsutomu; et al

Bioorganic & Medicinal Chemistry Letters (2009), 19(8), 2129-2132.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; overnight, 70 °C
- 1.2 **Reagents:** Hydrochloric acid, Stannous chloride
Solvents: Ethanol; 2 h, 50 °C

Scheme 354 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

Suppliers (104)

31-170-CAS-14863872

Steps: 1 Yield: 99%

Orally Active MMP-1 Sparing α -Tetrahydropyranyl and α -Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease

By: Becker, Daniel P.; et al

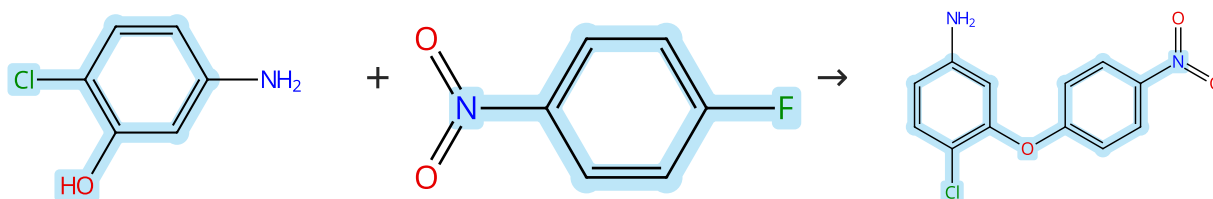
Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.

- 1.1 **Reagents:** Cesium carbonate
Solvents: Dimethylacetamide; 6 h, 95 °C

Experimental Protocols

Scheme 355 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (72)

Suppliers (100)

Supplier (1)

31-614-CAS-40338184

Steps: 1 Yield: 98%

Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors

By: Xu, Yue; et al

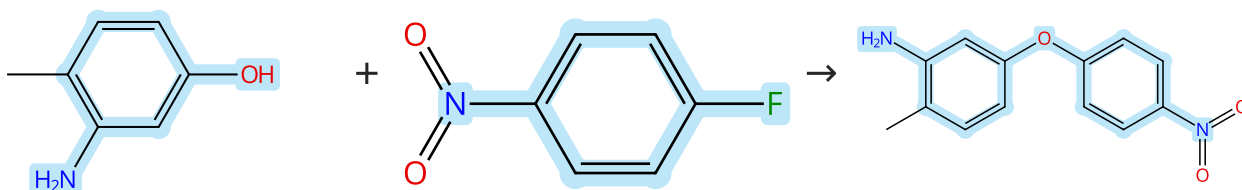
Bioorganic Chemistry (2023), 131, 106339.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C

Experimental Protocols

Scheme 356 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (72)

Suppliers (100)

31-614-CAS-40338182

Steps: 1 Yield: 98%

Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C

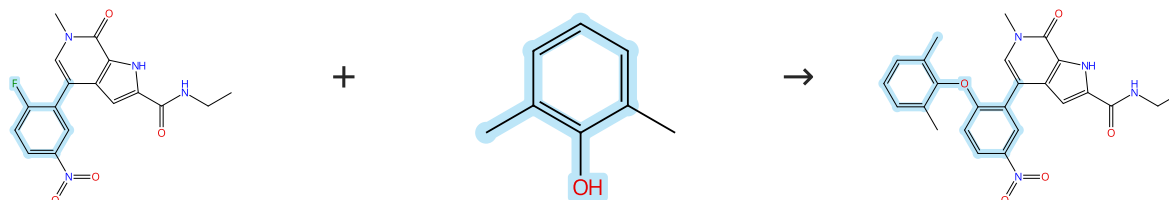
By: Xu, Yue; et al

Experimental Protocols

Bioorganic Chemistry (2023), 131, 106339.

Scheme 357 (1 Reaction)

Steps: 1 Yield: 98%



Supplier (1)

Suppliers (88)

Supplier (1)

31-614-CAS-24358170

Steps: 1 Yield: 98%

Discovery of N-Ethyl-4-[2-(4-fluoro-2,6-dimethyl-phenoxy)-5-(1-hydroxy-1-methyl-ethyl)phenyl]-6-methyl-7-oxo-1H-pyrrolo [2,3-c]pyridine-2-carboxamide (ABBV-744), a BET Bromodomain Inhibitor with Selectivity for the Second Bromodomain

1.1 **Reagents:** Cesium carbonate
Solvents: Dimethyl sulfoxide; 90 min, 50 °C

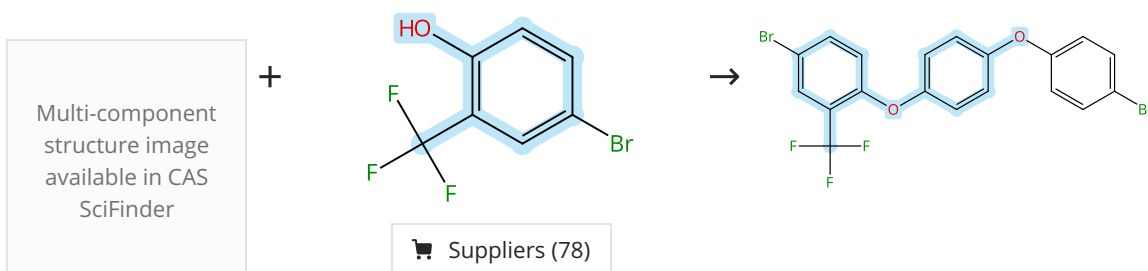
By: Sheppard, George S.; et al

Experimental Protocols

Journal of Medicinal Chemistry (2020), 63(10), 5585-5623.

Scheme 358 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (78)

31-614-CAS-41866187

Steps: 1 Yield: 98%

Synthesis of Diaryl Ethers via Hypervalent Iodine-Mediated C-H Functionalization

1.1 **Reagents:** Cesium carbonate
Solvents: Toluene; 8 h, 80 °C

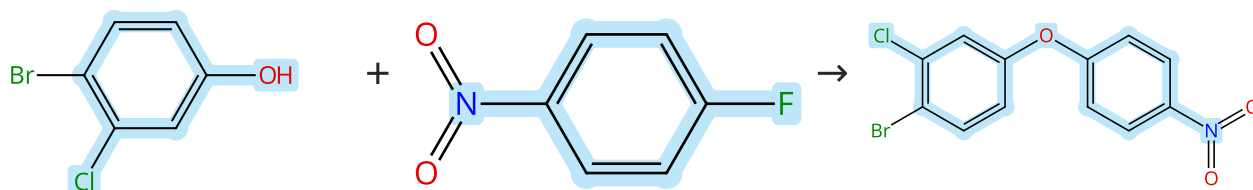
By: Hu, Chenghu; et al

Experimental Protocols

Organic Letters (2024), 26(37), 7864-7868.

Scheme 359 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

Suppliers (100)

Supplier (1)

31-614-CAS-37615376

Steps: 1 Yield: 98%

Discovery of N-Arylsulfonyl-Indole-2-Carboxamide Derivatives as Galectin-3 and Galectin-8 C-Terminal Domain Inhibitors

1.1 Reagents: Cesium carbonate

Solvents: Dimethylformamide; 6 h, 50 °C

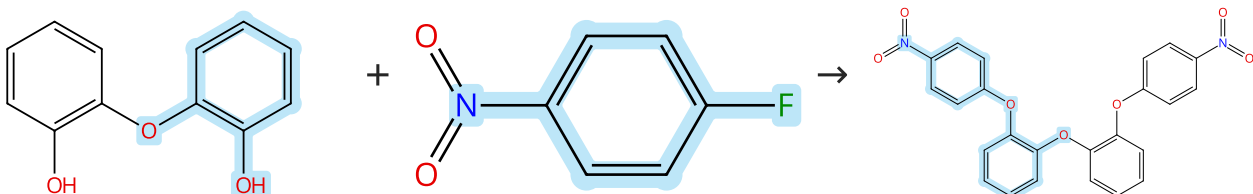
By: Zhang, Haoming; et al

Experimental Protocols

ACS Medicinal Chemistry Letters (2023), 14(9), 1257-1265.

Scheme 360 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (61)

Suppliers (100)

31-170-CAS-2680553

Steps: 1 Yield: 98%

Poly(ether imide)s from Diamines with Para-, Meta-, and Ortho-Arylene Substitutions: Synthesis, Characterization, and Liquid Crystalline Properties

1.1 Reagents: Potassium carbonate

Solvents: Toluene, Dimethylacetamide; 1.5 h, 135 °C; 135 °C → 175 °C; 175 °C → rt

By: Dingemans, Theo J.; et al

1.2 12 h, rt → 160 °C; 160 °C → rt

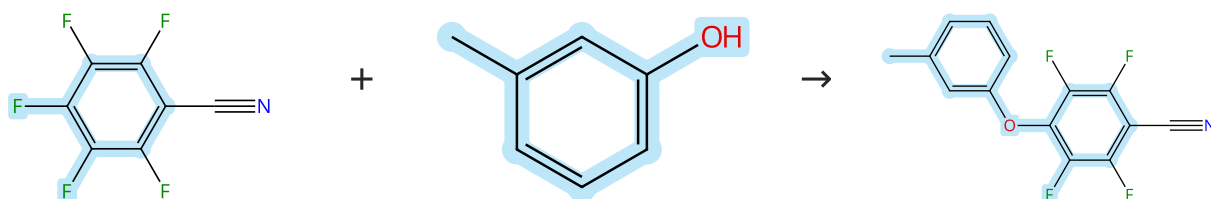
1.3 Reagents: Water; rt

Macromolecules (Washington, DC, United States) (2008), 41(7), 2474-2483.

Experimental Protocols

Scheme 361 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (71)

Suppliers (91)

31-170-CAS-13253162

Steps: 1 Yield: 98%

Semifluorinated multiple pendant group bearing poly (arylene ether) copolymers prepared at room temperature

1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 24 h, rt

By: Krishnan, Ranganathan; et al

1.2 Reagents: Hydrochloric acid

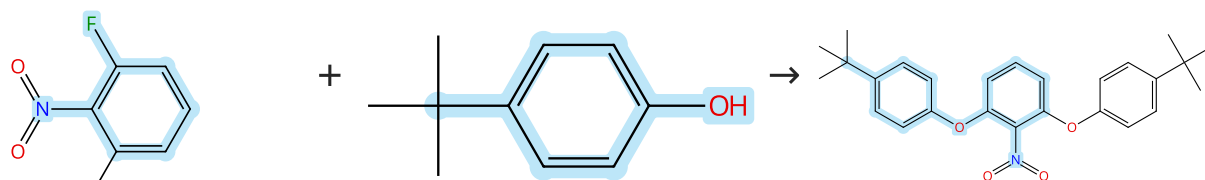
Solvents: Water; acidified, rt

Journal of Polymer Research (2013), 20(8), 1-8.

Experimental Protocols

Scheme 362 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (95)

Suppliers (88)

31-170-CAS-15895314

Steps: 1 Yield: 98%

Distance-Dependent Attractive and Repulsive Interactions of Bulky Alkyl Groups

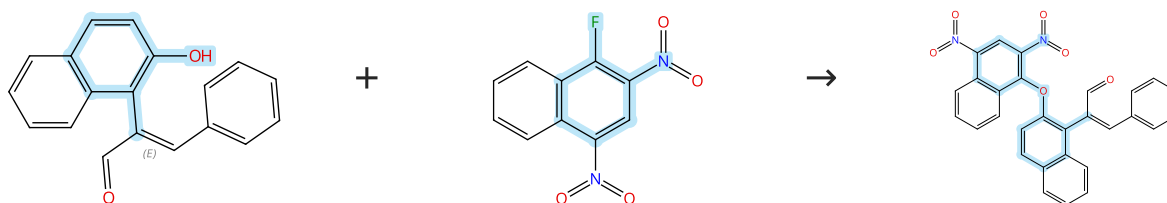
By: Hwang, Jungwon; et al

Angewandte Chemie, International Edition (2016), 55(28), 8086-8089.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; rt; overnight, 90 °C

Scheme 363 (1 Reaction)

Steps: 1 Yield: 98%



Double bond geometry shown

Suppliers (2)

31-614-CAS-37560123

Steps: 1 Yield: 98%

Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution

By: Guo, Fengyuan; et al

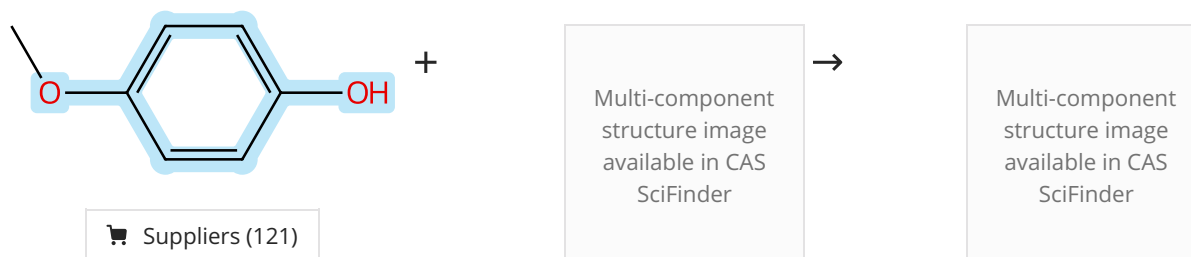
Nature Communications (2023), 14(1), 5050.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2*S*,3*R*)-2-[[[(2*R*,3*R*)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1)
Solvents: Chloroform; 24 h, -10 °C

Experimental Protocols

Scheme 364 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (121)

Multi-component structure image available in CAS SciFinder

Multi-component structure image available in CAS SciFinder

31-008-CAS-640838

Steps: 1 Yield: 98%

Studies on selective nucleophilic substitution reactions of [(cyclopentadienyl)(1,3-dichlorobenzene)M]⁺PF₆⁻ complexes (M = iron, ruthenium)

By: Pearson, Anthony J.; et al

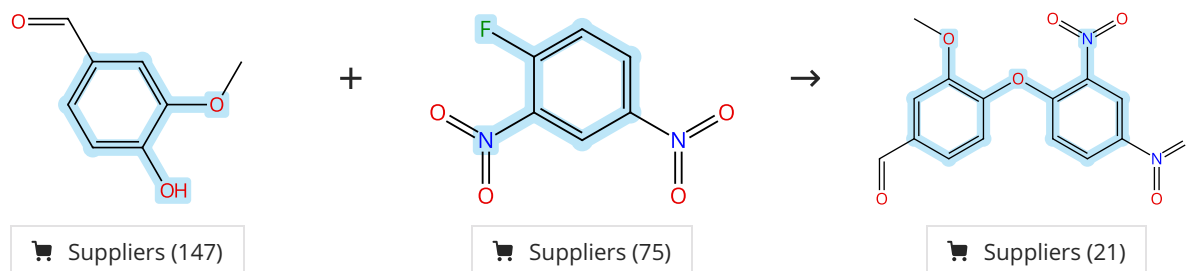
Journal of Organic Chemistry (1992), 57(13), 3583-9.

- 1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran

1.2 -

Scheme 365 (1 Reaction)

Steps: 1 Yield: 98%



31-170-CAS-1709626

Steps: 1 Yield: 98%

Synthesis of some new diaryl ethers

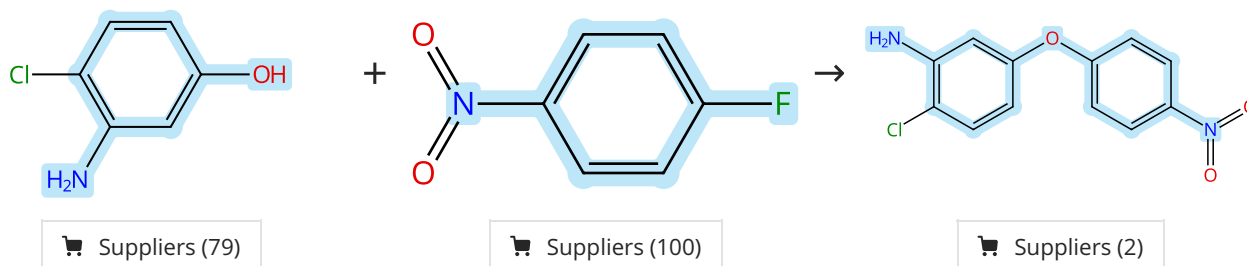
No Data Available

By: Buhts, R. E.; et al

Organic Preparations and Procedures International (1975), 7(4), 193-9.

Scheme 366 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-40338192

Steps: 1 Yield: 98%

Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors

1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C

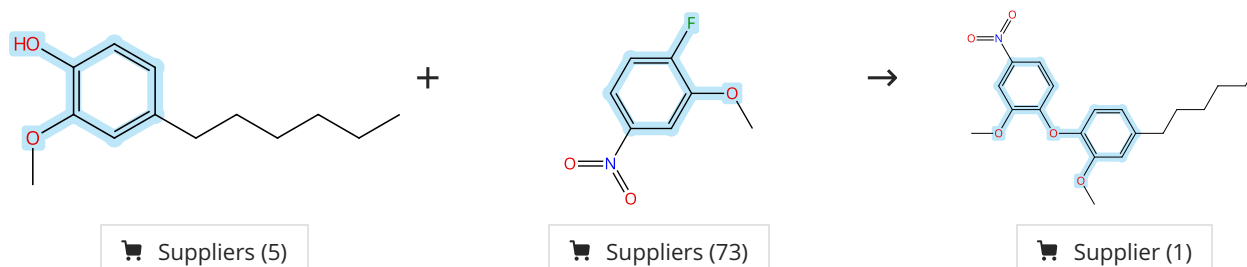
By: Xu, Yue; et al

Experimental Protocols

Bioorganic Chemistry (2023), 131, 106339.

Scheme 367 (1 Reaction)

Steps: 1 Yield: 98%



31-170-CAS-6351368

Steps: 1 Yield: 98%

Time-Dependent Diaryl Ether Inhibitors of InhA: Structure-Activity Relationship Studies of Enzyme Inhibition, Antibacterial Activity, and in vivo Efficacy

1.1 Reagents: Potassium carbonate
Catalysts: 18-Crown-6
Solvents: Dimethylformamide; rt; 3 h, 110 °C

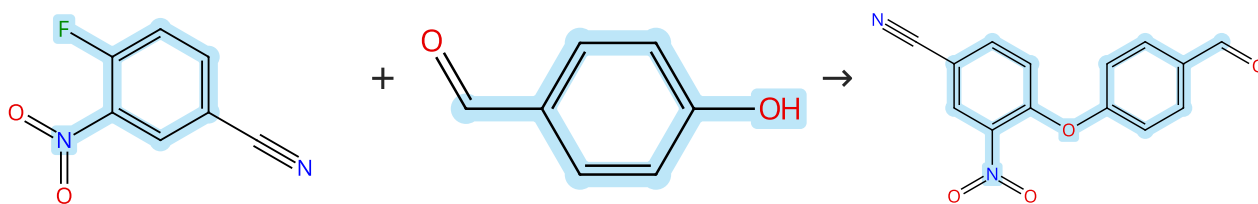
By: Pan, Pan; et al

Experimental Protocols

ChemMedChem (2014), 9(4), 776-791.

Scheme 368 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (93)

Suppliers (123)

Suppliers (3)

31-170-CAS-16594921

Steps: 1 Yield: 98%

Fluorine-18-Labeled Antagonist for PET Imaging of Kappa Opioid Receptors

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 70 °C

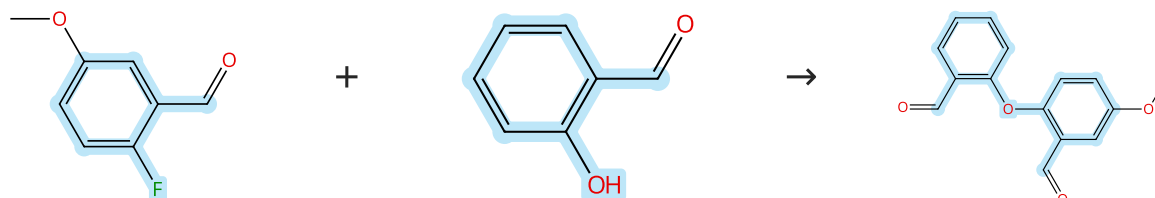
By: Cai, Zhengxin; et al

Experimental Protocols

ACS Chemical Neuroscience (2017), 8(1), 12-16.

Scheme 369 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (84)

Suppliers (95)

31-170-CAS-23309288

Steps: 1 Yield: 98%

N-Heterocyclic carbene (NHC)-catalyzed intramolecular benzoin condensation-oxidation

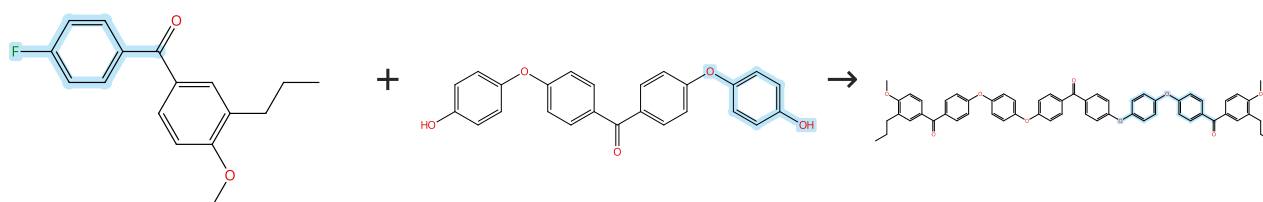
1.1 Reagents: Potassium carbonate
Solvents: Dimethylacetamide; 4 h, reflux

By: Satyam, Killari; et al

Organic & Biomolecular Chemistry (2021), 19(7), 1488-1492.

Scheme 370 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (2)

31-170-CAS-16188256

Steps: 1 Yield: 98%

Synthesis of Rigid-Flexible Polyetherketone-Based Copolymers

1.1 Reagents: Potassium carbonate
Solvents: Dimethylacetamide; 18 h, 180 °C

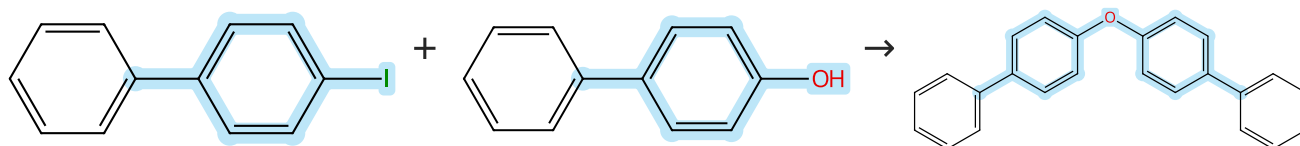
By: Benoit, Jean-Michel; et al

Experimental Protocols

Journal of Macromolecular Science, Part A: Pure and Applied Chemistry (2013), 50(6), 615-630.

Scheme 371 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (86)

Suppliers (105)

Suppliers (5)

31-170-CAS-1202261

Steps: 1 Yield: 98%

A Soluble Copper(I) Source and Stable Salts of Volatile Ligands for Copper-Catalyzed C-X Couplings

By: Maligres, Peter E.; et al

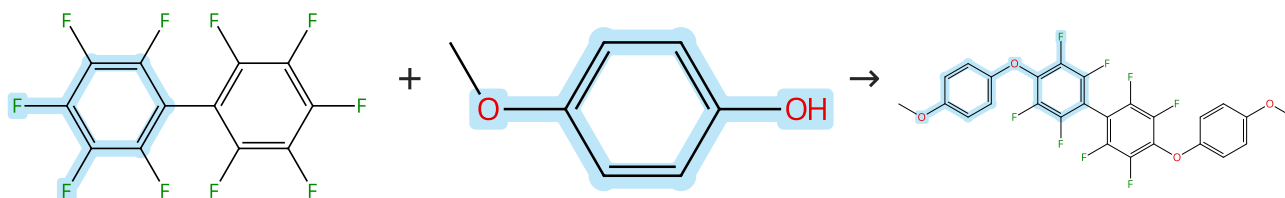
Journal of Organic Chemistry (2012), 77(17), 7646-7651.

1.1 **Reagents:** Cesium carbonate
Catalysts: Bis(2,2,6,6-tetramethyl-3,5-heptanedionato- $\kappa O^3, \kappa O^5$) copper
Solvents: Dimethylacetamide; 24 h, 130 °C

Experimental Protocols

Scheme 372 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (68)

Suppliers (121)

31-170-CAS-4850421

Steps: 1 Yield: 98%

Bisphenols from hexafluorobenzene and decafluorobiphenyl

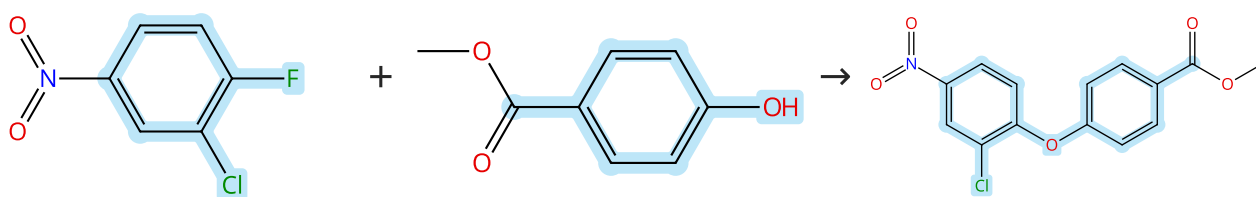
By: Kameneva, T. M.; et al

Zhurnal Organicheskoi Khimii (1989), 25(3), 576-82.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide

Scheme 373 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (86)

Suppliers (170)

Suppliers (6)

31-170-CAS-13439249

Steps: 1 Yield: 98%

A Fragment-Based Approach toward Substituted Trioxa[7] helicenes

By: Kelgtermans, Hans; et al

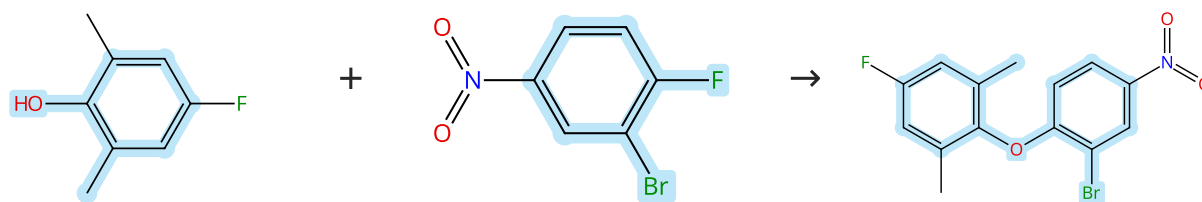
Organic Letters (2012), 14(20), 5200-5203.

1.1 **Reagents:** Potassium carbonate
Solvents: Acetonitrile; 3 h, reflux; cooled

Experimental Protocols

Scheme 374 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (53)

Suppliers (83)

Suppliers (6)

31-614-CAS-39886752

Steps: 1 Yield: 98%

1.1 Reagents: Cesium carbonate
Solvents: Dimethylformamide; 4 h, rt → 100 °C

Experimental Protocols

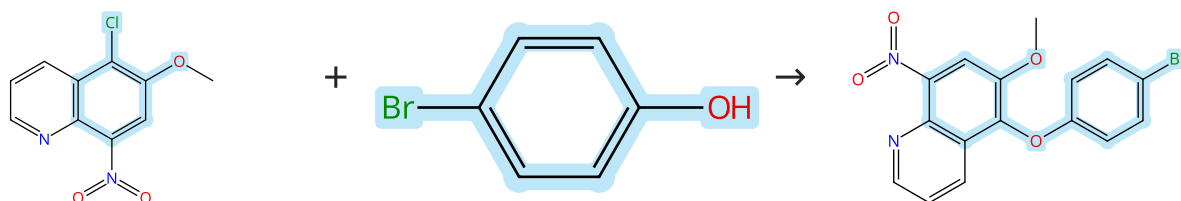
Structure-Based Discovery and Optimization of Furo[3,2-c]pyridin-4(5H)-one Derivatives as Potent and Second Bromodomain (BD2)-Selective Bromo and Extra Terminal Domain (BET) Inhibitors

By: Li, Junhua; et al

Journal of Medicinal Chemistry (2022), 65(7), 5760-5799.

Scheme 375 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (4)

Suppliers (82)

Supplier (1)

31-614-CAS-24716239

Steps: 1 Yield: 98%

1.1 Reagents: Lithium hydroxide
Solvents: Dimethyl sulfoxide; rt; 4 h, 100 °C

1.2 Reagents: Water

Experimental Protocols

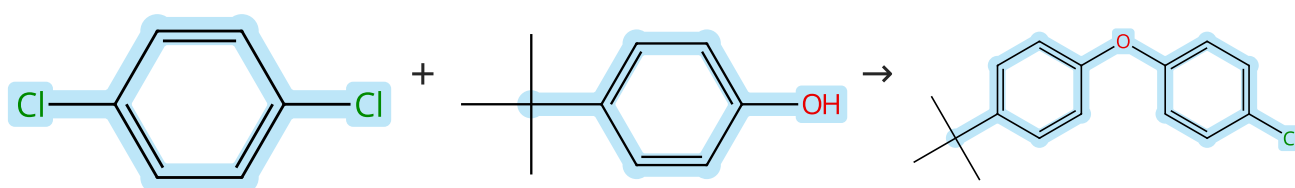
5-Phenoxy Primaquine Analogs and the Tetraoxane Hybrid as Antimalarial Agents

By: Jansongsaeng, Somrudee; et al

Molecules (2021), 26(13), 3991.

Scheme 376 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (107)

Suppliers (88)

Supplier (1)

31-170-CAS-3759700

Steps: 1 Yield: 98%

1.1 Reagents: Potassium carbonate
Catalysts: Cuprous iodide
Solvents: Dimethylformamide; 30 min, rt; 2 h, 110 °C; 110 °C → rt

Experimental Protocols

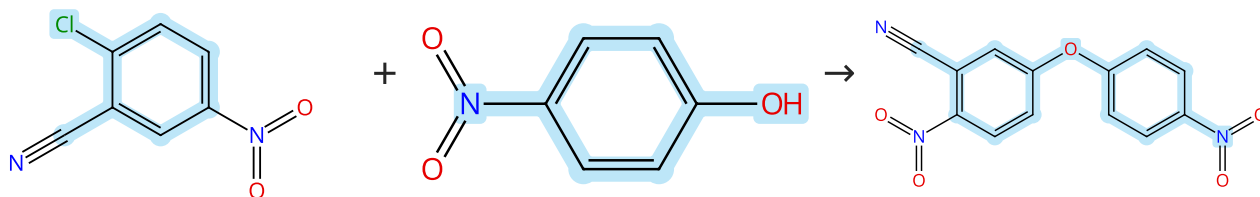
CuI Nanoparticles for C-N and C-O Cross Coupling of Heterocyclic Amines and Phenols with Chlorobenzenes

By: Sreedhar, B.; et al

Journal of Organic Chemistry (2009), 74(20), 7951-7954.

Scheme 377 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (86)

Suppliers (133)

Suppliers (2)

31-614-CAS-27808546

Steps: 1 Yield: 98%

Scale-up in microwave-accelerated organic synthesis

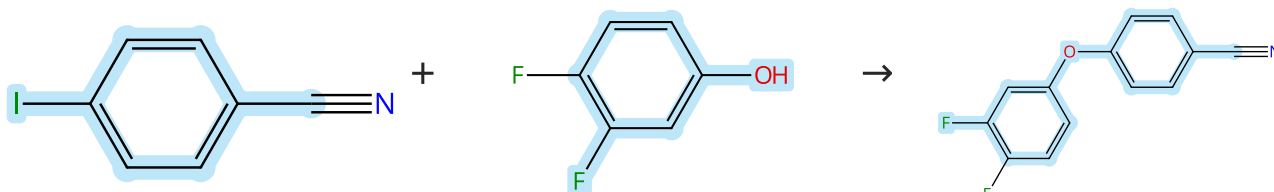
1.1 **Reagents:** Potassium carbonate
Solvents: Acetonitrile; 10 min, 150 °C

By: Lehmann, H.

Ernst Schering Foundation Symposium Proceedings (2007),
 (2006-03), 133-149.

Scheme 378 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (81)

Suppliers (77)

Suppliers (36)

31-614-CAS-36007621

Steps: 1 Yield: 98%

**Natural cellulose-based Cu(II) complex: An eco-friendly
 nanocatalyst for Ullmann condensations at room temper
 ature**

1.1 **Reagents:** Potassium carbonate
Catalysts: Hydroxylamine hydrochloride, Copper, Cellulose,
 polymer with 2-propenenitrile, graft (reaction products with
 hydroxylamine hydrochloride, copper complexes)
Solvents: Acetonitrile; 4.5 h, 40 °C

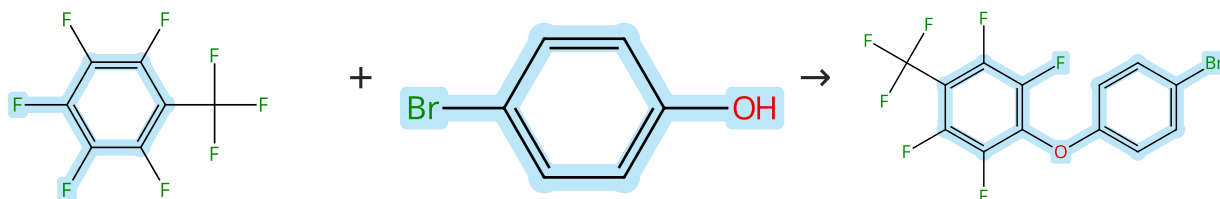
By: Rahman, Lutfor Md; et al

Journal of Cleaner Production (2023), 390, 136015.

Experimental Protocols

Scheme 379 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (69)

Suppliers (82)

Supplier (1)

31-170-CAS-250575

Steps: 1 Yield: 98%

**Influence of the Length and Positioning of the Antiestrogenic
 Side Chain of Endoxifen and 4-Hydroxytamoxifen on Gene
 Activation and Growth of Estrogen Receptor Positive Cancer
 Cells**

1.1 **Reagents:** Sodium hydroxide, Tetrabutylammonium hydrogen
 sulfate
Solvents: Dichloromethane, Water; 4 h, rt

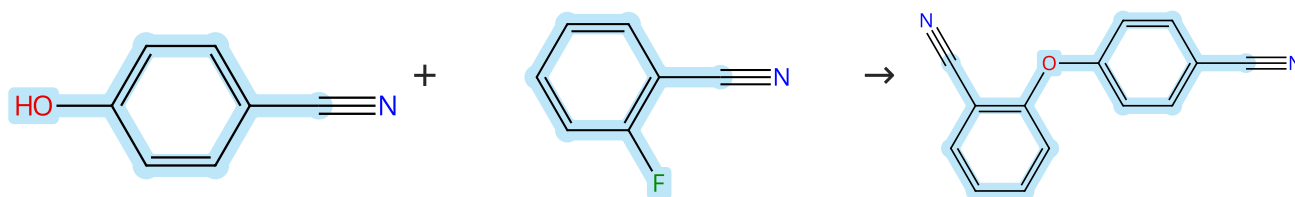
By: Maximov, Philipp Y.; et al

Journal of Medicinal Chemistry (2014), 57(11), 4569-4583.

Experimental Protocols

Scheme 380 (2 Reactions)

Steps: 1 Yield: 98%



Suppliers (90)

Suppliers (87)

Suppliers (4)

31-170-CAS-4043165

Steps: 1 Yield: 98%

Synthesis of diaryl ethers, diaryl sulfides, heteroaryl ethers and heteroaryl sulfides under microwave dielectric heating

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; 30 - 40 s, rt → reflux; 10 min, reflux; reflux → rt
 1.2 **Solvents:** Water; 30 min, < rt

By: Li, Feng; et al

Synthesis (2005), (8), 1305-1313.

Experimental Protocols

31-170-CAS-6827762

Steps: 1 Yield: 98%

Microwave-Assisted Synthesis of Diaryl Ethers without Catalyst

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethyl sulfoxide; 30 - 40 s, rt → reflux; 10 min, reflux

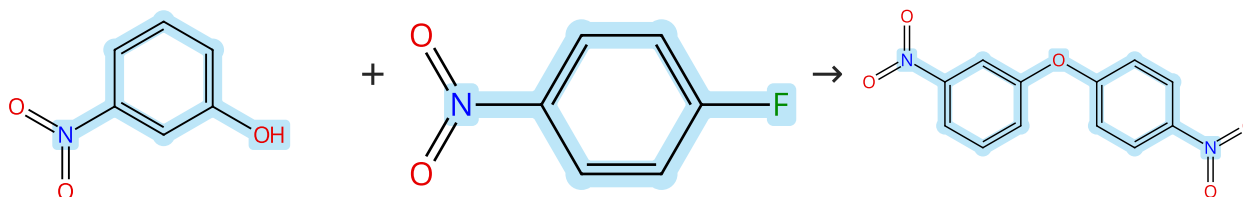
By: Li, Feng; et al

Organic Letters (2003), 5(12), 2169-2171.

Experimental Protocols

Scheme 381 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

Suppliers (100)

Suppliers (19)

31-170-CAS-11305433

Steps: 1 Yield: 98%

CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions

- 1.1 **Catalysts:** Cesium fluoride
Solvents: Dimethyl sulfoxide; 0.5 h, 110 °C

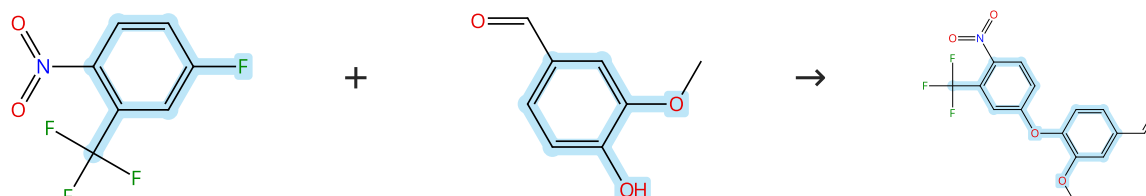
By: Keipour, Hoda; et al

Canadian Journal of Chemistry (2016), 94(1), 95-104.

Experimental Protocols

Scheme 382 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (83)

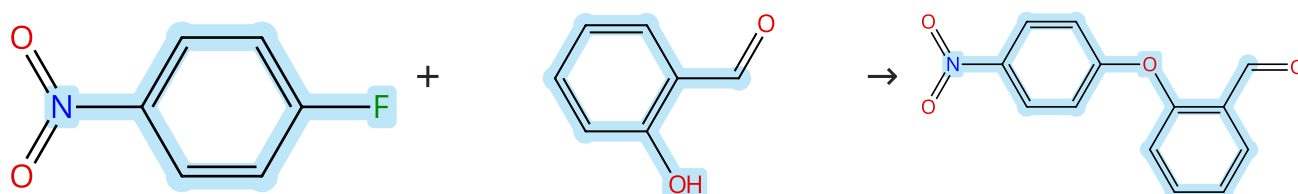
Suppliers (147)

Suppliers (5)

31-170-CAS-12994524	Steps: 1 Yield: 98%	Identification of Diaryl Ether-Based Ligands for Estrogen-Related Receptor α as Potential Antidiabetic Agents
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 5 h, 95 °C		By: Patch, Raymond J.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2011), 54(3), 788-808.

Scheme 383 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (100)

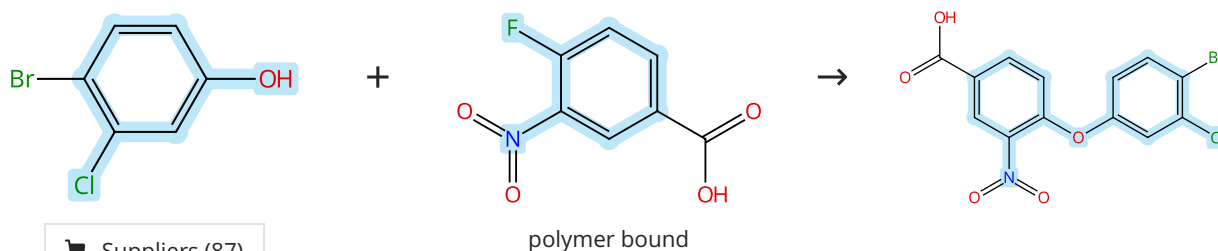
Suppliers (95)

Suppliers (27)

31-170-CAS-17069817	Steps: 1 Yield: 98%	NHC-Organocatalyzed C_{Ar}-O Bond Cleavage: Mild Access to 2-Hydroxybenzophenones
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 100 °C		By: Janssen-Mueller, Daniel; et al
		Angewandte Chemie, International Edition (2017), 56(22), 6276-6279.

Scheme 384 (1 Reaction)

Steps: 1 Yield: 98%

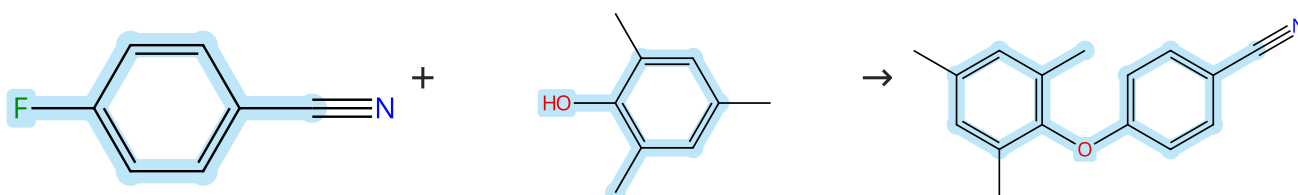


Suppliers (87)

31-170-CAS-11046612	Steps: 1 Yield: 98%	Solid phase synthesis of functionalized biaryl ethers: Versatile scaffolds for combinatorial chemistry
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide		By: Wijkman, Jac C. H. M.; et al
1.2 Reagents: Trifluoroacetic acid Solvents: Dichloromethane		Molecular Diversity (1998), 3(2), 117-120.
Experimental Protocols		

Scheme 385 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (92)

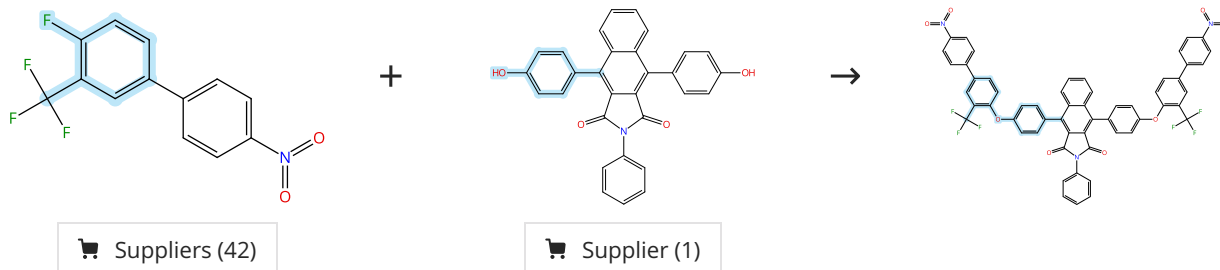
Suppliers (88)

Suppliers (7)

31-170-CAS-2851341	Steps: 1 Yield: 98%	Regioselective Chemical Modification of Fullerene by Destructive Electrophilic Reaction in Polyphosphoric Acid/Phosphorus Pentoxide
1.1 Reagents: Potassium carbonate, <i>N</i> -Methyl-2-pyrrolidone Solvents: Toluene; 8 h, 140 °C		By: Lim, Dae-Hyun; et al
Experimental Protocols		Journal of Physical Chemistry C (2008), 112(32), 12188-12194.

Scheme 386 (1 Reaction)

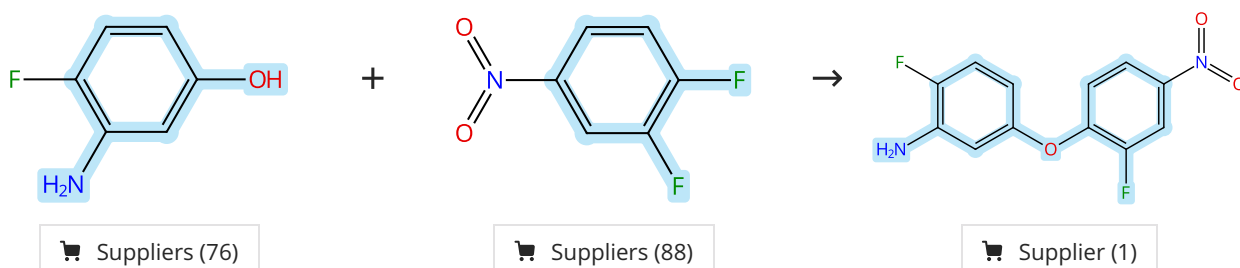
Steps: 1 Yield: 98%



31-170-CAS-11798108	Steps: 1 Yield: 98%	High T_g, processable fluorinated polyimides containing benzoisindoleone unit and evaluation of their gas transport properties
1.1 Reagents: Potassium carbonate Solvents: Toluene, <i>N</i> -Methyl-2-pyrrolidone; 4 h, 145 °C; 2 h, 160 °C		By: Kumar Sen, Suman; et al
Experimental Protocols		RSC Advances (2012), 2(15), 6274-6289.

Scheme 387 (1 Reaction)

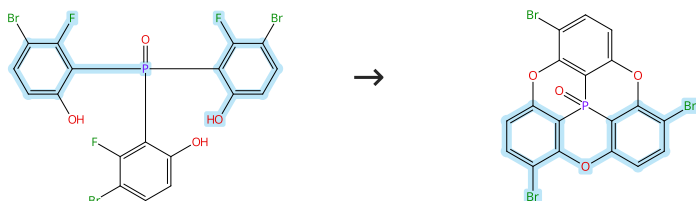
Steps: 1 Yield: 98%



31-614-CAS-36058197	Steps: 1 Yield: 98%	Targeting Receptor-Interacting Protein Kinase 1 by Novel Benzothiazole Derivatives: Treatment of Acute Lung Injury through the Necroptosis Pathway
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 2 h, 85 °C		By: Zhang, Xinqi; et al
Experimental Protocols		Journal of Medicinal Chemistry (2023), 66(7), 5261-5278.

Scheme 388 (1 Reaction)

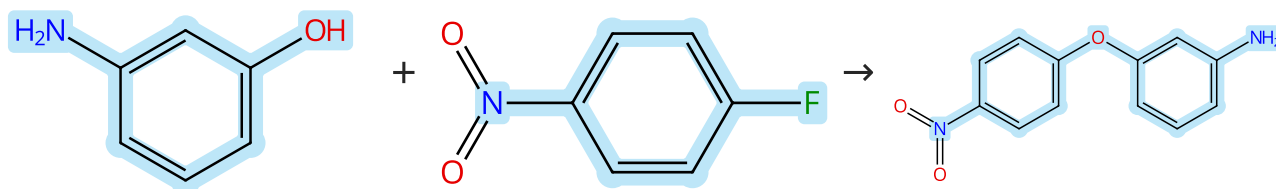
Steps: 1 Yield: 98%



31-170-CAS-9799163	Steps: 1 Yield: 98%	Phosphorus-Containing Chiral Molecule for Fullerene Recognition Based on Concave/Convex Interaction
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Dimethylformamide; 23 h, 140 °C		By: Yamamura, Masaki; et al
Experimental Protocols		Journal of the American Chemical Society (2014), 136(40), 14299-14306.

Scheme 389 (1 Reaction)

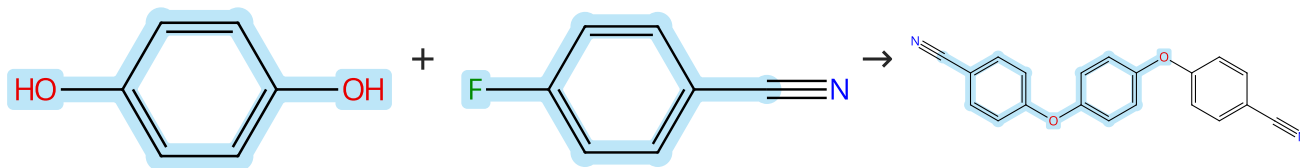
Steps: 1 Yield: 98%

 Suppliers (81) Suppliers (100) Suppliers (22)

31-614-CAS-40338183	Steps: 1 Yield: 98%	Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C		By: Xu, Yue; et al
Experimental Protocols		Bioorganic Chemistry (2023), 131, 106339.

Scheme 390 (1 Reaction)

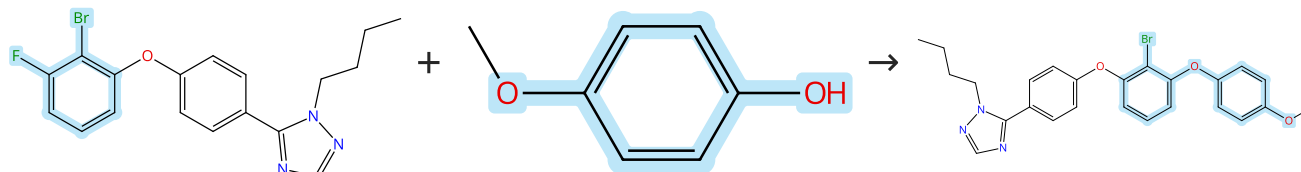
Steps: 1 Yield: 98%

 Suppliers (112) Suppliers (92) Suppliers (5)

31-170-CAS-15003807	Steps: 1 Yield: 98%	Anion conducting polymers based on ether linked polybenzimidazole (PBI-OO)
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 1 d, reflux		By: Henkensmeier, Dirk; et al
1.2 Solvents: Dimethylformamide; 20 h, reflux		International Journal of Hydrogen Energy (2014), 39(6), 2842-2853.

Scheme 391 (1 Reaction)

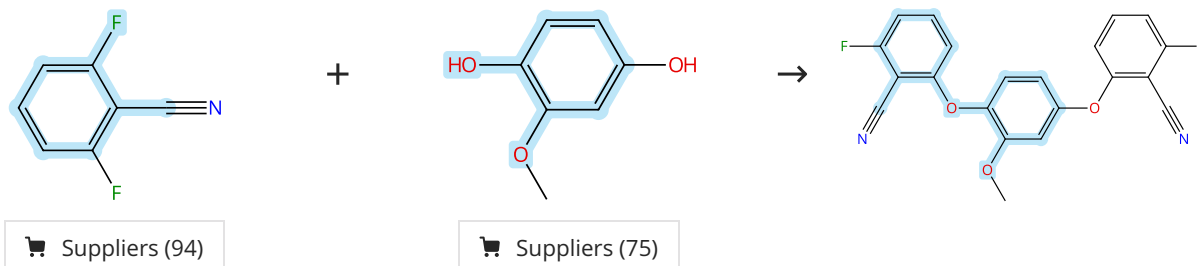
Steps: 1 Yield: 98%

 Suppliers (121)

31-170-CAS-22423904	Steps: 1 Yield: 98%	Triazole functionalized 5,9-dioxa-13b-boranaphtho[3,2,1-de]anthracene: a new family of multi-stimuli responsive materials
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; rt; 24 h, 150 °C		By: Meng, Guoyun; et al
Experimental Protocols		Journal of Materials Chemistry C: Materials for Optical and Electronic Devices (2020), 8(23), 7749-7754.

Scheme 392 (1 Reaction)

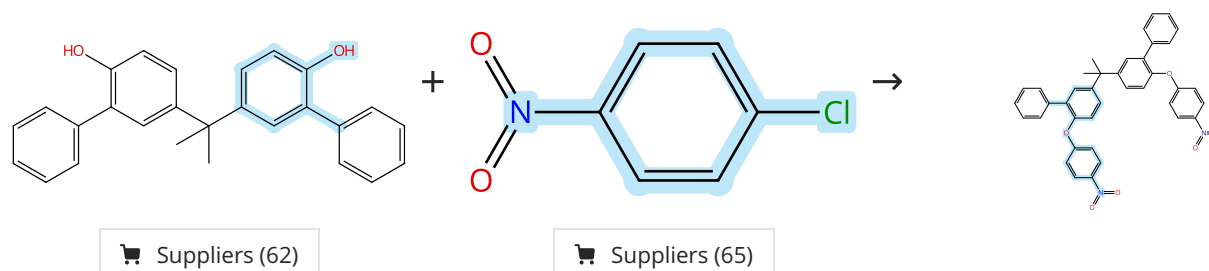
Steps: 1 Yield: 98%



31-170-CAS-16121165	Steps: 1 Yield: 98%	Synthesis of hyperbranched poly(ether nitrile)s as supporting polymers for palladium nanoparticles
1.1 Reagents: Cesium fluoride Solvents: Dimethylacetamide; 18 h, 60 °C		By: Jikei, Mitsutoshi; et al
Experimental Protocols		Polymer Journal (Tokyo, Japan) (2016), 48(9), 941-948.

Scheme 393 (1 Reaction)

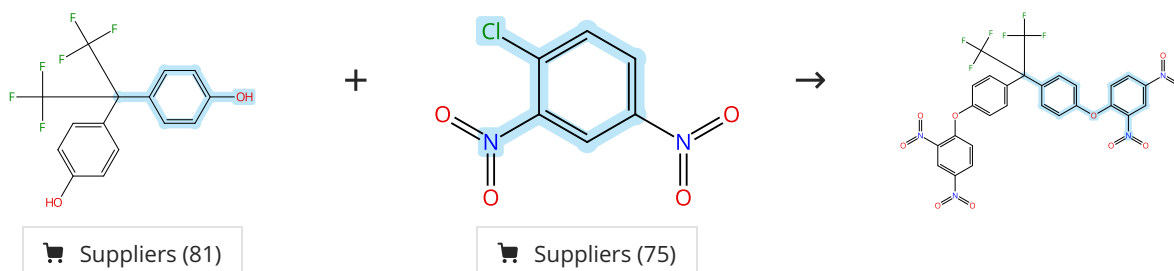
Steps: 1 Yield: 98%



31-170-CAS-14599375	Steps: 1 Yield: 98%	Synthesis of new diamine and related polyimide with bisphenoxy groups
1.1 Reagents: Potassium carbonate Catalysts: 18-Crown-6; 15 min, rt; rt → reflux; 4 - 5 h, reflux		By: Zhou, Penglai; et al
		Suliao Gongye (2009), 37(9), 1-4.

Scheme 394 (1 Reaction)

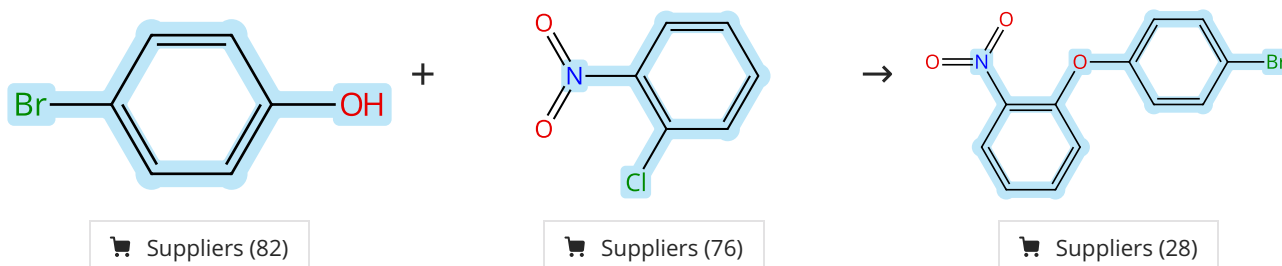
Steps: 1 Yield: 98%



31-170-CAS-23443414	Synthesis and characterization of novel star-branched polyimides derived from 2,2-bis[4-(2,4-diaminophenoxy)phenyl]hexafluoropropane
Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Toluene, Dimethylacetamide; rt → reflux; 2 h, reflux 1.2 5 h, 90 °C	By: Wu, Feng; et al RSC Advances (2017), 7(57), 35786-35794.
Experimental Protocols	

Scheme 395 (1 Reaction)

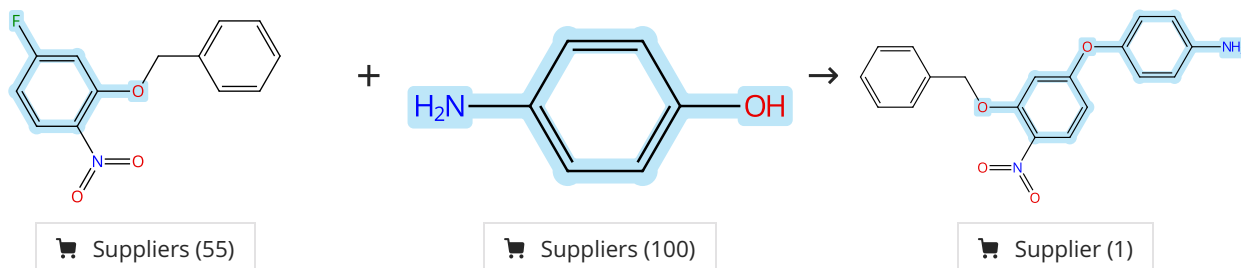
Steps: 1 Yield: 98%



31-170-CAS-1089674	Design, synthesis and antifungal/insecticidal evaluation of novel nicotinamide derivatives
Steps: 1 Yield: 98% 1.1 Reagents: Potassium hydroxide; 15 min, 90 °C 1.2 90 °C; 2 h, 150 °C	By: Wen, Fang; et al Pesticide Biochemistry and Physiology (2010), 98(2), 248-253.
Experimental Protocols	

Scheme 396 (1 Reaction)

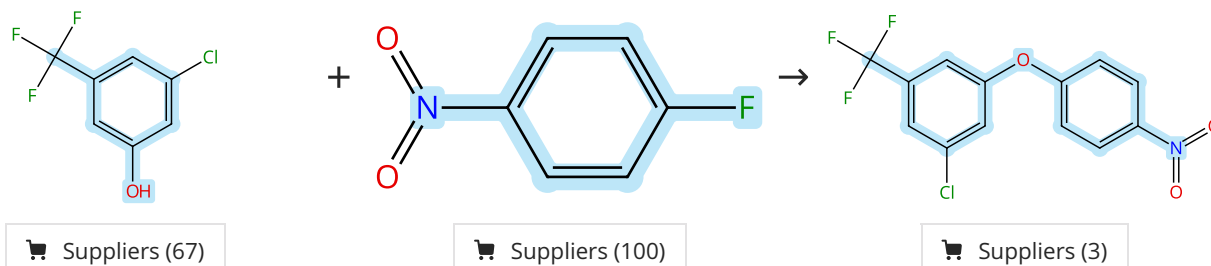
Steps: 1 Yield: 98%



31-170-CAS-4703958	Discovery of Novel Benzimidazoles as Potent Inhibitors of TIE-2 and VEGFR-2 Tyrosine Kinase Receptors
Steps: 1 Yield: 98% 1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 10 min, rt; overnight, 80 °C	By: Hasegawa, Masaichi; et al Journal of Medicinal Chemistry (2007), 50(18), 4453-4470.
Experimental Protocols	

Scheme 397 (1 Reaction)

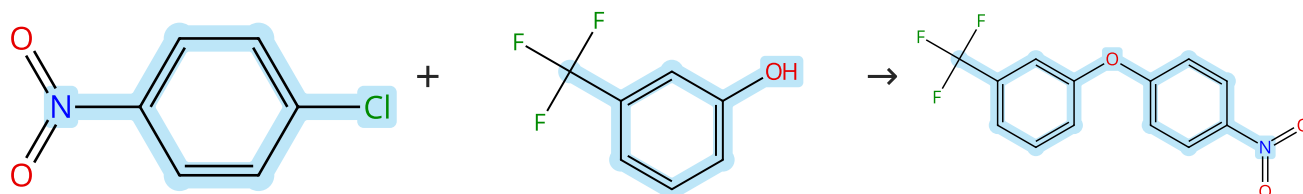
Steps: 1 Yield: 98%



31-170-CAS-23273059	Steps: 1 Yield: 98%	Discovery of Novel Small Molecule Dual Inhibitors Targeting Toll-Like Receptors 7 and 8
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 - 76 h, 90 °C		By: Padilla-Salinas, Rosaura; et al
Experimental Protocols		Journal of Medicinal Chemistry (2019), 62(22), 10221-10244.

Scheme 398 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (65)

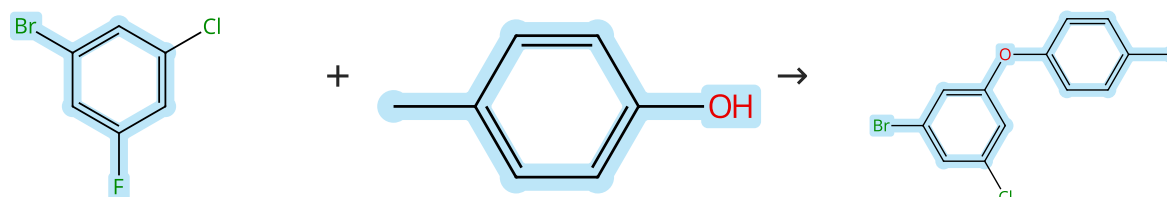
Suppliers (89)

Suppliers (23)

31-170-CAS-19244821	Steps: 1 Yield: 98%	Synthesis and evaluation of aminobenzothiazoles as blockers of N- and T-type calcium channels
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C		By: Sairaman, Anjali; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2018), 26(11), 3046-3059.

Scheme 399 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (72)

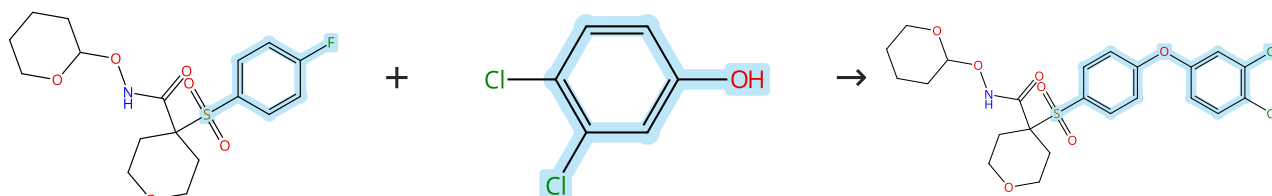
Suppliers (89)

Suppliers (3)

31-170-CAS-22561195	Steps: 1 Yield: 98%	Solution-Processable Pure Green Thermally Activated Delayed Fluorescence Emitter Based on the Multiple Resonance Effect
1.1 Reagents: Potassium carbonate Solvents: N-Methyl-2-pyrrolidone; rt; 17 h, 150 °C		By: Ikeda, Naoya; et al
Experimental Protocols		Advanced Materials (Weinheim, Germany) (2020), 32(40), 2004072.

Scheme 400 (1 Reaction)

Steps: 1 Yield: 98%



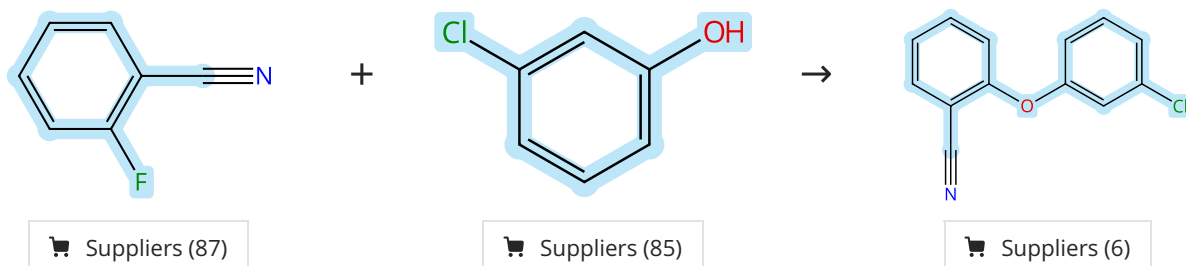
Supplier (1)

Suppliers (66)

31-170-CAS-10605875	Steps: 1 Yield: 98%	Orally Active MMP-1 Sparing α-Tetrahydropyranyl and α-Piperidinyl Sulfone Matrix Metalloproteinase (MMP) Inhibitors with Efficacy in Cancer, Arthritis, and Cardiovascular Disease By: Becker, Daniel P.; et al Journal of Medicinal Chemistry (2010), 53(18), 6653-6680.
1.1 Reagents: Cesium carbonate Solvents: Dimethylacetamide; 41 h, 95 °C		
Experimental Protocols		

Scheme 401 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-32528358	Steps: 1 Yield: 98%	Access to ortho-Hydroxyphenyl Ketimines via Imine Anion-Mediated Smiles Rearrangement By: Jinno, Shunki; et al Organic Letters (2022), 24(23), 4140-4144.
1.1 Reagents: Potassium hydroxide Solvents: Dimethyl sulfoxide; 14.5 h, 90 °C; 90 °C → rt		
1.2 Reagents: Water; rt		
Experimental Protocols		

Scheme 402 (1 Reaction)

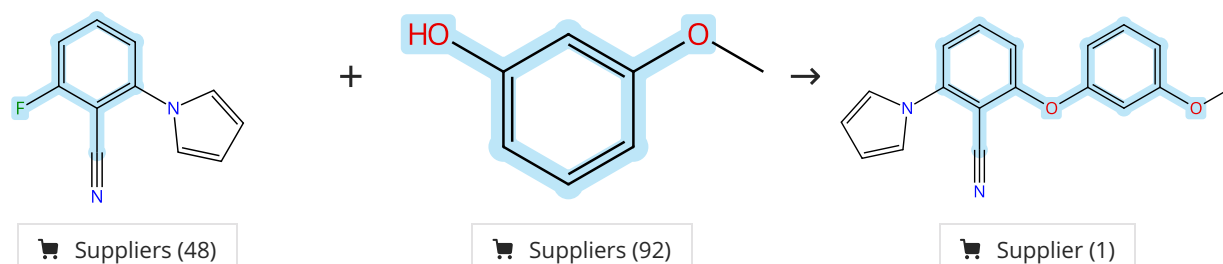
Steps: 1 Yield: 98%



31-614-CAS-40338188	Steps: 1 Yield: 98%	Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors By: Xu, Yue; et al Bioorganic Chemistry (2023), 131, 106339.
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C		
Experimental Protocols		

Scheme 403 (1 Reaction)

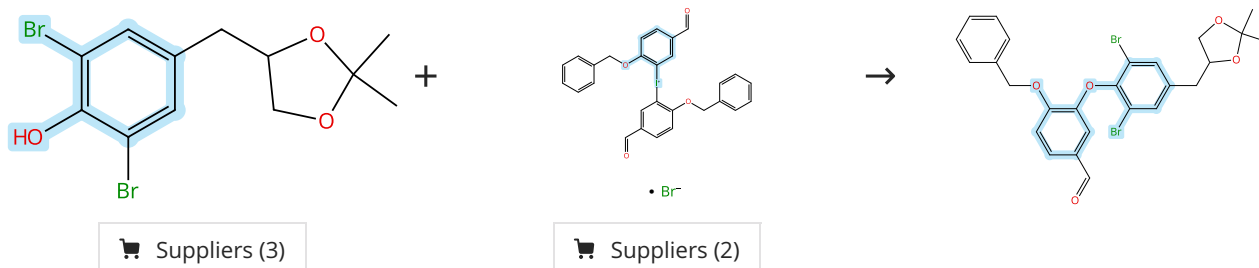
Steps: 1 Yield: 98%



31-170-CAS-13241350 Steps: 1 Yield: 98% 1.1 Catalysts: Alumina, Potassium fluoride, 18-Crown-6 Solvents: Acetonitrile	Synthesis of diaryl ethers, diaryl thioethers, and diaryl amines mediated by potassium fluoride-alumina and 18-crown-6: Expansion of scope and utility By: Sawyer, J. Scott; et al Journal of Organic Chemistry (1998), 63(18), 6338-6343.
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Scheme 404 (1 Reaction)

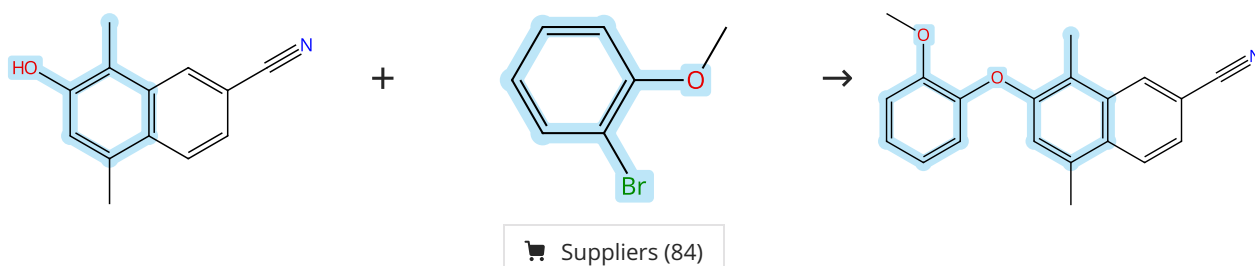
Steps: 1 Yield: 98%



31-170-CAS-7283722 Steps: 1 Yield: 98% 1.1 Reagents: Imidazole, Potassium carbonate Solvents: Dimethylformamide; rt; 1 h, rt 1.2 rt; rt → 80 °C; 6 h, 80 °C	A general method for the synthesis of bastaranes and isobastaranes: First total synthesis of bastadins 5, 10, 12, 16, 20, and 21 By: Couladouros, Elias A.; et al Chemistry - A European Journal (2005), 11(1), 406-421.
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Scheme 405 (1 Reaction)

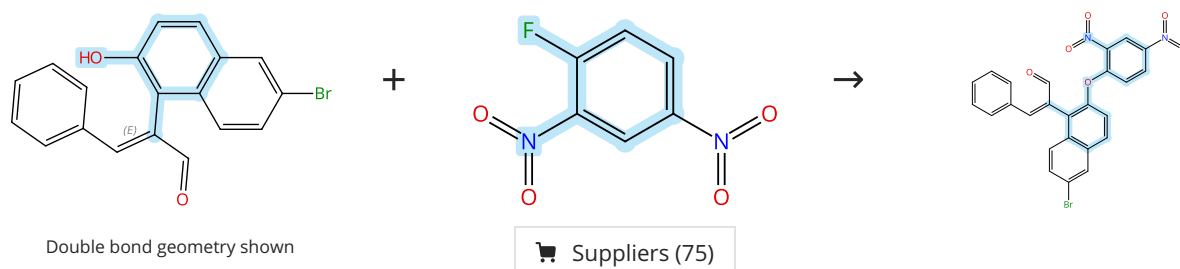
Steps: 1 Yield: 98%



31-170-CAS-16677925 Steps: 1 Yield: 98% 1.1 Reagents: Catechol, Cesium carbonate Catalysts: 2,2,6,6-Tetramethyl-3,5-heptanedione, Cuprous iodide Solvents: 1,4-Dioxane; overnight, 120 °C Experimental Protocols	Design, Conformation, and Crystallography of 2-Naphthyl Phenyl Ethers as Potent Anti-HIV Agents By: Lee, Won-Gil; et al ACS Medicinal Chemistry Letters (2016), 7(12), 1156-1160.
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Scheme 406 (1 Reaction)

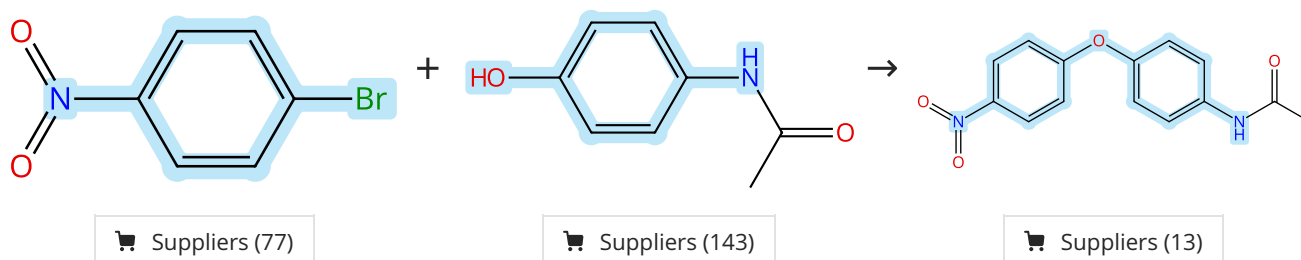
Steps: 1 Yield: 98%



31-614-CAS-37560118 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Catalysts: Phosphonium, [[3,5-bis(1,1-dimethylethyl)phenyl]methyl][[(2<i>S</i>,3<i>R</i>)-2-[[[(2<i>R</i>,3<i>R</i>)-2-[[[(1,1-dimethylethoxy)carbonyl]amino]-3-[[[(1,1-dimethylethyl)dimethylsilyl]oxy]-1-oxobutyl]amino]-3-hydroxybutyl]diphenyl-, bromide (1:1) Solvents: Chloroform; 24 h, -10 °C Experimental Protocols	Enantioselective organocatalytic synthesis of axially chiral aldehyde-containing styrenes via S_NAr reaction-guided dynamic kinetic resolution By: Guo, Fengyuan; et al Nature Communications (2023), 14(1), 5050.
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Scheme 407 (1 Reaction)

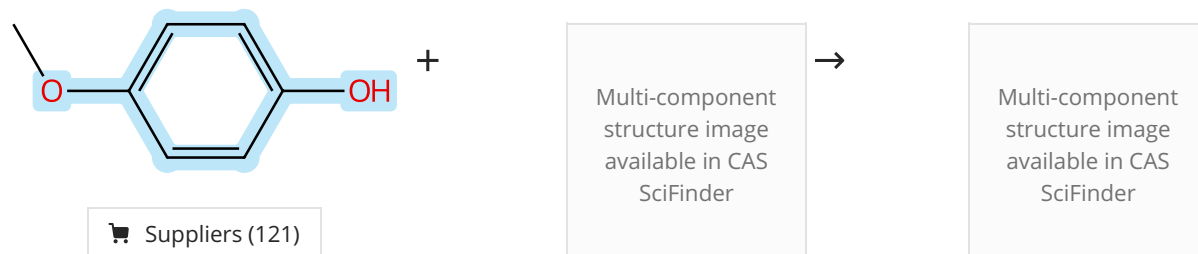
Steps: 1 Yield: 98%



31-170-CAS-15019276 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Catalysts: Cuprous iodide, 2-Quinoxalinecarboxylic acid, 3-hydroxy-, methyl ester, 1,4-dioxide Solvents: Dimethylformamide; 11 h, 110 °C Experimental Protocols	2-Carbomethoxy-3-hydroxyquinoxaline-di-N-oxide as a novel ligand for the copper-catalyzed coupling reaction of phenols and aryl halides By: Qiu, Yatao; et al Organic & Biomolecular Chemistry (2013), 11(9), 1502-1510.
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Scheme 408 (1 Reaction)

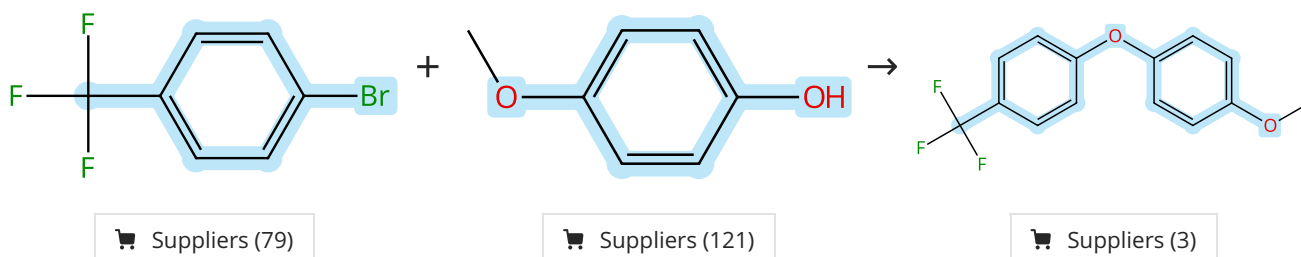
Steps: 1 Yield: 98%



31-008-CAS-14022095 Steps: 1 Yield: 98% 1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran 1.2 -	Studies on selective nucleophilic substitution reactions of [(cyclopentadienyl)(1,3-dichlorobenzene)M]⁺PF₆⁻ complexes (M = iron, ruthenium) By: Pearson, Anthony J.; et al Journal of Organic Chemistry (1992), 57(13), 3583-9.
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Scheme 409 (1 Reaction)

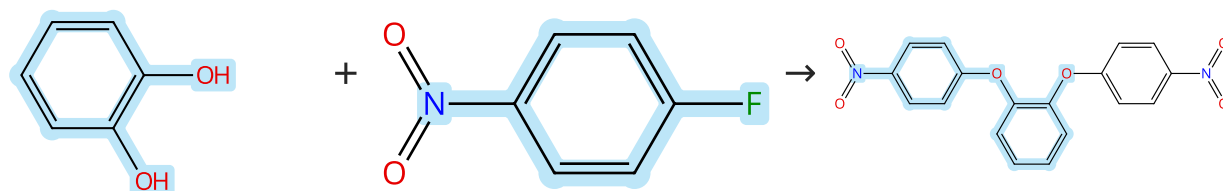
Steps: 1 Yield: 98%



31-614-CAS-43873487	Steps: 1 Yield: 98%	Electromagnetic milling promoted mechanochemical copper-catalyzed C-O coupling reaction for diaryl ether formation
1.1 Reagents: Cesium carbonate Catalysts: Copper(II) acetylacetonate, <i>N</i> ¹ , <i>N</i> ² -Di-1-naphthale nylethanediamide Solvents: Dimethylformamide; 3 h, 79 °C		By: Zeng, Zhaoxuan; et al Organic Chemistry Frontiers (2025), 12(5), 1528-1536.
Experimental Protocols		

Scheme 410 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (106)

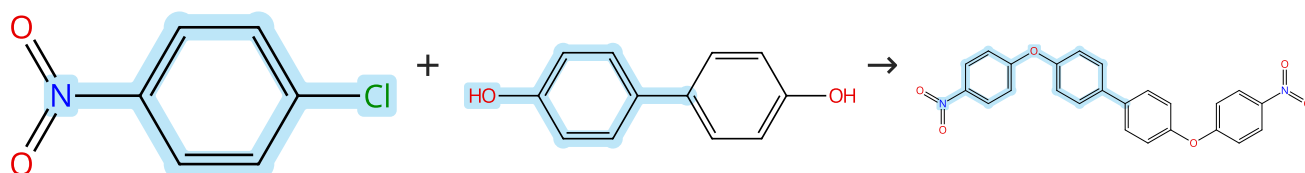
Suppliers (100)

Supplier (1)

31-170-CAS-13911419	Steps: 1 Yield: 98%	Methyl- and fluoro-substituted bis(4-aminophenoxy) benzenes. A convenient method of synthesis
1.1 Solvents: Dimethylformamide 1.2 Reagents: Potassium carbonate Solvents: Toluene 1.3 - 1.4 Solvents: Water 1.5 Solvents: Ethanol		By: Eastmond, G. C.; et al Synthesis (1998), (6), 894-898.
Experimental Protocols		

Scheme 411 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (65)

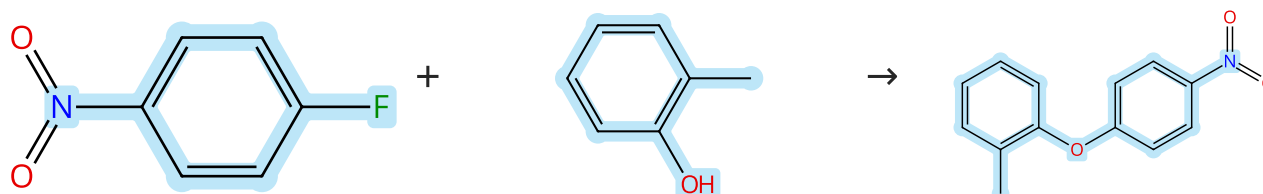
Suppliers (100)

Suppliers (15)

31-170-CAS-10583484	Steps: 1 Yield: 98%	Influence of the Water Content on the Diffusion Coefficients of Li ⁺ and Water across Naphthalenic Based Copolyimide Cation-Exchange Membranes
1.1 Reagents: Potassium carbonate Solvents: Toluene, Dimethylacetamide; rt → 140 °C; 2 h, 140 °C; 20 h, 160 °C		By: Garrido, Leoncio; et al Journal of Physical Chemistry B (2012), 116(38), 11754-11766.
Experimental Protocols		

Scheme 412 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (100)

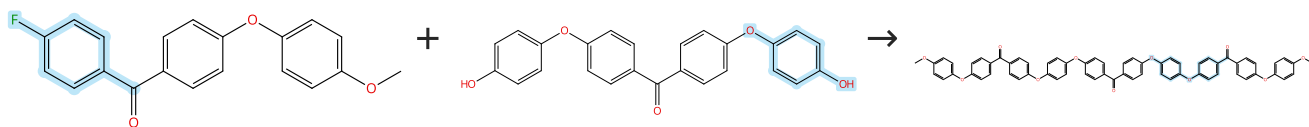
Suppliers (83)

Suppliers (10)

31-170-CAS-16917424	Steps: 1 Yield: 98%	Discovery of Pyrrolidine-Containing GPR40 Agonists: Stereochemistry Effects a Change in Binding Mode
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; overnight, 80 °C; 80 °C → rt		By: Jurica, Elizabeth A.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2017), 60(4), 1417-1431.

Scheme 413 (1 Reaction)

Steps: 1 Yield: 98%



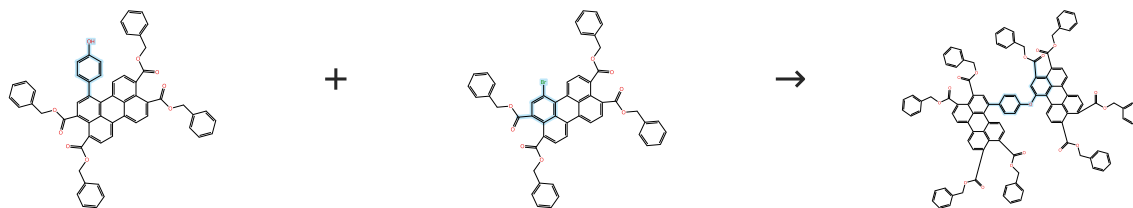
Suppliers (3)

Suppliers (2)

31-170-CAS-16188257	Steps: 1 Yield: 98%	Synthesis of Rigid-Flexible Polyetherketone-Based Copolymers
1.1 Reagents: Potassium carbonate Solvents: Dimethylacetamide; 18 h, 180 °C		By: Benoit, Jean-Michel; et al
Experimental Protocols		Journal of Macromolecular Science, Part A: Pure and Applied Chemistry (2013), 50(6), 615-630.

Scheme 414 (1 Reaction)

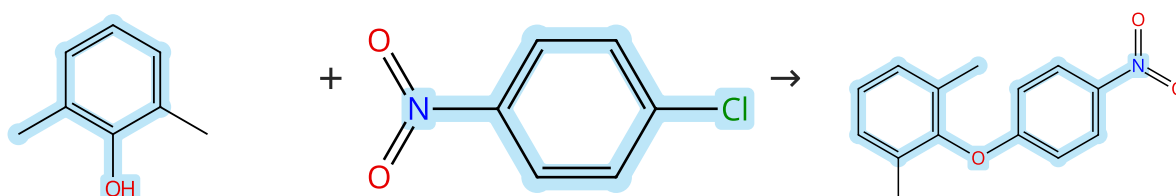
Steps: 1 Yield: 98%



31-170-CAS-18359074	Steps: 1 Yield: 98%	Controllable Monobromination of Perylene Ring System: Synthesis of Bay-Functionalized Perylene Dyes
1.1 Reagents: Potassium carbonate Catalysts: 18-Crown-6 Solvents: Dimethylformamide; rt; rt → 50 °C; 5 h, 50 °C; 50 °C → rt		By: Takahashi, Masaki; et al
1.2 Solvents: Water; rt		Journal of Organic Chemistry (2018), 83(2), 624-631.
Experimental Protocols		

Scheme 415 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (88)

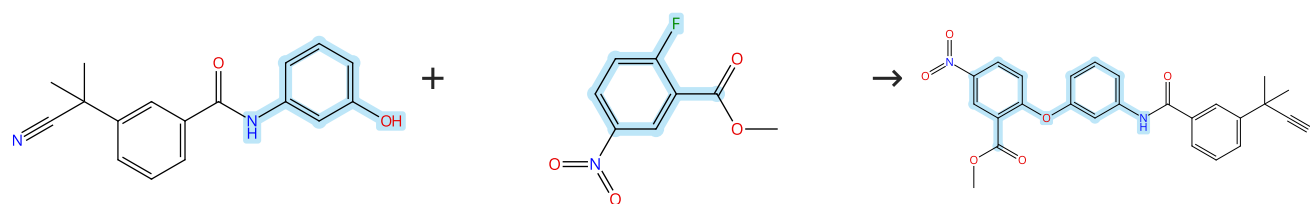
Suppliers (65)

Suppliers (16)

31-170-CAS-17492719	Steps: 1 Yield: 98%	The effect of O-nucleophiles structure on the features of halogen substitution in heterogeneous systems
1.1 Reagents: Potassium carbonate, Iron oxide (Fe ₂ O ₃) Solvents: Dimethylformamide; 270 min, 130 °C		By: Orlov, V. Yu.; et al
Experimental Protocols		Russian Journal of General Chemistry (2017), 87(3), 386-388.

Scheme 416 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (3)

Suppliers (76)

Supplier (1)

31-170-CAS-7045762

Steps: 1 Yield: 98%

Discovery of a Selective Kinase Inhibitor (TAK-632) Targeting Pan-RAF Inhibition: Design, Synthesis, and Biological Evaluation of C-7-Substituted 1,3-Benzothiazole Derivatives

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; rt; 18 h, 80 °C

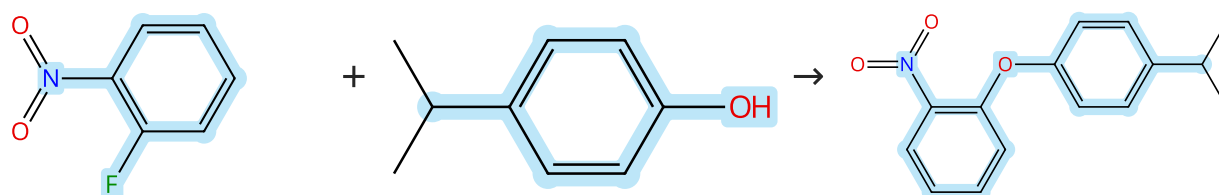
Experimental Protocols

By: Okaniwa, Masanori; et al

Journal of Medicinal Chemistry (2013), 56(16), 6478-6494.

Scheme 417 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (93)

Suppliers (71)

Supplier (1)

31-170-CAS-15921686

Steps: 1 Yield: 98%

Distance-Dependent Attractive and Repulsive Interactions of Bulky Alkyl Groups

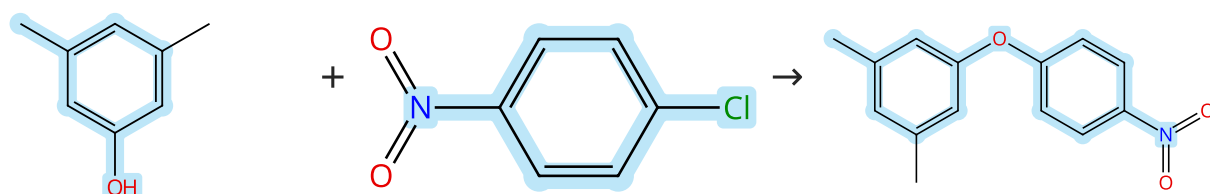
1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; rt; 3 h, 90 °C

By: Hwang, Jungwon; et al

Angewandte Chemie, International Edition (2016), 55(28), 8086-8089.

Scheme 418 (2 Reactions)

Steps: 1 Yield: 98%



Suppliers (89)

Suppliers (65)

Suppliers (19)

31-170-CAS-20090735

Steps: 1 Yield: 98%

Design of BNPs-TAPC Palladium Complex as a Reusable Heterogeneous Nanocatalyst for the O-Arylation of Phenols and N-Arylation of Amines

1.1 Reagents: Potassium hydroxide
Catalysts: Palladium, 2256068-46-7 (palladium complexes)
Solvents: Dimethylformamide; 1 h, 100 °C

By: Bahrami, Kiumars; et al

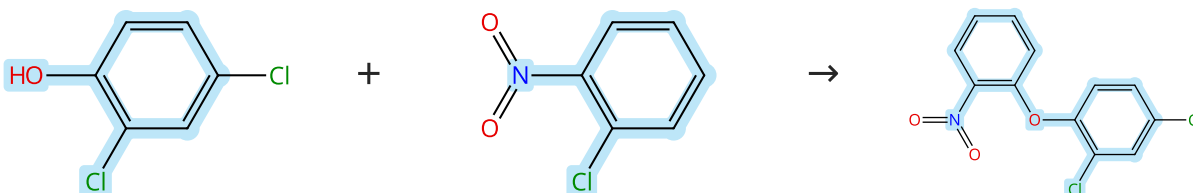
Catalysis Letters (2019), 149(3), 688-698.

Experimental Protocols

31-170-CAS-2340307	Steps: 1 Yield: 98%	On the frontier between nucleophilic aromatic substitution and catalysis
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; 40 h, 22 °C		By: Pichette Drapeau, Martin; et al
Experimental Protocols		Chemistry - A European Journal (2014), 20(18), 5231-5236.

Scheme 419 (3 Reactions)

Steps: 1 Yield: 98%



Suppliers (102)

Suppliers (76)

Suppliers (29)

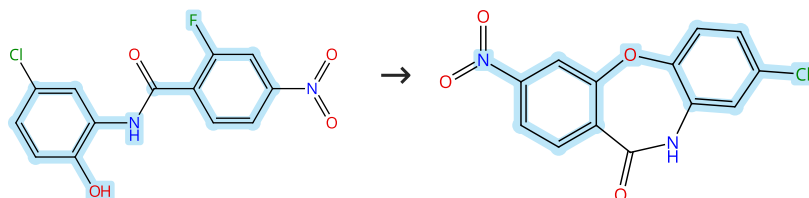
31-170-CAS-8414825	Steps: 1 Yield: 98%	Design, synthesis and antifungal activity of novel furancarboxamide derivatives
1.1 Reagents: Potassium hydroxide; 15 min, 90 °C		By: Wen, Fang; et al
1.2 90 °C; 2 h, 150 °C		
Experimental Protocols		European Journal of Medicinal Chemistry (2016), 120, 244-251.

31-170-CAS-10837643	Steps: 1 Yield: 98%	Synthesis and bioactivity evaluation of novel benzamide derivatives containing a diphenyl ether moiety
1.1 Reagents: Potassium hydroxide; 90 °C; 15 min, 90 °C		By: Chen, Mingjing; et al
1.2 90 °C; 2 h, 150 °C		
Experimental Protocols		Journal of Pesticide Science (Tokyo, Japan) (2014), 39(4), 187-192.

31-170-CAS-9609127	Steps: 1 Yield: 98%	Design, synthesis and antifungal/insecticidal evaluation of novel nicotinamide derivatives
1.1 Reagents: Potassium hydroxide; 15 min, 90 °C		By: Wen, Fang; et al
1.2 90 °C; 2 h, 150 °C		
Experimental Protocols		Pesticide Biochemistry and Physiology (2010), 98(2), 248-253.

Scheme 420 (1 Reaction)

Steps: 1 Yield: 98%

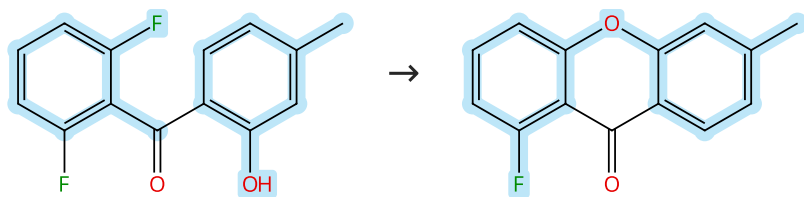


Suppliers (4)

31-170-CAS-19724791	Steps: 1 Yield: 98%	Dibenzo[b,f][1,4]oxazepines and dibenzo[b,e]oxepines: Influence of the chlorine substitution pattern on the pharmacology at the H₁R, H₄R, 5-HT_{2A}R and other selected GPCRs
1.1 Reagents: Sodium hydroxide Solvents: Dimethylformamide; 5 h, reflux		By: Naporra, Franziska; et al
Experimental Protocols		Pharmacological Research (2016), 113(Part_A), 610-625.

Scheme 421 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (3)

Suppliers (2)

31-614-CAS-26175026

Steps: 1 Yield: 98%

The synthesis and nucleophilic substitution of haloxanthenes

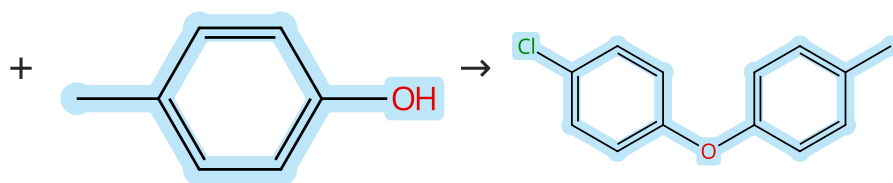
1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide

By: Sharghi, Hashem; et al
 Journal of Heterocyclic Chemistry (2001), 38(3), 617-622.

Scheme 422 (1 Reaction)

Steps: 1 Yield: 98%

Multi-component
 structure image
 available in CAS
 SciFinder



Suppliers (89)

Suppliers (10)

31-170-CAS-13740640

Steps: 1 Yield: 98%

Facile preparation of unsymmetrical diaryl ethers from unsymmetrical diaryliodonium tosylates and phenols with high regioselectivity

1.1 **Reagents:** Potassium carbonate
Solvents: Acetonitrile; 6 h, 50 °C

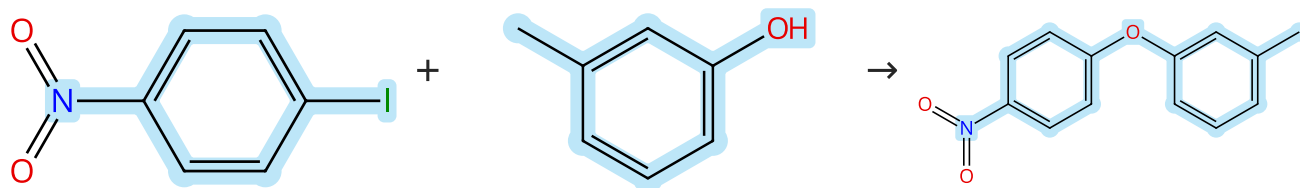
By: Kakinuma, Yohji; et al
 Synthesis (2013), 45(2), 183-188.

1.2 **Reagents:** Water

Experimental Protocols

Scheme 423 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (70)

Suppliers (91)

Suppliers (39)

31-170-CAS-20274082

Steps: 1 Yield: 98%

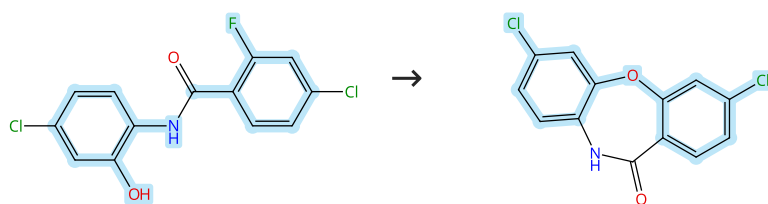
A Novel Ligand-free Manganese-catalyzed C-O Coupling Protocol for the Synthesis of Biaryl Ethers

1.1 **Reagents:** Cesium carbonate
Catalysts: Manganese dichloride
Solvents: Acetonitrile; 24 h, 110 °C

By: Rohit, K. R.; et al
 ChemistrySelect (2019), 4(17), 5150-5154.

Scheme 424 (1 Reaction)

Steps: 1 Yield: 98%



31-170-CAS-19724792

Steps: 1 Yield: 98%

1.1 Reagents: Sodium hydroxide
Solvents: Dimethylformamide; 5 h, reflux

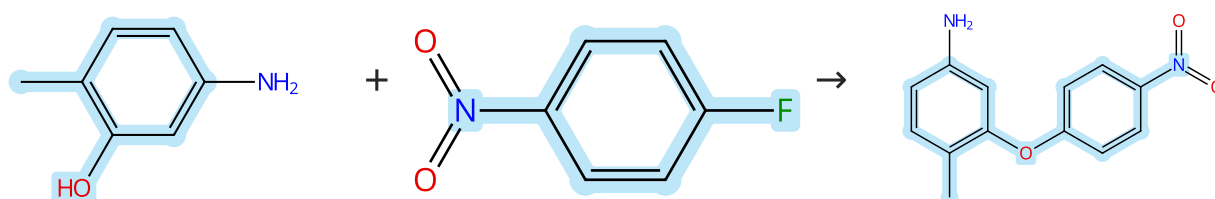
Dibenzo[b,f][1,4]oxazepines and dibenzo[b,e]oxepines:
Influence of the chlorine substitution pattern on the pharmacology at the H₁R, H₄R, 5-HT_{2A}R and other selected GPCRs

By: Naporra, Franziska; et al

Pharmacological Research (2016), 113(Part_A), 610-625.

Scheme 425 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (95)

Suppliers (100)

Suppliers (2)

31-614-CAS-40338207

Steps: 1 Yield: 98%

1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C

Experimental Protocols

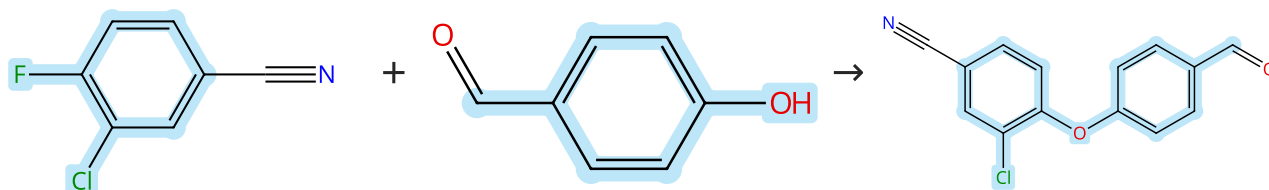
Profiling of the chemical space on the phenyl group of substituted benzothiazole RIPK3 inhibitors

By: Xu, Yue; et al

Bioorganic Chemistry (2023), 131, 106339.

Scheme 426 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (78)

Suppliers (123)

Suppliers (24)

31-614-CAS-35482771

Steps: 1 Yield: 98%

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 12 h, 100 °C

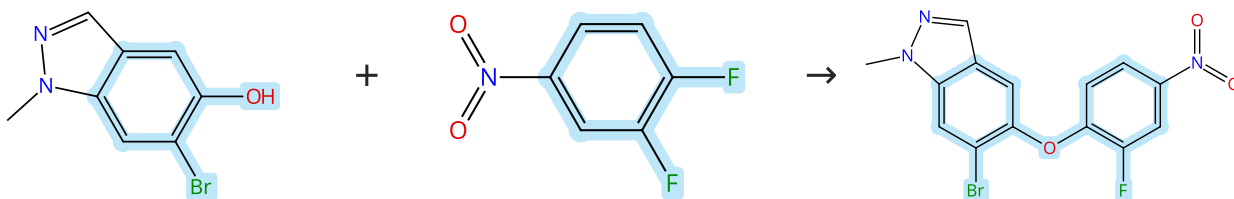
Strategies for the Production of [¹¹C]LY2795050 for Clinical Use

By: Kaur, Tanpreet; et al

Organic Process Research & Development (2023), 27(2), 373-381.

Scheme 427 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (51)

Suppliers (88)

Suppliers (45)

31-170-CAS-838583

Steps: 1 Yield: 98%

Route Design and Development of a MET Kinase Inhibitor: A Copper-Catalyzed Preparation of an N1-Methylindazole

1.1 Reagents: Sodium bicarbonate
Solvents: Dimethylformamide; 15 h, rt → 65 °C

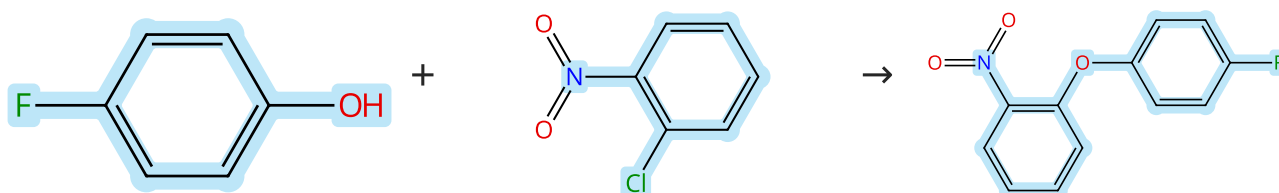
By: Kallman, Neil J.; et al

Experimental Protocols

Organic Process Research & Development (2014), 18(4), 501-510.

Scheme 428 (3 Reactions)

Steps: 1 Yield: 98%



Suppliers (86)

Suppliers (76)

Suppliers (17)

31-170-CAS-9073162

Steps: 1 Yield: 98%

Design, synthesis and antifungal activity of novel furancar boxamide derivatives

1.1 Reagents: Potassium hydroxide; 15 min, 90 °C
1.2 90 °C; 2 h, 150 °C

By: Wen, Fang; et al

Experimental Protocols

European Journal of Medicinal Chemistry (2016), 120, 244-251.

31-170-CAS-3256397

Steps: 1 Yield: 98%

Design, synthesis and antifungal/insecticidal evaluation of novel nicotinamide derivatives

1.1 Reagents: Potassium hydroxide; 15 min, 90 °C
1.2 90 °C; 2 h, 150 °C

By: Wen, Fang; et al

Pesticide Biochemistry and Physiology (2010), 98(2), 248-253.

31-170-CAS-14619383

Steps: 1 Yield: 98%

Synthesis of novel fluorophenyl aryl(or heteroaryl) ethers

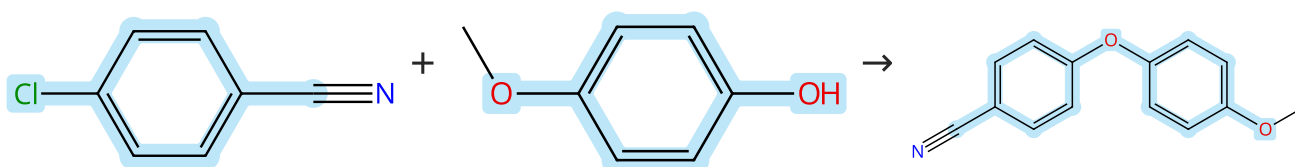
1.1 Reagents: Potassium hydroxide; 30 min, 110 °C
1.2 2 h, 120 °C

By: Das, Biplab K.; et al

ARKIVOC (Gainesville, FL, United States) (2007), (16), 73-82.

Scheme 429 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (90)

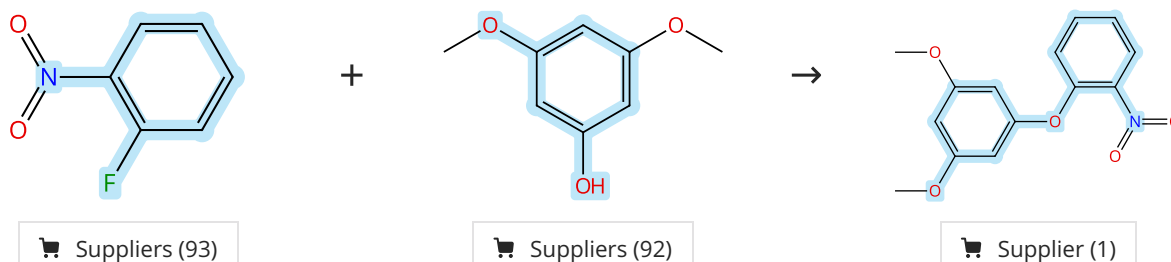
Suppliers (121)

Suppliers (38)

31-170-CAS-11888096 Steps: 1 Yield: 98% 1.1 Reagents: Tripotassium phosphate Catalysts: Bis(dibenzylideneacetone)palladium, Bis(1,1-dimethylethyl)[2'-(1-methylethoxy)[1,1'-binaphthalen]-2-yl] phosphine Solvents: Toluene; 15 min, rt 1.2 Solvents: Toluene; 18 h, 110 °C; 110 °C → rt 1.3 Solvents: Water; rt Experimental Protocols	The coupling reactions of aryl halides and phenols catalyzed by palladium and MOP-type ligands By: Zhang, Yi; et al Tetrahedron (2015), 71(30), 4927-4932.
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Scheme 430 (1 Reaction)

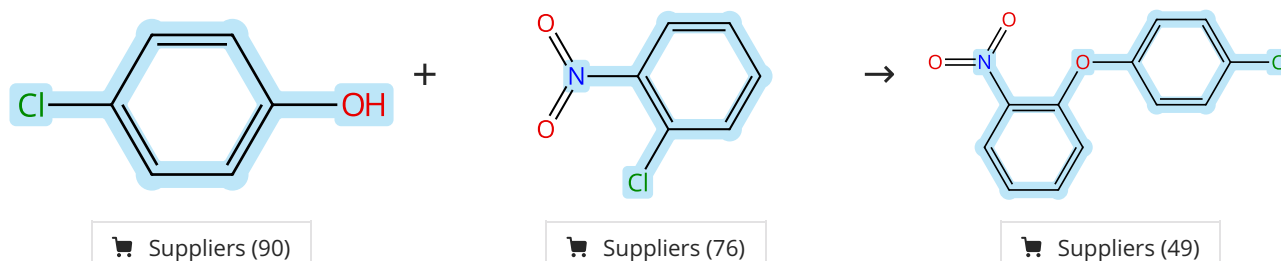
Steps: 1 Yield: 98%



31-170-CAS-22969462 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt → 95 °C; 26 h, 95 °C Experimental Protocols	Synthesis of Cycloheptatrienes, Oxepines, Thiepinines, and Silepinines: A Comparison between Brønsted Acid and Au-Catalysis By: Sprenger, Kristin; et al European Journal of Organic Chemistry (2020), 2020(39), 6245-6254.
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Scheme 431 (3 Reactions)

Steps: 1 Yield: 98%



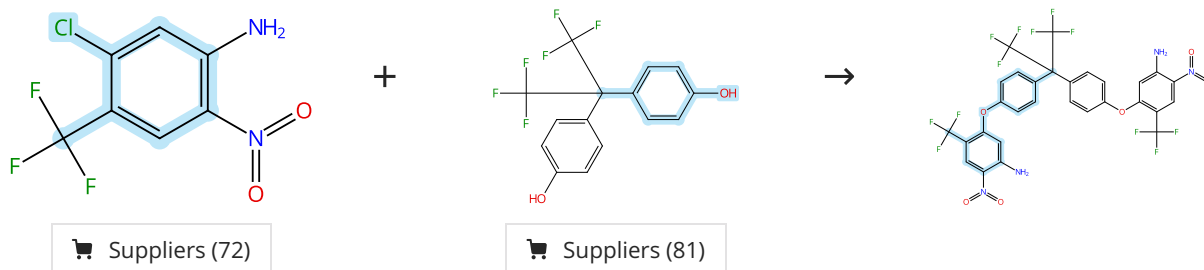
31-170-CAS-10588777 Steps: 1 Yield: 98% 1.1 Reagents: Potassium hydroxide; 15 min, 90 °C 1.2 90 °C; 2 h, 150 °C Experimental Protocols	Design, synthesis and antifungal activity of novel furancarboxamide derivatives By: Wen, Fang; et al European Journal of Medicinal Chemistry (2016), 120, 244-251.
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31-170-CAS-10889735 Steps: 1 Yield: 98% 1.1 Reagents: Potassium hydroxide; 90 °C; 15 min, 90 °C 1.2 90 °C; 2 h, 150 °C	Synthesis and bioactivity evaluation of novel benzamide derivatives containing a diphenyl ether moiety By: Chen, Mingjing; et al Journal of Pesticide Science (Tokyo, Japan) (2014), 39(4), 187-192.
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31-170-CAS-5338847 Steps: 1 Yield: 98% 1.1 Reagents: Potassium hydroxide; 15 min, 90 °C 1.2 90 °C; 2 h, 150 °C	Design, synthesis and antifungal/insecticidal evaluation of novel nicotinamide derivatives By: Wen, Fang; et al Pesticide Biochemistry and Physiology (2010), 98(2), 248-253.
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Scheme 432 (1 Reaction)

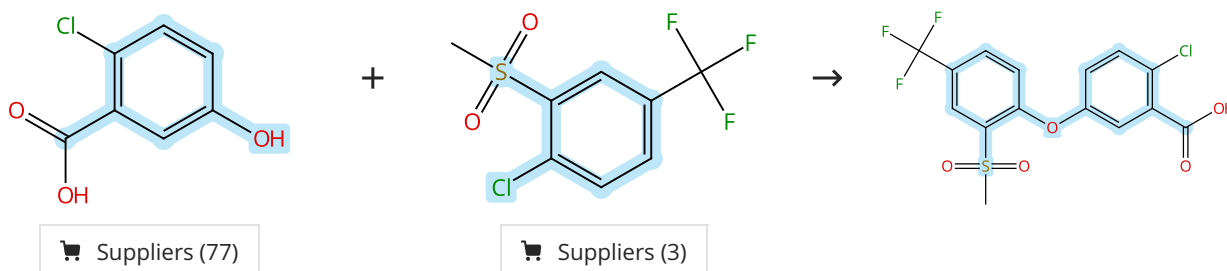
Steps: 1 Yield: 98%



31-170-CAS-19395973 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 5 min, 80 °C Experimental Protocols	An effective synthesis of polynuclear tetraamines, monomers for polyheteroarenes By: Begunov, R. S.; et al Russian Chemical Bulletin (2018), 67(6), 1072-1077.
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Scheme 433 (1 Reaction)

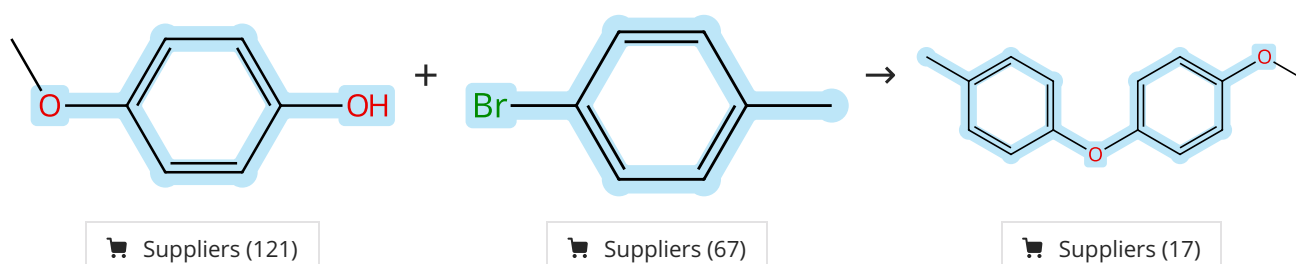
Steps: 1 Yield: 98%



31-614-CAS-24705959 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; rt; rt → 80 °C; 1 h, 70 - 80 °C 1.2 80 °C → 140 °C; 5 - 6 h, 130 - 140 °C; 140 °C → rt 1.3 Reagents: Hydrochloric acid Solvents: Water; pH 4 - 5, rt	Synthesis and herbicidal activities of diphenyl ethers with sulfonyl groups By: Cao, Li-dan; et al Jingxi Huagong Zhongjianti (2015), 45(1), 12-15.
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Scheme 434 (1 Reaction)

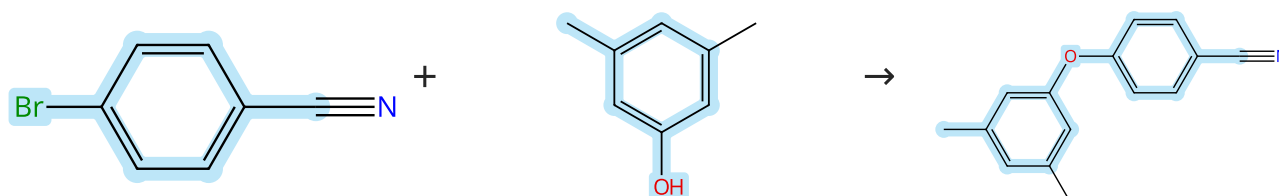
Steps: 1 Yield: 98%



31-614-CAS-24540884	Steps: 1 Yield: 98%	Chitosan@Cu ₂ O as a facile, efficient and reusable catalyst for ligand-free C-O and C-N coupling
1.1 Reagents: Tripotassium phosphate Catalysts: Copper oxide (Cu ₂ O) Solvents: Dimethylformamide; 18 h, 110 °C		By: Qian, Cunwei; et al Youji Huaxue (2019), 39(6), 1695-1703.
Experimental Protocols		

Scheme 435 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (93)

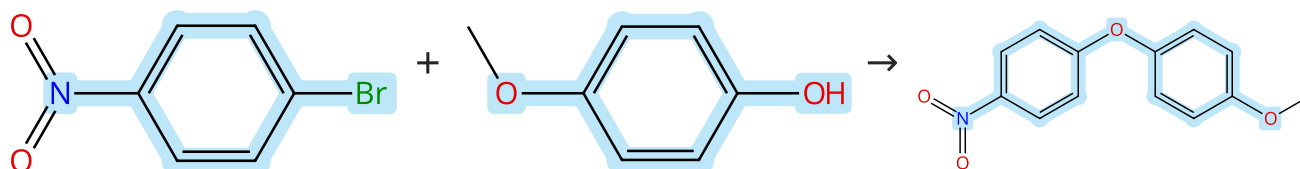
Suppliers (89)

Suppliers (22)

31-170-CAS-13277383	Steps: 1 Yield: 98%	Efficient and versatile sol-gel immobilized copper catalyst for Ullmann arylation of phenols
1.1 Reagents: Cesium carbonate Catalysts: Potassium iodide, Cuprous iodide, <i>N,N'</i> -[2,2'-Bipyridine]-4,4'-diylbis[<i>N</i> -[3-(trihydroxysilyl)propyl]urea] (Sol-Gel immobilized) Solvents: Dimethylformamide; rt; rt → 120 °C; 22 h, 120 °C; 120 °C → rt		By: Benyahya, Sofia; et al Advanced Synthesis & Catalysis (2008), 350(14+15), 2205-2208.
Experimental Protocols		

Scheme 436 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (77)

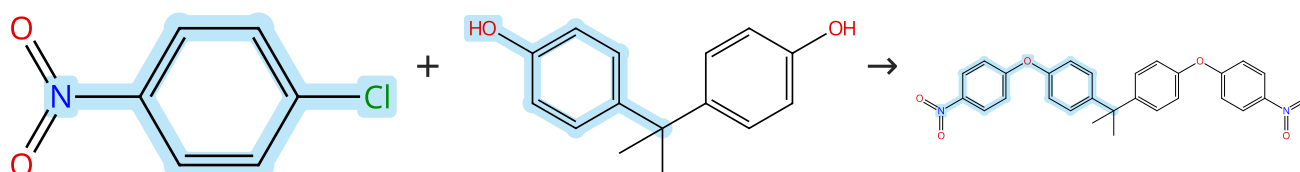
Suppliers (121)

Suppliers (34)

31-170-CAS-7961678	Steps: 1 Yield: 98%	Ligand free copper-catalyzed heterogeneous O-arylation reaction under green condition
1.1 Reagents: Cesium carbonate Catalysts: 2-Pyridinecarboxaldehyde, Cupric chloride, Tris[μ-[2-amino-1,4-benzenedicarboxylato(2-)-κO ¹ :κO ¹]]-μ ₄ -oxotetrazine (reaction products with 2-pyridinecarboxaldehyde, Cu complexes) Solvents: Ethanol; 20 h, 80 °C		By: Maity, Tanmoy; et al Catalysis Communications (2015), 58, 141-148.

Scheme 437 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (65)

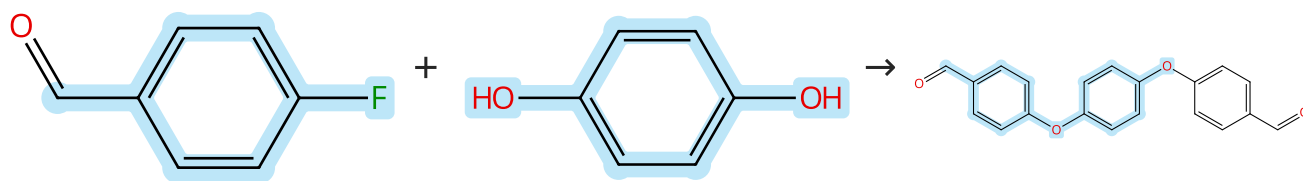
Suppliers (111)

Suppliers (13)

31-170-CAS-23828054	Steps: 1 Yield: 98%	Synthesis and characterization of bisphenol A-type polyimide monomer BAPP
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide, Toluene; 6 h, 120 - 150 °C		By: Ke, Changmei; et al Huagong Xinxing Cailiao (2012), 40(12), 94-96, 103.

Scheme 438 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (93)

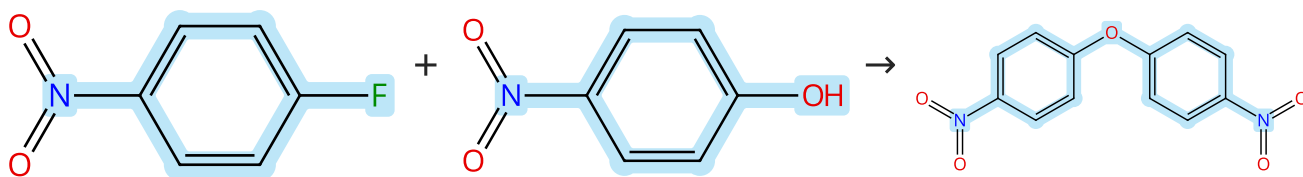
Suppliers (112)

Suppliers (23)

31-614-CAS-32693620	Steps: 1 Yield: 98%	Synthesis of poly(pyrazolopyridine)s by Hantzsch multicomponent polymerization
1.1 Reagents: Potassium carbonate Solvents: Dimethylacetamide; 22 h, rt → 150 °C		By: Saxer, Samantha; et al
Experimental Protocols		Polymer (2022), 252, 124935.

Scheme 439 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (100)

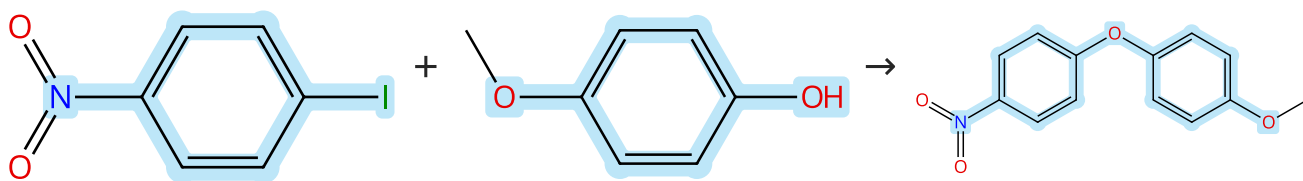
Suppliers (133)

Suppliers (52)

31-170-CAS-292666	Steps: 1 Yield: 98%	CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions
1.1 Catalysts: Cesium fluoride Solvents: Dimethyl sulfoxide; 0.75 h, 110 °C		By: Keipour, Hoda; et al
Experimental Protocols		Canadian Journal of Chemistry (2016), 94(1), 95-104.

Scheme 440 (2 Reactions)

Steps: 1 Yield: 98%



Suppliers (70)

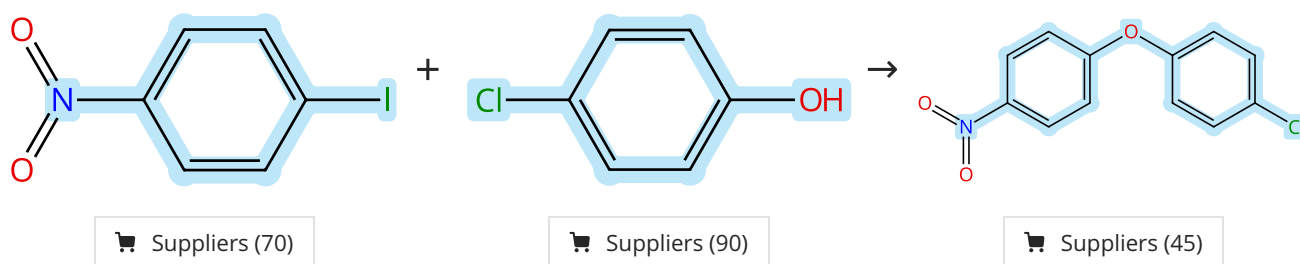
Suppliers (121)

Suppliers (34)

31-170-CAS-20960513 Steps: 1 Yield: 98% 1.1 Reagents: Potassium carbonate Catalysts: 2-(2-Aminophenyl)benzimidazole (reaction products with cellulose tosylate, complexes with palladium), Palladium Solvents: Dimethyl sulfoxide; 15 min, 80 °C Experimental Protocols	Cellulose nanocrystal supported palladium as a novel recyclable catalyst for Ullmann coupling reactions By: Seyednejhad, Sajedah; et al Cellulose (Dordrecht, Netherlands) (2019), 26(8), 5015-5031.
31-170-CAS-11804656 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Catalysts: Copper Solvents: Dimethylformamide; 24 h, 130 °C Experimental Protocols	Maghemite-Copper Nanocomposites: Applications for Ligand-Free Cross-Coupling (C-O, C-S, and C-N) Reactions By: Sharma, Rakesh K.; et al ChemCatChem (2015), 7(21), 3495-3502.

Scheme 441 (2 Reactions)

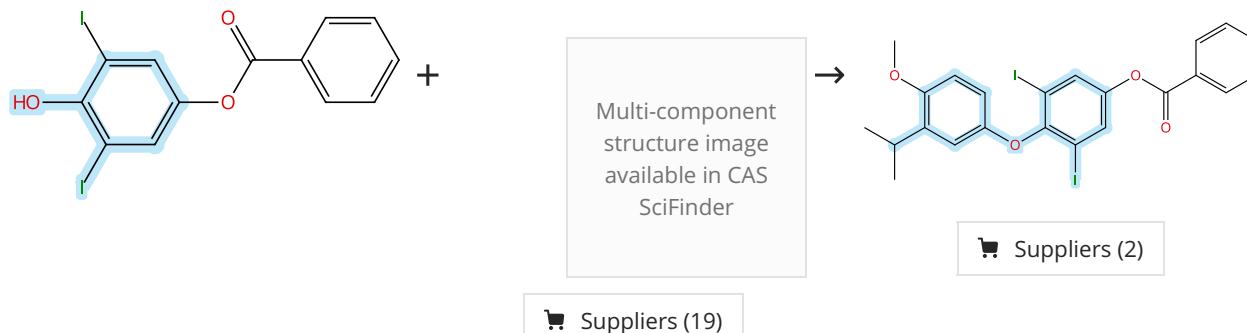
Steps: 1 Yield: 98%



31-170-CAS-20274079 Steps: 1 Yield: 98% 1.1 Reagents: Cesium carbonate Catalysts: Manganese dichloride Solvents: Acetonitrile; 24 h, 110 °C	A Novel Ligand-free Manganese-catalyzed C-O Coupling Protocol for the Synthesis of Biaryl Ethers By: Rohit, K. R.; et al ChemistrySelect (2019), 4(17), 5150-5154.
31-170-CAS-17776504 Steps: 1 Yield: 98% 1.1 Reagents: Potassium fluoride Catalysts: Alumina, Copper Solvents: Dimethyl sulfoxide; 16 h, 135 °C Experimental Protocols	A highly active and recyclable catalyst for the synthesis of indole and phenyl ether By: Qiao, Bo; et al RSC Advances (2015), 5(113), 93463-93469.

Scheme 442 (1 Reaction)

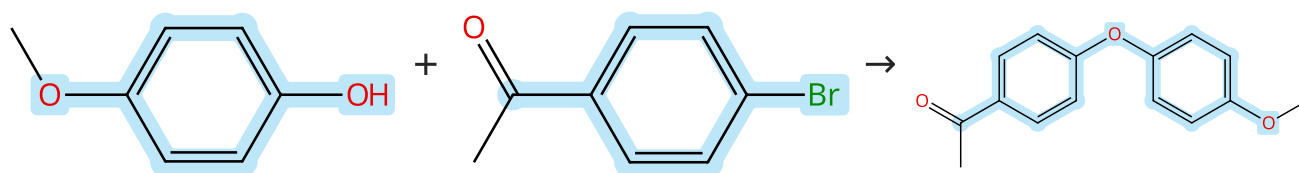
Steps: 1 Yield: 98%



31-170-CAS-9151878 Steps: 1 Yield: 98% 1.1 Reagents: Triethylamine, Copper Solvents: Dichloromethane; 0 °C; 24 h, rt Experimental Protocols	Synthesis and Biological Evaluation of a Series of Liver-Selective Phosphonic Acid Thyroid Hormone Receptor Agonists and Their Prodrugs By: Boyer, Serge H.; et al Journal of Medicinal Chemistry (2008), 51(22), 7075-7093.
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Scheme 443 (2 Reactions)

Steps: 1 Yield: 98%



Suppliers (121)

Suppliers (85)

Suppliers (62)

31-614-CAS-43873538

Steps: 1 Yield: 98%

Electromagnetic milling promoted mechanochemical copper-catalyzed C-O coupling reaction for diaryl ether formation

By: Zeng, Zhaoxuan; et al

Organic Chemistry Frontiers (2025), 12(5), 1528-1536.

1.1 **Reagents:** Cesium carbonate
Catalysts: Copper(II) acetylacetonate, *N*¹-[2-(1-Methylethyl)phenyl]-*N*²-(phenylmethyl)ethanediamide
Solvents: Dimethylformamide; 3 h, 79 °C

Experimental Protocols

31-614-CAS-43873491

Steps: 1 Yield: 98%

Electromagnetic milling promoted mechanochemical copper-catalyzed C-O coupling reaction for diaryl ether formation

By: Zeng, Zhaoxuan; et al

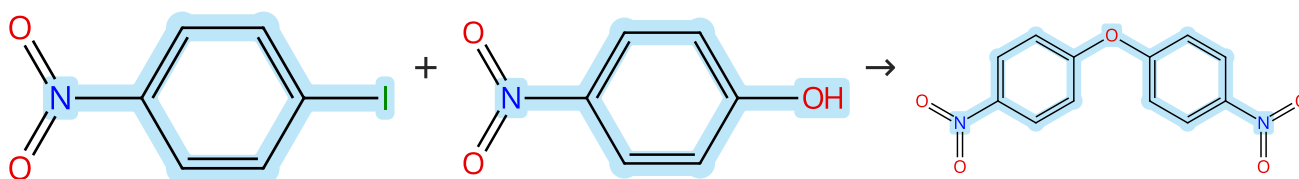
Organic Chemistry Frontiers (2025), 12(5), 1528-1536.

1.1 **Reagents:** Cesium carbonate
Catalysts: Copper(II) acetylacetonate, *N*¹,*N*²-Di-1-naphthyleneethanediamide
Solvents: Dimethylformamide; 3 h, 79 °C

Experimental Protocols

Scheme 444 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (70)

Suppliers (133)

Suppliers (52)

31-170-CAS-9983336

Steps: 1 Yield: 98%

A PEG₁₀₀₀-DAIL[CdCl₂]-toluene temperature-dependent biphasic system that regulates homogeneously catalyzed C-O coupling of organic halides with phenols and alcohols under ligand-free conditions

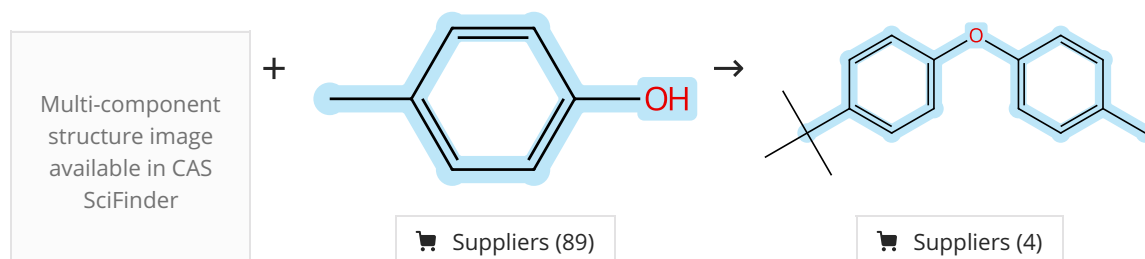
By: Hu, Yu Lin; et al

Canadian Journal of Chemistry (2011), 89(4), 471-480.

1.1 **Reagents:** Potassium carbonate
Catalysts: Poly(oxy-1,2-ethanediyl), α-[2-[1-(3-sulfopropyl)-1*H*-imidazolium-3-yl]ethyl]-ω-[2-[1-(3-sulfopropyl)-1*H*-imidazolium-3-yl]ethoxy]-, sulfate (1:2) (cadmium complexes)
Solvents: Toluene; 8 h, 95 °C

Scheme 445 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (12)

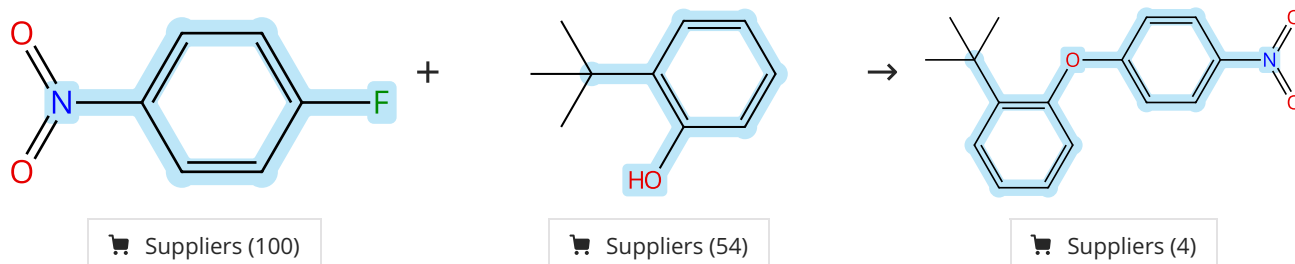
Suppliers (89)

Suppliers (4)

31-170-CAS-5194454	Steps: 1 Yield: 98%	Facile preparation of unsymmetrical diaryl ethers from unsymmetrical diaryliodonium tosylates and phenols with high regioselectivity
1.1 Reagents: Potassium carbonate Solvents: Acetonitrile; 2.5 h, 50 °C		By: Kakinuma, Yohji; et al
1.2 Reagents: Water		Synthesis (2013), 45(2), 183-188.
Experimental Protocols		

Scheme 446 (1 Reaction)

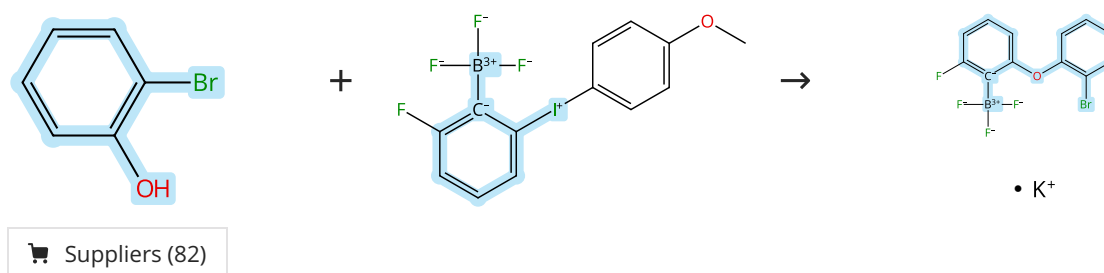
Steps: 1 Yield: 98%



31-170-CAS-7868488	Steps: 1 Yield: 98%	CsF/clinoptilolite: an efficient solid base in S_NAr and copper-catalyzed Ullmann reactions
1.1 Catalysts: Cesium fluoride Solvents: Dimethyl sulfoxide; 0.75 h, 110 °C		By: Keipour, Hoda; et al
Experimental Protocols		Canadian Journal of Chemistry (2016), 94(1), 95-104.

Scheme 447 (1 Reaction)

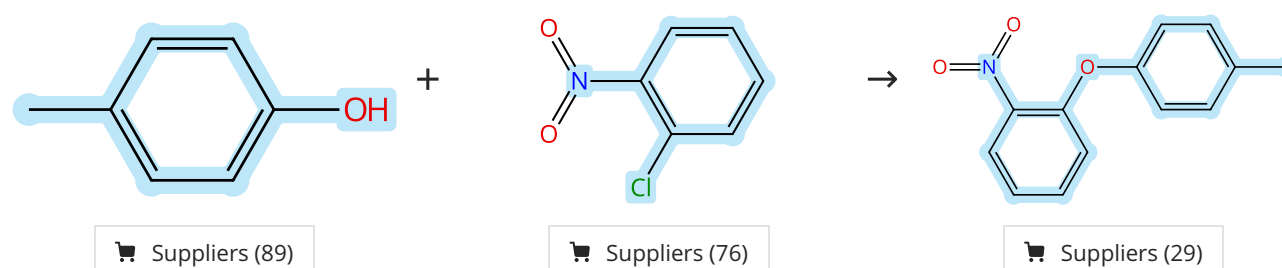
Steps: 1 Yield: 98%



31-008-CAS-18628946	Steps: 1 Yield: 98%	Investigation of Iodonium Trifluoroborate Zwitterions as Bifunctional Arene Reagents
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Tetrahydrofuran; 5 min, 50 °C		By: Robidas, Raphael; et al
1.2 Solvents: Tetrahydrofuran; 2 h, 50 °C		Organic Letters (2017), 19(23), 6420-6423.

Scheme 448 (1 Reaction)

Steps: 1 Yield: 98%



31-170-CAS-5636321	Steps: 1 Yield: 98%	Synthesis and bioactivity evaluation of novel benzamide derivatives containing a diphenyl ether moiety By: Chen, Mingjing; et al Journal of Pesticide Science (Tokyo, Japan) (2014), 39(4), 187-192.
1.1 Reagents: Potassium hydroxide; 90 °C; 15 min, 90 °C 1.2 90 °C; 2 h, 150 °C		

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